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Bertrand Russell and Alfred North Whitehead

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PRINCIPIA MATHEMATICA

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ALPHABETICAL LIST OF PROPOSITIONS REFERRED TO BY NAMES.

Name Number

Abs *2·01.

Add *1·3.

Ass *3·35.

Assoc *1·5.

Comm *2·04.

Comp *3·43.

Exp *3·3.

Fact *3·45.

Id *2·08.

Imp *3·31.

Perm *1·4.

Simp *2·02.

" *3·26.

" *3·27.

Sum *1·6.

Syll *2·05.

" *2·06.

" *3·33.

" *3·34.

Taut *1·2.

Transp *2·03.

" *2·15.

" *2·16.

" *2·17.

" *3·37.

" *4·1.

" *4·11.

ERRATA.

p. 14, line 2, for "states" read "allows us to infer."

p. 14, line 7, after "*3·03" insert "*1·7, *1·71, and *1·72."

p. 15, last line but one, for "function of " read "function ."

p. 34, line 15, for "" read "."

p. 68, line 20, for "classes" read "classes of classes."

p. 86, line 2, after "must" insert "neither be nor."

p. 91, line 8, delete "and in *3·03."

p. 103, line 7, for "assumption" read "assertion."

p. 103, line 25, at end of line, for "" read "."

p. 218, last line but one, for "" read "" [owing to brittleness of the

type, the same error is liable to occur elsewhere].

p. 382, last line but one, delete "in the theory of selections (*83·92) and."

p. 487, line 13, for "*95" read "*94."

p. 503, line 14, for "*88·38" read "*88·36."

[Pg v]

PREFACE

THE mathematical treatment of the principles of mathematics, which is the subject of the present work, has arisen from the conjunction of two different studies, both in the main very modern. On the one hand we have the work of analysts and geometers, in the way of formulating and systematising their axioms, and the work of Cantor and others on such matters as the theory of aggregates. On the other hand we have symbolic logic, which, after a necessary period of growth, has now, thanks to Peano and his followers, acquired the technical adaptability and the logical comprehensiveness that are essential to a mathematical instrument for dealing with what have hitherto been the beginnings of mathematics. From the combination of these two studies two results emerge, namely (1) that what were formerly taken, tacitly or explicitly, as axioms, are either unnecessary or demonstrable; (2) that the same methods by which supposed axioms are demonstrated will give valuable results in regions, such as infinite number, which had formerly been regarded as inaccessible to human knowledge. Hence the scope of mathematics is enlarged both by the addition of new subjects and by a backward extension into provinces hitherto abandoned to philosophy.

The present work was originally intended by us to be comprised in a second volume of The Principles of Mathematics. With that object in view, the writing of it was begun in 1900. But as we advanced, it became increasingly evident that the subject is a very much larger one than we had supposed; moreover on many fundamental questions which had been left obscure and doubtful in the former work, we have now arrived at what we believe to be satisfactory solutions. It therefore became necessary to make our book independent of The Principles of Mathematics. We have, however, avoided both controversy and general philosophy, and made our statements dogmatic in form. The justification for this is that the chief reason in favour of any theory on the principles of mathematics must always be inductive, i.e. it must lie in the fact that the theory in question enables us to deduce ordinary mathematics. In mathematics, the greatest degree of self-evidence is usually not to be found quite at the beginning, but at some later point; hence the early deductions, until they reach this point, give reasons rather[Pg vi] for believing the premisses because true consequences follow from them, than for believing the consequences

because they follow from the premisses.

In constructing a deductive system such as that contained in the present work, there are two opposite tasks which have to be concurrently performed. On the one hand, we have to analyse existing mathematics, with a view to discovering what premisses are employed, whether these premisses are mutually consistent, and whether they are capable of reduction to more fundamental premisses. On the other hand, when we have decided upon our premisses, we have to build up again as much as may seem necessary of the data previously analysed, and as many other consequences of our premisses as are of sufficient general interest to deserve statement. The preliminary labour of analysis does not appear in the final presentation, which merely sets forth the outcome of the analysis in certain undefined ideas and undemonstrated propositions. It is not claimed that the analysis could not have been carried farther: we have no reason to suppose that it is impossible to find simpler ideas and axioms by means of which those with which we start could be defined and demonstrated. All that is affirmed is that the ideas and axioms with which we start are sufficient, not that they are necessary.

In making deductions from our premisses, we have considered it essential to carry them up to the point where we have proved as much as is true in whatever would ordinarily be taken for granted. But we have not thought it desirable to limit ourselves too strictly to this task. It is customary to consider only particular cases, even when, with our apparatus, it is just as easy to deal with the general case. For example, cardinal arithmetic is usually conceived in connection with finite numbers, but its general laws hold equally for infinite numbers, and are most easily proved without any mention of the distinction between finite and infinite. Again, many of the properties commonly associated with series hold of arrangements which are not strictly serial, but have only some of the distinguishing properties of serial arrangements. In such cases, it is a defect in logical style to prove for a particular class of arrangements what might just as well have been proved more generally. An analogous process of generalization is involved, to a greater or less degree, in all our work. We have sought always the most general reasonably simple hypothesis from which any given conclusion could be reached. For this reason, especially in the later parts of the book, the importance of a proposition usually lies in its hypothesis. The conclusion will often be something which, in a certain class of cases, is familiar, but the hypothesis will, whenever possible, be wide enough to admit many cases besides those in which the conclusion is familiar.

We have found it necessary to give very full proofs, because otherwise it is scarcely possible to see what hypotheses are really required, or whether[Pg vii] our results follow from our explicit premisses. (It must be remembered that we are not affirming merely that such and such propositions are true, but also that the axioms stated by us are sufficient to prove them.) At the same time, though full proofs are necessary for the avoidance of errors, and for convincing those who may feel doubtful as to our correctness, yet the proofs of propositions may usually be omitted by a reader who is not specially interested in that part of the subject concerned, and who feels no doubt of our substantial accuracy on the matter in hand. The reader who is specially

interested in some particular portion of the book will probably find it sufficient, as regards earlier portions, to read the summaries of previous parts, sections, and numbers, since these give explanations of the ideas involved and statements of the principal propositions proved. The proofs in Part I, Section A, however, are necessary, since in the course of them the manner of stating proofs is explained. The proofs of the earliest propositions are given without the omission of any step, but as the work proceeds the proofs are gradually compressed, retaining however sufficient detail to enable the reader by the help of the references to reconstruct proofs in which no step is omitted.

The order adopted is to some extent optional. For example, we have treated cardinal arithmetic and relation-arithmetic before series, but we might have treated series first. To a great extent, however, the order is determined by logical necessities.

A very large part of the labour involved in writing the present work has been expended on the contradictions and paradoxes which have infected logic and the theory of aggregates. We have examined a great number of hypotheses for dealing with these contradictions; many such hypotheses have been advanced by others, and about as many have been invented by ourselves. Sometimes it has cost us several months' work to convince ourselves that a hypothesis was untenable. In the course of such a prolonged study, we have been led, as was to be expected, to modify our views from time to time; but it gradually became evident to us that some form of the doctrine of types must be adopted if the contradictions were to be avoided. The particular form of the doctrine of types advocated in the present work is not logically indispensable, and there are various other forms equally compatible with the truth of our deductions. We have particularized, both because the form of the doctrine which we advocate appears to us the most probable, and because it was necessary to give at least one perfectly definite theory which avoids the contradictions. But hardly anything in our book would be changed by the adoption of a different form of the doctrine of types. In fact, we may go farther, and say that, supposing some other way of avoiding the contradictions to exist, not very much of our book, except what explicitly deals with types, is dependent upon the adoption of the doctrine of types in any form, so soon as it has been shown (as we claim[Pg viii] that we have shown) that it is possible to construct a mathematical logic which does not lead to contradictions. It should be observed that the whole effect of the doctrine of types is negative: it forbids certain inferences which would otherwise be valid, but does not permit any which would otherwise be invalid. Hence we may reasonably expect that the inferences which the doctrine of types permits would remain valid even if the doctrine should be found to be invalid.

Our logical system is wholly contained in the numbered propositions, which are independent of the Introduction and the Summaries. The Introduction and the Summaries are wholly explanatory, and form no part of the chain of deductions. The explanation of the hierarchy of types in the Introduction differs slightly from that given in *12 of the body of the work. The later explanation is stricter and is that which is assumed throughout the rest of the book.

The symbolic form of the work has been forced upon us by necessity: without its help we should have been unable to perform the requisite reasoning. It has been developed as the result of actual practice, and is not an excrescence introduced for the mere purpose of exposition. The general method which guides our handling of logical symbols is due to Peano. His great merit consists not so much in his definite logical discoveries nor in the details of his notations (excellent as both are), as in the fact that he first showed how symbolic logic was to be freed from its undue obsession with the forms of ordinary algebra, and thereby made it a suitable instrument for research. Guided by our study of his methods, we have used great freedom in constructing, or reconstructing, a symbolism which shall be adequate to deal with all parts of the subject. No symbol has been introduced except on the ground of its practical utility for the immediate purposes of our reasoning.

A certain number of forward references will be found in the notes and explanations. Although we have taken every reasonable precaution to secure the accuracy of these forward references, we cannot of course guarantee their accuracy with the same confidence as is possible in the case of backward references.

Detailed acknowledgments of obligations to previous writers have not very often been possible, as we have had to transform whatever we have borrowed, in order to adapt it to our system and our notation. Our chief obligations will be obvious to every reader who is familiar with the literature of the subject. In the matter of notation, we have as far as possible followed Peano, supplementing his notation, when necessary, by that of Frege or by that of Schröder. A great deal of the symbolism, however, has had to be new, not so much through dissatisfaction with the symbolism of others, as through the fact that we deal with ideas not previously symbolised. In all[Pg ix] questions of logical analysis, our chief debt is to Frege. Where we differ from him, it is largely because the contradictions showed that he, in common with all other logicians ancient and modern, had allowed some error to creep into his premisses; but apart from the contradictions, it would have been almost impossible to detect this error. In Arithmetic and the theory of series, our whole work is based on that of Georg Cantor. In Geometry we have had continually before us the writings of v. Staudt, Pasch, Peano, Pieri, and Veblen.

We have derived assistance at various stages from the criticisms of friends, notably Mr G. G. Berry of the Bodleian Library and Mr R. G. Hawtrey.

We have to thank the Council of the Royal Society for a grant towards the expenses of printing of £200 from the Government Publication Fund, and also the Syndics of the University Press who have liberally undertaken the greater portion of the expense incurred in the production of the work. The technical excellence, in all departments, of the University Press, and the zeal and courtesy of its officials, have materially lightened the task of proof-correction.

The second volume is already in the press, and both it and the third will appear as soon as the printing can be completed.

A. N. W.

B. R.

CAMBRIDGE,

November, 1910.

[Pg 1]

INTRODUCTION.

THE mathematical logic which occupies Part I of the present work has been constructed under the guidance of three different purposes. In the first place, it aims at effecting the greatest possible analysis of the ideas with which it deals and of the processes by which it conducts demonstrations, and at diminishing to the utmost the number of the undefined ideas and undemonstrated propositions (called respectively primitive ideas and primitive propositions) from which it starts. In the second place, it is framed with a view to the perfectly precise expression, in its symbols, of mathematical propositions: to secure such expression, and to secure it in the simplest and most convenient notation possible, is the chief motive in the choice of topics. In the third place, the system is specially framed to solve the paradoxes which, in recent years, have troubled students of symbolic logic and the theory of aggregates; it is believed that the theory of types, as set forth in what follows, leads both to the avoidance of contradictions, and to the detection of the precise fallacy which has given rise to them.

Of the above three purposes, the first and third often compel us to adopt methods, definitions, and notations which are more complicated or more difficult than they would be if we had the second object alone in view. This applies especially to the theory of descriptive expressions (*14 and *30) and to the theory of classes and relations (*20 and *21). On these two points, and to a lesser degree on others, it has been found necessary to make some sacrifice of lucidity to correctness. The sacrifice is, however, in the main only temporary: in each case, the notation ultimately adopted, though its real meaning is very complicated, has an apparently simple meaning which, except at certain crucial points, can without danger be substituted in thought for the real meaning. It is therefore convenient, in a preliminary explanation of the notation, to treat these apparently simple meanings as primitive ideas, i.e. as ideas introduced without definition. When the notation has grown more or less familiar, it is easier to follow the more complicated explanations which we believe to be more correct. In the body of the work, where it is necessary to adhere rigidly to the strict logical order[Pg 2] the easier order of development could not be adopted; it is therefore given in the Introduction. The explanations given in Chapter I of the Introduction are such as place lucidity before correctness; the full explanations are partly supplied in succeeding Chapters of the Introduction, partly given in the body of the work.

The use of a symbolism, other than that of words, in all parts of the book which aim at embodying strictly accurate demonstrative reasoning, has been forced on us by the consistent pursuit of the above three purposes. The reasons for this extension of symbolism beyond the familiar regions of number and allied ideas are many:

(1) The ideas here employed are more abstract than those familiarly considered in language. Accordingly there are no words which are used mainly in the exact consistent senses which are required here. Any use of words would require unnatural limitations to their ordinary meanings, which would be in fact more difficult to remember consistently than are the definitions of entirely new symbols.

(2) The grammatical structure of language is adapted to a wide variety of usages. Thus it possesses no unique simplicity in representing the few simple, though highly abstract, processes and ideas arising in the deductive trains of reasoning employed here. In fact the very abstract simplicity of the ideas of this work defeats language. Language can represent complex ideas more easily. The proposition "a whale is big" represents language at its best, giving terse expression to a complicated fact; while the true analysis of "one is a number" leads, in language, to an intolerable prolixity. Accordingly terseness is gained by using a symbolism especially designed to represent the ideas and processes of deduction which occur in this work.

(3) The adaptation of the rules of the symbolism to the processes of deduction aids the intuition in regions too abstract for the imagination readily to present to the mind the true relation between the ideas employed. For various collocations of symbols become familiar as representing important collocations of ideas; and in turn the possible relations—according to the rules of the symbolism—between these collocations of symbols become familiar, and these further collocations represent still more complicated relations between the abstract ideas. And thus the mind is finally led to construct trains of reasoning in regions of thought in which the imagination would be entirely unable to sustain itself without symbolic help. Ordinary language yields no such help. Its grammatical structure does not represent uniquely the relations between the ideas involved. Thus, "a whale is big" and "one is a number" both look alike, so that the eye gives no help to the imagination.

[Pg 3]

(4) The terseness of the symbolism enables a whole proposition to be represented to the eyesight as one whole, or at most in two or three parts divided where the natural breaks, represented in the symbolism, occur. This is a humble property, but is in fact very important in connection with the advantages enumerated under the heading (3).

(5) The attainment of the first-mentioned object of this work, namely the complete enumeration of all the ideas and steps in reasoning employed in mathematics, necessitates both terseness and the presentation of each proposition with the maximum of formality in a form as characteristic of itself as possible.

Further light on the methods and symbolism of this book is thrown by a slight consideration of the limits to their useful employment:

(Most mathematical investigation is concerned not with the analysis of the complete process of reasoning, but with the presentation of such an abstract of the proof as is sufficient to convince a properly instructed mind. For such investigations the detailed presentation of the steps in reasoning is of course unnecessary, provided that the detail is carried far enough to guard against error. In this connection it may

be remembered that the investigations of Weierstrass and others of the same school have shown that, even in the common topics of mathematical thought, much more detail is necessary than previous generations of mathematicians had anticipated.

(In proportion as the imagination works easily in any region of thought, symbolism (except for the express purpose of analysis) becomes only necessary as a convenient shorthand writing to register results obtained without its help. It is a subsidiary object of this work to show that, with the aid of symbolism, deductive reasoning can be extended to regions of thought not usually supposed amenable to mathematical treatment. And until the ideas of such branches of knowledge have become more familiar, the detailed type of reasoning, which is also required for the analysis of the steps, is appropriate to the investigation of the general truths concerning these subjects.

[Pg 4]

CHAPTER I.

PRELIMINARY EXPLANATIONS OF IDEAS AND NOTATIONS.

THE notation adopted in the present work is based upon that of Peano, and the following explanations are to some extent modelled on those which he prefixes to his Formulario Mathematico. His use of dots as brackets is adopted, and so are many of his symbols.

Variables. The idea of a variable, as it occurs in the present work, is more general than that which is explicitly used in ordinary mathematics. In ordinary mathematics, a variable generally stands for an undetermined number or quantity. In mathematical logic, any symbol whose meaning is not determinate is called a variable, and the various determinations of which its meaning is susceptible are called the values of the variable. The values may be any set of entities, propositions, functions, classes or relations, according to circumstances. If a statement is made about "Mr A and Mr B," "Mr A" and "Mr B" are variables whose values are confined to men. A variable may either have a conventionally-assigned range of values, or may (in the absence of any indication of the range of values) have as the range of its values all determinations which render the statement in which it occurs significant. Thus when a text-book of logic asserts that " is ," without any indication as to what may be, what is meant is that any statement of the form " is " is true. We may call a variable restricted when its values are confined to some only of those of which it is capable; otherwise, we shall call it unrestricted. Thus when an unrestricted variable occurs, it represents any object such that the statement concerned can be made significantly (i.e. either truly or falsely) concerning that object. For the purposes of logic, the unrestricted variable is more convenient than the restricted variable, and we shall always employ it. We shall find that the unrestricted variable is still subject to limitations imposed by the manner of its occurrence, i.e. things which can be said significantly concerning a proposition cannot be said significantly concerning a class or a relation, and so on. But the limitations to which the unrestricted variable is subject do not need to be explicitly indicated, since they are the limits of significance of the statement in which

the variable occurs, and are therefore intrinsically determined by this statement. This will be more fully explained later[1].[Pg 5] To sum up, the three salient facts connected with the use of the variable are: (1) that a variable is ambiguous in its denotation and accordingly undefined: (2) that a variable preserves a recognizable identity in various occurrences throughout the same context, so that many variables can occur together in the same context each with its separate identity: and (3) that either the range of possible determinations of two variables may be the same, so that a possible determination of one variable is also a possible determination of the other, or the ranges of two variables may be different, so that, if a possible determination of one variable is given to the other, the resulting complete phrase is meaningless instead of becoming a complete unambiguous proposition (true or false) as would be the case if all variables in it had been given any suitable determinations.

The uses of various letters. Variables will be denoted by single letters, and so will certain constants; but a letter which has once been assigned to a constant by a definition must not afterwards be used to denote a variable. The small letters of the ordinary alphabet will all be used for variables, except and after *40, in which constant meanings are assigned to these two letters. The following capital letters will receive constant meanings: , , , , , and . Among small Greek letters, we shall give constant meanings to , and (at a later stage) to , and . Certain Greek capitals will from time to time be introduced for constants, but Greek capitals will not be used for variables. Of the remaining letters, , , will be called propositional letters, and will stand for variable propositions (except that, from *40 onwards, must not be used for a variable); , , , , , and (until *33) will be called functional letters, and will be used for variable functions.

The small Greek letters not already mentioned will be used for variables whose values are classes, and will be referred to simply as Greek letters. Ordinary capital letters not already mentioned will be used for variables whose values are relations, and will be referred to simply as capital letters. Ordinary small letters other than , , , , , will be used for variables whose values are not known to be functions, classes, or relations; these letters will be referred to simply as small Latin letters.

After the early part of the work, variable propositions and variable functions will hardly ever occur. We shall then have three main kinds of variables: variable classes, denoted by small Greek letters; variable relations, denoted by capitals; and variables not given as necessarily classes or relations, which will be denoted by small Latin letters.

In addition to this usage of small Greek letters for variable classes, capital letters for variable relations, small Latin letters for variables of type wholly undetermined by the context (these arise from the possibility of[Pg 6] "systematic ambiguity," explained later in the explanations of the theory of types), the reader need only remember that all letters represent variables, unless they have been defined as constants in some previous place in the book. In general the structure of the context determines the scope of the variables contained in it; but the special indication of the nature of the variables employed, as here proposed, saves considerable labour of thought.

The fundamental functions of propositions. An aggregation of propositions, considered as wholes not necessarily unambiguously determined, into a single proposition more complex than its constituents, is a function with propositions as arguments. The general idea of such an aggregation of propositions, or of variables representing propositions, will not be employed in this work. But there are four special cases which are of fundamental importance, since all the aggregations of subordinate propositions into one complex proposition which occur in the sequel are formed out of them step by step.

They are (1) the Contradictory Function, (2) the Logical Sum, or Disjunctive Function, (3) the Logical Product, or Conjunctive Function, (4) the Implicative Function. These functions in the sense in which they are required in this work are not all independent; and if two of them are taken as primitive undefined ideas, the other two can be defined in terms of them. It is to some extent—though not entirely—arbitrary as to which functions are taken as primitive. Simplicity of primitive ideas and symmetry of treatment seem to be gained by taking the first two functions as primitive ideas.

The Contradictory Function with argument , where is any proposition, is the proposition which is the contradictory of , that is, the proposition asserting that is not true. This is denoted by . Thus is the contradictory function with as argument and means the negation of the proposition . It will also be referred to as the proposition not-. Thus means not-, which means the negation of .

The Logical Sum is a propositional function with two arguments and , and is the proposition asserting or disjunctively, that is, asserting that at least one of the two and is true. This is denoted by . Thus is the logical sum with and as arguments. It is also called the logical sum of and . Accordingly means that at least or is true, not excluding the case in which both are true.

The Logical Product is a propositional function with two arguments and , and is the proposition asserting and conjunctively, that is, asserting that both and are true. This is denoted by , or—in order to make the dots act as brackets in a way to be explained immediately—by , or by , or by . Thus is the logical product with[Pg 7] and as arguments. It is also called the logical product of and . Accordingly means that both and are true. It is easily seen that this function can be defined in terms of the two preceding functions. For when and are both true it must be false that either is true. Hence in this book is merely a shortened form of symbolism for

If any further idea attaches to the proposition "both and are true," it is not required here.

The Implicative Function is a propositional function with two arguments and , and is the proposition that either not- or is true, that is, it is the proposition . Thus if is true, is false, and accordingly the only alternative left by the proposition is that is true. In other words if and are both true, then is true. In this sense the proposition will be quoted as stating that implies . The idea contained in this propositional function is so important that it requires a symbolism which with direct simplicity represents the proposition as connecting and without the intervention of . But

"implies" as used here expresses nothing else than the connection between and also expressed by the disjunction "not- or ." The symbol employed for " implies ," i.e. for "" is "." This symbol may also be read "if , then ." The association of implication with the use of an apparent variable produces an extension called "formal implication." This is explained later: it is an idea derivative from "implication" as here defined. When it is necessary explicitly to discriminate "implication" from "formal implication," it is called "material implication." Thus "material implication" is simply "implication" as here defined. The process of inference, which in common usage is often confused with implication, is explained immediately.

These four functions of propositions are the fundamental constant (i.e. definite) propositional functions with propositions as arguments, and all other constant propositional functions with propositions as arguments, so far as they are required in the present work, are formed out of them by successive steps. No variable propositional functions of this kind occur in this work.

Equivalence. The simplest example of the formation of a more complex function of propositions by the use of these four fundamental forms is furnished by "equivalence." Two propositions and are said to be "equivalent" when implies and implies . This relation between and is denoted by "" Thus "" stands for "." It is easily seen that two propositions are equivalent when, and only when, they are both true or are both false. Equivalence rises in the scale of importance when we come to "formal implication" and thus to "formal equivalence." It must not be supposed that two propositions which are equivalent are in[Pg 8] any sense identical or even remotely concerned with the same topic. Thus "Newton was a man" and "the sun is hot" are equivalent as being both true, and "Newton was not a man" and "the sun is cold" are equivalent as being both false. But here we have anticipated deductions which follow later from our formal reasoning. Equivalence in its origin is merely mutual implication as stated above.

Truth-values. The "truth-value" of a proposition is truth if it is true, and falsehood if it is false[2]. It will be observed that the truth-values of , , , , depend only upon those of and , namely the truth-value of "" is truth if the truth-value of either or is truth, and is falsehood otherwise; that of "" is truth if that of both and is truth, and is falsehood otherwise; that of "" is truth if either that of is falsehood or that of is truth; that of "" is the opposite of that of ; and that of "" is truth if and have the same truth-value, and is falsehood otherwise. Now the only ways in which propositions will occur in the present work are ways derived from the above by combinations and repetitions. Hence it is easy to see (though it cannot be formally proved except in each particular case) that if a proposition occurs in any proposition which we shall ever have occasion to deal with, the truth-value of will depend, not upon the particular proposition , but only upon its truth-value; i.e. if , we shall have . Thus whenever two propositions are known to be equivalent, either may be substituted for the other in any formula with which we shall have occasion to deal.

We may call a function a "truth-function" when its argument is a proposition, and the truth-value of depends only upon the truth-value of . Such functions are by no means the only common functions of propositions. For example, " believes " is a

function of which will vary its truth-value for different arguments having the same truth-value: may believe one true proposition without believing another, and may believe one false proposition without believing another. Such functions are not excluded from our consideration, and are included in the scope of any general propositions we may make about functions; but the particular functions of propositions which we shall have occasion to construct or to consider explicitly are all truth-functions. This fact is closely connected with a characteristic of mathematics, namely, that mathematics is always concerned with extensions rather than intensions. The connection, if not now obvious, will become more so when we have considered the theory of classes and relations.

Assertion-sign. The sign "," called the "assertion-sign," means that what follows is asserted. It is required for distinguishing a complete proposition, which we assert, from any subordinate propositions contained in it but[Pg 9] not asserted. In ordinary written language a sentence contained between full stops denotes an asserted proposition, and if it is false the book is in error. The sign "" prefixed to a proposition serves this same purpose in our symbolism. For example, if "" occurs, it is to be taken as a complete assertion convicting the authors of error unless the proposition "" is true (as it is). Also a proposition stated in symbols without this sign "" prefixed is not asserted, and is merely put forward for consideration, or as a subordinate part of an asserted proposition.

Inference. The process of inference is as follows: a proposition "" is asserted, and a proposition " implies " is asserted, and then as a sequel the proposition "" is asserted. The trust in inference is the belief that if the two former assertions are not in error, the final assertion is not in error. Accordingly whenever, in symbols, where and have of course special determinations,

have occurred, then "" will occur if it is desired to put it on record. The process of the inference cannot be reduced to symbols. Its sole record is the occurrence of "." It is of course convenient, even at the risk of repetition, to write "" and "" in close juxtaposition before proceeding to "" as the result of an inference. When this is to be done, for the sake of drawing attention to the inference which is being made, we shall write instead

which is to be considered as a mere abbreviation of the threefold statement

Thus "" may be read ", therefore ," being in fact the same abbreviation, essentially, as this is; for ", therefore " does not explicitly state, what is part of its meaning, that implies . An inference is the dropping of a true premiss; it is the dissolution of an implication.

The use of dots. Dots on the line of the symbols have two uses, one to bracket off propositions, the other to indicate the logical product of two propositions. Dots immediately preceded or followed by "" or "" or "" or "," or by "," "," "" ... or "," "," "" ... or "" or "" or analogous expressions, serve to bracket off a proposition; dots occurring otherwise serve to mark a logical product. The general principle is that a larger number of dots indicates an outside bracket, a smaller number indicates an inside bracket. The exact rule as to the scope of the bracket indicated by dots is arrived at

by dividing the occurrences of dots into three groups which we will name I, II, and III. Group I consists of dots adjoining a sign of implication () or of equivalence () or of disjunction () or of equality by definition (Df). Group II consists of dots following brackets indicative of an apparent variable, such as or or or[Pg 10] or or analogous expressions[3]. Group III consists of dots which stand between propositions in order to indicate a logical product. Group I is of greater force than Group II, and Group II than Group III. The scope of the bracket indicated by any collection of dots extends backwards or forwards beyond any smaller number of dots, or any equal number from a group of less force, until we reach either the end of the asserted proposition or a greater number of dots or an equal number belonging to a group of equal or superior force. Dots indicating a logical product have a scope which works both backwards and forwards; other dots only work away from the adjacent sign of disjunction, implication, or equivalence, or forward from the adjacent symbol of one of the other kinds enumerated in Group II.

Some examples will serve to illustrate the use of dots.

"" means the proposition "' or ' implies ' or .'" When we assert this proposition, instead of merely considering it, we write

where the two dots after the assertion-sign show that what is asserted is the whole of what follows the assertion-sign, since there are not as many as two dots anywhere else. If we had written "," that would mean the proposition "either is true, or implies ' or .'" If we wished to assert this, we should have to put three dots after the assertion-sign. If we had written "," that would mean the proposition "either ' or ' implies q, or p is true." The forms "" and "" have no meaning.

"" will mean "if implies , then if implies implies ." If we wish to assert this (which is true) we write

Again "" will mean "if ' implies ' implies ' implies ,' then implies ." This is in general untrue. (Observe that "" is sometimes most conveniently read as " implies ," and sometimes as "if , then .") "" will mean "if implies , and implies , then implies ." In this formula, the first dot indicates a logical product; hence the scope of the second dot extends backwards to the beginning of the proposition. "" will mean " implies ; and if implies , then implies ." (This is not true in general.) Here the two dots indicate a logical product; since two dots do not occur anywhere else, the scope of these two dots extends backwards to the beginning of the proposition, and forwards to the end.

"" will mean "if either or is true, then if either or ' implies ' is true, it follows that either or is true."[Pg 11] If this is to be asserted, we must put four dots after the assertion-sign, thus:

(This proposition is proved in the body of the work; it is *2·75.) If we wish to assert (what is equivalent to the above) the proposition: "if either or is true, and either or ' implies ' is true, then either or is true," we write

Here the first pair of dots indicates a logical product, while the second pair does not. Thus the scope of the second pair of dots passes over the first pair, and back until we reach the three dots after the assertion-sign.

Other uses of dots follow the same principles, and will be explained as they are introduced. In reading a proposition, the dots should be noticed first, as they show its structure. In a proposition containing several signs of implication or equivalence, the one with the greatest number of dots before or after it is the principal one: everything that goes before this one is stated by the proposition to imply or be equivalent to everything that comes after it.

Definitions. A definition is a declaration that a certain newly-introduced symbol or combination of symbols is to mean the same as a certain other combination of symbols of which the meaning is already known. Or, if the defining combination of symbols is one which only acquires meaning when combined in a suitable manner with other symbols[4], what is meant is that any combination of symbols in which the newly-defined symbol or combination of symbols occurs is to have that meaning (if any) which results from substituting the defining combination of symbols for the newly-defined symbol or combination of symbols wherever the latter occurs. We will give the names of definiendum and definiens respectively to what is defined and to that which it is defined as meaning. We express a definition by putting the definiendum to the left and the definiens to the right, with the sign "=" between, and the letters "Df" to the right of the definiens. It is to be understood that the sign "=" and the letters "Df" are to be regarded as together forming one symbol. The sign "=" without the letters "Df" will have a different meaning, to be explained shortly.

An example of a definition is

It is to be observed that a definition is, strictly speaking, no part of the subject in which it occurs. For a definition is concerned wholly with the symbols, not with what they symbolise. Moreover it is not true or false, being the expression of a volition, not of a proposition. (For this reason,[Pg 12] definitions are not preceded by the assertion-sign.) Theoretically, it is unnecessary ever to give a definition: we might always use the definiens instead, and thus wholly dispense with the definiendum. Thus although we employ definitions and do not define "definition," yet "definition" does not appear among our primitive ideas, because the definitions are no part of our subject, but are, strictly speaking, mere typographical conveniences. Practically, of course, if we introduced no definitions, our formulae would very soon become so lengthy as to be unmanageable; but theoretically, all definitions are superfluous.

In spite of the fact that definitions are theoretically superfluous, it is nevertheless true that they often convey more important information than is contained in the propositions in which they are used. This arises from two causes. First, a definition usually implies that the definiens is worthy of careful consideration. Hence the collection of definitions embodies our choice of subjects and our judgment as to what is most important. Secondly, when what is defined is (as often occurs) something already familiar, such as cardinal or ordinal numbers, the definition contains an analysis of a common idea, and may therefore express a notable advance. Cantor's definition of the continuum illustrates this: his definition amounts to the statement that what he is defining is the object which has the properties commonly associated with the word "continuum," though what precisely constitutes these properties had not before been known. In such cases, a definition is a "making definite": it gives

definiteness to an idea which had previously been more or less vague.

For these reasons, it will be found, in what follows, that the definitions are what is most important, and what most deserves the reader's prolonged attention.

Some important remarks must be made respecting the variables occurring in the definiens and the definiendum. But these will be deferred till the notion of an "apparent variable" has been introduced, when the subject can be considered as a whole.

Summary of preceding statements. There are, in the above, three primitive ideas which are not "defined" but only descriptively explained. Their primitiveness is only relative to our exposition of logical connection and is not absolute; though of course such an exposition gains in importance according to the simplicity of its primitive ideas. These ideas are symbolised by "" and "," and by "" prefixed to a proposition.

Three definitions have been introduced:

[Pg 13]

Primitive propositions. Some propositions must be assumed without proof, since all inference proceeds from propositions previously asserted. These, as far as they concern the functions of propositions mentioned above, will be found stated in *1, where the formal and continuous exposition of the subject commences. Such propositions will be called "primitive propositions." These, like the primitive ideas, are to some extent a matter of arbitrary choice; though, as in the previous case, a logical system grows in importance according as the primitive propositions are few and simple. It will be found that owing to the weakness of the imagination in dealing with simple abstract ideas no very great stress can be laid upon their obviousness. They are obvious to the instructed mind, but then so are many propositions which cannot be quite true, as being disproved by their contradictory consequences. The proof of a logical system is its adequacy and its coherence. That is: (1) the system must embrace among its deductions all those propositions which we believe to be true and capable of deduction from logical premisses alone, though possibly they may require some slight limitation in the form of an increased stringency of enunciation; and (2) the system must lead to no contradictions, namely in pursuing our inferences we must never be led to assert both and not-, i.e. both "" and "" cannot legitimately appear.

The following are the primitive propositions employed in the calculus of propositions. The letters "Pp" stand for "primitive proposition."

(1) Anything implied by a true premiss is true Pp.

This is the rule which justifies inference.

(2) ,

i.e. if or is true, then is true.

(3) ,

i.e. if is true, then or is true.

(4) ,

i.e. if or is true, then or is true.

(5) ,

i.e. if either is true or " or " is true, then either is true or " or " is true.

(6) ,

i.e. if implies , then " or " implies " or ."

(7) Besides the above primitive propositions, we require a primitive proposition called "the axiom of identification of real variables." When we have separately asserted two different functions of , where is undetermined, it is often important to know whether we can identify the in one[Pg 14] assertion with the in the other. This will be the case—so our axiom allow us to infer—if both assertions present x as the argument to some one function, that is to say, if is a constituent in both assertions (whatever propositional function may be), or, more generally, if is a constituent in one assertion, and is a constituent in the other. This axiom introduces notions which have not yet been explained; for a fuller account, see the remarks accompanying *3·03, *1·7, *1·71, and *1·72 (which is the statement of this axiom) in the body of the work, as well as the explanation of propositional functions and ambiguous assertion to be given shortly.

Some simple propositions. In addition to the primitive propositions we have already mentioned, the following are among the most important of the elementary properties of propositions appearing among the deductions.

The law of excluded middle:

This is *2·11 below. We shall indicate in brackets the numbers given to the following propositions in the body of the work.

The law of contradiction (*3·24):

The law of double negation (*4·13):

The principle of transposition, i.e. "if implies , then not- implies not-," and vice versa: this principle has various forms, namely

as well as others which are variants of these.

The law of tautology, in the two forms:

i.e. " is true" is equivalent to " is true and is true," as well as to " is true or is true." From a formal point of view, it is through the law of tautology and its consequences that the algebra of logic is chiefly distinguished from ordinary algebra.

The law of absorption :

i.e. " implies " is equivalent to " is equivalent to ." This is called the law of absorption because it shows that the factor in the product is[Pg 15] absorbed by the factor , if implies . This principle enables us to replace an implication () by an equivalence (whenever it is convenient to do so.

An analogous and very important principle is the following:

Logical addition and multiplication of propositions obey the associative and commutative laws, and the distributive law in two forms, namely

The second of these distinguishes the relations of logical addition and multiplication from those of arithmetical addition and multiplication.

Propositional functions. Let be a statement containing a variable and such that it becomes a proposition when is given any fixed determined meaning. Then is called a "propositional function"; it is not a proposition, since owing to the ambiguity of it really makes no assertion at all. Thus " is hurt" really makes no assertion at all, till we have settled who is. Yet owing to the individuality retained by the ambiguous variable , it is an ambiguous example from the collection of propositions arrived at by giving all possible determinations to in " is hurt" which yield a proposition, true or false. Also if " is hurt" and " is hurt" occur in the same context, where is another variable, then according to the determinations given to and , they can be settled to be (possibly) the same proposition or (possibly) different propositions. But apart from some determination given to and , they retain in that context their ambiguous differentiation. Thus " is hurt" is an ambiguous "value" of a propositional function. When we wish to speak of the propositional function corresponding to " is hurt," we shall write " is hurt." Thus " is hurt" is the propositional function and " is hurt" is an ambiguous value of that function. Accordingly though " is hurt" and " is hurt" occurring in the same context can be distinguished, " is hurt" and " is hurt" convey no distinction of meaning at all. More generally, is an ambiguous value of the propositional function , and when a definite signification is substituted for , is an unambiguous value of .

Propositional functions are the fundamental kind from which the more usual kinds of function, such as "" or "" or "the father of ," are derived. These derivative functions are considered later, and are called "descriptive functions." The functions of propositions considered above are a particular case of propositional functions.

The range of values and total variation. Thus corresponding to any propositional function , there is a range, or collection, of values, consisting of all the propositions (true or false) which can be obtained by giving[Pg 16] every possible determination to in . A value of for which is true will be said to "satisfy" . Now in respect to the truth or falsehood of propositions of this range three important cases must be noted and symbolised. These cases are given by three propositions of which one at least must be true. Either (1) all propositions of the range are true, or (2) some propositions of the range are true, or (3) no proposition of the range is true. The statement (1) is symbolised by "," and (2) is symbolised by "." No definition is given of these two symbols, which accordingly embody two new primitive ideas in our system. The symbol "" may be read " always," or " is always true," or " is true for all possible values of ." The symbol "" may be read "there exists an for which is true," or "there exists an satisfying ," and thus conforms to the natural form of the expression of thought.

Proposition (3) can be expressed in terms of the fundamental ideas now on hand. In order to do this, note that "" stands for the contradictory of . Accordingly is another propositional function such that each value of contradicts a value of and vice versa. Hence "" symbolises the proposition that every value of is untrue. This is number (3) as stated above.

It is an obvious error, though one easy to commit, to assume that cases (1) and (3) are each other's contradictories. The symbolism exposes this fallacy at once, for (1) is , and (3) is , while the contradictory of (1) is . For the sake of brevity of symbolism a definition is made, namely

Definitions of which the object is to gain some trivial advantage in brevity by a slight adjustment of symbols will be said to be of "merely symbolic import," in contradistinction to those definitions which invite consideration of an important idea.

The proposition is called the "total variation" of the function .

For reasons which will be explained in Chapter II, we do not take negation as a primitive idea when propositions of the forms and are concerned, but we define the negation of , i.e. of " is always true," as being " is sometimes false," i.e. "," and similarly we define the negation of as being . Thus we put

In like manner we define a disjunction in which one of the propositions is of the form "" or "(" in terms of a disjunction of propositions not of this form, putting [Pg 17] i.e. "either is always true, or is true" is to mean "' or ' is always true," with similar definitions in other cases. This subject is resumed in Chapter II, and in *9 in the body of the work.

Apparent variables. The symbol "(" denotes one definite proposition, and there is no distinction in meaning between "(" and "" when they occur in the same context. Thus the "" in "" is not an ambiguous constituent of any expression in which "(" occurs; and such an expression does not cease to convey a determinate meaning by reason of the ambiguity of the x in the "." The symbol "(" has some analogy to the symbol

for definite integration, since in neither case is the expression a function of .

The range of in "(" or "(" extends over the complete field of the values of for which "" has meaning, and accordingly the meaning of "(" or "(" involves the supposition that such a field is determinate. The which occurs in "(" or "(" is called (following Peano) an "apparent variable." It follows from the meaning of "(" that the in this expression is also an apparent variable. A proposition in which occurs as an apparent variable is not a function of . Thus e.g. "(" will mean "everything is equal to itself." This is an absolute constant, not a function of a variable . This is why the is called an apparent variable in such cases.

Besides the "range" of in "(" or "(," which is the field of the values that may have, we shall speak of the "scope" of , meaning the function of which all values or some value are being affirmed. If we are asserting all values (or some value) of "," "" is the scope of ; if we are asserting all values (or some value) of "," "" is the scope of ; if we are asserting all values (or some value) of "," "" will be the scope of , and so on. The

scope of is indicated by the number of dots after the "()" or ""; that is to say, the scope extends forwards until we reach an equal number of dots not indicating a logical product, or a greater number indicating a logical product, or the end of the asserted proposition in which the "()" or "" occurs, whichever of these happens first[5]. Thus e.g.

will mean " always implies ," but

will mean "if is always true, then is true for the argument ."

Note that in the proposition [Pg 18] the two 's have no connection with each other. Since only one dot follows the in brackets, the scope of the first is limited to the "" immediately following the in brackets. It usually conduces to clearness to write

since the use of different letters emphasises the absence of connection between the two variables; but there is no logical necessity to use different letters, and it is sometimes convenient to use the same letter.

Ambiguous assertion and the real variable. Any value "" of the function can be asserted. Such an assertion of an ambiguous member of the values of is symbolised by

Ambiguous assertion of this kind is a primitive idea, which cannot be defined in terms of the assertion of propositions. This primitive idea is the one which embodies the use of the variable. Apart from ambiguous assertion, the consideration of "," which is an ambiguous member of the values of , would be of little consequence. When we are considering or asserting "," the variable is called a "real variable." Take, for example, the law of excluded middle in the form which it has in traditional formal logic:

Here and are real variables: as they vary, different propositions are expressed, though all of them are true. While and are undetermined, as in the above enunciation, no one definite proposition is asserted, but what is asserted is any value of the propositional function in question. This can only be legitimately asserted if, whatever value may be chosen, that value is true, i.e. if all the values are true. Thus the above form of the law of excluded middle is equivalent to

i.e. to "it is always true that is either or not ." But these two, though equivalent, are not identical, and we shall find it necessary to keep them distinguished.

When we assert something containing a real variable, as in e.g.

we are asserting any value of a propositional function. When we assert something containing an apparent variable, as in

we are asserting, in the first case all values, in the second case some value (undetermined), of the propositional function in question. It is plain that[Pg 19] we can only legitimately assert "any value" if all values are true; for otherwise, since the value of the variable remains to be determined, it might be so determined as to give a false proposition. Thus in the above instance, since we have

And generally, given an assertion containing a real variable , we may transform the real variable into an apparent one by placing the in brackets at the beginning, followed by as many dots as there are after the assertion-sign.

When we assert something containing a real variable, we cannot strictly be said to be asserting a proposition, for we only obtain a definite proposition by assigning a value to the variable, and then our assertion only applies to one definite case, so that it has not at all the same force as before. When what we assert contains a real variable, we are asserting a wholly undetermined one of all the propositions that result from giving various values to the variable. It will be convenient to speak of such assertions as asserting a propositional function. The ordinary formulae of mathematics contain such assertions; for example

does not assert this or that particular case of the formula, nor does it assert that the formula holds for all possible values of , though it is equivalent to this latter assertion; it simply asserts that the formula holds, leaving wholly undetermined; and it is able to do this legitimately, because, however may be determined, a true proposition results.

Although an assertion containing a real variable does not, in strictness, assert a proposition, yet it will be spoken of as asserting a proposition except when the nature of the ambiguous assertion involved is under discussion.

Definition and real variables. When the definiens contains one or more real variables, the definiendum must also contain them. For in this case we have a function of the real variables, and the definiendum must have the same meaning as the definiens for all values of these variables, which requires that the symbol which is the definiendum should contain the letters representing the real variables. This rule is not always observed by mathematicians, and its infringement has sometimes caused important confusions of thought, notably in geometry and the philosophy of space.

In the definitions given above of "" and "" and "," and are real variables, and therefore appear on both sides of the definition. In the definition of "" only the function considered, namely , is a real variable; thus so far as concerns the rule in question, need not appear on the left. But when a real variable is a function, it is necessary to indicate[Pg 20] how the argument is to be supplied, and therefore there are objections to omitting an apparent variable where (as in the case before us) this is the argument to the function which is the real variable. This appears more plainly if, instead of a general function , we take some particular function, say "," and consider the definition of . Our definition gives

But if we had adopted a notation in which the ambiguous value "," containing the apparent variable , did not occur in the definiendum, we should have had to construct a notation employing the function itself, namely "." This does not involve an apparent variable, but would be clumsy in practice. In fact we have found it convenient and possible—except in the explanatory portions—to keep the explicit use of symbols of the type "," either as constants [e.g.] or as real variables, almost entirely out of this work.

Propositions connecting real and apparent variables. The most important propositions connecting real and apparent variables are the following:

(1) "When a propositional function can be asserted, so can the proposition that all values of the function are true." More briefly, if less exactly, "what holds of any, however chosen, holds of all." This translates itself into the rule that when a real variable occurs in an assertion, we may turn it into an apparent variable by putting the letter representing it in brackets immediately after the assertion-sign.

(2) "What holds of all, holds of any," i.e.

This states "if is always true, then is true."

(3) "If is true, then is sometimes true," i.e.

An asserted proposition of the form "" expresses an "existence-theorem," namely "there exists an for which is true." The above proposition gives what is in practice the only way of proving existence-theorems: we always have to find some particular for which holds, and thence to infer "." If we were to assume what is called the multiplicative axiom, or the equivalent axiom enunciated by Zermelo, that would, in an important class of cases, give an existence-theorem where no particular instance of its truth can be found.

In virtue of "" and "," we have "," i.e. "what is always true is sometimes true." This would not be the case if nothing existed; thus our assumptions contain the assumption that there is something. This is involved in the principle[Pg 21] that what holds of all, holds of any; for this would not be true if there were no "any."

(4) "If is always true, and is always true, then '' is always true," i.e.

(This requires that and should be functions which take arguments of the same type. We shall explain this requirement at a later stage.) The converse also holds; i.e. we have

It is to some extent optional which of the propositions connecting real and apparent variables are taken as primitive propositions. The primitive propositions assumed, on this subject, in the body of the work (*9), are the following:

i.e. if either is true, or is true, then (is true. (On the necessity for this primitive proposition, see remarks on *9·11 in the body of the work.)

(3) If we can assert , where y is a real variable, then we can assert ; i.e. what holds of any, however chosen, holds of all.

[Pg 22]

Formal implication and formal equivalence. When an implication, say , is said to hold always, i.e. when , we shall say that formally implies ; and propositions of the form "" will be said to state formal implications. In the usual instances of implication, such as "'Socrates is a man' implies 'Socrates is mortal,'" we have a proposition of the form "" in a case in which "" is true. In such a case, we feel the implication as a particular case of a formal implication. Thus it has come about that implications which are not particular cases of formal implications have not been regarded as

implications at all. There is also a practical ground for the neglect of such implications, for, speaking generally, they can only be known when it is already known either that their hypothesis is false or that their conclusion is true; and in neither of these cases do they serve to make us know the conclusion, since in the first case the conclusion need not be true, and in the second it is known already. Thus such implications do not serve the purpose for which implications are chiefly useful, namely that of making us know, by deduction, conclusions of which we were previously ignorant. Formal implications, on the contrary, do serve this purpose, owing to the psychological fact that we often know "" and , in cases where (which follows from these premisses) cannot easily be known directly.

These reasons, though they do not warrant the complete neglect of implications that are not instances of formal implications, are reasons which make formal implication very important. A formal implication states that, for all possible values of , if the hypothesis is true, the conclusion is true. Since "" will always be true when is false, it is only the values of that make true that are important in a formal implication; what is effectively stated is that, for all these values, is true. Thus propositions of the form "all is ," "no is " state formal implications, since the first (as appears by what has just been said) states

while the second states

And any formal implication "(" may be interpreted as: "All values of which satisfy[6] satisfy ," while the formal implication "(" may be interpreted as: "No values of which satisfy satisfy ."

We have similarly for "some is " the formula

and for "some is not " the formula

Two functions , are called formally equivalent when each always implies the other, i.e. when

and a proposition of this form is called a formal equivalence. In virtue of what was said about truth-values, if and are formally equivalent, either may replace the other in any truth-function. Hence for all the purposes of mathematics or of the present work, may replace or vice versa in any proposition with which we shall be concerned. Now to say that and are formally equivalent is the same thing as to say that and have the same extension, i.e. that any value of which satisfies either satisfies the other. Thus whenever a constant function occurs in our work, the truth-value of the proposition in which it occurs depends only upon the extension of the function. A proposition containing a function and having this property (i.e. that its truth-value depends only upon the extension of will be called an extensional function of . Thus the functions of functions with which we shall be specially concerned will all be extensional functions of functions.

What has just been said explains the connection (noted above) between the fact that the functions of propositions with which mathematics is specially[Pg 23] concerned are all truth-functions and the fact that mathematics is concerned with extensions rather than intensions.

Convenient abbreviation. The following definitions give alternative and often more convenient notations:

This notation "" is due to Peano, who, however, has no notation for the general idea "." It may be noticed as an exercise in the use of dots as brackets that we might have written

In practice however, when and are special functions, it is not possible to employ fewer dots than in the first form, and often more are required.

The following definitions give abbreviated notations for functions of two or more variables:

and so on for any number of variables;

and so on for any number of variables.

Identity. The propositional function " is identical with " is expressed by

This will be defined (cf. *13·01), but, owing to certain difficult points involved in the definition, we shall here omit it (cf. Chapter II). We have, of course,

The first of these expresses the reflexive property of identity: a relation is called reflexive when it holds between a term and itself, either universally, or whenever it holds between that term and some term. The second of the above propositions expresses that identity is a symmetrical relation: a relation is called symmetrical if, whenever it holds between and , it also holds between and . The third proposition expresses that identity is a transitive relation: a relation is called transitive if, whenever it holds between and and between and , it holds also between and .

We shall find that no new definition of the sign of equality is required in mathematics: all mathematical equations in which the sign of equality is[Pg 24] used in the ordinary way express some identity, and thus use the sign of equality in the above sense.

If and are identical, either can replace the other in any proposition without altering the truth-value of the proposition; thus we have

This is a fundamental property of identity, from which the remaining properties mostly follow.

It might be thought that identity would not have much importance, since it can only hold between and if and are different symbols for the same object. This view, however, does not apply to what we shall call "descriptive phrases," i.e. "the so-and-so." It is in regard to such phrases that identity is important, as we shall shortly explain. A proposition such as "Scott was the author of Waverley" expresses an identity in which there is a descriptive phrase (namely "the author of Waverley"); this illustrates how, in such cases, the assertion of identity may be important. It is essentially the same case when the newspapers say "the identity of the criminal has not transpired." In such a case, the criminal is known by a descriptive phrase, namely "the man who did the deed," and we wish to find an of whom it is true that " = the man who did the deed." When such an has been found, the identity of the criminal has

transpired.

Classes and relations. A class (which is the same as a manifold or aggregate) is all the objects satisfying some propositional function. If is the class composed of the objects satisfying , we shall say that is the class determined by . Every propositional function thus determines a class, though if the propositional function is one which is always false, the class will be null, i.e. will have no members. The class determined by the function will be represented by [7]. Thus for example if is an equation, will be the class of its roots; if is " has two legs and no feathers," will be the class of men; if is "," will be the class of proper fractions, and so on.

It is obvious that the same class of objects will have many determining functions. When it is not necessary to specify a determining function of a class, the class may be conveniently represented by a single Greek letter. Thus Greek letters, other than those to which some constant meaning is assigned, will be exclusively used for classes.

There are two kinds of difficulties which arise in formal logic; one kind arises in connection with classes and relations and the other in connection with descriptive functions. The point of the difficulty for classes and relations, so far as it concerns classes, is that a class cannot be an object suitable as an argument to any of its determining functions. If represents[Pg 25] a class and one of its determining functions [so that], it is not sufficient that be a false proposition, it must be nonsense. Thus a certain classification of what appear to be objects into things of essentially different types seems to be rendered necessary. This whole question is discussed in Chapter II, on the theory of types, and the formal treatment in the systematic exposition, which forms the main body of this work, is guided by this discussion. The part of the systematic exposition which is specially concerned with the theory of classes is *20, and in this Introduction it is discussed in Chapter III. It is sufficient to note here that, in the complete treatment of *20, we have avoided the decision as to whether a class of things has in any sense an existence as one object. A decision of this question in either way is indifferent to our logic, though perhaps, if we had regarded some solution which held classes and relations to be in some real sense objects as both true and likely to be universally received, we might have simplified one or two definitions and a few preliminary propositions. Our symbols, such as "" and and others, which represent classes and relations, are merely defined in their use, just as , standing for

has no meaning apart from a suitable function of , , on which to operate. The result of our definitions is that the way in which we use classes corresponds in general to their use in ordinary thought and speech; and whatever may be the ultimate interpretation of the one is also the interpretation of the other. Thus in fact our classification of types in Chapter II really performs the single, though essential, service of justifying us in refraining from entering on trains of reasoning which lead to contradictory conclusions. The justification is that what seem to be propositions are really nonsense.

The definitions which occur in the theory of classes, by which the idea of a class (at least in use) is based on the other ideas assumed as primitive, cannot be understood without a fuller discussion than can be given now (cf. Chapter II of this Introduction and also *20). Accordingly, in this preliminary survey, we proceed to state the more important simple propositions which result from those definitions, leaving the reader to employ in his mind the ordinary unanalysed idea of a class of things. Our symbols in their usage conform to the ordinary usage of this idea in language. It is to be noticed that in the systematic exposition our treatment of classes and relations requires no new primitive ideas and only two new primitive propositions, namely the two forms of the "Axiom of Reducibility" (cf. next Chapter) for one and two variables respectively.

The propositional function " is a member of the class " will be expressed, following Peano, by the notation [Pg 26] Here is chosen as the initial of the word ἐστί. "" may be read " is an ." Thus "" will mean " is a man," and so on. For typographical convenience we shall put

For "class" we shall write "Cls"; thus "" means " is a class."

We have

i.e. "' is a member of the class determined by ' is equivalent to ' satisfies ,' or to ' is true.'"

A class is wholly determinate when its membership is known, that is, there cannot be two different classes having the same membership. Thus if , are formally equivalent functions, they determine the same class; for in that case, if is a member of the class determined by , and therefore satisfies , it also satisfies , and is therefore a member of the class determined by . Thus we have

The following propositions are obvious and important:

i.e. is identical with the class determined by when, and only when, " is an " is formally equivalent to ;

i.e. two classes and are identical when, and only when, they have the same membership;

i.e. the class whose determining function is " is an " is , in other words, is the class of objects which are members of ;

i.e. the class determined by the function is a class.

It will be seen that, according to the above, any function of one variable can be replaced by an equivalent function of the form "." Hence any extensional function of functions which holds when its argument is a function of the form "," whatever possible value may have, will hold also when its argument is any function . Thus variation of classes can replace variation of functions of one variable in all the propositions of the sort with which we are concerned.

In an exactly analogous manner we introduce dual or dyadic relations, i.e. relations between two terms. Such relations will be called simply "relations"; relations between

more than two terms will be distinguished[Pg 27] as multiple relations, or (when the number of their terms is specified) as triple, quadruple, ... relations, or as triadic, tetradic, ... relations. Such relations will not concern us until we come to Geometry. For the present, the only relations we are concerned with are dual relations.

Relations, like classes, are to be taken in extension, i.e. if and are relations which hold between the same pairs of terms, and are to be identical. We may regard a relation, in the sense in which it is required for our purposes, as a class of couples; i.e. the couple (is to be one of the class of couples constituting the relation if has the relation to [8]. This view of relations as classes of couples will not, however, be introduced into our symbolic treatment, and is only mentioned in order to show that it is possible so to understand the meaning of the word relation that a relation shall be determined by its extension.

Any function determines a relation between and . If we regard a relation as a class of couples, the relation determined by is the class of couples for which is true. The relation determined by the function will be denoted by

We shall use a capital letter for a relation when it is not necessary to specify the determining function. Thus whenever a capital letter occurs, it is to be understood that it stands for a relation.

The propositional function " has the relation to " will be expressed by the notation

This notation is designed to keep as near as possible to common language, which, when it has to express a relation, generally mentions it between its terms, as in " loves ," " equals ," " is greater than ," and so on. For "relation" we shall write "Rel" thus "" means " is a relation."

Owing to our taking relations in extension, we shall have

i.e. two functions of two variables determine the same relation when, and only when, the two functions are formally equivalent.

i.e. " has to the relation determined by the function " is equivalent to ;

[Pg 28]

These propositions are analogous to those previously given for classes. It results from them that any function of two variables is formally equivalent to some function of the form ; hence, in extensional functions of two variables, variation of relations can replace variation of functions of two variables.

Both classes and relations have properties analogous to most of those of propositions that result from negation and the logical sum. The logical product of two classes and is their common part, i.e. the class of terms which are members of both. This is represented by . Thus we put

This gives us

i.e. " is a member of the logical product of and " is equivalent to the logical product of " is a member of " and " is a member of ."

Similarly the logical sum of two classes and is the class of terms which are members of either; we denote it by . The definition is

and the connection with the logical sum of propositions is given by

The negation of a class consists of those terms for which "" can be significantly and truly denied. We shall find that there are terms of other types for which "" is neither true nor false, but nonsense. These terms are not members of the negation of .

Thus the negation of a class is the class of terms of suitable type which are not members of it, i.e. the class . We call this class "" (read "not"); thus the definition is

and the connection with the negation of propositions is given by

In place of implication we have the relation of inclusion. A class is said to be included or contained in a class if all members of are members of , i.e. if . We write "" for " is contained in ." Thus we put

Most of the formulae concerning , , , remain true if we substitute , , -, . In place of equivalence, we substitute identity; for "" was defined as "," but "" gives "," whence .

[Pg 29]

The following are some propositions concerning classes which are analogues of propositions previously given concerning propositions:

i.e. the common part of and is the negation of "not or not";

i.e., " is a member of or not";

i.e. " is not a member of both and not";

The two last are the two forms of the law of tautology.

The law of absorption holds in the form

Thus for example "all Cretans are liars" is equivalent to "Cretans are identical with lying Cretans."

Just as we have

so we have

This expresses the ordinary syllogism in Barbara (with the premisses interchanged); for "" means the same as "all 's are 's," so that the above proposition states: "If all 's are 's, and all 's are 's, then all 's are 's." (It should be observed that syllogisms are traditionally expressed with "therefore," as if they asserted both premisses and conclusion. This is, of course, merely a slipshod way of speaking, since what is really asserted is only the connection of premisses with conclusion.)

The syllogism in Barbara when the minor premiss has an individual subject is

e.g. "if Socrates is a man, and all men are mortals, then Socrates is a mortal." This, as was pointed out by Peano, is not a particular case of "," since "" is not a particular case of "." This point is important, since traditional logic is here mistaken. The nature and magnitude of its mistake will become clearer at a later stage.

For relations, we have precisely analogous definitions and propositions. We put [Pg 30]

Generally, when we require analogous but different symbols for relations and for classes, we shall choose for relations the symbol obtained by adding a dot, in some convenient position, to the corresponding symbol for classes. (The dot must not be put on the line, since that would cause confusion with the use of dots as brackets.) But such symbols require and receive a special definition in each case.

A class is said to exist when it has at least one member: " exists" is denoted by "" Thus we put

The class which has no members is called the "null-class," and is denoted by "." Any propositional function which is always false determines the null-class. One such function is known to us already, namely " is not identical with ," which we denote by "." Thus we may use this function for defining , and put

The class determined by a function which is always true is called the universal class, and is represented by ; thus

Thus is the negation of . We have

i.e. "' is a member of ' is always true"; and

i.e. "' is a member of ' is always false." Also

i.e. " is the null-class" is equivalent to " does not exist."

For relations we use similar notations. We put

i.e. "" means that there is at least one couple , between which the relation holds. will be the relation which never holds, and the relation which always holds. is practically never required; will be the relation . We have

[Pg 31]

There are no classes which contain objects of more than one type. Accordingly there is a universal class and a null-class proper to each type of object. But these symbols need not be distinguished, since it will be found that there is no possibility of confusion. Similar remarks apply to relations.

Descriptions. By a "description" we mean a phrase of the form "the so-and-so" or of some equivalent form. For the present, we confine our attention to the in the singular. We shall use this word strictly, so as to imply uniqueness; e.g. we should not say " is the son of " if had other sons besides . Thus a description of the form "the so-and-so" will only have an application in the event of there being one so-and-so and no more. Hence a description requires some propositional function which is satisfied by one value of and by no other values; then "the which satisfies " is a description which definitely describes a certain object, though we may not know what object it describes. For example, if is a man, " is the father of " must be true for one, and only one, value of . Hence "the father of " is a description of a certain man, though we may not know what man it describes. A phrase containing "the" always presupposes some initial propositional function not containing "the"; thus instead of " is the father of "

we ought to take as our initial function " begot "; then "the father of " means the one value of which satisfies this propositional function.

If is a propositional function, the symbol "" is used in our symbolism in such a way that it can always be read as "the which satisfies ." But we do not define "" as standing for "the which satisfies ," thus treating this last phrase as embodying a primitive idea. Every use of "," where it apparently occurs as a constituent of a proposition in the place of an object, is defined in terms of the primitive ideas already on hand. An example of this definition in use is given by the proposition "" which is considered immediately. The whole subject is treated more fully in Chapter III.

[Pg 32]

The symbol should be compared and contrasted with "" which in use can always be read as "the 's which satisfy ." Both symbols are incomplete symbols defined only in use, and as such are discussed in Chapter III. The symbol "" always has an application, namely to the class determined by ; but "" only has an application when is only satisfied by one value of , neither more nor less. It should also be observed that the meaning given to the symbol by the definition, given immediately below, of does not presuppose that we know the meaning of "one." This is also characteristic of the definition of any other use of .

We now proceed to define "" so that it can be read "the satisfying exists." (It will be observed that this is a different meaning of existence from that which we express by ""). Its definition is

i.e. "the satisfying exists" is to mean "there is an object such that is true when is but not otherwise."

The following are equivalent forms:

The last of these states that "the satisfying exists" is equivalent to "there is an object satisfying , and every object other than does not satisfy ."

The kind of existence just defined covers a great many cases. Thus for example "the most perfect Being exists" will mean:

which, taking the last of the above equivalences, is equivalent to

A proposition such as "Apollo exists" is really of the same logical form, although it does not explicitly contain the word the. For "Apollo" means really "the object having such-and-such properties," say "the object having the properties enumerated in the Classical Dictionary[9]." If these properties make up the propositional function , then "Apollo" means "," and "Apollo exists" means "." To take another illustration, "the author of Waverley" means "the man who (or rather, the object which) wrote Waverley." Thus "Scott is the author of Waverley" is

Here (as we observed before) the importance of identity in connection with descriptions plainly appears.

The notation "," which is long and inconvenient, is seldom used, being chiefly required to lead up to another notation, namely "" meaning "the object having the

relation to ." That is, we put

The inverted comma may be read "of." Thus "" is read "the of ." Thus if is the relation of father to son, "" means "the father of "; if is the relation of son to father, "" means "the son of ," which will[Pg 33] only "exist" if has one son and no more. is a function of , but not a propositional function; we shall call it a descriptive function. All the ordinary functions of mathematics are of this kind, as will appear more fully in the sequel. Thus in our notation, "" would be written "," and "sin" would stand for the relation which has to . Instead of a variable descriptive function , we put , where the variable relation takes the place of the variable function . A descriptive function will in general exist while belongs to a certain domain, but not outside that domain; thus if we are dealing with positive rationals, will be significant if is a perfect square, but not otherwise; if we are dealing with real numbers, and agree that "" is to mean the positive square root (or, is to mean the negative square root), will be significant provided is positive, but not otherwise; and so on. Thus every descriptive function has what we may call a "domain of definition" or a "domain of existence," which may be thus defined: If the function in question is , its domain of definition or of existence will be the class of those arguments for which we have , i.e. for which , i.e. for which there is one , and no more, having the relation to .

If is any relation, we will speak of as the "associated descriptive function." A great many of the constant relations which we shall have occasion to introduce are only or chiefly important on account of their associated descriptive functions. In such cases, it is easier (though less correct) to begin by assigning the meaning of the descriptive function, and to deduce the meaning of the relation from that of the descriptive function. This will be done in the following explanations of notation.

Various descriptive functions of relations. If is any relation, the converse of is the relation which holds between and whenever holds between and . Thus greater is the converse of less, before of after, cause of effect, husband of wife, etc. The converse of is written[10] or . The definition is

The second of these is not a formally correct definition, since we ought to define "Cnv" and deduce the meaning of . But it is not worth while to adopt this plan in our present introductory account, which aims at simplicity rather than formal correctness.

A relation is called symmetrical if , i.e. if it holds between and whenever it holds between and (and therefore vice versa). Identity,[Pg 34] diversity, agreement or disagreement in any respect, are symmetrical relations. A relation is called asymmetrical when it is incompatible with its converse, i.e. when , or, what is equivalent,

Before and after, greater and less, ancestor and descendant, are asymmetrical, as are all other relations of the sort that lead to series. But there are many asymmetrical relations which do not lead to series, for instance, that of wife's brother[11]. A relation maybe neither symmetrical nor asymmetrical; for example, this holds of the relation of inclusion between classes: and will both be true if , but otherwise only one of them, at most, will be true. The relation brother is neither symmetrical nor

asymmetrical, for if is the brother of , may be either the brother or the sister of .

In the propositional function , we call the referent and the relatum. The class , consisting of all the 's which have the relation to , is called the class of referents of with respect to ; the class , consisting of all the 's to which has the relation , is called the class of relata of with respect to . These two classes are denoted respectively by and . Thus

The arrow runs towards in the first case, to show that we are concerned with things having the relation to ; it runs away from in the second case to show that the relation goes from to the members of . It runs in fact from a referent and towards a relatum.

The notations , are very important, and are used constantly. If is the relation of parent to child, = the parents of , = the children of . We have

These equivalences are often embodied in common language. For example, we say indiscriminately " is an inhabitant of London" or " inhabits London." If we put "" for "inhabits," " inhabits London" is " London," while " is an inhabitant of London" is "."

[Pg 35]

Instead of and we sometimes use , , where "" stands for "sagitta," and "" is "" backwards. Thus we put

These notations are sometimes more convenient than an arrow when the relation concerned is represented by a combination of letters, instead of a single letter such as . Thus e.g. we should write , rather than put an arrow over the whole length of ().

The class of all terms that have the relation to something or other is called the domain of . Thus if is the relation of parent and child, the domain of will be the class of parents. We represent the domain of by "." Thus we put

Similarly the class of all terms to which something or other has the relation is called the converse domain of ; it is the same as the domain of the converse of . The converse domain of is represented by ""; thus

The sum of the domain and the converse domain is called the field, and is represented by : thus

The field is chiefly important in connection with series. If is the ordering relation of a series, will be the class of terms of the series, will be all the terms except the last (if any), and will be all the terms except the first (if any). The first term, if it exists, is the only member of , since it is the only term which is a predecessor but not a follower. Similarly the last term (if any) is the only member of . The condition that a series should have no end is , i.e. "every follower is a predecessor"; the condition for no beginning is . These conditions are equivalent respectively to and .

The relative product of two relations and is the relation which holds between and when there is an intermediate term such that has the relation to and has the relation to . The relative product of and is represented by ; thus we put

Thus "paternal aunt" is the relative product of sister and father; "paternal grandmother" is the relative product of mother and father; "maternal[Pg 36] grandfather" is the relative product of father and mother. The relative product is not commutative, but it obeys the associative law, i.e.

It also obeys the distributive law with regard to the logical addition of relations, i.e. we have

But with regard to the logical product, we have only

The relative product does not obey the law of tautology, i.e. we do not have in general . We put

Thus paternal grandfather = (,

maternal grandmother = (.

A relation is called transitive when , i.e. when, if and , we always have , i.e. when

Relations which generate series are always transitive; thus e.g.

If is a relation which generates a series, may conveniently be read "precedes"; thus "" becomes "if precedes and precedes , then always precedes ." The class of relations which generate series are partially characterized by the fact that they are transitive and asymmetrical, and never relate a term to itself.

If is a relation which generates a series, and if we have not merely , but , then generates a series which is compact (überall dicht), i.e. such that there are terms between any two. For in this case we have

i.e. if precedes , there is a term such that precedes and precedes , i.e. there is a term between and . Thus among relations which generate series, those which generate compact series are those for which .

[Pg 37]

Many relations which do not generate series are transitive, for example, identity, or the relation of inclusion between classes. Such cases arise when the relations are not asymmetrical. Relations which are transitive and symmetrical are an important class: they may be regarded as consisting in the possession of some common property.

Plural descriptive functions. The class of terms which have the relation to some member of a class is denoted by or . The definition is

Thus for example let be the relation of inhabiting, and the class of towns; then = inhabitants of towns. Let be the relation "less than" among rationals, and the class of those rationals which are of the form , for integral values of ; then will be all rationals less than some member of , i.e. all rationals less than . If is the generating relation of a series, and is any class of members of , will be predecessors of 's, i.e. the segment defined by . If is a relation such that always exists when , will be the class of all terms of the form for values of which are members of ; i.e.

Thus a member of the class "fathers of great men" will be the father of , where is some great man. In other cases, this will not hold; for instance, let be the relation of a number to any number of which it is a factor; then , but this class is not composed of terms of the form "the factor of ," where is an even number, because numbers do not have only one factor apiece.

Unit classes. The class whose only member is might be thought to be identical with , but Peano and Frege have shown that this is not the case. (The reasons why this is not the case will be explained in a preliminary way in Chapter II of the Introduction.) We denote by "" the class whose only member is : thus

i.e. "" means "the class of objects which are identical with ."

The class consisting of and will be ; the class got by adding to a class will be ; the class got by taking away from a class will be . (We write as an abbreviation for .)

It will be observed that unit classes have been defined without reference to the number ; in fact, we use unit classes to define the number . This number is defined as the class of unit classes, i.e.

This leads to

From this it appears further that

i.e. " is a unit class" is equivalent to "the satisfying exists."

[Pg 38]

If , is the only member of , for the only member of is the only term to which has the relation . Thus "" takes the place of "(," if stands for . In practice, "" is a more convenient notation than "(," and is generally used instead of "(."

The above account has explained most of the logical notation employed in the present work. In the applications to various parts of mathematics, other definitions are introduced; but the objects defined by these later definitions belong, for the most part, rather to mathematics than to logic. The reader who has mastered the symbols explained above will find that any later formulae can be deciphered by the help of comparatively few additional definitions.

FOOTNOTES:

[1] Cf. Chapter II of the Introduction.

[2] This phrase is due to Frege.

[3] The meaning of these expressions will be explained later, and examples of the use of dots in connection with them will be given on pp. 17, 18.

[4] This case will be fully considered in Chapter III of the Introduction. It need not further concern us at present.

[5] This agrees with the rules for the occurrences of dots of the type of Group II as explained above, pp. 9 and 10.

[6] A value of is said to satisfy or when is true for that value of .

[7] Any other letter may be used instead of .

[8] Such a couple has a sense, i.e. the couple is different from the couple , unless . We shall call it a "couple with sense," to distinguish it from the class consisting of and . It may also be called an ordered couple.

[9] The same principle applies to many uses of the proper names of existent objects, e.g. to all uses of proper names for objects known to the speaker only by report, and not by personal acquaintance.

[10] The second of these notations is taken from Schröder's Algebra und Logik der Relative.

[11] This relation is not strictly asymmetrical, but is so except when the wife's brother is also the sister's husband. In the Greek Church the relation is strictly asymmetrical.

[Pg 39]

CHAPTER II.

THE THEORY OF LOGICAL TYPES.

THE theory of logical types, to be explained in the present Chapter, recommended itself to us in the first instance by its ability to solve certain contradictions, of which the one best known to mathematicians is Burali-Forti's concerning the greatest ordinal. But the theory in question is not wholly dependent upon this indirect recommendation: it has also a certain consonance with common sense which makes it inherently credible. In what follows, we shall therefore first set forth the theory on its own account, and then apply it to the solution of the contradictions.

I. The Vicious-Circle Principle.

An analysis of the paradoxes to be avoided shows that they all result from a certain kind of vicious circle[12]. The vicious circles in question arise from supposing that a collection of objects may contain members which can only be defined by means of the collection as a whole. Thus, for example, the collection of propositions will be supposed to contain a proposition stating that "all propositions are either true or false." It would seem, however, that such a statement could not be legitimate unless "all propositions" referred to some already definite collection, which it cannot do if new propositions are created by statements about "all propositions." We shall, therefore, have to say that statements about "all propositions" are meaningless. More generally, given any set of objects such that, if we suppose the set to have a total, it will contain members which presuppose this total, then such a set cannot have a total. By saying that a set has "no total," we mean, primarily, that no significant statement can be made about "all its members." Propositions, as the above illustration shows, must be a set having no total. The same is true, as we shall shortly see, of propositional functions, even when these are restricted to such as can significantly have as argument a given object . In such cases, it is necessary to break

up our set into smaller sets, each of which is capable of a total. This is what the theory of types aims at effecting.

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The principle which enables us to avoid illegitimate totalities may be stated as follows: "Whatever involves all of a collection must not be one of the collection"; or, conversely: "If, provided a certain collection had a total, it would have members only definable in terms of that total, then the said collection has no total." We shall call this the "vicious-circle principle," because it enables us to avoid the vicious circles involved in the assumption of illegitimate totalities. Arguments which are condemned by the vicious-circle principle will be called "vicious-circle fallacies." Such arguments, in certain circumstances, may lead to contradictions, but it often happens that the conclusions to which they lead are in fact true, though the arguments are fallacious. Take, for example, the law of excluded middle, in the form "all propositions are true or false." If from this law we argue that, because the law of excluded middle is a proposition, therefore the law of excluded middle is true or false, we incur a vicious-circle fallacy. "All propositions" must be in some way limited before it becomes a legitimate totality, and any limitation which makes it legitimate must make any statement about the totality fall outside the totality. Similarly, the imaginary sceptic, who asserts that he knows nothing, and is refuted by being asked if he knows that he knows nothing, has asserted nonsense, and has been fallaciously refuted by an argument which involves a vicious-circle fallacy. In order that the sceptic's assertion may become significant, it is necessary to place some limitation upon the things of which he is asserting his ignorance, because the things of which it is possible to be ignorant form an illegitimate totality. But as soon as a suitable limitation has been placed by him upon the collection of propositions of which he is asserting his ignorance, the proposition that he is ignorant of every member of this collection must not itself be one of the collection. Hence any significant scepticism is not open to the above form of refutation.

The paradoxes of symbolic logic concern various sorts of objects: propositions, classes, cardinal and ordinal numbers, etc. All these sorts of objects, as we shall show, represent illegitimate totalities, and are therefore capable of giving rise to vicious-circle fallacies. But by means of the theory (to be explained in Chapter III) which reduces statements that are verbally concerned with classes and relations to statements that are concerned with propositional functions, the paradoxes are reduced to such as are concerned with propositions and propositional functions. The paradoxes that concern propositions are only indirectly relevant to mathematics, while those that more nearly concern the mathematician are all concerned with propositional functions. We shall therefore proceed at once to the consideration of propositional functions.

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II. The Nature of Propositional Functions.

By a "propositional function" we mean something which contains a variable , and expresses a proposition as soon as a value is assigned to . That is to say, it differs

from a proposition solely by the fact that it is ambiguous: it contains a variable of which the value is unassigned. It agrees with the ordinary functions of mathematics in the fact of containing an unassigned variable: where it differs is in the fact that the values of the function are propositions. Thus e.g. " is a man" or "" is a propositional function. We shall find that it is possible to incur a vicious-circle fallacy at the very outset, by admitting as possible arguments to a propositional function terms which presuppose the function. This form of the fallacy is very instructive, and its avoidance leads, as we shall see, to the hierarchy of types.

The question as to the nature of a function[13] is by no means an easy one. It would seem, however, that the essential characteristic of a function is ambiguity. Take, for example, the law of identity in the form " is ," which is the form in which it is usually enunciated. It is plain that, regarded psychologically, we have here a single judgment. But what are we to say of the object of the judgment? We are not judging that Socrates is Socrates, nor that Plato is Plato, nor any other of the definite judgments that are instances of the law of identity. Yet each of these judgments is, in a sense, within the scope of our judgment. We are in fact judging an ambiguous instance of the propositional function " is ." We appear to have a single thought which does not have a definite object, but has as its object an undetermined one of the values of the function " is ." It is this kind of ambiguity that constitutes the essence of a function. When we speak of "" where is not specified, we mean one value of the function, but not a definite one. We may express this by saying that "" ambiguously denotes , , , etc., where , , , etc., are the various values of "."

When we say that "" ambiguously denotes , , , etc., we mean that "" means one of the objects , , , etc., though not a definite one, but an undetermined one. It follows that "" only has a well-defined meaning (well-defined, that is to say, except in so far as it is of its essence to be ambiguous) if the objects , , , etc., are well-defined. That is to say, a function is not a well-defined function unless all its values are already well-defined. It follows from this that no function can have among its values anything which presupposes the function, for if it had, we could not regard the objects ambiguously denoted by the function as definite until the function was definite, while conversely, as we have just seen, the function cannot be[Pg 42] definite until its values are definite. This is a particular case, but perhaps the most fundamental case, of the vicious-circle principle. A function is what ambiguously denotes some one of a certain totality, namely the values of the function; hence this totality cannot contain any members which involve the function, since, if it did, it would contain members involving the totality, which, by the vicious-circle principle, no totality can do.

It will be seen that, according to the above account, the values of a function are presupposed by the function, not vice versa. It is sufficiently obvious, in any particular case, that a value of a function does not presuppose the function. Thus for example the proposition "Socrates is human" can be perfectly apprehended without regarding it as a value of the function " is human." It is true that, conversely, a function can be apprehended without its being necessary to apprehend its values severally and individually. If this were not the case, no function could be apprehended at all, since the number of values (true and false) of a function is

necessarily infinite and there are necessarily possible arguments with which we are unacquainted. What is necessary is not that the values should be given individually and extensionally, but that the totality of the values should be given intensionally, so that, concerning any assigned object, it is at least theoretically determinate whether or not the said object is a value of the function.

It is necessary practically to distinguish the function itself from an undetermined value of the function. We may regard the function itself as that which ambiguously denotes, while an undetermined value of the function is that which is ambiguously denoted. If the undetermined value is written "," we will write the function itself "." (Any other letter may be used in place of .) Thus we should say " is a proposition," but " is a propositional function." When we say " is a proposition," we mean to state something which is true for every possible value of , though we do not decide what value is to have. We are making an ambiguous statement about any value of the function. But when we say " is a function," we are not making an ambiguous statement. It would be more correct to say that we are making a statement about an ambiguity, taking the view that a function is an ambiguity. The function itself, , is the single thing which ambiguously denotes its many values; while , where is not specified, is one of the denoted objects, with the ambiguity belonging to the manner of denoting.

We have seen that, in accordance with the vicious-circle principle, the values of a function cannot contain terms only definable in terms of the function. Now given a function , the values for the function[14] are all propositions[Pg 43] of the form . It follows that there must be no propositions, of the form , in which has a value which involves . (If this were the case, the values of the function would not all be determinate until the function was determinate, whereas we found that the function is not determinate unless its values are previously determinate.) Hence there must be no such thing as the value for with the argument , or with any argument which involves . That is to say, the symbol "" must not express a proposition, as "" does if is a value for . In fact "" must be a symbol which does not express anything: we may therefore say that it is not significant. Thus given any function , there are arguments with which the function has no value, as well as arguments with which it has a value. We will call the arguments with which has a value "possible values of ." We will say that is "significant with the argument " when has a value with the argument .

When it is said that e.g. "" is meaningless, and therefore neither true nor false, it is necessary to avoid a misunderstanding. If "" were interpreted as meaning "the value for with the argument is true," that would be not meaningless, but false. It is false for the same reason for which "the King of France is bald" is false, namely because there is no such thing as "the value for with the argument ." But when, with some argument , we assert , we are not meaning to assert "the value for with the argument is true"; we are meaning to assert the actual proposition which is the value for with the argument . Thus for example if is " is a man," (Socrates) will be "Socrates is a man," not "the value for the function ' is a man,' with the argument Socrates, is true." Thus in accordance with our principle that "" is meaningless, we cannot legitimately deny "the function ' is a man' is a man," because this is nonsense, but we can legitimately

deny "the value for the function ' is a man' with the argument ' is a man' is true," not on the ground that the value in question is false, but on the ground that there is no such value for the function.

We will denote by the symbol "(" the proposition " always[15]," i.e. the proposition which asserts all the values for . This proposition involves the function , not merely an ambiguous value of the function. The assertion of , where is unspecified, is a different assertion from the one which asserts all values for , for the former is an ambiguous assertion, whereas the latter is in no sense ambiguous. It will be observed that "(" does not assert " with all values of ," because, as we have seen, there must be values of with which "" is meaningless. What is asserted by "(" is all propositions which are values for ; hence it is[Pg 44] only with such values of as make "" significant, i.e. with all possible arguments, that is asserted when we assert "(." Thus a convenient way to read "(" is " is true with all possible values of ." This is, however, a less accurate reading than " always," because the notion of truth is not part of the content of what is judged. When we judge "all men are mortal," we judge truly, but the notion of truth is not necessarily in our minds, any more than it need be when we judge "Socrates is mortal."

III. Definition and Systematic Ambiguity of Truth and Falsehood.

Since "(" involves the function , it must, according to our principle, be impossible as an argument to . That is to say, the symbol "" must be meaningless. This principle would seem, at first sight, to have certain exceptions. Take, for example, the function " is false," and consider the proposition "(is false." This should be a proposition asserting all propositions of the form " is false." Such a proposition, we should be inclined to say, must be false, because " is false" is not always true. Hence we should be led to the proposition

i.e. we should be led to a proposition in which "(is false" is the argument to the function " is false," which we had declared to be impossible. Now it will be seen that "(is false," in the above, purports to be a proposition about all propositions, and that, by the general form of the vicious-circle principle, there must be no propositions about all propositions. Nevertheless, it seems plain that, given any function, there is a proposition (true or false) asserting all its values. Hence we are led to the conclusion that " is false" and " is false" must not always be the values, with the arguments and , for a single function " is false." This, however, is only possible if the word "false" really has many different meanings, appropriate to propositions of different kinds.

That the words "true" and "false" have many different meanings, according to the kind of proposition to which they are applied, is not difficult to see. Let us take any function , and let be one of its values. Let us call the sort of truth which is applicable to "first truth." (This is not to assume that this would be first truth in another context: it is merely to indicate that it is the first sort of truth in our context.) Consider now the proposition (. If this has truth of the sort appropriate to it, that will mean that every value has "first truth." Thus if we call the sort of truth that is appropriate to ("second truth," we may define " has second truth" as meaning "every value for has first truth," i.e. "(.(has first truth)." Similarly, if we denote by "(" the proposition "

sometimes," i.e. as we may less accurately express it, " with some value of ," we find that (has second truth if there is an with[Pg 45] which has first truth; thus we may define " has second truth" as meaning "some value for has first truth," i.e. "(." Similar remarks apply to falsehood. Thus " has second falsehood" will mean "some value for has first falsehood," i.e. "(," while " has second falsehood" will mean "all values for have first falsehood," i.e. "(." Thus the sort of falsehood that can belong to a general proposition is different from the sort that can belong to a particular proposition.

Applying these considerations to the proposition "(is false," we see that the kind of falsehood in question must be specified. If, for example, first falsehood is meant, the function " has first falsehood" is only significant when is the sort of proposition which has first falsehood or first truth. Hence "(is false" will be replaced by a statement which is equivalent to "all propositions having either first truth or first falsehood have first falsehood." This proposition has second falsehood, and is not a possible argument to the function " has first falsehood." Thus the apparent exception to the principle that "" must be meaningless disappears.

Similar considerations will enable us to deal with "not-" and with " or ." It might seem as if these were functions in which any proposition might appear as argument. But this is due to a systematic ambiguity in the meanings of "not" and "or," by which they adapt themselves to propositions of any order. To explain fully how this occurs, it will be well to begin with a definition of the simplest kind of truth and falsehood.

The universe consists of objects having various qualities and standing in various relations. Some of the objects which occur in the universe are complex. When an object is complex, it consists of interrelated parts. Let us consider a complex object composed of two parts and standing to each other in the relation . The complex object "-in-the-relation--to-" may be capable of being perceived; when perceived, it is perceived as one object. Attention may show that it is complex; we then judge that and stand in the relation . Such a judgment, being derived from perception by mere attention, may be called a "judgment of perception." This judgment of perception, considered as an actual occurrence, is a relation of four terms, namely and and and the percipient. The perception, on the contrary, is a relation of two terms, namely "-in-the-relation--to-," and the percipient. Since an object of perception cannot be nothing, we cannot perceive "-in-the-relation--to-" unless a is in the relation to . Hence a judgment of perception, according to the above definition, must be true. This does not mean that, in a judgment which appears to us to be one of perception, we are sure of not being in error, since we may err in thinking that our judgment has really been derived merely by analysis of[Pg 46] what was perceived. But if our judgment has been so derived, it must be true. In fact, we may define truth, where such judgments are concerned, as consisting in the fact that there is a complex corresponding to the discursive thought which is the judgment. That is, when we judge " has the relation to ," our judgment is said to be true when there is a complex "-in-the-relation--to-," and is said to be false when this is not the case. This is a definition of truth and falsehood in relation to judgments of this kind.

It will be seen that, according to the above account, a judgment does not have a single object, namely the proposition, but has several interrelated objects. That is to

say, the relation which constitutes judgment is not a relation of two terms, namely the judging mind and the proposition, but is a relation of several terms, namely the mind and what are called the constituents of the proposition. That is, when we judge (say) "this is red," what occurs is a relation of three terms, the mind, and "this," and red. On the other hand, when we perceive "the redness of this," there is a relation of two terms, namely the mind and the complex object "the redness of this." When a judgment occurs, there is a certain complex entity, composed of the mind and the various objects of the judgment. When the judgment is true, in the case of the kind of judgments we have been considering, there is a corresponding complex of the objects of the judgment alone. Falsehood, in regard to our present class of judgments, consists in the absence of a corresponding complex composed of the objects alone. It follows from the above theory that a "proposition," in the sense in which a proposition is supposed to be the object of a judgment, is a false abstraction, because a judgment has several objects, not one. It is the severalness of the objects in judgment (as opposed to perception) which has led people to speak of thought as "discursive," though they do not appear to have realized clearly what was meant by this epithet.

Owing to the plurality of the objects of a single judgment, it follows that what we call a "proposition" (in the sense in which this is distinguished from the phrase expressing it) is not a single entity at all. That is to say, the phrase which expresses a proposition is what we call an "incomplete" symbol[16]; it does not have meaning in itself, but requires some supplementation in order to acquire a complete meaning. This fact is somewhat concealed by the circumstance that judgment in itself supplies a sufficient supplement, and that judgment in itself makes no verbal addition to the proposition. Thus "the proposition 'Socrates is human'" uses "Socrates is human" in a way which requires a supplement of some kind before it acquires a complete meaning; but when I judge "Socrates is human," the meaning is completed by the act of judging, and we no longer have an incomplete symbol. The fact that propositions are "incomplete symbols"[Pg 47] is important philosophically, and is relevant at certain points in symbolic logic.

The judgments we have been dealing with hitherto are such as are of the same form as judgments of perception, i.e. their subjects are always particular and definite. But there are many judgments which are not of this form. Such are "all men are mortal," "I met a man," "some men are Greeks." Before dealing with such judgments, we will introduce some technical terms.

We will give the name of "a complex" to any such object as " in the relation to " or " having the quality ," or " and and standing in the relation ." Broadly speaking, a complex is anything which occurs in the universe and is not simple. We will call a judgment elementary when it merely asserts such things as " has the relation to ," " has the quality " or " and and stand in the relation ." Then an elementary judgment is true when there is a corresponding complex, and false when there is no corresponding complex.

But take now such a proposition as "all men are mortal." Here the judgment does not correspond to one complex, but to many, namely "Socrates is mortal," "Plato is mortal," "Aristotle is mortal," etc. (For the moment, it is unnecessary to inquire

whether each of these does not require further treatment before we reach the ultimate complexes involved. For purposes of illustration, "Socrates is mortal" is here treated as an elementary judgment, though it is in fact not one, as will be explained later. Truly elementary judgments are not very easily found.) We do not mean to deny that there may be some relation of the concept man to the concept mortal which may be equivalent to "all men are mortal," but in any case this relation is not the same thing as what we affirm when we say that all men are mortal. Our judgment that all men are mortal collects together a number of elementary judgments. It is not, however, composed of these, since (e.g.) the fact that Socrates is mortal is no part of what we assert, as may be seen by considering the fact that our assertion can be understood by a person who has never heard of Socrates. In order to understand the judgment "all men are mortal," it is not necessary to know what men there are. We must admit, therefore, as a radically new kind of judgment, such general assertions as "all men are mortal." We assert that, given that is human, is always mortal. That is, we assert " is mortal" of every which is human. Thus we are able to judge (whether truly or falsely) that all the objects which have some assigned property also have some other assigned property. That is, given any propositional functions and , there is a judgment asserting with every for which we have . Such judgments we will call general judgments.

It is evident (as explained above) that the definition of truth is different[Pg 48] in the case of general judgments from what it was in the case of elementary judgments. Let us call the meaning of truth which we gave for elementary judgments "elementary truth." Then when we assert that it is true that all men are mortal, we shall mean that all judgments of the form " is mortal," where is a man, have elementary truth. We may define this as "truth of the second order" or "second-order truth." Then if we express the proposition "all men are mortal" in the form

and call this judgment , then " is true" must be taken to mean " has second-order truth," which in turn means

In order to avoid the necessity for stating explicitly the limitation to which our variable is subject, it is convenient to replace the above interpretation of "all men are mortal" by a slightly different interpretation. The proposition "all men are mortal" is equivalent to "' is a man' implies ' is mortal,' with all possible values of ." Here is not restricted to such values as are men, but may have any value with which "' is a man' implies ' is mortal'" is significant, i.e. either true or false. Such a proposition is called a "formal implication." The advantage of this form is that the values which the variable may take are given by the function to which it is the argument: the values which the variable may take are all those with which the function is significant.

We use the symbol "(" to express the general judgment which asserts all judgments of the form "." Then the judgment "all men are mortal" is equivalent to

i.e. (in virtue of the definition of implication) to

As we have just seen, the meaning of truth which is applicable to this proposition is not the same as the meaning of truth which is applicable to " is a man" or to " is mortal." And generally, in any judgment (, the sense in which this judgment is or may

be true is not the same as that in which is or may be true. If is an elementary judgment, it is true when it points to a corresponding complex. But (does not point to a single corresponding complex: the corresponding complexes are as numerous as the possible values of .

It follows from the above that such a proposition as "all the judgments made by Epimenides are true" will only be prima facie capable of truth if all his judgments are of the same order. If they are of varying orders, of which the nth is the highest, we may make n assertions of the form "all the judgments of order made by Epimenides are true," where has all values[Pg 49] up to . But no such judgment can include itself in its own scope, since such a judgment is always of higher order than the judgments to which it refers.

Let us consider next what is meant by the negation of a proposition of the form "." We observe, to begin with, that " in some cases," or " sometimes," is a judgment which is on a par with " in all cases," or " always." The judgment " sometimes" is true if one or more values of exist for which is true. We will express the proposition " sometimes" by the notation "," where "" stands for "there exists," and the whole symbol may be read "there exists an such that ." We take the two kinds of judgment expressed by "" and "" as primitive ideas. We also take as a primitive idea the negation of an elementary proposition. We can then define the negations of and (. The negation of any proposition will be denoted by the symbol "." Then the negation of will be defined as meaning

and the negation of will be defined as meaning "." Thus, in the traditional language of formal logic, the negation of a universal affirmative is to be defined as the particular negative, and the negation of the particular affirmative is to be defined as the universal negative. Hence the meaning of negation for such propositions is different from the meaning of negation for elementary propositions.

An analogous explanation will apply to disjunction. Consider the statement "either , or always." We will denote the disjunction of two propositions , by "." Then our statement is "." We will suppose that is an elementary proposition, and that is always an elementary proposition. We take the disjunction of two elementary propositions as a primitive idea, and we wish to define the disjunction

This may be defined as "," i.e. "either is true, or is always true" is to mean "' or ' is always true." Similarly we will define

as meaning "," i.e. we define "either is true or there is an for which is true" as meaning "there is an for which either or is true." Similarly we can define a disjunction of two universal propositions: "" will be defined as meaning "," i.e. "either is always true or is always true" is to mean "' or ' is always true." By this method we obtain definitions of disjunctions containing propositions of the form or (in terms of disjunctions of elementary propositions; but the meaning of "disjunction" is not the[Pg 50] same for propositions of the forms (, (, as it was for elementary propositions.

Similar explanations could be given for implication and conjunction, but this is unnecessary, since these can be defined in terms of negation and disjunction.

IV. Why a Given Function requires Arguments of a Certain Type.

The considerations so far adduced in favour of the view that a function cannot significantly have as argument anything defined in terms of the function itself have been more or less indirect. But a direct consideration of the kinds of functions which have functions as arguments and the kinds of functions which have arguments other than functions will show, if we are not mistaken, that not only is it impossible for a function to have itself or anything derived from it as argument, but that, if is another function such that there are arguments with which both "" and "" are significant, then and anything derived from it cannot significantly be argument to . This arises from the fact that a function is essentially an ambiguity, and that, if it is to occur in a definite proposition, it must occur in such a way that the ambiguity has disappeared, and a wholly unambiguous statement has resulted. A few illustrations will make this clear. Thus "," which we have already considered, is a function of ; as soon as is assigned, we have a definite proposition, wholly free from ambiguity. But it is obvious that we cannot substitute for the function something which is not a function: "" means " in all cases," and depends for its significance upon the fact that there are "cases" of , i.e. upon the ambiguity which is characteristic of a function. This instance illustrates the fact that, when a function can occur significantly as argument, something which is not a function cannot occur significantly as argument. But conversely, when something which is not a function can occur significantly as argument, a function cannot occur significantly. Take, e.g. " is a man," and consider " is a man." Here there is nothing to eliminate the ambiguity which constitutes ; there is thus nothing definite which is said to be a man. A function, in fact, is not a definite object, which could be or not be a man; it is a mere ambiguity awaiting determination, and in order that it may occur significantly it must receive the necessary determination, which it obviously does not receive if it is merely substituted for something determinate in a proposition[17]. This argument does not, however, apply directly as against such a statement as " is a man." Common sense would pronounce such a statement to be meaningless, but it cannot be condemned on the ground of ambiguity in its subject. We need[Pg 51] here a new objection, namely the following: A proposition is not a single entity, but a relation of several; hence a statement in which a proposition appears as subject will only be significant if it can be reduced to a statement about the terms which appear in the proposition. A proposition, like such phrases as "the so-and-so," where grammatically it appears as subject, must be broken up into its constituents if we are to find the true subject or subjects[18]. But in such a statement as " is a man," where is a proposition, this is not possible. Hence " is a man" is meaningless.

V. The Hierarchy of Functions and Propositions.

We are thus led to the conclusion, both from the vicious-circle principle and from direct inspection, that the functions to which a given object can be an argument are incapable of being arguments to each other, and that they have no term in common with the functions to which they can be arguments. We are thus led to construct a hierarchy. Beginning with and the other terms which can be arguments to the same functions to which can be argument, we come next to functions to which is a possible

argument, and then to functions to which such functions are possible arguments, and so on. But the hierarchy which has to be constructed is not so simple as might at first appear. The functions which can take as argument form an illegitimate totality, and themselves require division into a hierarchy of functions. This is easily seen as follows. Let be a function of the two variables and . Then if, keeping fixed for the moment, we assert this with all possible values of , we obtain a proposition:

Here, if is variable, we have a function of ; but as this function involves a totality of values of [19], it cannot itself be one of the values included in the totality, by the vicious-circle principle. It follows that the totality of values of concerned in (is not the totality of all functions in which can occur as argument, and that there is no such totality as that of all functions in which can occur as argument.

It follows from the above that a function in which appears as argument requires that "" should not stand for any function which is capable of a given argument, but must be restricted in such a way that none of the functions which are possible values of "" should involve any reference to the totality of such functions. Let us take as an illustration the definition of identity. We might attempt to define " is identical with " as meaning "whatever is true of is true of ," i.e. " always implies ." But here,[Pg 52] since we are concerned to assert all values of " implies " regarded as a function of , we shall be compelled to impose upon some limitation which will prevent us from including among values of values in which "all possible values of " are referred to. Thus for example " is identical with " is a function of ; hence, if it is a legitimate value of in " always implies " we shall be able to infer, by means of the above definition, that if is identical with , and is identical with , then is identical with . Although the conclusion is sound, the reasoning embodies a vicious-circle fallacy, since we have taken " implies " as a possible value of , which it cannot be. If, however, we impose any limitation upon , it may happen, so far as appears at present, that with other values of we might have true and false, so that our proposed definition of identity would plainly be wrong. This difficulty is avoided by the "axiom of reducibility," to be explained later. For the present, it is only mentioned in order to illustrate the necessity and the relevance of the hierarchy of functions of a given argument.

Let us give the name "-functions" to functions that are significant for a given argument . Then suppose we take any selection of -functions, and consider the proposition " satisfies all the functions belonging to the selection in question." If we here replace by a variable, we obtain an -function; but by the vicious-circle principle this -function cannot be a member of our selection, since it refers to the whole of the selection. Let the selection consist of all those functions which satisfy . Then our new function is

where is the argument. It thus appears that, whatever selection of -functions we may make, there will be other -functions that lie outside our selection. Such -functions, as the above instance illustrates, will always arise through taking a function of two arguments, and , and asserting all or some of the values resulting from varying . What is necessary, therefore, in order to avoid vicious-circle fallacies, is to divide our -functions into "types," each of which contains no functions which refer to the whole of that type.

When something is asserted or denied about all possible values or about some (undetermined) possible values of a variable, that variable is called apparent, after Peano. The presence of the words all or some in a proposition indicates the presence of an apparent variable; but often an apparent variable is really present where language does not at once indicate its presence. Thus for example " is mortal" means "there is a time at which will die." Thus a variable time occurs as apparent variable.

The clearest instances of propositions not containing apparent variables are such as express immediate judgments of perception, such as "this is red" or "this is painful," where "this" is something immediately given. In other[Pg 53] judgments, even where at first sight no variable appears to be present, it often happens that there really is one. Take (say) "Socrates is human." To Socrates himself, the word "Socrates" no doubt stood for an object of which he was immediately aware, and the judgment "Socrates is human" contained no apparent variable. But to us, who only know Socrates by description, the word "Socrates" cannot mean what it meant to him; it means rather "the person having such-and-such properties," (say) "the Athenian philosopher who drank the hemlock." Now in all propositions about "the so-and-so" there is an apparent variable, as will be shown in Chapter III. Thus in what we have in mind when we say "Socrates is human" there is an apparent variable, though there was no apparent variable in the corresponding judgment as made by Socrates, provided we assume that there is such a thing as immediate awareness of oneself.

Whatever may be the instances of propositions not containing apparent variables, it is obvious that propositional functions whose values do not contain apparent variables are the source of propositions containing apparent variables, in the sense in which the function is the source of the proposition . For the values for do not contain the apparent variable , which appears in ; if they contain an apparent variable , this can be similarly eliminated, and so on. This process must come to an end, since no proposition which we can apprehend can contain more than a finite number of apparent variables, on the ground that whatever we can apprehend must be of finite complexity. Thus we must arrive at last at a function of as many variables as there have been stages in reaching it from our original proposition, and this function will be such that its values contain no apparent variables. We may call this function the matrix of our original proposition and of any other propositions and functions to be obtained by turning some of the arguments to the function into apparent variables. Thus for example, if we have a matrix-function whose values are , we shall derive from it

, which is a function of ,

, which is a function of ,

, meaning " is true with all possible values of and ." This last is a proposition containing no real variable, i.e. no variable except apparent variables.

It is thus plain that all possible propositions and functions are obtainable from matrices by the process of turning the arguments to the matrices into apparent variables. In order to divide our propositions and functions into types, we shall,

therefore, start from matrices, and consider how they are to be divided with a view to the avoidance of vicious-circle fallacies in the definitions of the functions concerned. For this purpose, we will use such letters as , , , , , , , to denote objects which are neither propositions nor functions. Such objects we shall call individuals. Such objects will be[Pg 54] constituents of propositions or functions, and will be genuine constituents, in the sense that they do not disappear on analysis, as (for example) classes do, or phrases of the form "the so-and-so."

The first matrices that occur are those whose values are of the forms

i.e. where the arguments, however many there may be, are all individuals. The functions , , ..., since (by definition) they contain no apparent variables, and have no arguments except individuals, do not presuppose any totality of functions. From the functions , ... we may proceed to form other functions of , such as (, (, (, (, and so on. All these presuppose no totality except that of individuals. We thus arrive at a certain collection of functions of , characterized by the fact that they involve no variables except individuals. Such functions we will call "first-order functions."

We may now introduce a notation to express "any first-order function." We will denote any first-order function by "" and any value for such a function by "." Thus "" stands for any value for any function which involves no variables except individuals. It will be seen that "" is itself a function of two variables, namely and . Thus involves a variable which is not an individual, namely . Similarly "(" is a function of the variable , and thus involves a variable other than an individual. Again, if is a given individual,

is a function of , but it is not a function of the form , because it involves an (apparent) variable which is not an individual. Let us give the name "predicate" to any first-order function (This use of the word "predicate" is only proposed for the purposes of the present discussion.) Then the statement " implies with all possible values of " may be read "all the predicates of are predicates of ." This makes a statement about , but does not attribute to a predicate in the special sense just defined.

Owing to the introduction of the variable first-order function , we now have a new set of matrices. Thus "" is a function which contains no apparent variables, but contains the two real variables and . (It should be observed that when is assigned, we may obtain a function whose values do involve individuals as apparent variables, for example if is (. But so long as is variable, contains no apparent variables.) Again, if is a definite individual, is a function of the one variable . If and are definite individuals, " implies " is a function of the two variables , , and so on. We are thus led to a whole set of new matrices,

These matrices contain individuals and first-order functions as arguments, but[Pg 55] (like all matrices) they contain no apparent variables. Any such matrix, if it contains more than one variable, gives rise to new functions of one variable by turning all its arguments except one into apparent variables. Thus we obtain the functions

We will give the name of second-order matrices to such matrices as have first-order functions among their arguments, and have no arguments except

first-order functions and individuals. (It is not necessary that they should have individuals among their arguments.) We will give the name of second-order functions to such as either are second-order matrices or are derived from such matrices by turning some of the arguments into apparent variables. It will be seen that either an individual or a first-order function may appear as argument to a second-order function. Second-order functions are such as contain variables which are first-order functions, but contain no other variables except (possibly) individuals.

We now have various new classes of functions at our command. In the first place, we have second-order functions which have one argument which is a first-order function. We will denote a variable function of this kind by the notation , and any value of such a function by . Like , is a function of two variables, namely and . Among possible values of will be (where is constant), , , and so on. (These result from assigning a value to , leaving to be assigned.) We will call such functions "predicative functions of first-order functions."

In the second place, we have second-order functions of two arguments, one of which is a first-order function while the other is an individual. Let us denote undetermined values of such functions by the notation

As soon as is assigned, we shall have a predicative function of . If our function contains no first-order function as apparent variable, we shall obtain a predicative function of if we assign a value to . Thus, to take the simplest possible case, if is , the assignment of a value to gives us a predicative function of , in virtue of the definition of "." But if contains a first-order function as apparent variable, the assignment of a value to gives us a second-order function of .

In the third place, we have second-order functions of individuals. These will all be derived from functions of the form by turning into an apparent variable. We do not, therefore, need a new notation for them.

[Pg 56]

We have also second-order functions of two first-order functions, or of two such functions and an individual, and so on.

We may now proceed in exactly the same way to third-order matrices, which will be functions containing second-order functions as arguments, and containing no apparent variables, and no arguments except individuals and first-order functions and second-order functions. Thence we shall proceed, as before, to third-order functions; and so we can proceed indefinitely. If the highest order of variable occurring in a function, whether as argument or as apparent variable, is a function of the th order, then the function in which it occurs is of the th order. We do not arrive at functions of an infinite order, because the number of arguments and of apparent variables in a function must be finite, and therefore every function must be of a finite order. Since the orders of functions are only defined step by step, there can be no process of "proceeding to the limit," and functions of an infinite order cannot occur.

We will define a function of one variable as predicative when it is of the next order above that of its argument, i.e. of the lowest order compatible with its having that

argument. If a function has several arguments, and the highest order of function occurring among the arguments is the th, we call the function predicative if it is of the th order, i.e. again, if it is of the lowest order compatible with its having the arguments it has. A function of several arguments is predicative if there is one of its arguments such that, when the other arguments have values assigned to them, we obtain a predicative function of the one undetermined argument.

It is important to observe that all possible functions in the above hierarchy can be obtained by means of predicative functions and apparent variables. Thus, as we saw, second-order functions of an individual are of the form

where is a second-order predicative function. And speaking generally, a non-predicative function of the th order is obtained from a predicative function of the th order by turning all the arguments of the th order into apparent variables. (Other arguments also may be turned into apparent variables.) Thus we need not introduce as variables any functions except predicative functions. Moreover, to obtain any function of one variable , we need not go beyond predicative functions of two variables. For the function (, where is given, is a function of and , and is predicative. Thus it is of the form , and therefore (is of the form (. Thus speaking generally, by a succession of steps we find that, if is a predicative function of a sufficiently high order, any assigned non-predicative function of will be of one of the two forms

where is a predicative function of and .

[Pg 57]

The nature of the above hierarchy of functions may be restated as follows. A function, as we saw at an earlier stage, presupposes as part of its meaning the totality of its values, or, what comes to the same thing, the totality of its possible arguments. The arguments to a function may be functions or propositions or individuals. (It will be remembered that individuals were defined as whatever is neither a proposition nor a function.) For the present we neglect the case in which the argument to a function is a proposition. Consider a function whose argument is an individual. This function presupposes the totality of individuals; but unless it contains functions as apparent variables, it does not presuppose any totality of functions. If, however, it does contain a function as apparent variable, then it cannot be defined until some totality of functions has been defined. It follows that we must first define the totality of those functions that have individuals as arguments and contain no functions as apparent variables. These are the predicative functions of individuals. Generally, a predicative function of a variable argument is one which involves no totality except that of the possible values of the argument, and those that are presupposed by any one of the possible arguments. Thus a predicative function of a variable argument is any function which can be specified without introducing new kinds of variables not necessarily presupposed by the variable which is the argument.

A closely analogous treatment can be developed for propositions. Propositions which contain no functions and no apparent variables may be called elementary propositions. Propositions which are not elementary, which contain no functions, and no apparent variables except individuals, may be called first-order propositions. (It

should be observed that no variables except apparent variables can occur in a proposition, since whatever contains a real variable is a function, not a proposition.) Thus elementary and first-order propositions will be values of first-order functions. (It should be remembered that a function is not a constituent in one of its values: thus for example the function " is human" is not a constituent of the proposition "Socrates is human.") Elementary and first-order propositions presuppose no totality except (at most) the totality of individuals. They are of one or other of the three forms

where is a predicative function of an individual. It follows that, if represents a variable elementary proposition or a variable first-order proposition, a function is either or or . Thus a function of an elementary or a first-order proposition may always be reduced to a function of a first-order function. It follows that a proposition involving the totality of first-order propositions may be reduced to one involving the totality of first-order functions; and this obviously applies equally to higher[Pg 58] orders. The propositional hierarchy can, therefore, be derived from the functional hierarchy, and we may define a proposition of the nth order as one which involves an apparent variable of the th order in the functional hierarchy. The propositional hierarchy is never required in practice, and is only relevant for the solution of paradoxes; hence it is unnecessary to go into further detail as to the types of propositions.

VI. The Axiom of Reducibility.

It remains to consider the "axiom of reducibility." It will be seen that, according to the above hierarchy, no statement can be made significantly about "all -functions," where is some given object. Thus such a notion as "all properties of ," meaning "all functions which are true with the argument ," will be illegitimate. We shall have to distinguish the order of function concerned. We can speak of "all predicative properties of ," "all second-order properties of ," and so on. (If is not an individual, but an object of order , "second-order properties of " will mean "functions of order satisfied by .") But we cannot speak of "all properties of ." In some cases, we can see that some statement will hold of "all th-order properties of ," whatever value may have. In such cases, no practical harm results from regarding the statement as being about "all properties of ," provided we remember that it is really a number of statements, and not a single statement which could be regarded as assigning another property to , over and above all properties. Such cases will always involve some systematic ambiguity, such as that involved in the meaning of the word "truth," as explained above. Owing to this systematic ambiguity, it will be possible, sometimes, to combine into a single verbal statement what are really a number of different statements, corresponding to different orders in the hierarchy. This is illustrated in the case of the liar, where the statement "all 's statements are false" should be broken up into different statements referring to his statements of various orders, and attributing to each the appropriate kind of falsehood.

The axiom of reducibility is introduced in order to legitimate a great mass of reasoning, in which, prima facie, we are concerned with such notions as "all properties of " or "all -functions," and in which, nevertheless, it seems scarcely possible to suspect any substantial error. In order to state the axiom, we must first

define what is meant by "formal equivalence." Two functions , are said to be "formally equivalent" when, with every possible argument , is equivalent to , i.e. and are either both true or both false. Thus two functions are formally equivalent when they are satisfied by the same set of arguments. The axiom of reducibility is the assumption that, given any function , there is a formally equivalent predicative function,[Pg 59] i.e. there is a predicative function which is true when is true and false when is false. In symbols, the axiom is:

For two variables, we require a similar axiom, namely: Given any function , there is a formally equivalent predicative function, i.e.

In order to explain the purposes of the axiom of reducibility, and the nature of the grounds for supposing it true, we shall first illustrate it by applying it to some particular cases.

If we call a predicate of an object a predicative function which is true of that object, then the predicates of an object are only some among its properties. Take for example such a proposition as "Napoleon had all the qualities that make a great general." We may interpret this as meaning "Napoleon had all the predicates that make a great general." Here there is a predicate which is an apparent variable. If we put "" for " is a predicate required in a great general," our proposition is

Since this refers to a totality of predicates, it is not itself a predicate of Napoleon. It by no means follows, however, that there is not some one predicate common and peculiar to great generals. In fact, it is certain that there is such a predicate. For the number of great generals is finite, and each of them certainly possessed some predicate not possessed by any other human being—for example, the exact instant of his birth. The disjunction of such predicates will constitute a predicate common and peculiar to great generals[20]. If we call this predicate , the statement we made about Napoleon was equivalent to . And this equivalence holds equally if we substitute any other individual for Napoleon. Thus we have arrived at a predicate which is always equivalent to the property we ascribed to Napoleon, i.e. it belongs to those objects which have this property, and to no others. The axiom of reducibility states that such a predicate always exists, i.e. that any property of an object belongs to the same collection of objects as those that possess some predicate.

We may next illustrate our principle by its application to identity. In this connection, it has a certain affinity with Leibniz's identity of indiscernibles. It is plain that, if and are identical, and is true, then is true. Here it cannot matter what sort of function may be: the statement must hold for any function. But we cannot say, conversely: "If, with all values of , implies , then and are identical"; because "all values of " is inadmissible. If we wish to speak of "all values of ," we must confine ourselves to functions of one order. We may confine to predicates, or to[Pg 60] second-order functions, or to functions of any order we please. But we must necessarily leave out functions of all but one order. Thus we shall obtain, so to speak, a hierarchy of different degrees of identity. We may say "all the predicates of belong to ," "all second-order properties of belong to ," and so on. Each of these statements implies all its predecessors: for example, if all second-order properties of belong to ,

then all predicates of belong to , for to have all the predicates of is a second-order property, and this property belongs to . But we cannot, without the help of an axiom, argue conversely that if all the predicates of belong to , all the second-order properties of must also belong to . Thus we cannot, without the help of an axiom, be sure that and are identical if they have the same predicates. Leibniz's identity of indiscernibles supplied this axiom. It should be observed that by "indiscernibles" he cannot have meant two objects which agree as to all their properties, for one of the properties of is to be identical with , and therefore this property would necessarily belong to if and agreed in all their properties. Some limitation of the common properties necessary to make things indiscernible is therefore implied by the necessity of an axiom. For purposes of illustration (not of interpreting Leibniz) we may suppose the common properties required for indiscernibility to be limited to predicates. Then the identity of indiscernibles will state that if and agree as to all their predicates, they are identical. This can be proved if we assume the axiom of reducibility. For, in that case, every property belongs to the same collection of objects as is defined by some predicate. Hence there is some predicate common and peculiar to the objects which are identical with . This predicate belongs to , since is identical with itself; hence it belongs to , since y has all the predicates of ; hence y is identical with . It follows that we may define and as identical when all the predicates of belong to , i.e. when (. We therefore adopt the following definition of identity[21]:

But apart from the axiom of reducibility, or some axiom equivalent in this connection, we should be compelled to regard identity as indefinable, and to admit (what seems impossible) that two objects may agree in all their predicates without being identical.

The axiom of reducibility is even more essential in the theory of classes. It should be observed, in the first place, that if we assume the existence of classes, the axiom of reducibility can be proved. For in that case, given any function of whatever order, there is a class consisting of just those objects which satisfy . Hence "" is equivalent to " belongs to ." But " belongs to " is a statement containing no apparent variable, and is therefore a predicative function of . Hence if we assume the existence of[Pg 61] classes, the axiom of reducibility becomes unnecessary. The assumption of the axiom of reducibility is therefore a smaller assumption than the assumption that there are classes. This latter assumption has hitherto been made unhesitatingly. However, both on the ground of the contradictions, which require a more complicated treatment if classes are assumed, and on the ground that it is always well to make the smallest assumption required for proving our theorems, we prefer to assume the axiom of reducibility rather than the existence of classes. But in order to explain the use of the axiom in dealing with classes, it is necessary first to explain the theory of classes, which is a topic belonging to Chapter III. We therefore postpone to that Chapter the explanation of the use of our axiom in dealing with classes.

It is worth while to note that all the purposes served by the axiom of reducibility are equally well served if we assume that there is always a function of the th order (where is fixed) which is formally equivalent to , whatever may be the order of . Here we shall mean by "a function of the th order" a function of the th order relative to the

arguments to ; thus if these arguments are absolutely of the th order, we assume the existence of a function formally equivalent to whose absolute order is the th. The axiom of reducibility in the form assumed above takes , but this is not necessary to the use of the axiom. It is also unnecessary that should be the same for different values of ; what is necessary is that should be constant so long as is constant. What is needed is that, where extensional functions of functions are concerned, we should be able to deal with any -function by means of some formally equivalent function of a given type, so as to be able to obtain results which would otherwise require the illegitimate notion of "all -functions"; but it does not matter what the given type is. It does not appear, however, that the axiom of reducibility is rendered appreciably more plausible by being put in the above more general but more complicated form.

The axiom of reducibility is equivalent to the assumption that "any combination or disjunction of predicates[22] is equivalent to a single predicate," i.e. to the assumption that, if we assert that has all the predicates that satisfy a function there is some one predicate which will have whenever our assertion is true, and will not have whenever it is false, and similarly if we assert that has some one of the predicates that satisfy a function). For by means of this assumption, the order of a non-predicative function can be lowered by one; hence, after some finite number of steps, we shall be able to get from any non-predicative function to a formally equivalent predicative function. It does not seem probable that[Pg 62] the above assumption could be substituted for the axiom of reducibility in symbolic deductions, since its use would require the explicit introduction of the further assumption that by a finite number of downward steps we can pass from any function to a predicative function, and this assumption could not well be made without developments that are scarcely possible at an early stage. But on the above grounds it seems plain that in fact, if the above alternative axiom is true, so is the axiom of reducibility. The converse, which completes the proof of equivalence, is of course evident.

VII. Reasons for Accepting the Axiom of Reducibility.

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That the axiom of reducibility is self-evident is a proposition which can hardly be maintained. But in fact self-evidence is never more than a part of the reason for accepting an axiom, and is never indispensable. The reason for accepting an axiom, as for accepting any other proposition, is always largely inductive, namely that many propositions which are nearly indubitable can be deduced from it, and that no equally plausible way is known by which these propositions could be true if the axiom were false, and nothing which is probably false can be deduced from it. If the axiom is apparently self-evident, that only means, practically, that it is nearly indubitable; for things have been thought to be self-evident and have yet turned out to be false. And if the axiom itself is nearly indubitable, that merely adds to the inductive evidence derived from the fact that its consequences are nearly indubitable: it does not provide new evidence of a radically different kind. Infallibility is never attainable, and therefore some element of doubt should always attach to every axiom and to all its consequences. In formal logic, the element of doubt is less than in most sciences, but it is not absent, as appears from the fact that the paradoxes followed from premisses

which were not previously known to require limitations. In the case of the axiom of reducibility, the inductive evidence in its favour is very strong, since the reasonings which it permits and the results to which it leads are all such as appear valid. But although it seems very improbable that the axiom should turn out to be false, it is by no means improbable that it should be found to be deducible from some other more fundamental and more evident axiom. It is possible that the use of the vicious-circle principle, as embodied in the above hierarchy of types, is more drastic than it need be, and that by a less drastic use the necessity for the axiom might be avoided. Such changes, however, would not render anything false which had been asserted on the basis of the principles explained above: they would merely provide easier proofs of the same theorems. There would seem, therefore, to be but the slenderest ground for fearing that the use of the axiom of reducibility may lead us into error.

VIII. The Contradictions.

We are now in a position to show how the theory of types affects the solution of the contradictions which have beset mathematical logic. For this purpose, we shall begin by an enumeration of some of the more important and illustrative of these contradictions, and shall then show how they all embody vicious-circle fallacies, and are therefore all avoided by the theory of types. It will be noticed that these paradoxes do not relate exclusively to the ideas of number and quantity. Accordingly no solution can be adequate which seeks to explain them merely as the result of some illegitimate use of these ideas. The solution must be sought in some such scrutiny of fundamental logical ideas as has been attempted in the foregoing pages.

(1) The oldest contradiction of the kind in question is the Epimenides. Epimenides the Cretan said that all Cretans were liars, and all other statements made by Cretans were certainly lies. Was this a lie? The simplest form of this contradiction is afforded by the man who says "I am lying"; if he is lying, he is speaking the truth, and vice versa.

(2) Let be the class of all those classes which are not members of themselves. Then, whatever class may be, " is a " is equivalent to " is not an ." Hence, giving to the value , " is a " is equivalent to " is not a ."

(3) Let be the relation which subsists between two relations and whenever does not have the relation to . Then, whatever relations and may be, " has the relation to " is equivalent to " does not have the relation to ." Hence, giving the value to both and , " has the relation to " is equivalent to " does not have the relation to ."

(4) Burali-Forti's contradiction[23] may be stated as follows: It can be shown that every well-ordered series has an ordinal number, that the series of ordinals up to and including any given ordinal exceeds the given ordinal by one, and (on certain very natural assumptions) that the series of all ordinals (in order of magnitude) is well-ordered. It follows that the series of all ordinals has an ordinal number, say. But in that case the series of all ordinals including has the ordinal number , which must be greater than . Hence is not the ordinal number of all ordinals.

(5) The number of syllables in the English names of finite integers tends to increase as the integers grow larger, and must gradually increase indefinitely, since

only a finite number of names can be made with a given finite number of syllables. Hence the names of some integers must consist of at least nineteen syllables, and among these there must be a least. Hence "the least integer not nameable in fewer than nineteen syllables"[Pg 64] must denote a definite integer; in fact, it denotes 111,777. But "the least integer not nameable in fewer than nineteen syllables" is itself a name consisting of eighteen syllables; hence the least integer not nameable in fewer than nineteen syllables can be named in eighteen syllables, which is a contradiction[24].

(6) Among transfinite ordinals some can be defined, while others can not; for the total number of possible definitions is [25], while the number of transfinite ordinals exceeds . Hence there must be indefinable ordinals, and among these there must be a least. But this is defined as "the least indefinable ordinal," which is a contradiction[26].

(7) Richard's paradox[27] is akin to that of the least indefinable ordinal. It is as follows: Consider all decimals that can be defined by means of a finite number of words; let be the class of such decimals. Then has terms; hence its members can be ordered as the 1st, 2nd, 3rd,.... Let be a number defined as follows. If the th figure in the th decimal is , let the th figure in be (or 0, if). Then is different from all the members of , since, whatever finite value may have, the th figure in is different from the th figure in the nth of the decimals composing , and therefore is different from the th decimal. Nevertheless we have defined in a finite number of words, and therefore ought to be a member of . Thus both is and is not a member of .

In all the above contradictions (which are merely selections from an indefinite number) there is a common characteristic, which we may describe as self-reference or reflexiveness. The remark of Epimenides must include itself in its own scope. If all classes, provided they are not members of themselves, are members of , this must also apply to ; and similarly for the analogous relational contradiction. In the cases of names and definitions, the paradoxes result from considering non-nameability and indefinability as elements in names and definitions. In the case of Burali-Forti's paradox, the series whose ordinal number causes the difficulty is the series of all ordinal numbers. In each contradiction something is said about all cases of some kind, and from what is said a new case seems to be generated,[Pg 65] which both is and is not of the same kind as the cases of which all were concerned in what was said. But this is the characteristic of illegitimate totalities, as we defined them in stating the vicious-circle principle. Hence all our contradictions are illustrations of vicious-circle fallacies. It only remains to show, therefore, that the illegitimate totalities involved are excluded by the hierarchy of types which we have constructed.

(1) When a man says "I am lying," we may interpret his statement as: "There is a proposition which I am affirming and which is false." That is to say, he is asserting the truth of some value of the function "I assert , and is false." But we saw that the word "false" is ambiguous, and that, in order to make it unambiguous, we must specify the order of falsehood, or, what comes to the same thing, the order of the proposition to which falsehood is ascribed. We saw also that, if is a proposition of the th order, a proposition in which occurs as an apparent variable is not of the th order,

but of a higher order. Hence the kind of truth or falsehood which can belong to the statement "there is a proposition which I am affirming and which has falsehood of the th order" is truth or falsehood of a higher order than the th. Hence the statement of Epimenides does not fall within its own scope, and therefore no contradiction emerges.

If we regard the statement "I am lying" as a compact way of simultaneously making all the following statements: "I am asserting a false proposition of the first order," "I am asserting a false proposition of the second order," and so on, we find the following curious state of things: As no proposition of the first order is being asserted, the statement "I am asserting a false proposition of the first order" is false. This statement is of the second order, hence the statement "I am making a false statement of the second order" is true. This is a statement of the third order, and is the only statement of the third order which is being made. Hence the statement "I am making a false statement of the third order" is false. Thus we see that the statement "I am making a false statement of order " is false, while the statement "I am making a false statement of order " is true. But in this state of things there is no contradiction.

(2) In order to solve the contradiction about the class of classes which are not members of themselves, we shall assume, what will be explained in the next Chapter, that a proposition about a class is always to be reduced to a statement about a function which defines the class, i.e. about a function which is satisfied by the members of the class and by no other arguments. Thus a class is an object derived from a function and presupposing the function, just as, for example, presupposes the function . Hence a class cannot, by the vicious-circle principle, significantly be the argument to its defining function, that is to say, if we denote[Pg 66] by "" the class defined by , the symbol "" must be meaningless. Hence a class neither satisfies nor does not satisfy its defining function, and therefore (as will appear more fully in Chapter III) is neither a member of itself nor not a member of itself. This is an immediate consequence of the limitation to the possible arguments to a function which was explained at the beginning of the present Chapter. Thus if is a class, the statement " is not a member of " is always meaningless, and there is therefore no sense in the phrase "the class of those classes which are not members of themselves." Hence the contradiction which results from supposing that there is such a class disappears.

(3) Exactly similar remarks apply to "the relation which holds between and whenever does not have the relation to ." Suppose the relation is defined by a function , i.e. holds between and whenever is true, but not otherwise. Then in order to interpret " has the relation to " we shall have to suppose that and can significantly be the arguments to . But (assuming, as will appear in Chapter III, that presupposes its defining function) this would require that should be able to take as argument an object which is defined in terms of , and this no function can do, as we saw at the beginning of this Chapter. Hence " has the relation to " is meaningless, and the contradiction ceases.

(4) The solution of Burali-Forti's contradiction requires some further developments for its solution. At this stage, it must suffice to observe that a series is a relation, and

an ordinal number is a class of series. (These statements are justified in the body of the work.) Hence a series of ordinal numbers is a relation between classes of relations, and is of higher type than any of the series which are members of the ordinal numbers in question. Burali-Forti's "ordinal number of all ordinals" must be the ordinal number of all ordinals of a given type, and must therefore be of higher type than any of these ordinals. Hence it is not one of these ordinals, and there is no contradiction in its being greater than any of them[28].

(5) The paradox about "the least integer not nameable in fewer than nineteen syllables" embodies, as is at once obvious, a vicious-circle fallacy. For the word "nameable" refers to the totality of names, and yet is allowed to occur in what professes to be one among names. Hence there can be no such thing as a totality of names, in the sense in which the paradox speaks of "names." It is easy to see that, in virtue of the hierarchy of functions, the theory of types renders a totality of "names" impossible. We may, in fact, distinguish names of different orders as follows: (a) Elementary names will be such as are true "proper names," i.e. conventional[Pg 67] appellations not involving any description. (b) First-order names will be such as involve a description by means of a first-order function; that is to say, if is a first-order function, "the term which satisfies " will be a first-order name, though there will not always be an object named by this name. (c) Second-order names will be such as involve a description by means of a second-order function; among such names will be those involving a reference to the totality of first-order names. And so we can proceed through a whole hierarchy. But at no stage can we give a meaning to the word "nameable" unless we specify the order of names to be employed; and any name in which the phrase "nameable by names of order " occurs is necessarily of a higher order than the th. Thus the paradox disappears.

The solutions of the paradox about the least indefinable ordinal and of Richard's paradox are closely analogous to the above. The notion of "definable," which occurs in both, is nearly the same as "nameable," which occurs in our fifth paradox: "definable" is what "nameable" becomes when elementary names are excluded, i.e. "definable" means "nameable by a name which is not elementary." But here there is the same ambiguity as to type as there was before, and the same need for the addition of words which specify the type to which the definition is to belong. And however the type may be specified, "the least ordinal not definable by definitions of this type" is a definition of a higher type; and in Richard's paradox, when we confine ourselves, as we must, to decimals that have a definition of a given type, the number , which causes the paradox, is found to have a definition which belongs to a higher type, and thus not to come within the scope of our previous definitions.

An indefinite number of other contradictions, of similar nature to the above seven, can easily be manufactured. In all of them, the solution is of the same kind. In all of them, the appearance of contradiction is produced by the presence of some word which has systematic ambiguity of type, such as truth, falsehood, function, property, class, relation, cardinal, ordinal, name, definition. Any such word, if its typical ambiguity is overlooked, will apparently generate a totality containing members defined in terms of itself, and will thus give rise to vicious-circle fallacies. In most

cases, the conclusions of arguments which involve vicious-circle fallacies will not be self-contradictory, but wherever we have an illegitimate totality, a little ingenuity will enable us to construct a vicious-circle fallacy leading to a contradiction, which disappears as soon as the typically ambiguous words are rendered typically definite, i.e. are determined as belonging to this or that type.

[Pg 68]

Thus the appearance of contradiction is always due to the presence of words embodying a concealed typical ambiguity, and the solution of the apparent contradiction lies in bringing the concealed ambiguity to light.

In spite of the contradictions which result from unnoticed typical ambiguity, it is not desirable to avoid words and symbols which have typical ambiguity. Such words and symbols embrace practically all the ideas with which mathematics and mathematical logic are concerned: the systematic ambiguity is the result of a systematic analogy. That is to say, in almost all the reasonings which constitute mathematics and mathematical logic, we are using ideas which may receive any one of an infinite number of different typical determinations, any one of which leaves the reasoning valid. Thus by employing typically ambiguous words and symbols, we are able to make one chain of reasoning applicable to any one of an infinite number of different cases, which would not be possible if we were to forego the use of typically ambiguous words and symbols.

Among propositions wholly expressed in terms of typically ambiguous notions practically the only ones which may differ, in respect of truth or falsehood, according to the typical determination which they receive, are existence-theorems. If we assume that the total number of individuals is , then the total number of classes of individuals is , the total number of classes of classes of individuals is , and so on. Here may be either finite or infinite, and in either case . Thus cardinals greater than but not greater than exist as applied to classes of classes, but not as applied to classes of individuals, so that whatever may be supposed to be the number of individuals, there will be existence-theorems which hold for higher types but not for lower types. Even here, however, so long as the number of individuals is not asserted, but is merely assumed hypothetically, we may replace the type of individuals by any other type, provided we make a corresponding change in all the other types occurring in the same context. That is, we may give the name "relative individuals" to the members of an arbitrarily chosen type , and the name "relative classes of individuals" to classes of "relative individuals," and so on. Thus so long as only hypotheticals are concerned, in which existence-theorems for one type are shown to be implied by existence-theorems for another, only relative types are relevant even in existence-theorems. This applies also to cases where the hypothesis (and therefore the conclusion) is asserted, provided the assertion holds for any type, however chosen. For example, any type has at least one member; hence any type which consists of classes, of whatever order, has at least two members. But the further pursuit of these topics must be left to the body of the work.

FOOTNOTES:

[12] See the last section of the present Chapter. Cf. also H. Poincaré, "Les mathématiques et la logique," Revue de Métaphysique et de Morale, Mai 1906, p. 307.

[13] When the word "function" is used in the sequel, "propositional function" is always meant. Other functions will not be in question in the present Chapter.

[14] We shall speak in this Chapter of "values for " and of "values of ," meaning in each case the same thing, namely , , , etc. The distinction of phraseology serves to avoid ambiguity where several variables are concerned, especially when one of them is a function.

[15] We use "always" as meaning "in all cases," not "at all times." Similarly "sometimes" will mean "in some cases."

[16] See Chapter III.

[17] Note that statements concerning the significance of a phrase containing "" concern the symbol "," and therefore do not fall under the rule that the elimination of the functional ambiguity is necessary to significance. Significance is a property of signs. Cf. p. 43.

[18] Cf. Chapter III.

[19] When we speak of "values of " it is , not , that is to be assigned. This follows from the explanation in the note on p. 42. When the function itself is the variable, it is possible and simpler to write rather than , except in positions where it is necessary to emphasize that an argument must be supplied to secure significance.

[20] When a (finite) set of predicates is given by actual enumeration, their disjunction is a predicate, because no predicate occurs as apparent variable in the disjunction.

[21] Note that in this definition the second sign of equality is to be regarded as combining with "Df" to form one symbol; what is defined is the sign of equality not followed by the letters "Df."

[22] Here the combination or disjunction is supposed to be given intensionally. If given extensionally (i.e. by enumeration), no assumption is required; but in this case the number of predicates concerned must be finite.

[23] "Una questione sui numeri transfiniti," Rendiconti del circolo matematico di Palermo, Vol. XI. (1897). See *256.

[24] This contradiction was suggested to us by Mr G. G. Berry of the Bodleian Library.

[25] is the number of finite integers. See *123.

[26] Cf. König, "Ueber die Grundlagen der Mengenlehre und das Kontinuumproblem," Math. Annalen, Vol. LXI. (1905); A. C. Dixon, "On 'well-ordered' aggregates," Proc. London Math. Soc. Series 2, Vol. IV. Part i. (1906); and E. W. Hobson, "On the Arithmetic Continuum," ibid. The solution offered in the last of these

papers depends upon the variation of the "apparatus of definition," and is thus in outline in agreement with the solution adopted here. But it does not invalidate the statement in the text, if "definition" is given a constant meaning.

[27] Cf. Poincaré, "Les mathématiques et la logique," Revue de Métaphysique et de Morale, Mai 1906, especially sections VII. and IX.; also Peano, Revista de Mathematica, Vol. VIII. No. 5 (1906), p. 149 ff.

[28] The solution of Burali-Forti's paradox by means of the theory of types is given in detail in *256.

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CHAPTER III.

INCOMPLETE SYMBOLS.

(1) Descriptions. By an "incomplete" symbol we mean a symbol which is not supposed to have any meaning in isolation, but is only defined in certain contexts. In ordinary mathematics, for example, and are incomplete symbols: something has to be supplied before we have anything significant. Such symbols have what may be called a "definition in use." Thus if we put

we define the use of , but by itself remains without meaning. This distinguishes such symbols from what (in a generalized sense) we may call proper names: "Socrates," for example, stands for a certain man, and therefore has a meaning by itself, without the need of any context. If we supply a context, as in "Socrates is mortal," these words express a fact of which Socrates himself is a constituent: there is a certain object, namely Socrates, which does have the property of mortality, and this object is a constituent of the complex fact which we assert when we say "Socrates is mortal." But in other cases, this simple analysis fails us. Suppose we say: "The round square does not exist." It seems plain that this is a true proposition, yet we cannot regard it as denying the existence of a certain object called "the round square." For if there were such an object, it would exist: we cannot first assume that there is a certain object, and then proceed to deny that there is such an object. Whenever the grammatical subject of a proposition can be supposed not to exist without rendering the proposition meaningless, it is plain that the grammatical subject is not a proper name, i.e. not a name directly representing some object. Thus in all such cases, the proposition must be capable of being so analysed that what was the grammatical subject shall have disappeared. Thus when we say "the round square does not exist," we may, as a first attempt at such analysis, substitute "it is false that there is an object which is both round and square." Generally, when "the so-and-so" is said not to exist, we have a proposition of the form[29]

[Pg 70] or some equivalent. Here the apparent grammatical subject has completely disappeared; thus in "," is an incomplete symbol.

By an extension of the above argument, it can easily be shown that is always an incomplete symbol. Take, for example, the following proposition: "Scott is the author of Waverley." [Here "the author of Waverley" is .] This proposition expresses an

identity; thus if "the author of Waverley" could be taken as a proper name, and supposed to stand for some object , the proposition would be "Scott is ." But if is any one except Scott, this proposition is false; while if is Scott, the proposition is "Scott is Scott," which is trivial, and plainly different from "Scott is the author of Waverley." Generalizing, we see that the proposition

is one which may be true or may be false, but is never merely trivial, like ; whereas, if were a proper name, would necessarily be either false or the same as the trivial proposition . We may express this by saying that is not a value of the propositional function , from which it follows that is not a value of . But since may be anything, it follows that (is nothing. Hence, since in use it has meaning, it must be an incomplete symbol.

It might be suggested that "Scott is the author of Waverley" asserts that "Scott" and "the author of Waverley" are two names for the same object. But a little reflection will show that this would be a mistake. For if that were the meaning of "Scott is the author of Waverley," what would be required for its truth would be that Scott should have been called the author of Waverley: if he had been so called, the proposition would be true, even if some one else had written Waverley; while if no one called him so, the proposition would be false, even if he had written Waverley. But in fact he was the author of Waverley at a time when no one called him so, and he would not have been the author if every one had called him so but some one else had written Waverley. Thus the proposition "Scott is the author of Waverley" is not a proposition about names, like "Napoleon is Bonaparte"; and this illustrates the sense in which "the author of Waverley" differs from a true proper name.

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Thus all phrases (other than propositions) containing the word the (in the singular) are incomplete symbols: they have a meaning in use, but not in isolation. For "the author of Waverley" cannot mean the same as "Scott," or "Scott is the author of Waverley" would mean the same as "Scott is Scott," which it plainly does not; nor can "the author of Waverley" mean anything other than "Scott," or "Scott is the author of Waverley" would be false. Hence "the author of Waverley" means nothing.

It follows from the above that we must not attempt to define "(," but must define the uses of this symbol, i.e. the propositions in whose symbolic expression it occurs. Now in seeking to define the uses of this symbol, it is important to observe the import of propositions in which it occurs. Take as an illustration: "The author of Waverley was a poet." This implies (1) that Waverley was written, (2) that it was written by one man, and not in collaboration, (3) that the one man who wrote it was a poet. If any one of these fails, the proposition is false. Thus "the author of 'Slawkenburgius on Noses' was a poet" is false, because no such book was ever written; "the author of 'The Maid's Tragedy' was a poet" is false, because this play was written by Beaumont and Fletcher jointly. These two possibilities of falsehood do not arise if we say "Scott was a poet." Thus our interpretation of the uses of must be such as to allow for them. Now taking to replace " wrote Waverley," it is plain that any statement apparently about (requires (1) (and (2) ; here (1) states that at least one object satisfies , while

(2) states that at most one object satisfies . The two together are equivalent to

Thus "" must be part of what is affirmed by any proposition about (. If our proposition is , what is further affirmed is , if . Thus we have

i.e. "the satisfying satisfies " is to mean: "There is an object such that is true when, and only when, is , and is true," or, more exactly: "There is a such that '' is always equivalent to ' is ,' and ." In this, "" has completely disappeared; thus "" is merely symbolic, and does not directly represent an object, as single small Latin letters are assumed to do[30].

The proposition "" is easily shown to be equivalent to "." For, by the definition, it is

i.e. "there is a for which , and this is ," which is equivalent to "." Thus "Scott is the author of Waverley" is equivalent to:

i.e. " wrote Waverley" is true when is Scott and false when is not Scott.

Thus although "" has no meaning by itself, it may be substituted for in any propositional function , and we get a significant proposition, though not a value of .

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When , as above defined, forms part of some other proposition, we shall say that (has a secondary occurrence. When (has a secondary occurrence, a proposition in which it occurs may be true even when (does not exist. This applies, e.g. to the proposition: "There is no such person as the King of France." We may interpret this as

if "" stands for " is King of France." In either case, what is asserted is that a proposition in which (occurs is false, and this proposition is thus part of a larger proposition. The same applies to such a proposition as the following: "If France were a monarchy, the King of France would be of the House of Orleans."

It should be observed that such a proposition as

is ambiguous; it may deny , in which case it will be true if (does not exist, or it may mean

in which case it can only be true if (exists. In ordinary language, the latter interpretation would usually be adopted. For example, the proposition "the King of France is not bald" would usually be rejected as false, being held to mean "the King of France exists and is not bald," rather than "it is false that the King of France exists and is bald." When (exists, the two interpretations of the ambiguity give equivalent results; but when (does not exist, one interpretation is true and one is false. It is necessary to be able to distinguish these in our notation; and generally, if we have such propositions as

[Pg 73]and so on, we must be able by our notation to distinguish whether the whole or only part of the proposition concerned is to be treated as the "" of our definition. For this purpose, we will put "[(]" followed by dots at the beginning of the part (or whole) which is to be taken as , the dots being sufficiently numerous to bracket off the ; i.e. is to be everything following the dots until we reach an equal number of dots not signifying a logical product, or a greater number signifying a

logical product, or the end of the sentence, or the end of a bracket enclosing "[(]." Thus

It is important to distinguish these two, for if (does not exist, the first is true and the second false. Again

Here again, when (does not exist, the first is false and the second true.

In order to avoid this ambiguity in propositions containing (, we amend our definition, or rather our notation, putting

By means of this definition, we avoid any doubt as to the portion of our whole asserted proposition which is to be treated as the "" of the definition. This portion will be called the scope of (. Thus in

the scope of (is ; but in

the scope is ; but in

It will be seen that when (has the whole of the proposition concerned for its scope, the proposition concerned cannot be true unless ; but when (has only part of the proposition concerned for its scope, it may often be true even when (does not exist. It will be seen further that when , we may enlarge or diminish the scope of (as much as we please without altering the truth-value of any proposition in which it occurs.

If a proposition contains two descriptions, say (and (, we have to distinguish which of them has the larger scope, i.e. we have to distinguish

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The first of these, eliminating (, becomes

which, eliminating (becomes

and the same proposition results if, in (1), we eliminate first (and then (. Similarly (2) becomes, when (and (are eliminated,

(4) and (5) are equivalent, so that the truth-value of a proposition containing two descriptions is independent of the question which has the larger scope.

It will be found that, in most cases in which descriptions occur, their scope is, in practice, the smallest proposition enclosed in dots or other brackets in which they are contained. Thus for example

will occur much more frequently than

For this reason it is convenient to decide that, when the scope of an occurrence of (is the smallest proposition, enclosed in dots or other brackets, in which the occurrence in question is contained, the scope need not be indicated by "[(]." Thus e.g.

This convention enables us, in the vast majority of cases that actually occur, to dispense with the explicit indication of the scope of a descriptive symbol; and it will be found that the convention agrees very closely with the tacit conventions of ordinary language on this subject. Thus for example, if "(" is "the so-and-so," "" is to

be read " is not the so-and-so," which would ordinarily be regarded as implying that "the so-and-so" exists; but "" is to be read "it is not true that is the so-and-so," which would generally be allowed to hold if "the so-and-so" does not exist. Ordinary language is, of course, rather loose and fluctuating in its implications on this matter; but subject to the requirement of definiteness, our convention seems to keep as near to ordinary language as possible.

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In the case when the smallest proposition enclosed in dots or other brackets contains two or more descriptions, we shall assume, in the absence of any indication to the contrary, that one which typographically occurs earlier has a larger scope than one which typographically occurs later. Thus

These two propositions are easily shown to be equivalent.

(2) Classes. The symbols for classes, like those for descriptions, are, in our system, incomplete symbols: their uses are defined, but they themselves are not assumed to mean anything at all. That is to say, the uses of such symbols are so defined that, when the definiens is substituted for the definiendum, there no longer remains any symbol which could be supposed to represent a class. Thus classes, so far as we introduce them, are merely symbolic or linguistic conveniences, not genuine objects as their members are if they are individuals.

It is an old dispute whether formal logic should concern itself mainly with intensions or with extensions. In general, logicians whose training was mainly philosophical have decided for intensions, while those whose training was mainly mathematical have decided for extensions. The facts seem to be that, while mathematical logic requires extensions, philosophical logic refuses to supply anything except intensions. Our theory of classes recognizes and reconciles these two apparently opposite facts, by showing that an extension (which is the same as a class) is an incomplete symbol, whose use always acquires its meaning through a reference to intension.

In the case of descriptions, it was possible to prove that they are incomplete symbols. In the case of classes, we do not know of any equally definite proof, though arguments of more or less cogency can be elicited from the ancient problem of the One and the Many[31]. It is not necessary for our purposes, however, to assert dogmatically that there are no such things as classes. It is only necessary for us to show that the incomplete symbols which we introduce as representatives of classes yield all the propositions for the sake of which classes might be thought essential. When this has been shown, the mere principle of economy of primitive ideas leads to the non-introduction of classes except as incomplete symbols.

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To explain the theory of classes, it is necessary first to explain the distinction between extensional and intensional functions. This is effected by the following definitions:

The truth-value of a proposition is truth if it is true, and falsehood if it is false. (This expression is due to Frege.)

Two propositions are said to be equivalent when they have the same truth-value, i.e. when they are both true or both false.

Two propositional functions are said to be formally equivalent when they are equivalent with every possible argument, i.e. when any argument which satisfies the one satisfies the other, and vice versa. Thus " is a man" is formally equivalent to " is a featherless biped"; " is an even prime" is formally equivalent to " is identical with 2."

A function of a function is called extensional when its truth-value with any argument is the same as with any formally equivalent argument. That is to say, is an extensional function of if, provided is formally equivalent to , is equivalent to . Here the apparent variables and are necessarily of the type from which arguments can significantly be supplied to . We find no need to use as apparent variables any functions of non-predicative types; accordingly in the sequel all extensional functions considered are in fact functions of predicative functions[32].

A function of a function is called intensional when it is not extensional.

The nature and importance of the distinction between intensional and extensional functions will be made clearer by some illustrations. The proposition "' is a man' always implies ' is a mortal'" is an extensional function of the function " is a man," because we may substitute, for " is a man," " is a featherless biped," or any other statement which applies to the same objects to which " is a man" applies, and to no others. But the proposition " believes that ' is a man' always implies ' is a mortal'" is an intensional function of " is a man," because may never have considered the question whether featherless bipeds are mortal, or may believe wrongly that there are featherless bipeds which are not mortal. Thus even if " is a featherless biped" is formally equivalent to " is a man," it by no means follows that a person who believes that all men are mortal must believe that all featherless bipeds are mortal, since he may have never thought about featherless bipeds, or have supposed that featherless bipeds were not always men. Again the proposition "the number of arguments that satisfy the function is " is an extensional function of , because its truth or falsehood is unchanged if we substitute for any other function which is true whenever is true, and false whenever is false. But the proposition " asserts that the number of arguments satisfying is n" is an intensional function of ,[Pg 77] since, if asserts this concerning , he certainly cannot assert it concerning all predicative functions that are equivalent to , because life is too short. Again, consider the proposition "two white men claim to have reached the North Pole." This proposition states "two arguments satisfy the function ' is a white man who claims to have reached the North Pole.'" The truth or falsehood of this proposition is unaffected if we substitute for " is a white man who claims to have reached the North Pole" any other statement which holds of the same arguments, and of no others. Hence it is an extensional function. But the proposition "it is a strange coincidence that two white men should claim to have reached the North Pole," which states "it is a strange coincidence that two arguments should satisfy the function ' is a white man who claims to have reached the North Pole,'" is

not equivalent to "it is a strange coincidence that two arguments should satisfy the function ' is Dr Cook or Commander Peary.'" Thus "it is a strange coincidence that should be satisfied by two arguments" is an intensional function of .

The above instances illustrate the fact that the functions of functions with which mathematics is specially concerned are extensional, and that intensional functions of functions only occur where non-mathematical ideas are introduced, such as what somebody believes or affirms, or the emotions aroused by some fact. Hence it is natural, in a mathematical logic, to lay special stress on extensional functions of functions.

When two functions are formally equivalent, we may say that they have the same extension. In this definition, we are in close agreement with usage. We do not assume that there is such a thing as an extension: we merely define the whole phrase "having the same extension." We may now say that an extensional function of a function is one whose truth or falsehood depends only upon the extension of its argument. In such a case, it is convenient to regard the statement concerned as being about the extension. Since extensional functions are many and important, it is natural to regard the extension as an object, called a class, which is supposed to be the subject of all the equivalent statements about various formally equivalent functions. Thus e.g. if we say "there were twelve Apostles," it is natural to regard this statement as attributing the property of being twelve to a certain collection of men, namely those who were Apostles, rather than as attributing the property of being satisfied by twelve arguments to the function " was an Apostle." This view is encouraged by the feeling that there is something which is identical in the case of two functions which "have the same extension." And if we take such simple problems as "how many combinations can be made of things?" it seems at first sight necessary that each "combination" should be a single object which can be counted as one. This, however, is certainly not necessary technically, and we see no reason to suppose that it is true[Pg 78] philosophically. The technical procedure by which the apparent difficulty is overcome is as follows.

We have seen that an extensional function of a function may be regarded as a function of the class determined by the argument-function, but that an intensional function cannot be so regarded. In order to obviate the necessity of giving different treatment to intensional and extensional functions of functions, we construct an extensional function derived from any function of a predicative function , and having the property of being equivalent to the function from which it is derived, provided this function is extensional, as well as the property of being significant (by the help of the systematic ambiguity of equivalence) with any argument whose arguments are of the same type as those of . The derived function, written "," is defined as follows: Given a function), our derived function is to be "there is a predicative function which is formally equivalent to and satisfies ." If is a predicative function, our derived function will be true whenever) is true. If) is an extensional function, and is a predicative function, our derived function will not be true unless) is true; thus in this case, our derived function is equivalent to). If) is not an extensional function, and if is a predicative function, our derived function may sometimes be true when the

original function is false. But in any case the derived function is always extensional.

In order that the derived function should be significant for any function , of whatever order, provided it takes arguments of the right type, it is necessary and sufficient that) should be significant, where is any predicative function. The reason of this is that we only require, concerning an argument , the hypothesis that it is formally equivalent to some predicative function , and formal equivalence has the same kind of systematic ambiguity as to type that belongs to truth and falsehood, and can therefore hold between functions of any two different orders, provided the functions take arguments of the same type. Thus by means of our derived function we have not merely provided extensional functions everywhere in place of intensional functions, but we have practically removed the necessity for considering differences of type among functions whose arguments are of the same type. This effects the same kind of simplification in our hierarchy as would result from never considering any but predicative functions.

If) can be built up by means of the primitive ideas of disjunction, negation, (, and (, as is the case with all the functions of functions that explicitly occur in the present work, it will be found that, in virtue of the systematic ambiguity of the above primitive ideas, any function whose arguments are of the same type as those of can significantly be substituted for in f without any other symbolic change. Thus in[Pg 79] such a case what is symbolically, though not really, the same function can receive as arguments functions of various different types. If, with a given argument , the function), so interpreted, is equivalent to) whenever is formally equivalent to , then is equivalent to) provided there is any predicative function formally equivalent to . At this point, we make use of the axiom of reducibility, according to which there always is a predicative function formally equivalent to .

As was explained above, it is convenient to regard an extensional function of a function as having for its argument not the function, but the class determined by the function. Now we have seen that our derived function is always extensional. Hence if our original function was), we write the derived function , where ")" may be read "the class of arguments which satisfy ," or more simply "the class determined by ." Thus "" will mean: "There is a predicative function which is formally equivalent to and is such that) is true." This is in reality a function of , but we treat it symbolically as if it had an argument). By the help of the axiom of reducibility, we find that the usual properties of classes result. For example, two formally equivalent functions determine the same class, and conversely, two functions which determine the same class are formally equivalent. Also to say that is a member of), i.e. of the class determined by , is true when is true, and false when is false. Thus all the mathematical purposes for which classes might seem to be required are fulfilled by the purely symbolic objects) provided we assume the axiom of reducibility.

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In virtue of the axiom of reducibility, if is any function, there is a formally equivalent predicative function ; then the class) is identical with the class), so that every class can be defined by a predicative function. Hence the totality of the classes

to which a given term can be significantly said to belong or not to belong is a legitimate totality, although the totality of functions which a given term can be significantly said to satisfy or not to satisfy is not a legitimate totality. The classes to which a given term belongs or does not belong are the classes defined by -functions; they are also the classes defined by predicative -functions. Let us call them -classes. Then "-classes" form a legitimate totality, derived from that of predicative -functions. Hence many kinds of general statements become possible which would otherwise involve vicious-circle paradoxes. These general statements are none of them such as lead to contradictions, and many of them such as it is very hard to suppose illegitimate. The fact that they are rendered possible by the axiom of reducibility, and that they would otherwise be excluded by the vicious-circle principle, is to be regarded as an argument in favour of the axiom of reducibility.

The above definition of "the class defined by the function ," or rather, of any proposition in which this phrase occurs, is, in symbols, as follows:

In order to recommend this definition, we shall enumerate five requisites which a definition of classes must satisfy, and we shall then show that the above definition satisfies these five requisites.

We require of classes, if they are to serve the purposes for which they are commonly employed, that they shall have certain properties, which may be enumerated as follows. (1) Every propositional function must determine a class, which may be regarded as the collection of all the arguments satisfying the function in question. This principle must hold when the function is satisfied by an infinite number of arguments as well as when it is satisfied by a finite number. It must hold also when no arguments satisfy the function; i.e. the "null-class" must be just as good a class as any other. (2) Two propositional functions which are formally equivalent, i.e. such that any argument which satisfies either satisfies the other, must determine the same class; that is to say, a class must be something wholly determined by its membership, so that e.g. the class "featherless bipeds" is identical with the class "men," and the class "even primes" is identical with the class "numbers identical with 2." (3) Conversely, two propositional functions which determine the same class must be formally equivalent; in other words, when the class is given, the membership is determinate: two different sets of objects cannot yield the same class. (4) In the same sense in which there are classes (whatever this sense may be), or in some closely analogous sense, there must also be classes of classes. Thus for example "the combinations of things at a time," where the things form a given class, is a class of classes; each combination of things is a class, and each such class is a member of the specified set of combinations, which set is therefore a class whose members are classes. Again, the class of unit classes, or of couples, is absolutely indispensable; the former is the number 1, the latter the number 2. Thus without classes of classes, arithmetic becomes impossible. (5) It must under all circumstances be meaningless to suppose a class identical with one of its own members. For if such a supposition had any meaning, "" would be a significant propositional function[33], and so would "." Hence, by (1) and (4), there would be a class of all classes satisfying the function "." If we call this class , we shall have

Since, by our hypothesis, "" is supposed significant, the above equivalence, which holds with all possible values of , holds with the value , i.e. [Pg 81] But this is a contradiction[34]. Hence "" and "" must always be meaningless. In general, there is nothing surprising about this conclusion, but it has two consequences which deserve special notice. In the first place, a class consisting of only one member must not be identical with that one member, i.e. we must not have . For we have , and therefore, if , we have , which, we saw, must be meaningless. It follows that "" must be absolutely meaningless, not simply false. In the second place, it might appear as if the class of all classes were a class, i.e. as if (writing "Cls" for "class") "" were a true proposition. But this combination of symbols must be meaningless; unless, indeed, an ambiguity exists in the meaning of "Cls," so that, in "," the first "Cls" can be supposed to have a different meaning from the second.

As regards the above requisites, it is plain, to begin with, that, in accordance with our definition, every propositional function determines a class). Assuming the axiom of reducibility, there must always be true propositions about), i.e. true propositions of the form . For suppose is formally equivalent to , and suppose satisfies some function . Then) also satisfies . Hence, given any function , there are true propositions of the form , i.e. true propositions in which "the class determined by " is grammatically the subject. This shows that our definition fulfils the first of our five requisites.

The second and third requisites together demand that the classes) and) should be identical when, and only when, their defining functions are formally equivalent, i.e. that we should have

Here the meaning of " = " is to be derived, by means of a two-fold application of the definition of , from the definition of

by the general definition of identity.

In interpreting "," we will adopt the convention which we adopted in regard to () and (, namely that the incomplete symbol which occurs first is to have the larger scope. Thus becomes, by our definition,

which, by eliminating), becomes

which is equivalent to [Pg 82] which, again, is equivalent to

which, in virtue of the axiom of reducibility, is equivalent to

Thus our definition of the use of is such as to satisfy the conditions (2) and (3) which we laid down for classes, i.e. we have

Before considering classes of classes, it will be well to define membership of a class, i.e. to define the symbol "," which may be read " is a member of the class determined by ." Since this is a function of the form , it must be derived, by means of our general definition of such functions, from the corresponding function . We therefore put

This definition is only needed in order to give a meaning to ""; the meaning it gives is, in virtue of the definition of ,

It thus appears that "" implies , since it implies , and is equivalent to ; also, in virtue of the axiom of reducibility, implies "," since there is a predicative function formally equivalent to , and must satisfy , since (ex hypothesi) satisfies . Thus in virtue of the axiom of reducibility we have

i.e. is a member of the class when, and only when, satisfies the function which defines the class.

We have next to consider how to interpret a class of classes. As we have defined , we shall naturally regard a class of classes as consisting of those values of which satisfy . Let us write for ; then we may write for the class of values of which satisfy [35]. We shall apply the same definition, and put

where "" stands for any expression of the form .

Let us take "" as an instance of . Then

Just as we put

so we put

Thus we find

[Pg 83]

If we now extend the axiom of reducibility so as to apply to functions of functions, i.e. if we assume

we easily deduce

Thus every function which can take classes as arguments, i.e. every function of functions, determines a class of classes, whose members are those classes which satisfy the determining function. Thus the theory of classes of classes offers no difficulty.

We have next to consider our fifth requisite, namely that ")" is to be meaningless. Applying our definition of , we find that if this collection of symbols had a meaning, it would mean

i.e. in virtue of the definition

But here the symbol "" occurs, which assigns a function as argument to itself. Such a symbol is always meaningless, for the reasons explained at the beginning of Chapter II (pp. 41-43). Hence "" is meaningless, and our fifth and last requisite is fulfilled.

As in the case of), so in that of , there is an ambiguity as to the scope of if it occurs in a proposition which itself is part of a larger proposition. But in the case of classes, since we always have the axiom of reducibility, namely

which takes the place of , it follows that the truth-value of any proposition in which occurs is the same whatever scope we may give to , provided the proposition is an extensional function of whatever functions it may contain. Hence we may adopt the convention that the scope is to be always the smallest proposition enclosed in dots or brackets in which occurs. If at any time a larger scope is required, we may indicate it

by "[]" followed by dots, in the same way as we did for .

[Pg 84]

Similarly when two class symbols occur, e.g. in a proposition of the form , we need not remember rules for the scopes of the two symbols, since all choices give equivalent results, as it is easy to prove. For the preliminary propositions a rule is desirable, so we can decide that the class symbol which occurs first in the order of writing is to have the larger scope.

The representation of a class by a single letter can now be understood. For the denotation of is ambiguous, in so far as it is undecided as to which of the symbols , , , etc. it is to stand for, where , , , etc. are the various determining functions of the class. According to the choice made, different propositions result. But all the resulting propositions are equivalent by virtue of the easily proved proposition:

Hence unless we wish to discuss the determining function itself, so that the notion of a class is really not properly present, the ambiguity in the denotation of is entirely immaterial, though, as we shall see immediately, we are led to limit ourselves to predicative determining functions. Thus "," where is a variable class, is really "," where is a variable function, that is, it is

where is a variable function. But here a difficulty arises which is removed by a limitation to our practice and by the axiom of reducibility. For the determining functions , , etc. will be of different types, though the axiom of reducibility secures that some are predicative functions. Then, in interpreting as a variable in terms of the variation of any determining function, we shall be led into errors unless we confine ourselves to predicative determining functions. These errors especially arise in the transition to total variation (cf. pp. 15, 16). Accordingly

It is the peculiarity of a definition of the use of a single letter [viz.] for a variable incomplete symbol that it, though in a sense a real variable, occurs only in the definiendum, while "," though a real variable, occurs only in the definiens.

Thus "" stands for

and "(" stands for

Accordingly, in mathematical reasoning, we can dismiss the whole apparatus of functions and think only of classes as "quasi-things," capable of immediate representation by a single name. The advantages are two-fold: (1) classes are determined by their membership, so that to one set of members there is one class, (2) the "type" of a class is entirely defined by the type of its members.

Also a predicative function of a class can be defined thus

Thus a predicative function of a class is always a predicative function of any predicative determining function of the class, though the converse does not hold.

[Pg 85]

(3) Relations. With regard to relations, we have a theory strictly analogous to that which we have just explained as regards classes. Relations in extension, like classes,

are incomplete symbols. We require a division of functions of two variables into predicative and non-predicative functions, again for reasons which have been explained in Chapter II. We use the notation "" for a predicative function of and .

We use "" for the function as opposed to its values; and we use "" for the relation (in extension) determined by . We put

Thus even when is not an extensional function of , is an extensional function of . Hence, just as in the case of classes, we deduce

i.e. a relation is determined by its extension, and vice versa.

On the analogy of the definition of "," we put[36]

This definition, like that of "," is not introduced for its own sake, but in order to give a meaning to

This meaning, in virtue of our definitions, is

and this, in virtue of the axiom of reducibility

is equivalent to .

Thus we have always

Whenever the determining function of a relation is not relevant, we may replace by a single capital letter. In virtue of the propositions given above,

Classes of relations, and relations of relations, can be dealt with as classes of classes were dealt with above.

[Pg 86]

Just as a class must not be capable of being or not being a member of itself, so a relation must neither be nor not be referent or relatum with respect to itself. This turns out to be equivalent to the assertion that cannot significantly be either of the arguments or in . This principle, again, results from the limitation to the possible arguments to a function explained at the beginning of Chapter II.

We may sum up this whole discussion on incomplete symbols as follows.

The use of the symbol "(" as if in "" it directly represented an argument to the function is rendered possible by the theorems

The use of the symbol "" (or of a single letter, such as , to represent such a symbol) as if, in "," it directly represented an argument to a function , is rendered possible by the theorems

Throughout these propositions the types must be supposed to be properly adjusted, where ambiguity is possible.

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[Pg 87]

It follows from these three groups of theorems that these incomplete symbols are obedient to the same formal rules of identity as symbols which directly represent objects, so long as we only consider the equivalence of the resulting variable (or constant) values of propositional functions and not their identity. This consideration of the identity of propositions never enters into our formal reasoning.

Similarly the limitations to the use of these symbols can be summed up as follows. In the case of (, the chief way in which its incompleteness is relevant is that we do not always have

i.e. a function which is always true may nevertheless not be true of (. This is possible because is not a value of , so that even when all values of are true, may not be true. This happens when (does not exist. Thus for example we have (, but we do not have

is only valid when . As soon as we know , the fact that () is an incomplete symbol becomes irrelevant so long as we confine ourselves to truth-functions[37] of whatever proposition is its scope. But even when , the incompleteness of (may be relevant when we pass outside truth-functions. For example, George IV wished to know whether Scott was the author of Waverley, i.e. he wished to know whether a proposition of the form "" was true. But there was no proposition of the form "" concerning which he wished to know if it was true.

In regard to classes, the relevance of their incompleteness is somewhat different. It may be illustrated by the fact that we may have

For, by a direct application of the definitions, we find that

Thus we shall have

but we shall not necessarily have under these circumstances, for two functions may well be formally equivalent without being identical; for example, [Pg 88] but the function "=the author of Waverley" has the property that George IV wished to know whether its value with the argument "Scott" was true, whereas the function "=Scott" has no such property, and therefore the two functions are not identical. Hence there is a propositional function, namely

which holds without any exception, and yet does not hold when for we substitute a class, and for and we substitute functions. This is only possible because a class is an incomplete symbol, and therefore "" is not a value of "."

It will be observed that "" is not an extensional function of . Thus the scope of) is relevant in interpreting the product

If we take the whole of the product as the scope of , the product is equivalent to

[Pg 89]

We may say generally that the fact that) is an incomplete symbol is not relevant so long as we confine ourselves to extensional functions of functions, but is apt to become relevant for other functions of functions.

FOOTNOTES:

[29] Cf. pp. 31, 32.

[30] We shall generally write "" rather than "" in future.

[31] Briefly, these arguments reduce to the following: If there is such an object as a class, it must be in some sense one object. Yet it is only of classes that many can be predicated. Hence, if we admit classes as objects, we must suppose that the same object can be both one and many, which seems impossible.

[32] Cf. p. 56.

[33] As explained in Chapter I (pp. 25, 26), "" means " is a member of the class ," or, more shortly, " is an ." The definition of this expression in terms of our theory of classes will be given shortly.

[34] This is the second of the contradictions discussed at the end of Chapter II.

[35] The use of a single letter, such as or , to represent a variable class, will be further explained shortly.

[36] This definition raises certain questions as to the two senses of a relation, which are dealt with in *21.

[37] Cf. p. 8.

PART I.

MATHEMATICAL LOGIC.

[Pg 91]

SUMMARY OF PART I.

IN this Part, we shall deal with such topics as belong traditionally to symbolic logic, or deserve to belong to it in virtue of their generality. We shall, that is to say, establish such properties of propositions, propositional functions, classes and relations as are likely to be required in any mathematical reasoning, and not merely in this or that branch of mathematics.

The subjects treated in Part I may be viewed in two aspects: (1) as a deductive chain depending on the primitive propositions, (2) as a formal calculus. Taking the first view first: We begin, in *1, with certain axioms as to deduction of one proposition or asserted propositional function from another. From these primitive propositions, in Section A, we deduce various propositions which are all concerned with four ways of obtaining new propositions from given propositions, namely negation, disjunction, joint assertion and implication, of which the last two can be defined in terms of the

first two. Throughout this first section, although, as will be shown at the beginning of Section B, our propositions, symbolically unchanged, will apply to any propositions as values of our variables, yet it will be supposed that our variable propositions are all what we shall call elementary propositions, i.e. such as contain no reference, explicit or implicit, to any totality. This restriction is imposed on account of the distinction between different types of propositions, explained in Chapter II of the Introduction. Its importance and purpose, however, are purely philosophical, and so long as only mathematical purposes are considered, it is unnecessary to remember this preliminary restriction to elementary propositions, which is symbolically removed at the beginning of the next section.

Section B deals, to begin with, with the relations of propositions containing apparent variables (i.e. involving the notions of "all" or "some") to each other and to propositions not containing apparent variables. We show that, where propositions containing apparent variables are concerned, we can define negation, disjunction, joint assertion and implication in such a way that their properties shall be exactly analogous to the properties of the corresponding ideas as applied to elementary propositions. We show also that formal implication, i.e. "" considered as a relation of to , has many properties analogous to those of material implication, i.e. ""[Pg 92] considered as a relation of and . We then consider predicative functions and the axiom of reducibility, which are vital in the employment of functions as apparent variables. An example of such employment is afforded by identity, which is the next topic considered in Section B. Finally, this section deals with descriptions, i.e. phrases of the form "the so-and-so" (in the singular). It is shown that the appearance of a grammatical subject "the so-and-so" is deceptive, and that such propositions, fully stated, contain no such subject, but contain instead an apparent variable.

Section C deals with classes, and with relations in so far as they are analogous to classes. Classes and relations, like descriptions, are shown to be "incomplete symbols" (cf. Introduction, Chapter III), and it is shown that a proposition which is grammatically about a class is to be regarded as really concerned with a propositional function and an apparent variable whose values are predicative propositional functions (with a similar result for relations). The remainder of Section C deals with the calculus of classes, and with the calculus of relations in so far as it is analogous to that of classes.

Section D deals with those properties of relations which have no analogues for classes. In this section, a number of ideas and notations are introduced which are constantly needed throughout the rest of the work. Most of the properties of relations which have analogues in the theory of classes are comparatively unimportant, while those that have no such analogues are of the very greatest utility. It is partly for this reason that emphasis on the calculus-aspect of symbolic logic has proved a hindrance, hitherto, to the proper development of the theory of relations.

Section E, finally, extends the notions of the addition and multiplication of classes or relations to cases where the summands or factors are not individually given, but are given as the members of some class. The advantage obtained by this extension is that it enables us to deal with an infinite number of summands or factors.

Considered as a formal calculus, mathematical logic has three analogous branches, namely (1) the calculus of propositions, (2) the calculus of classes, (3) the calculus of relations. Of these, (1) is dealt with in Section A, while (2) and (3), in so far as they are analogous, are dealt with in Section C. We have, for each of the three, the four analogous ideas of negation, addition, multiplication, and implication or inclusion. Of these, negation is analogous to the negative in ordinary algebra, and implication or inclusion is analogous to the relation "less than or equal to" in ordinary algebra. But the analogy must not be pressed, as it has important limitations. The sum of two propositions is their disjunction, the sum of two classes is the class of terms belonging to one or other, the sum of two relations is the relation consisting in the fact that one or other of the two relations holds. The sum of a class of classes is the class of all terms belonging to some one or other of the classes, and the sum of a class of relations is the relation consisting in the fact that some one relation of the class holds. The product of two propositions is their joint assertion, the product of two classes is their common part, the product of two relations is the relation consisting in the fact that both the relations hold. The product of a class of classes is the part common to all of them, and the product of a class of relations is the relation consisting in the fact that all relations of the class in question hold. The inclusion of one class in another consists in the fact that all members of the one are members of the other, while the inclusion of one relation in another consists in the fact that every pair of terms which has the one relation also has the other relation. It is then shown that the properties of negation, addition, multiplication and inclusion are exactly analogous for classes and relations, and are, with certain exceptions, analogous to the properties of negation, addition, multiplication and implication for propositions. (The exceptions arise chiefly from the fact that " implies " is itself a proposition, and can therefore imply and be implied, while " is contained in ," where and are classes, is not a class, and can therefore neither contain nor be contained in another class .) But classes have certain properties not possessed by propositions: these arise from the fact that classes have not a two-fold division corresponding to the division of propositions into true and false, but a threefold division, namely into (1) the universal class, which contains the whole of a certain type, (2) the null-class, which has no members, (3) all other classes, which neither contain nothing nor contain everything of the appropriate type. The resulting properties of classes, which are not analogous to properties of propositions, are dealt with in *24. And just as classes have properties not analogous to any properties of propositions, so relations have properties not analogous to any properties of classes, though all the properties of classes have analogues among relations. The special properties of relations are much more numerous and important than the properties belonging to classes but not to propositions. These special properties of relations therefore occupy a whole section, namely Section D.

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SECTION A.

THE THEORY OF DEDUCTION.

THE purpose of the present section is to set forth the first stage of the deduction of pure mathematics from its logical foundations. This first stage is necessarily concerned with deduction itself, i.e. with the principles by which conclusions are inferred from premisses. If it is our purpose to make all our assumptions explicit, and to effect the deduction of all our other propositions from these assumptions, it is obvious that the first assumptions we need are those that are required to make deduction possible. Symbolic logic is often regarded as consisting of two coordinate parts, the theory of classes and the theory of propositions. But from our point of view these two parts are not coordinate; for in the theory of classes we deduce one proposition from another by means of principles belonging to the theory of propositions, whereas in the theory of propositions we nowhere require the theory of classes. Hence, in a deductive system, the theory of propositions necessarily precedes the theory of classes.

But the subject to be treated in what follows is not quite properly described as the theory of propositions. It is in fact the theory of how one proposition can be inferred from another. Now in order that one proposition may be inferred from another, it is necessary that the two should have that relation which makes the one a consequence of the other. When a proposition is a consequence of a proposition , we say that implies . Thus deduction depends upon the relation of implication, and every deductive system must contain among its premisses as many of the properties of implication as are necessary to legitimate the ordinary procedure of deduction. In the present section, certain propositions will be stated as premisses, and it will be shown that they are sufficient for all common forms of inference. It will not be shown that they are all necessary, and it is possible that the number of them might be diminished. All that is affirmed concerning the premisses is (1) that they are true, (2) that they are sufficient for the theory of deduction, (3) that we do not know how to diminish their number. But with regard to (2), there must always be some element of doubt, since it is hard to be sure that one never uses some principle unconsciously. The habit of being rigidly guided by formal symbolic rules is a safeguard against unconscious assumptions; but even this safeguard is not always adequate.

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1. PRIMITIVE IDEAS AND PROPOSITIONS.

Since all definitions of terms are effected by means of other terms, every system of definitions which is not circular must start from a certain apparatus of undefined terms. It is to some extent optional what ideas we take as undefined in mathematics; the motives guiding our choice will be (1) to make the number of undefined ideas as small as possible, (2) as between two systems in which the number is equal, to choose the one which seems the simpler and easier. We know no way of proving that such and such a system of undefined ideas contains as few as will give such and such results[38]. Hence we can only say that such and such ideas are undefined in such and such a system, not that they are indefinable. Following Peano, we shall call the undefined ideas and the undemonstrated propositions primitive ideas and primitive propositions respectively. The primitive ideas are explained by means of descriptions intended to point out to the reader what is meant; but the explanations do not

constitute definitions, because they really involve the ideas they explain.

In the present number, we shall first enumerate the primitive ideas required in this section; then we shall define implication; and then we shall enunciate the primitive propositions required in this section. Every definition or proposition in the work has a number, for purposes of reference. Following Peano, we use numbers having a decimal as well as an integral part, in order to be able to insert new propositions between any two. A change in the integral part of the number will be used to correspond to a new chapter. Definitions will generally have numbers whose decimal part is less than ·1, and will be usually put at the beginning of chapters. In references, the integral parts of the numbers of propositions will be distinguished by being preceded by a star; thus "*1·01" will mean the definition or proposition so numbered, and "*1" will mean the chapter in which propositions have numbers whose integral part is 1, i.e. the present chapter. Chapters will generally be called "numbers."

PRIMITIVE IDEAS.

(1) Elementary propositions. By an "elementary" proposition we mean one which does not involve any variables, or, in other language, one which does not involve such words as "all," "some," "the" or equivalents for such words. A proposition such as "this is red," where "this" is something given[Pg 96] in sensation, will be elementary. Any combination of given elementary propositions by means of negation, disjunction or conjunction (see below) will be elementary. In the primitive propositions of the present number, and therefore in the deductions from these primitive propositions in *2—*5, the letters , , , will be used to denote elementary propositions.

(2) Elementary propositional functions. By an "elementary propositional function" we shall mean an expression containing an undetermined constituent, i.e. a variable, or several such constituents, and such that, when the undetermined constituent or constituents are determined, i.e. when values are assigned to the variable or variables, the resulting value of the expression in question is an elementary proposition. Thus if is an undetermined elementary proposition, "not-" is an elementary propositional function.

We shall show in *9 how to extend the results of this and the following numbers (*1—*5) to propositions which are not elementary.

(3) Assertion. Any proposition may be either asserted or merely considered. If I say "Caesar died," I assert the proposition "Caesar died," if I say "'Caesar died' is a proposition," I make a different assertion, and "Caesar died" is no longer asserted, but merely considered. Similarly in a hypothetical proposition, e.g. "if , then ," we have two unasserted propositions, namely "" and "," while what is asserted is that the first of these implies the second. In language, we indicate when a proposition is merely considered by "if so-and-so" or "that so-and-so" or merely by inverted commas. In symbols, if is a proposition, by itself will stand for the unasserted proposition, while the asserted proposition will be designated by

The sign "" is called the assertion-sign[39]; it may be read "it is true that" (although philosophically this is not exactly what it means). The dots after the

assertion-sign indicate its range; that is to say, everything following is asserted until we reach either an equal number of dots preceding a sign of implication or the end of the sentence. Thus "" means "it is true that implies ," whereas "" means " is true; therefore is true[40]." The first of these does not necessarily involve the truth either of or of , while the second involves the truth of both.

(4) Assertion of a propositional function. Besides the assertion of definite propositions, we need what we shall call "assertion of a propositional function." The general notion of asserting any propositional function is not used until *9, but we use at once the notion of asserting various special elementary propositional functions. Let be a propositional function whose argument is ; then we may assert without assigning a value to .[Pg 97] This is done, for example, when the law of identity is asserted in the form " is ." Here is left undetermined, because, however may be determined, the result will be true. Thus when we assert , leaving undetermined, we are asserting an ambiguous value of our function. This is only legitimate if, however the ambiguity may be determined, the result will be true. Thus take, as an illustration, the primitive proposition *1·2 below, namely

i.e. "' or ' implies ." Here p may be any elementary proposition: by leaving undetermined, we obtain an assertion which can be applied to any particular elementary proposition. Such assertions are like the particular enunciations in Euclid: when it is said "let be an isosceles triangle; then the angles at the base will be equal," what is said applies to any isosceles triangle; it is stated concerning one triangle, but not concerning a definite one. All the assertions in the present work, with a very few exceptions, assert propositional functions, not definite propositions.

As a matter of fact, no constant elementary proposition will occur in the present work, or can occur in any work which employs only logical ideas. The ideas and propositions of logic are all general: an assertion (for example) which is true of Socrates but not of Plato, will not belong to logic[41], and if an assertion which is true of both is to occur in logic, it must not be made concerning either, but concerning a variable . In order to obtain, in logic, a definite proposition instead of a propositional function, it is necessary to take some propositional function and assert that it is true always or sometimes, i.e. with all possible values of the variable or with some possible value. Thus, giving the name "individual" to whatever there is that is neither a proposition nor a function, the proposition "every individual is identical with itself" or the proposition "there are individuals" will be a proposition belonging to logic. But these propositions are not elementary.

(5) Negation. If is any proposition, the proposition "not-," or " is false," will be represented by "." For the present, must be an elementary proposition.

(6) Disjunction. If and are any propositions, the proposition " or ," i.e. "either is true or is true," where the alternatives are to be not mutually exclusive, will be represented by

This is called the disjunction or the logical sum of and . Thus "" will mean " is false or is true"; will mean "it is false that either or is true," which is equivalent to " and are both false";[Pg 98]) will mean "it is false that either is false or is false," which is

equivalent to " and are both true"; and so on. For the present, and must be elementary propositions.

The above are all the primitive ideas required in the theory of deduction. Other primitive ideas will be introduced in Section B.

Definition of Implication. When a proposition follows from a proposition , so that if is true, must also be true, we say that implies . The idea of implication, in the form in which we require it, can be defined. The meaning to be given to implication in what follows may at first sight appear somewhat artificial; but although there are other legitimate meanings, the one here adopted is very much more convenient for our purposes than any of its rivals. The essential property that we require of implication is this: "What is implied by a true proposition is true." It is in virtue of this property that implication yields proofs. But this property by no means determines whether anything, and if so what, is implied by a false proposition. What it does determine is that, if implies , then it cannot be the case that is true and is false, i.e. it must be the case that either is false or is true. The most convenient interpretation of implication is to say, conversely, that if either is false or is true, then " implies " is to be true. Hence " implies " is to be defined to mean: "Either is false or is true." Hence we put:

*1·01.

Here the letters "Df" stand for "definition." They and the sign of equality together are to be regarded as forming one symbol, standing for "is defined to mean[42]." Whatever comes to the left of the sign of equality is defined to mean the same as what comes to the right of it. Definition is not among the primitive ideas, because definitions are concerned solely with the symbolism, not with what is symbolised; they are introduced for practical convenience, and are theoretically unnecessary.

In virtue of the above definition, when "" holds, then either is false or is true; hence if is true, must be true. Thus the above definition preserves the essential characteristic of implication; it gives, in fact, the most general meaning compatible with the preservation of this characteristic.

PRIMITIVE PROPOSITIONS.

*1·1. Anything implied by a true elementary proposition is true. Pp[43].

The above principle will be extended in *9 to propositions which are not elementary. It is not the same as "if is true, then if implies , is[Pg 99] true." This is a true proposition, but it holds equally when is not true and when does not imply . It does not, like the principle we are concerned with, enable us to assert simply, without any hypothesis. We cannot express the principle symbolically, partly because any symbolism in which is variable only gives the hypothesis that is true, not the fact that it is true[44].

The above principle is used whenever we have to deduce a proposition from a proposition. But the immense majority of the assertions in the present work are assertions of propositional functions, i.e. they contain an undetermined variable. Since the assertion of a propositional function is a different primitive idea from the assertion of a proposition, we require a primitive proposition different from *1·1,

though allied to it, to enable us to deduce the assertion of a propositional function "" from the assertions of the two propositional functions "" and "." This primitive proposition is as follows:

*1·11. When can be asserted, where is a real variable, and can be asserted, where is a real variable, then can be asserted, where is a real variable. Pp.

This principle is also to be assumed for functions of several variables.

Part of the importance of the above primitive proposition is due to the fact that it expresses in the symbolism a result following from the theory of types, which requires symbolic recognition. Suppose we have the two assertions of propositional functions "" and ""; then the "" in is not absolutely anything, but anything for which as argument the function "" is significant; similarly in "" the is anything for which "" is significant. Apart from some axiom, we do not know that the 's for which "" is significant are the same as those for which "" is significant. The primitive proposition *1·11, by securing that, as the result of the assertions of the propositional functions "" and "" the propositional function can also be asserted, secures partial symbolic recognition, in the form most useful in actual deductions, of an important principle which follows from the theory of types, namely that, if there is any one argument a for which both "" and "" are significant, then the range of arguments for which "" is significant is the same as the range of arguments for which "" is significant. It is obvious that, if the propositional function "" can be asserted, there must be arguments a for which "" is significant, and for which, therefore, "" and "" must be significant. Hence, by our principle, the values of for which "" is significant are the same as those for which "" is significant, i.e. the type of possible arguments for (cf. p. 15) is the same as that of possible arguments for . The[Pg 100] primitive proposition *1·11, since it states a practically important consequence of this fact, is called the "axiom of identification of type."

Another consequence of the principle that, if there is an argument for which both and are significant, then is significant whenever is significant, and vice versa, will be given in the "axiom of identification of real variables," introduced in **1·72. These two propositions, *1·11 and *1·72, give what is symbolically essential to the conduct of demonstrations in accordance with the theory of types.

The above proposition *1·11 is used in every inference from one asserted propositional function to another. We will illustrate the use of this proposition by setting forth at length the way in which it is first used, in the proof of *2·06. That proposition is

We have already proved, in *2·05, the proposition

It is obvious that *2·06 results from *2·05 by means of *2·04, which is

For if, in this proposition, we replace by , by , and by , we obtain, as an instance of *2·04, the proposition

and here the hypothesis is asserted by *2·05. Thus our primitive proposition *1·11 enables us to assert the conclusion.

*1·2.

This proposition states: "If either is true or is true, then is true." It is called the "principle of tautology," and will be quoted by the abbreviated title of "Taut." It is convenient, for purposes of reference, to give names to a few of the more important propositions; in general, propositions will be referred to by their numbers.

*1·3.

This principle states: "If is true, then ' or ' is true." Thus e.g. if is "to-day is Wednesday" and is "to-day is Tuesday," the principle states: "If to-day is Wednesday, then to-day is either Tuesday or Wednesday." It is called the "principle of addition," because it states that if a proposition is true, any alternative may be added without making it false. The principle will be referred to as "Add."

*1·4.

[Pg 101]

This principle states that " or " implies " or ." It states the permutative law for logical addition of propositions, and will be called the "principle of permutation." It will be referred to as "Perm."

1·5. .

This principle states: "If either is true, or ' or ' is true, then either is true, or ' or ' is true." It is a form of the associative law for logical addition, and will be called the "associative principle." It will be referred to as "Assoc." The proposition

which would be the natural form for the associative law, has less deductive power, and is therefore not taken as a primitive proposition.

*1·6 .

This principle states: "If implies , then ' or ' implies ' or .'" In other words, in an implication, an alternative may be added to both premiss and conclusion without impairing the truth of the implication. The principle will be called the "principle of summation," and will be referred to as "Sum."

*1·7. If is an elementary proposition, is an elementary proposition. Pp.

*1·71. If and are elementary propositions, is an elementary proposition. Pp.

*1·72. If and are elementary propositional functions which take elementary propositions as arguments, is an elementary propositional function. Pp.

This axiom is to apply also to functions of two or more variables. It is called the "axiom of identification of real variables." It will be observed that if and are functions which take arguments of different types, there is no such function as "," because and cannot significantly have the same argument. A more general form of the above axiom will be given in *9.

The use of the above axioms will generally be tacit. It is only through them and the axioms of *9 that the theory of types explained in the Introduction becomes relevant, and any view of logic which justifies these axioms justifies such subsequent reasoning

as employs the theory of types.

This completes the list of primitive propositions required for the theory of deduction as applied to elementary propositions.

[Pg 102]

*2. IMMEDIATE CONSEQUENCES OF THE PRIMITIVE PROPOSITIONS.

Summary of *2.

The proofs of the earlier of the propositions of this number consist simply in noticing that they are instances of the general rules given in *1. In such cases, these rules are not premisses, since they assert any instance of themselves, not something other than their instances. Hence when a general rule is adduced in early proofs, it will be adduced in brackets[45], with indications, when required, as to the changes of letters from those given in the rule to those in the case considered. Thus "Taut " will mean what "Taut" becomes when is written in place of . If "Taut " is enclosed in square brackets before an asserted proposition, that means that, in accordance with "Taut," we are asserting what "Taut" becomes when is written in place of . The recognition that a certain proposition is an instance of some general proposition previously proved or assumed is essential to the process of deduction from general rules, but cannot itself be erected into a general rule, since the application required is particular, and no general rule can explicitly include a particular application.

Again, when two different sets of symbols express the same proposition in virtue of a definition, say *1·01, and one of these, which we will call (1), has been asserted, the assertion of the other is made by writing "[(1).(*1·01)]" before it, meaning that, in virtue of *1·01, the new set of symbols asserts the same proposition as was asserted in (1). A reference to a definition is distinguished from a reference to a previous proposition by being enclosed in round brackets.

The propositions in this number are all, or nearly all, actually needed in deducing mathematics from our primitive propositions. Although certain abbreviating processes will be gradually introduced, proofs will be given very fully, because the importance of the present subject lies, not in the propositions themselves, but (1) in the fact that they follow from the primitive propositions, (2) in the fact that the subject is the easiest, simplest, and most elementary example of the symbolic method of dealing with the principles of mathematics generally. Later portions—the theories of classes, relations, cardinal numbers, series, ordinal numbers, geometry, etc.—all employ the same method, but with an increasing complexity in the entities and functions considered.

[Pg 103]

The most important propositions proved in the present number are the following:

*2·02.

I.e. implies that implies , i.e. a true proposition is implied by any proposition. This proposition is called the "principle of simplification" (referred to as "Simp"), because, as will appear later, it enables us to pass from the joint assertion of and to the

assertion of simply. When the special meaning which we have given to implication is remembered, it will be seen that this proposition is obvious.

*2·03.

*2·15.

*2·16.

*2·17.

These four analogous propositions constitute the "principle of transposition," referred to as "Transp." They lead to the rule that in an implication the two sides may be interchanged by turning negative into positive and positive into negative. They are thus analogous to the algebraical rule that the two sides of an equation may be interchanged by changing the signs.

*2·04.

This is called the "commutative principle" and referred to as "Comm." It states that, if follows from provided is true, then follows from provided is true.

*2·05.

*2·06.

These two propositions are the source of the syllogism in Barbara (as will be shown later) and are therefore called the "principle of the syllogism" (referred to as "Syll"). The first states that, if follows from , then if follows from , follows from . The second states the same thing with the premisses interchanged.

*2·08.

I.e. any proposition implies itself. This is called the "principle of identity" and referred to as "Id." It is not the same as the "law of identity" (" is identical with "), but the law of identity is inferred from it (cf. *13·15).

*2·21.

I.e. a false proposition implies any proposition.

[Pg 104]

The later propositions of the present number are mostly subsumed under propositions in *3 or *4, which give the same results in more compendious forms. We now proceed to formal deductions.

*2·01.

This proposition states that, if implies its own falsehood, then is false. It is called the "principle of the reductio ad absurdum," and will be referred to as "Abs."[46] The proof is as follows (where "Dem." is short for demonstration"):

Dem.

*2·02.

Dem.

*2·03.

Dem.

*2·04.

Dem.

*2·05.

Dem.

*2·06.

Dem.

[Pg 105]

In the last line of this proof, "(1).(2).*1·11" means that we are inferring in accordance with *1·11, having before us a proposition, namely , which, by (1), is implied by , which, by (2), is true. In general, in such cases, we shall omit the reference to *1·11.

The above two propositions will both be referred to as the "principle of the syllogism" (shortened to "Syll"), because, as will appear later, the syllogism in Barbara is derived from them.

*2·07

Here we put nothing beyond "*1·3 ," because the proposition to be proved is what *1·3 becomes when is written in place of .

*2·08

Dem.

*2·1

*2·11

Dem.

This is the law of excluded middle.

*2·12

Dem.

[Pg 106]

*2·13.

This proposition is a lemma for *2·14, which, with *2·12, constitutes the principle of double negation.

Dem.

*2·14.

Dem.

*2·15.

Dem.

[Pg 107]

Note on the proof of *2·15. In the above proof, it will be seen that (3), (4), (6) are respectively of the forms , , , where is the proposition to be proved. From , , the proposition results by repeated applications of *2·05 or *2·06 (both of which are called "Syll"). It is tedious and unnecessary to repeat this process every time it is used; it will therefore be abbreviated into

where (a) is of the form , (b) of the form , (c) of the form , and (d) of the form . The same abbreviation will be applied to a sorites of any length.

Also where we have "" and "," and is the proposition to be proved, it is convenient to write simply

where "etc." will be a reference to the previous propositions in virtue of which the implication "" holds. This form embodies the use of *1·11 or *1·1, and makes many proofs at once shorter and easier to follow. It is used in the first two lines of the following proof.

*2·16.

Dem.

[Pg 108]

Note. The proposition to be proved will be called "Prop," and when a proof ends, like that of *2·16, by an implication between asserted propositions, of which the consequent is the proposition to be proved, we shall write "". Thus "" ends a proof, and more or less corresponds to "Q.E.D."

*2·17.

Dem.

*2·15, *2·16 and *2·17 are forms of the principle of transposition, and will be all referred to as "Transp."

*2·18.

Dem.

This is the complement of the principle of the reductio ad absurdum. It states that a proposition which follows from the hypothesis of its own falsehood is true.

*2·2.

Dem.

*2·21.

The above two propositions are very frequently used.

*2·24.

[Pg 109]

*2·25.

Dem.

*2·26.

*2·27.

*2·3.

Dem.

*2·31.

This proposition and *2·32 together constitute the associative law for logical addition of propositions. In the proof, the following abbreviation (constantly used hereafter) will be employed[47]: When we have a series of propositions of the form , , , all asserted, and "" is the proposition to be proved, the proof in full is as follows:

It is tedious to write out this process in full; we therefore write simply

where "" is the proposition to be proved. We indicate on the left by references in square brackets the propositions in virtue of which the successive implications hold. We put one dot (not two) after "," to show[Pg 110] that it is , not "," that implies . But we put two dots after , to show that now the whole proposition "" is concerned. If "" is not the proposition to be proved, but is to be used subsequently in the proof, we put

and then "(1)" means "" The proof of *2·31 is as follows:

Dem.

*2·32.

Dem.

*2·33.

This definition serves only for the avoidance of brackets.

*2·36.

Dem.

*2·37.

*2·38.

[Pg 111]

The proofs of *2·37·38 are exactly analogous to that of *2·36. (We use "*2·37·38" as an abbreviation for "*2·37 and *2·38." Such abbreviations will be used throughout.)

The use of a general principle of deduction, such as either form of "Syll," in a proof, is different from the use of the particular premisses to which the principle of deduction is applied. The principle of deduction gives the general rule according to

which the inference is made, but is not itself a premiss in the inference. If we treated it as a premiss, we should need either it or some other general rule to enable us to infer the desired conclusion, and thus we should gradually acquire an increasing accumulation of premisses without ever being able to make any inference. Thus when a general rule is adduced in drawing an inference, as when we write "" the mention of "Syll" is only required in order to remind the reader how the inference is drawn.

The rule of inference may, however, also occur as one of the ordinary premisses, that is to say, in the case of "Syll" for example, the proposition "" may be one of those to which our rules of deduction are applied, and it is then an ordinary premiss. The distinction between the two uses of principles of deduction is of some philosophical importance, and in the above proofs we have indicated it by putting the rule of inference in square brackets. It is, however, practically inconvenient to continue to distinguish in the manner of the reference. We shall therefore henceforth both adduce ordinary premisses in square brackets where convenient, and adduce rules of inference, along with other propositions, in asserted premisses, i.e. we shall write e.g.

*2·4.

Dem.

*2·41.

Dem.

*2·42.

*2·43.

*2·45.

*2·46.

[Pg 112]

*2·47.

*2·48.

*2·49.

*2·5.

*2·51.

*2·52.

*2·521.

*2·53.

Dem.

*2·54.

*2·55.

*2·56.

*2·6.

Dem.

*2·61.

*2·62.

*2·621.

*2·63.

*2·64.

*2·65.

*2·67.

Dem.

[Pg 113]

*2·68.

Dem.

*2·69.

*2·73.

*2·74.

*2·75.

*2·76.

*2·77.

*2·8.

Dem.

*2·81.

Dem.

*2·82.

*2·83.

*2·85.

Dem.

*2·86.

FOOTNOTES:

[38] The recognized methods of proving independence are not applicable, without reserve, to fundamentals. Cf. Principles of Mathematics, § 17. What is there said concerning primitive propositions applies with even greater force to primitive ideas.

[39] We have adopted both the idea and the symbol of assertion from Frege.

[40] Cf. Principles of Mathematics, § 38.

[41] When we say that a proposition "belongs to logic," we mean that it can be expressed in terms of the primitive ideas of logic. We do not mean that logic applies to it, for that would of course be true of any proposition.

[42] The sign of equality not followed by the letters "Df" will have a different meaning, to be defined later.

[43] The letters "Pp" stand for "primitive proposition," as with Peano.

[44] For further remarks on this principle, cf. Principles of Mathematics, § 38.

[45] Later on we shall cease to mark the distinction between a premiss and a rule according to which an inference is conducted. It is only in early proofs that this distinction is important.

[46] There is an interesting historical article on this principle by Vailati, "A proposito d'un passo del Teeteto e di una dimostrazione di Euclide," Rivista di Filosofia e scienze affine, 1904.

[47] This abbreviation applies to the same type of cases as those concerned in the note to *2·15, but is often more convenient than the abbreviation explained in that note.

[Pg 114]

*3. THE LOGICAL PRODUCT OF TWO PROPOSITIONS.

Summary of *3.

The logical product of two propositions and is practically the proposition " and are both true." But this as it stands would have to be a new primitive idea. We therefore take as the logical product the proposition , i.e. "it is false that either is false or is false," which is obviously true when and only when and are both true. Thus we put

*3·01.

where "" is the logical product of and .

*3·02.

This definition serves merely to abbreviate proofs.

When we are given two asserted propositional functions "" and "," we shall have "" whenever and take arguments of the same type. This will be proved for any functions in *9; for the present, we are confined to elementary propositional functions of elementary propositions. In this case, the result is proved as follows:

By *1·7, and are elementary propositional functions, and therefore, by *1·72, is an elementary propositional function. Hence by *2·11,

Hence by *2·32 and *1·01,

i.e. by *3·01,

Hence by *1·11, when we have "" and "" we have "." This proposition is *3·03. It is to be understood, like *1·72, as applying also to functions of two or more variables.

The above is the practically most useful form of the axiom of identification of real variables (cf. *1·72). In practice, when the restriction to elementary propositions and propositional functions has been removed, a convenient means by which two functions can often be recognized as taking arguments of the same type is the following:

If contains, in any way, a constituent and contains, in any way, a constituent , then both and take arguments[Pg 115] of the type of the argument in , and therefore both and take arguments of the same type. Hence, in such a case, if both and can be asserted, so can .

As an example of the use of this proposition, take the proof of *3·47. We there prove

and what we wish to prove is

which is *3·47. Now in (1) and (2), , , , are elementary propositions (as everywhere in Section A); hence by *1·7·71, applied repeatedly, "" and "" are elementary propositional functions. Hence by *3·03, we have

whence the result follows by *3·43 and *3·33.

The principal propositions of the present number are the following:

*3·2

I.e. " implies that implies ," i.e. if each of two propositions is true, so is their logical product.

*3·26

*3·27

I.e. if the logical product of two propositions is true, then each of the two propositions severally is true.

*3·3

I.e. if and jointly imply , then implies that implies . This principle (following Peano) will be called "exportation," because is "exported" from the hypothesis. It will be referred to as "Exp."

*3·31

This is the correlative of the above, and will be called (following Peano) "importation" (referred to as "Imp").

*3·35

I.e. "if is true, and follows from it, then is true." This will be called the "principle of assertion" (referred to as "Ass"). It differs from *1·1 by the fact that it does not apply only when really is true, but requires merely the hypothesis that p is true.

*3·43

[Pg 116]

I.e. if a proposition implies each of two propositions, then it implies their logical product. This is called by Peano the "principle of composition." It will be referred to as "Comp."

*3·45.

I.e. both sides of an implication may be multiplied by a common factor. This is called by Peano the "principle of the factor." It will be referred to as "Fact."

*3·47.

I.e. if implies and implies , then and jointly imply and jointly. The law of contradiction, ")," is proved in this number (*3·24); but in spite of its fame we have found few occasions for its use.

*3·01.

*3·02.

*3·03. Given two asserted elementary propositional functions "" and "" whose arguments are elementary propositions, we have .

Dem.

*3·1.

*3·11.

*3·12.

*3·13.

*3·14.

*3·2.

*3·21.

*3·22.

This is one form of the commutative law for logical multiplication. A more complete form is given in *4·3.

Dem.

[Pg 117]

Note that, in the above proof, "(1)" stands for the proposition

as was explained in the proof of *2·31.

*3·24.

Dem.

The above is the law of contradiction.

*3·26.

Dem.

*3·27.

Dem.

*3·26·27 will both be called the "principle of simplification," like *2·02, from which they are deduced. They will be referred to as "Simp."

*3·3.

Dem.

*3·31.

Dem.

[Pg 118]

*3·33.

*3·34.

These two propositions will hereafter be referred to as "Syll"; they are usually more convenient than either *2·05 or *2·06.

*3·35.

*3·37.

Dem.

This is another form of transposition.

*3·4.

*3·41.

*3·42.

*3·43.

Dem.

*3·44.

This principle is analogous to *3·43. The analogy between *3·43 and *3·44 is of a sort which generally subsists between formulae concerning products and formulae concerning sums.

Dem.

[Pg 119]

*3·45.

This principle shows that we may multiply both sides of an implication by a common factor; hence it is called by Peano the "principle of the factor." We shall refer to it as "Fact." It is the analogue, for multiplication, of the primitive proposition *1·6.

Dem.

*3·47.

This proposition, or rather its analogue for classes, was proved by Leibniz, and evidently pleased him, since he calls it "præclarum theorema[48]."

Dem.

*3·48.

This theorem is the analogue of *3·47.

Dem.

FOOTNOTES:

[48] Philosophical works, Gerhardt's edition, Vol. VII. p. 223.

[Pg 120]

*4. EQUIVALENCE AND FORMAL RULES.

Summary of *4.

In this number, we shall be concerned with rules analogous, more or less, to those of ordinary algebra. It is from these rules that the usual "calculus of formal logic" starts. Treated as a "calculus," the rules of deduction are capable of many other interpretations. But all other interpretations depend upon the one here considered, since in all of them we deduce consequences from our rules, and thus presuppose the theory of deduction. One very simple interpretation of the "calculus" is as follows: The entities considered are to be numbers which are all either or ; "" is to have the value if is and is ; otherwise it is to have the value ; is to be if is , and if is ; is to be if and are both , and is to be in any other case; is to be if and are both , and is to be in any other case; and the assertion-sign is to mean that what follows has the value . Symbolic logic considered as a calculus has undoubtedly much interest on its own account; but in our opinion this aspect has hitherto been too much emphasized, at the expense of the aspect in which symbolic logic is merely the most elementary part of mathematics, and the logical prerequisite of all the rest. For this reason, we shall only deal briefly with what is required for the algebra of symbolic logic.

When each of two propositions implies the other, we say that the two are equivalent, which we write "." We put

*4·01.

It is obvious that two propositions are equivalent when, and only when, both are true or both are false. Following Frege, we shall call the truth-value of a proposition truth if it is true, and falsehood if it is false. Thus two propositions are equivalent when they have the same truth-value.

It should be observed that, if , may be substituted for without altering the truth-value of any function of which involves no primitive ideas except those enumerated in *1. This can be proved in each separate case, but not generally, because we have no means of specifying (with our apparatus of primitive ideas) that a function is one which can be built up out[Pg 121] of these ideas alone. We shall give the name of a truth-function to a function whose argument is a proposition, and whose truth-value depends only upon the truth-value of its argument. All the functions of propositions with which we shall be specially concerned will be truth-functions, i.e. we shall have

The reason of this is, that the functions of propositions with which we deal are all built up by means of the primitive ideas of *1. But it is not a universal characteristic of functions of propositions to be truth-functions. For example, " believes " may be true for one true value of and false for another.

The principal propositions of this number are the following:

*4·1.

*4·11.

These are both forms of the "principle of transposition."

*4·13.

This is the principle of double negation, i.e. a proposition is equivalent to the falsehood of its negation.

*4·2.

*4·21.

*4·22.

These propositions assert that equivalence is reflexive, symmetrical and transitive.

*4·24.

*4·25.

I.e. is equivalent to " and " and to " or ," which are two forms of the law of tautology, and are the source of the principal differences between the algebra of symbolic logic and ordinary algebra.

*4·3.

This is the commutative law for the product of propositions.

*4·31.

This is the commutative law for the sum of propositions.

The associative laws for multiplication and addition of propositions, namely

*4·32.

*4·33.

The distributive law in the two forms

[Pg 122]

*4·4.

*4·41.

The second of these forms has no analogue in ordinary algebra.

*4·71.

I.e. implies when, and only when, is equivalent to . This proposition is used constantly; it enables us to replace any implication by an equivalence.

*4·73.

I.e. a true factor may be dropped from or added to a proposition without altering the truth-value of the proposition.

*4·01.

*4·02.

This definition serves merely to provide a convenient abbreviation.

*4·1.

*4·11.

*4·12.

*4·13.

*4·14.

*4·15.

*4·2.

*4·21.

*4·22.

Dem.

[Pg 123]

Note. The above three propositions show that the relation of equivalence is reflexive (*4·2), symmetrical (*4·21), and transitive (*4·22). Implication is reflexive and transitive, but not symmetrical. The properties of being symmetrical, transitive, and (at least within a certain field) reflexive are essential to any relation which is to have the formal characters of equality.

*4·24.

Dem.

*4·25.

Note. *4·24·25 are two forms of the law of tautology, which is what chiefly distinguishes the algebra of symbolic logic from ordinary algebra.

*4·3.

Note. Whenever we have, whatever values and may have,

we have also

For

*4·31.

*4·32.

Dem.

Note. Here "(1)" stands for "," which is obtained from the above steps by *4·22. The use of *4·22 will often be tacit, as above. The principle is the same as that explained in respect of implication in *2·31.

*4·33.

[Pg 124]

The above are the associative laws for multiplication and addition. To avoid brackets, we introduce the following definition:

*4·34.

*4·36.

*4·37.

*4·38.

*4·39.

*4·4.

This is the first form of the distributive law.

Dem.

*4·41.

This is the second form of the distributive law—a form to which there is nothing analogous in ordinary algebra. By the conventions as to dots, "" means "."

Dem.

*4·42.

Dem.

[Pg 125]

*4·43.

Dem.

*4·44.

Dem.

*4·45.

The following formulae are due to De Morgan, or rather, are the propositional analogues of formulae given by De Morgan for classes. The first of them, it will be observed, merely embodies our definition of the logical product.

*4·5.

*4·51.

*4·52.

*4·53.

*4·54.

*4·55.

*4·56.

*4·57.

[Pg 126]

The following formulae are obtained immediately from the above. They are important as showing how to transform implications into sums or into denials of products, and vice versa. It will be observed that the first of them merely embodies the definition *1·01.

*4·6.

*4·61.

*4·62.

*4·63.

*4·64.

*4·65.

*4·66.

*4·67.

*4·7.

Dem.

*4·71.

Dem.

The above proposition is constantly used. It enables us to transform every implication into an equivalence, which is an advantage if we wish to assimilate symbolic logic as far as possible to ordinary algebra. But when symbolic logic is regarded as an instrument of proof, we need implications, and it is usually inconvenient to substitute equivalences. Similar remarks apply to the following proposition.

*4·72.

Dem.

[Pg 127]

*4·73.

This proposition is very useful, since it shows that a true factor may be omitted from a product without altering its truth or falsehood, just as a true hypothesis may be omitted from an implication.

*4·74.

*4·76.

*4·77.

*4·78.

Dem.

*4·79.

Dem.

Note. The analogues, for classes, of *4·78·79 are false. Take, e.g. *4·78, and put = English people, = men, = women. Then is contained in or , but is not contained in and is not contained in .

*4·8.

*4·81.

*4·82.

*4·83.

Note. *4·82·83 may also be obtained from *4·43, of which they are virtually other forms.

*4·84.

*4·85.

*4·86.

*4·87.

*4·87 embodies in one proposition the principles of exportation and importation and the commutative principle.

[Pg 128]

*5. MISCELLANEOUS PROPOSITIONS.

Summary of *5.

The present number consists chiefly of propositions of two sorts: (1) those which will be required as lemmas in one or more subsequent proofs, (2) those which are on their own account illustrative, or would be important in other developments than those that we wish to make. A few of the propositions of this number, however, will be used very frequently. These are:

*5·1.

I.e. two propositions are equivalent if they are both true. (The statement that two propositions are equivalent if they are both false is *5·21.)

*5·32.

I.e. to say that, on the hypothesis , and are equivalent, is equivalent to saying that the joint assertion of and is equivalent to the joint assertion of and . This is a very useful rule in inference.

*5·6.

I.e. " and not- imply " is equivalent to " implies or ."

Among propositions never subsequently referred to, but inserted for their intrinsic interest, are the following: *5·11·12·13·14, which state that, given any two propositions , , either or must imply , and must imply either or not-, and either implies or implies ; and given any third proposition , either implies or implies [49].

Other propositions not subsequently referred to are *5·22·23·24; in these it is shown that two propositions are not equivalent when, and only when, one is true and the other false, and that two propositions are equivalent when, and only when, both are true or both false. It follows (*5·24) that the negation of "" is equivalent to "." *5·54·55 state that both the product and the sum of and are equivalent, respectively, either to or to .

The proofs of the following propositions are all easy, and we shall therefore often merely indicate the propositions used in the proofs.

[Pg 129]

*5·1.

*5·11.

*5·12.

*5·13.

*5·14.

*5·15.

Dem.

*5·16.

Dem.

*5·17.

Dem.

[Pg 130]

*5·18.

*5·19.

*5·21.

*5·22.

*5·23.

*5·24.

*5·25.

From *5·25 it appears that we might have taken implication, instead of disjunction, as a primitive idea, and have defined "" as meaning "." This course, however, requires more primitive propositions than are required by the method we have adopted.

*5·3.

*5·31.

*5·32.

This proposition is constantly required in subsequent proofs.

*5·33.

*5·35.

*5·36.

*5·4.

*5·41.

*5·42.

*5·44.

*5·5.

*5·501.

*5·53.

*5·54.

*5·55.

*5·6.

*5·61.

*5·62.

[Pg 131]

*5·63.

*5·7.

*5·71.

In the following proof, as always henceforth, "" means the hypothesis of the proposition to be proved.

Dem.

*5·74.

Dem.

*5·75.

Dem.

FOOTNOTES:

[49] Cf. Schröder, Vorlesungen über Algebra der Logik, Zweiter Band (Leipzig, 1891), pp. 270-271, where the apparent oddity of the above proposition is explained.

[Pg 132]

SECTION B.

THEORY OF APPARENT VARIABLES.

*9. EXTENSION OF THE THEORY OF DEDUCTION FROM LOWER TO HIGHER TYPES OF PROPOSITIONS.

Summary of *9.

In the present number, we introduce two new primitive ideas, which may be expressed as " is always[50] true" and " is sometimes[51] true," or, more correctly, as " always" and " sometimes." When we assert " always," we are asserting all values of , where "" means the function itself, as opposed to an ambiguous value of the function (cf. pp. 15, 42); we are not asserting that is true for all values of , because, in accordance with the theory of types, there are values of for which "" is meaningless; for example, the function itself must be such a value. We shall denote " always" by the notation

where the "" will be followed by a sufficiently large number of dots to cover the function of which "all values" are concerned. The form in which such propositions most frequently occur is the "formal implication," i.e. such a proposition as

i.e. " always implies ." This is the form in which we express the universal affirmative "all objects having the property have the property ."

We shall denote " sometimes" by the notation

Here "" stands for "there exists," and the whole symbol may be read "there exists an such that ."

In a proposition of either of the two forms , (, the is called an apparent variable. A proposition which contains no apparent variables is called "elementary," and a function, all whose values are[Pg 133] elementary propositions, is called an elementary function. For reasons explained in Chapter II of the Introduction, it would seem that negation and disjunction and their derivatives must have a different meaning when applied to elementary propositions from that which they have when applied to such propositions as or . If is an elementary function, we will in this number call and "first-order propositions." Then in virtue of the fact that disjunction and negation do not have the same meanings as applied to elementary or to first-order propositions, it follows that, in asserting the primitive propositions of *1, we must either confine them, in their application, to propositions of a single type, or we must regard them as the simultaneous assertion of a number of different primitive propositions, corresponding to the different meanings of "disjunction" and "negation." Likewise in regard to the primitive ideas of disjunction and negation, we must either, in the primitive propositions of *1, confine them to disjunctions and negations of elementary propositions, or we must regard them as really each multiple, so that in regard to each type of propositions we shall need a new primitive idea of negation and a new primitive idea of disjunction. In the present number, we shall show how, when the primitive ideas of negation and disjunction are restricted to elementary propositions, and the , , of *1—*5 are therefore necessarily elementary propositions, it is possible to obtain definitions of the negation and disjunction of first-order propositions, and proofs of the analogues, for first-order propositions, of the primitive propositions *1·2—·6. (*1·1 and *1·11 have to be assumed afresh for first-order propositions, and the analogues of *1·7·71·72 require a fresh treatment.) It follows that the analogues of the propositions of *2—*5 follow by merely repeating previous proofs. It follows also that the theory of deduction can be extended from first-order propositions to such as contain two apparent variables, by merely repeating the process which extends the theory of deduction from elementary to first-order propositions. Thus by merely repeating the process set forth in the present number, propositions of any order can be reached. Hence negation and disjunction may be treated in practice as if there were no difference in these ideas as applied to different types; that is to say, when "" or "" occurs, it is unnecessary in practice to know what is the type of or , since the properties of negation and disjunction assumed in *1 (which are alone used in proving other properties) can be asserted, without formal change, of propositions of any order or, in the case of , of any two orders. The limitation, in practice, to the treatment of negation or disjunction as single ideas, the same in all types, would only arise if we ever wished to assume that there is some one function of whose value is always , whatever may be the order of , or that there is some one function of and whose value is always , whatever may be the orders of and .

Such an assumption is not involved so long as (and) remain real variables,[Pg 134] since, in that case, there is no need to give the same meaning to negation and disjunction for different values of (and), when these different values are of different types. But if (or) is going to be turned into an apparent variable, then, since our two primitive ideas and both demand some definite function , and restrict the apparent variable to possible arguments for , it follows that negation and disjunction must, wherever they occur in the expression in which (or) is an apparent variable, be restricted to the kind of negation or disjunction appropriate to a given type or pair of types. Thus, to take an instance, if we assert the law of excluded middle in the form

there is no need to place any restriction upon : we may give to a value of any order, and then give to the negation and disjunction involved those meanings which are appropriate to that order. But if we assert

it is necessary, if our symbol is to be significant, that "" should be the value, for the argument , of a function ; and this is only possible if the negation and disjunction involved have meanings fixed in advance, and if, therefore, is limited to one type. Thus the assertion of the law of excluded middle in the form involving a real variable is more general than in the form involving an apparent variable. Similar remarks apply generally where the variable is the argument to a typically ambiguous function.

In what follows the single letters p and q will represent elementary propositions, and so will "," "," etc. We shall show how, assuming the primitive ideas and propositions of *1 as applied to elementary propositions, we can define and prove analogous ideas and propositions as applied to propositions of the forms and . By mere repetition of the analogous process, it will then follow that analogous ideas and propositions can be defined and proved for propositions of any order; whence, further, it follows that, in all that concerns disjunction and negation, so long as propositions do not appear as apparent variables, we may wholly ignore the distinction between different types of propositions and between different meanings of negation and disjunction. Since we never have occasion, in practice, to consider propositions as apparent variables, it follows that the hierarchy of propositions (as opposed to the hierarchy of functions) will never be relevant in practice after the present number.

The purpose and interest of the present number are purely philosophical, namely to show how, by means of certain primitive propositions, we can deduce the theory of deduction for propositions containing apparent variables from the theory of deduction for elementary propositions. From the purely technical point of view, the distinction between elementary and other propositions may be ignored, so long as propositions do not appear as apparent variables; we may then regard the primitive propositions of *1 as applying[Pg 135] to propositions of any type, and proceed as in *10, where the purely technical development is resumed.

It should be observed that although, in the present number, we prove that the analogues of the primitive propositions of *1, if they hold for propositions containing apparent variables, also hold for such as contain , yet we must not suppose that mathematical induction may be used to infer that the analogues of the primitive

propositions of *1 hold for propositions containing any number of apparent variables. Mathematical induction is a method of proof which is not yet applicable, and is (as will appear) incapable of being used freely until the theory of propositions containing apparent variables has been established. What we are enabled to do, by means of the propositions in the present number, is to prove our desired result for any assigned number of apparent variables—say ten—by ten applications of the same proof. Thus we can prove, concerning any assigned proposition, that it obeys the analogues of the primitive propositions of *1, but we can only do this by proceeding step by step, not by any such compendious method as mathematical induction would afford. The fact that higher types can only be reached step by step is essential, since to proceed otherwise we should need an apparent variable which would wander from type to type, which would contradict the principle upon which types are built up.

Definition of Negation. We have first to define the negations of (and (. We define the negation of as , i.e. "it is not the case that is always true" is to mean "it is the case that is sometimes true." Similarly the negation of (is to be defined as . Thus we put

*9·01.

*9·02.

To avoid brackets, we shall write in place of , and in place of . Thus:

*9·011.

*9·021.

Definition of Disjunction. To define disjunction when one or both of the propositions concerned is of the first order, we have to distinguish six cases, as follows:

*9·03. (

*9·04.

*9·05. (

*9·06.

*9·07. (

*9·08. (

[Pg 136]

(The definitions *9·07·08 are to apply also when and are not both elementary functions.)

In virtue of these definitions, the true scope of an apparent variable is always the whole of the asserted proposition in which it occurs, even when, typographically, its scope appears to be only part of the asserted proposition. Thus when (or appears as part of an asserted proposition, it does not really occur, since the scope of the apparent variable really extends to the whole asserted proposition. It will be shown, however, that, so far as the theory of deduction is concerned, (and (behave like propositions not containing apparent variables.

The definitions of implication, the logical product, and equivalence are to be transferred unchanged to (and (.

The above definitions can be repeated for successive types, and thus reach propositions of any type.

Primitive Propositions. The primitive propositions required are six in number, and may be divided into three sets of two. We have first two propositions which effect the passage from elementary to first-order propositions, namely

*9·1.

*9·11.

Of these, the first states that, if is true, then there is a value of which is true; i.e. if we can find an instance of a function which is true, then the function is "sometimes true." (When we speak of a function as "sometimes" true, we do not mean to assert that there is more than one argument for which it is true, but only that there is at least one.) Practically, the above primitive proposition gives the only method of proving "existence-theorems": in order to prove such theorems, it is necessary (and sufficient) to find some instance in which an object possesses the property in question. If we were to assume what may be called "existence-axioms," i.e. axioms stating (for some particular , these axioms would give other methods of proving existence. Instances of such axioms are the multiplicative axiom (*88) and the axiom of infinity (defined in *120·03). But we have not assumed any such axioms in the present work.

The second of the above primitive propositions is only used once, in proving (, which is the analogue of *1·2 (namely when is replaced by (. The effect of this primitive proposition is to emphasize the ambiguity of the required in order to secure (. We have, of course, in virtue of *9·1,

But if we try to infer from these that , we must use the[Pg 137] proposition , where is (. Now it will be found, on referring to *4·77 and the propositions used in its proof, that this proposition depends upon *1·2, i.e. . Hence it cannot be used by us to prove (, and thus we are compelled to assume the primitive proposition *9·11.

We have next two propositions concerned with inference to or from propositions containing apparent variables, as opposed to implication. First, we have, for the new meaning of implication resulting from the above definitions of negation and disjunction, the analogue of *1·1, namely

*9·12.

That is to say, given "" and "," we may proceed to "," even when the propositions and are not elementary. Also, as in *1·11, we may proceed from "" and "" to "" where is a real variable, and and are not necessarily elementary functions. It is in this latter form that the axiom is usually needed. It is to be assumed for functions of several variables as well as for functions of one variable.

We have next the primitive proposition which permits the passage from a real to an apparent variable, namely "when may be asserted, where may be any possible

argument, then (may be asserted." In other words, when is true however may be chosen among possible arguments, then (is true, i.e. all values of are true. That is to say, if we can assert a wholly ambiguous value , that must be because all values are true. We may express this primitive proposition by the words: "What is true in any case, however the case may be selected, is true in all cases." We cannot symbolise this proposition, because if we put

that means: "However may be chosen, implies (," which is in general false. What we mean is: "If is true however may be chosen, then (is true." But we have not supplied a symbol for the mere hypothesis of what is asserted in "," where is a real variable, and it is not worth while to supply such a symbol, because it would be very rarely required. If, for the moment, we use the symbol to express this hypothesis, then our primitive proposition is

In practice, this primitive proposition is only used for inference, not for implication; that is to say, when we actually have an assertion containing a real variable, it enables us to turn this real variable into an apparent variable by placing it in brackets immediately after the assertion-sign, followed by enough dots to reach to the end of the assertion. This process will be called "turning a real variable into an apparent variable." Thus we may assert our primitive proposition, for technical use, in the form:

[Pg 138]

*9·13. In any assertion containing a real variable, this real variable may be turned into an apparent variable of which all possible values are asserted to satisfy the function in question. Pp.

We have next two primitive propositions concerned with types. These require some preliminary explanations.

Primitive Idea: Individual. We say that is "individual" if is neither a proposition nor a function (cf. pp. 53, 54).

*9·131. Definition of "being of the same type." The following is a step-by-step definition, the definition for higher types presupposing that for lower types. We say that and "are of the same type" if (1) both are individuals, (2) both are elementary functions taking arguments of the same type, (3) is a function and is its negation, (4) is or and is , where and are elementary functions, (5) is and is (, where , are of the same type, (6) both are elementary propositions, (7) is a proposition and is , or (8) is (and is (, where and are of the same type.

Our primitive propositions are:

*9·14. If "" is significant, then if is of the same type as , "" is significant, and vice versa. Pp. (Cf. note on *10·121, p. 146.)

*9·15. If, for some , there is a proposition , then there is a function , and vice versa. Pp.

It will be seen that, in virtue of the definitions,

In order to prove that (and (obey the same rules of deduction as , we have to prove that propositions of the forms (and (may replace one or more of the propositions , , in *1·2—·6. When this has been proved, the previous proofs of subsequent propositions in *2—*5 become applicable. These proofs are given below. Certain other propositions, required in the proofs, are also proved.

*9·2.

The above proposition states the principle of deduction from the general to the particular, i.e. "what holds in all cases, holds in any one case."

Dem.

[Pg 139]

In the second line of the above proof, "" is taken as the value, for the argument , of the function "," where is the argument. A similar method of using *9·1 is employed in most of the following proofs.

*1·11 is used, as in the third line of the above proof, in almost all steps except such as are mere applications of definitions. Hence it will not be further referred to, unless in cases where its employment is obscure or specially important.

*9·21.

I.e. if always implies , then " always" implies " always." The use of this proposition is constant throughout the remainder of this work.

Dem.

This is the proposition to be proved, since "(" is the same proposition as "(," and "(" is the same proposition as "(."

*9·22.

I.e. if always implies , then if is sometimes true, so is . This proposition, like *9·21, is constantly used in the sequel.

Dem.

[Pg 140]

This is the proposition to be proved, because (is the same proposition as (, and (is the same proposition as (.

*9·23.

*9·24.

*9·25.

We are now in a position to prove the analogues of *1·2—·6, replacing one of the letters , , in those propositions by (or (. The proofs are given below.

*9·3.

Dem.

*9·31.

This is the only proposition which employs *9·11.

Dem.

*9·32.

Dem.

*9·33.

[Pg 141]

*9·34.

Dem.

*9·35.

*9·36.

Dem.

*9·361.

*9·37.

*9·371.

*9·4.

Dem.

*9·401.

*9·41.

*9·411.

*9·42.

*9·421.

*9·5.

Dem.

*9·501.

*9·51.

Dem.

[Pg 142]

*9·511.

*9·52.

Dem.

*9·521.

*9·6.

*9·61. If and are elementary functions of the same type, there is a function .

Dem.

By *9·14·15, there is an for which "," and therefore "," are significant, and therefore so is "," by the primitive idea of disjunction. Hence the result by *9·15.

The same proof holds for functions of any number of variables.

*9·62. If and are elementary functions, and the -argument to is of the same type as the argument to , there are functions

Dem.

By *9·15, there are propositions and , where by hypothesis and are of the same type. Hence by *9·14 there is a proposition , and therefore, by the primitive idea of disjunction, there is a proposition , and therefore, by *9·15 and *9·03, there is a proposition (. Similarly there is a proposition (. Hence the result, by *9·15.

*9·63. If , are elementary functions of the same type, there are functions (, etc. [Proof as above]

We have now completed the proof that, in the primitive propositions of *1, any one of the propositions that occur may be replaced by (or (. It follows that, by merely repeating the proofs, we can show that any other of the propositions that occur in these propositions can be simultaneously replaced by (or (. Thus all the primitive propositions of *1, and therefore all the propositions of *2—*5, hold equally when some or all of the propositions concerned are of one of the forms (, (, which was to be proved.

It follows, by mere repetition of the proofs, that the propositions of *1—*5 hold when , , are replaced by propositions containing any number of apparent variables.

FOOTNOTES:

[50] We use "always" as meaning "in all cases," not "at all times." A similar remark applies to "sometimes."

[51] As above.

[Pg 143]

*10. THEORY OF PROPOSITIONS CONTAINING ONE APPARENT VARIABLE.

Summary of *10.

The chief purpose of the propositions of this number is to extend to formal implications (i.e. to propositions of the form) as many as possible of the propositions proved previously for material implications, i.e. for propositions of the form . Thus e.g. we have proved in *3·33 that

Then we have "if 'Socrates is a Greek' implies 'Socrates is a man,' and 'Socrates is a man' implies 'Socrates is a mortal,' it follows that 'Socrates is a Greek' implies 'Socrates is a mortal.'" But this does not of itself prove that if all Greeks are men, and all men are mortals, then all Greeks are mortals.

we have to prove

It is such propositions that have to be proved in the present number. It will be seen that formal implication) is a relation of two functions and . Many of the formal properties of this relation are analogous to properties of the relation "" which expresses material implication; it is such analogues that are to be proved in this number.

We shall assume in this number, what has been proved in *9, that the propositions of *1—*5 can be applied to such propositions as (and (. Instead of the method adopted in *9, it is possible to take negation and disjunction as new primitive ideas, as applied to propositions containing apparent variables, and to assume that, with the new meanings of negation and disjunction, the primitive propositions of *1 still hold. If this[Pg 144] method is adopted, we need not take as a primitive idea, but may put

*10·01. (

In order to make it clear how this alternative method can be developed, we shall, in the present number, assume nothing of what has been proved in *9 except certain propositions which, in the alternative method, will be primitive propositions, and (what in part characterizes the alternative method) the applicability to propositions containing apparent variables of analogues of the primitive ideas and propositions of *1, and therefore of their consequences as set forth in *2—*5.

The two following definitions merely serve to introduce a notation which is often more convenient than the notation or .

*10·02.

*10·03.

The first of these notations is due to Peano, who, however, has no notation for (except in the special case of a formal implication.

The following propositions (*10·1·11·12·121·122) have already been given in *9. *10·1 is *9·2*10·11 is *9·13, *10·12 is *9·25, *10·121 is *9·14, and *10·122 is *9·15. These five propositions must all be taken as primitive propositions in the alternative method; on the other hand, *9·1 and *9·11 are not required as primitive propositions in the alternative method.

The propositions of the present number are very much used throughout the rest of the work. The propositions most used are the following:

*10·1.

I.e. what is true in all cases is true in any one case.

*10·11. If is true whatever possible argument may be, then (is true. In other words, whenever the propositional function can be asserted, so can the proposition (.

*10·21.

*10·22.

The conditions of significance in this proposition demand that and should take arguments of the same type.

*10·23.

I.e. if always implies , then if is ever true, is true.

*10·24.

[Pg 145]

I.e. if is true, then there is an for which is true. This is the sole method of proving existence-theorems.

*10·27.

I.e. if always implies , then " always" implies " always." The three following propositions, which are equally useful, are analogous to *10·27.

*10·271.

*10·28.

*10·281.

*10·35.

*10·42.

*10·5.

It should be noticed that whereas *10·42 expresses an equivalence, *10·5 only expresses an implication. This is the source of many subsequent differences between formulae concerning addition and formulae concerning multiplication.

*10·51.

This proposition is analogous to

which results from *4·63 by transposition.

Of the remaining propositions of this number, some are employed fairly often, while others are lemmas which are used only once or twice, sometimes at a much later stage.

*10·01. (

This definition is only to be used when we discard the method of *9 in favour of the alternative method already explained. In either case we have

*10·02.

*10·03.

*10·1.

*10·11. If is true whatever possible argument may be, then (is true. [*9·13]

[Pg 146]

This proposition is, in a sense, the converse of *10·1. *10·1 may be stated: "What is true of all is true of any," while *10·11 may be stated: "What is true of any, however chosen, is true of all."

*10·12.

According to the definitions in *9, this proposition is a mere example of "," since by definition the two sides of the implication are different symbols for the same proposition. According to the alternative method, on the contrary, *10·12 is a substantial proposition.

*10·121. If "" is significant, then if is of the same type as , "" is significant, and vice versa. [*9·14]

It follows from this proposition that two arguments to the same function must be of the same type; for if and a are arguments to , "" and "" are significant, and therefore and are of the same type. Thus the above primitive proposition embodies the outcome of our discussion of the vicious-circle paradoxes in Chapter II of the Introduction.

*10·122. If, for some , there is a proposition , then there is a function , and vice versa. [*9·15]

*10·13. If and take arguments of the same type, and we have "" and "," we shall have "."

Dem.

By repeated use of 9·61·62·63·131 (3), there is a function . Hence by *2·11 and *3·01,

*10·14.

This proposition is true whenever it is significant, but it is not always significant when its hypothesis is significant. For the thesis demands that and should take arguments of the same type, while the hypothesis does not demand this. Hence, if it is to be applied when and are given, or when is given as a function of or vice versa, we must not argue from the hypothesis to the thesis unless, in the supposed case, and take arguments of the same type.

Dem.

[Pg 147]

*10·2.

Dem.

*10·21.

This proposition is much more used than *10·2.

*10·22.

Dem.

The above proposition is true whenever it is significant; but, as was pointed out in connexion with *10·14, it is not always significant when "(" is significant.

*10·221. If contains a constituent and contains a constituent , where is an elementary function and are either constants or apparent variables, then and take arguments of the same type. This can be proved in each particular case, though not generally, provided that, in obtaining and from , is only submitted to negations, disjunctions and generalizations. The process may be illustrated by an example. Suppose is (, and is . By the definitions of *9, is (, and is (. Hence since the primitive ideas (and (only apply to functions, there are functions , . Hence there is a proposition . Hence, since "" and "" are only significant[Pg 148] when and are propositions, there is a proposition . Similarly, for some and , there are propositions and . Hence by *9·14, and , and are respectively of the same type, and (again by *9·14) there is a proposition . Hence (*9·15) there are functions , , therefore there are propositions

i.e. there are propositions , , which was to be proved. This process can be applied similarly in any other instance.

*10·23.

Dem.

In the above proof, we employ the definitions of *9. In the alternative method, in which (is defined in accordance with *10·01, the proof proceeds as follows.

*10·23.

Dem.

[Pg 149]

Whenever we have an asserted proposition of the form , we can pass by *10·11·21 to an asserted proposition . This passage is constantly required, as in the last line but one of the above proof. It will be indicated merely by the reference "*10·11·21," and the two steps which it requires will not be separately put down.

*10·24.

This is *9·1. In the alternative method, the proof is as follows.

Dem.

*10·25.

*10·251.

*10·252.

*10·253.

In the alternative method, in which (is defined as in *10·01, the proofs of *10·252·253 are as follows.

*10·252.

*10·253.

Dem.

*10·26.

This is one form of the syllogism in Barbara. E.g. put is a man, is mortal, = Socrates. Then the proposition becomes:

"If all men are mortal, and Socrates is a man, then Socrates is mortal."

Another form of the syllogism in Barbara is given in *10·3. The two forms, formerly wrongly identified, were first distinguished by Peano and Frege.

*10·27.

This is *9·21. In the alternative method, the proof is as follows.

[Pg 150]

Dem.

*10·271.

Dem.

*10·28.

This is *9·22. In the alternative method, the proof is as follows.

Dem.

*10·281.

*10·29.

Dem.

This is an extension of the principle of composition.

*10·3.

This is the second form of the syllogism in Barbara.

Dem.

[Pg 151]

*10·301.

Dem.

In the second line of the proofs of *10·3 and *10·301, we abbreviate the process of proof in a way which is often convenient. In *10·3, the full process would be as follows:

The above two propositions show that formal implication and formal equivalence are transitive relations between functions.

*10·31.

Dem.

*10·311.

Dem.

The above two propositions are extensions of the principle of the factor.

*10·32.

Dem.

This proposition shows that formal equivalence is symmetrical.

*10·321.

Dem.

*10·322.

Dem.

[Pg 152]

*10·33.

Dem.

*10·34.

This follows immediately from *9·05·01 and *1·01. In the alternative method, the proof is as follows.

Dem.

*10·35.

Dem.

[Pg 153]

*10·36.

This follows immediately from *9·05. In the alternative method, the proof is as follows.

Dem.

The above proposition is only required in order to lead to the following:

*10·37.

*10·39.

Dem.

This proposition is only true when the conclusion is significant; the significance of the hypothesis does not insure that of the conclusion. On the conditions of significance, see the remarks on *10·4, below.

*10·4.

Dem.

[Pg 154]

In *10·4 and many later propositions, as in *10·39, the conclusion may be not significant when the hypothesis is true. Hence, in order that it may be legitimate to use *10·4 in inference, i.e. to pass from the assertion of the hypothesis to the assertion of the conclusion, the functions , , , must be such as to have overlapping ranges of significance. In virtue of *10·221, this is secured if they are of the forms , , , . It is also secured if and or and or and or and are of such forms, for and must have overlapping ranges of significance if the hypothesis is to be significant, and so must and .

*10·41.

Dem.

Observe that in the above proof the uses of *2·2 and *1·3 are only legitimate if and have overlapping ranges of significance, for otherwise, if is such that there is a proposition , it is such that there is no proposition , and conversely.

*10·411.

Dem.

*10·412.

*10·413.

Dem.

*10·414.

Dem.

The propositions *10·413·414 are chiefly used in cases where either is replaced by or is replaced by , in which case half the hypothesis becomes superfluous, being true by *4·2.

*10·42.

Dem.

[Pg 155]

This proposition is very frequently used. It should be contrasted with *10·5, in which we have only an implication, not an equivalence.

*10·43.

Dem.

*10·5.

Dem.

The converse of the above proposition is false. The fact that this proposition states an implication, while *10·42 states an equivalence, is the source of many subsequent differences between formulae concerning logical addition and formulae concerning logical multiplication.

*10·51.

Dem.

*10·52.

Dem.

*10·53.

Dem.

*10·541.

Dem.

[Pg 156]

The above proposition is only needed in order to lead to the following:

*10·542.

This proposition is a lemma for *84·43.

*10·55.

Dem.

This proposition is a lemma for *117·12·121.

*10·56.

Dem.

This proposition and *10·57 are used in the theory of series (Part V).

*10·57.

Dem.

[Pg 157]

*11. THEORY OF TWO APPARENT VARIABLES.

Summary of *11.

In this number, the propositions proved for one variable in *10 are to be extended to two variables, with the addition of a few propositions having no analogues for one variable, such as *11·2·21·23·24 and *11·53·55·6·7. "" stands for a proposition containing and containing ; when and are unassigned, is a propositional function of and . The definition *11·01 shows that "the truth of all values of " does not need to be

taken as a new primitive idea, but is definable in terms of "the truth of all values of ." The reason is that, when is assigned, becomes a function of one variable, namely , whence it follows that, for every possible value of , "(" embodies merely the primitive idea introduced in *9. But "(" is again only a function of one variable, namely , since has here become an apparent variable. Hence the definition *11·01 below is legitimate. We put:

*11·01. (

*11·02. (

*11·03. (

*11·04. (

*11·05.

*11·06.

All the above definitions are supposed extended to any number of variables that may occur.

The propositions of this section can all be extended to any finite number of variables; as the analogy is exact, it is not necessary to carry the process beyond two variables in our proofs.

In addition to the definition *11·01, we need the primitive proposition that "whatever possible argument may be, is true whatever possible argument may be" implies the corresponding statement with and interchanged. Either may be taken as the meaning of " is true whatever possible arguments and may be."

[Pg 158]

The propositions of the present number are somewhat less used than those of *10, but some of them are used frequently. Such are the following:

*11·1.

*11·11. If is true whatever possible arguments and may be, then (is true.

These two propositions are the analogues of *10·1·11.

*11·2.

I.e. to say that "for all possible values of , is true for all possible values of " is equivalent to saying "for all possible values of , is true for all possible values of ."

*11·3.

This is the analogue of *10·21.

*11·32.

I.e. "if always implies , then ' always' implies ' always.'" This is the analogue of *10·21. *11·33·34·341 are respectively the analogues of *10·271·28·281, and are also much used.

*11·35.

I.e. if always implies , then if is ever true, is true. This is the analogue of *10·23.

*11·45.

This is the analogue of *10·35.

*11·54.

This proposition is useful because it analyses a proposition containing two apparent variables into two propositions which each contain only one. "" is a function of two variables, but is compounded of two functions of one variable each. Such a function is like a conic which is two straight lines: it may be called an "analysable" function.

*11·55.

I.e. to say "there are values of and for which is true" is equivalent to saying "there is a value of for which is true and for which there is a value of such that is true."

*11·6.

This gives a transformation which is useful in many proofs.

*11·62.

This transformation also is often useful.

[Pg 159]

*11·01. (

*11·02. (

*11·03. (

*11·04. (

*11·05.

*11·06.

with similar definitions for any number of variables.

*11·07. "Whatever possible argument may be, is true whatever possible argument may be" implies the corresponding statement with and interchanged. Pp.

*11·1.

Dem.

*11·11. If is true whatever possible arguments and may be, then (is true.

Dem.

By *10·11, the hypothesis implies that (is true whatever possible argument may be; and this, by *10·11, implies (.

*11·12.

Dem.

This proposition is only used for proving *11·2.

*11·13. If , take their first and second arguments respectively of the same type, and we have "" and "," we shall have "." [Proof as in *10·13]

*11·14.

Dem.

[Pg 160]

This proposition, like *10·14, is not always significant when its hypothesis is true. *11·13, on the contrary, is always significant when its hypothesis is true. For this reason, *11·13 may always be safely used in inference, whereas *11·14 can only be used in inference (i.e. for the actual assertion of the conclusion when the hypothesis is asserted) if it is known that the conclusion is significant.

*11·2.

Dem.

Note that "(" is the same proposition as "("; a proposition is not a function of any apparent variable which occurs in it.

*11·21.

Dem.

*11·22.

Dem.

*11·23.

Dem.

*11·24.

Dem.

*11·25.

[Pg 161]

*11·26.

Dem.

Note that the converse of this proposition is false. E.g. let be the propositional function "if is a proper fraction, then is a proper fraction greater than ." Then for all values of we have (, so that (is satisfied. In fact "(" expresses the proposition: "If is a proper fraction, then there is always a proper fraction greater than ." But "(" expresses the proposition: "There is a proper fraction which is greater than any proper fraction," which is false.

*11·27.

Dem.

All the propositions of *10 have analogues which hold for two or more variables. The more important of these are proved in what follows.

*11·3.

Dem.

*11·31.

Here the conditions of significance on the right-hand side require that and should take arguments of the same types.

Dem.

The proofs of most of the following propositions are conducted exactly as those of *11·3·31 are conducted: the analogous proposition in *10 is used[Pg 162] twice, together with *10·27 or *10·271 or *10·28 or *10·281 as the case may be. When proofs conform to this pattern we shall merely give references to the propositions used.

*11·311. If , take arguments of the same type, and we have "" and "," we shall have "." [Proof as in *10·13.]

*11·32.

*11·33.

*11·34.

*11·341.

*11·35.

*11·36.

Dem.

*11·37.

Dem.

In the following demonstration, "" means the hypothesis of the proposition to be proved. We shall employ this abbreviation, whenever convenient, in all cases where the proposition to be proved is a hypothetical, i.e. is of the form "." Similarly "" will mean "the hypothesis of (1)," and so on.

[Pg 163]

The above is a type of proof which recurs frequently in what follows. Proofs conforming to this pattern will be indicated only by the numbers of the propositions used.

*11·371.

*11·38.

*11·39.

*11·391.

Dem.

*11·4.

Dem.

*11·401.

*11·41.

*11·42.

*11·421.

*11·43.

*11·44.

[Pg 164]

*11·45.

*11·46.

*11·47.

*11·5.

Dem.

*11·51.

Dem.

*11·52.

Dem.

*11.521.

*11.53.

Dem.

*11·54.

Dem.

This proposition is very often used.

[Pg 165]

*11·55.

Dem.

This proposition is very often used.

*11·56.

Dem.

*11·57.

The use of *4·24 here depends upon the fact that (and (are the same proposition.

*11·58.

*11·59.

Dem.

*11·6.

This proposition is very frequently employed in subsequent proofs.

Dem.

[Pg 166]

*11·61.

Dem.

*11·62.

Dem.

*11·63.

Dem.

*11·7.

Dem.

In the last line of the above proof, use is made of the fact that

are the same proposition.

The first use of the following proposition occurs in the proof of *234·12. Its utility lies in its enabling us to pass from a hypothesis

containing two apparent variables, to the product of two hypotheses each containing only one.

[Pg 167]

*11·71.

Dem.

[Pg 168]

*12. THE HIERARCHY OF TYPES AND THE AXIOM OF REDUCIBILITY.

The primitive idea "" has been explained to mean " is always true," i.e. "all values of are true." But whatever function may be, there will be arguments with which is meaningless, i.e. with which as arguments does not have any value. The arguments with which has values form what we will call the "range of significance" of . A "type" is defined as the range of significance of some function. In virtue of *9·14, if , , and

are significant, i.e. either true or false, so is . From this it follows that two types which have a common member coincide, and that two different types are mutually exclusive. Any proposition of the form , i.e. any proposition containing an apparent variable, determines some type as the range of the apparent variable, the type being fixed by the function .

The division of objects into types is necessitated by the vicious-circle fallacies which otherwise arise[52]. These fallacies show that there must be no totalities which, if legitimate, would contain members defined in terms of themselves. Hence any expression containing an apparent variable must not be in the range of that variable, i.e. must belong to a different type. Thus the apparent variables contained or presupposed in an expression are what determines its type. This is the guiding principle in what follows.

As explained in *9, propositions containing variables are generated from propositional functions which do not contain these apparent variables, by the process of asserting all or some values of such functions. Suppose is a proposition containing ; we will give the name of generalization to the process which turns into or , and we will give the name of generalized propositions to all such as contain apparent variables. It is plain that propositions containing apparent variables presuppose others not containing apparent variables, from which they can be derived by generalization. Propositions which contain no apparent variables we call elementary propositions[53], and the terms of such propositions, other than functions, we call individuals. Then individuals form the first type.

[Pg 169]

It is unnecessary, in practice, to know what objects belong to the lowest type, or even whether the lowest type of variable occurring in a given context is that of individuals or some other. For in practice only the relative types of variables are relevant; thus the lowest type occurring in a given context may be called that of individuals, so far as that context is concerned. It follows that the above account of individuals is not essential to the truth of what follows; all that is essential is the way in which other types are generated from individuals, however the type of individuals may be constituted.

By applying the process of generalization to individuals occurring in elementary propositions, we obtain new propositions. The legitimacy of this process requires only that no individuals should be propositions. That this is so, is to be secured by the meaning we give to the word individual. We may explain an individual as something which exists on its own account; it is then obviously not a proposition, since propositions, as explained in Chapter II of the Introduction (p. 46), are incomplete symbols, having no meaning except in use. Hence in applying the process of generalization to individuals we run no risk of incurring reflexive fallacies. We will give the name of first-order propositions to such as contain one or more apparent variables whose possible values are individuals, but contain no other apparent variables. First-order propositions are not all of the same type, since, as was explained in *9, two propositions which do not contain the same number of apparent

variables cannot be of the same type. But owing to the systematic ambiguity of negation and disjunction, their differences of type may usually be ignored in practice. No reflexive fallacies will result, since no first-order proposition involves any totality except that of individuals.

Let us denote by "" or "" or etc. an elementary function whose argument or arguments are individual. We will call such a function a predicative function of an individual. Such functions, together with those derived from them by generalization, will be called first-order functions. In practice we may without risk of reflexive fallacies treat first-order functions as a type, since the only totality they involve is that of individuals, and, by means of the systematic ambiguity of negation and disjunction, any function of a first-order function which will concern us will be significant whatever first-order function is taken as argument, provided the right meanings are given to the negations and disjunctions involved.

For the sake of clearness, we will repeat in somewhat different terms our account of what is meant by a first-order function. Let us give the name of matrix to any function, of however many variables, which does not involve any apparent variables. Then any possible function other than a matrix is derived from a matrix by means of generalization, i.e. by considering the proposition which asserts that the function in question is true with all[Pg 170] possible values or with some value of one of the arguments, the other argument or arguments remaining undetermined. Thus e.g. from the function we shall be able to derive the four functions

of which the two first are functions of , while the two last are functions of . (All propositions, with the exception of such as are values of matrices, are also derived from matrices by the above process of generalization. In order to obtain a proposition from a matrix containing variables, without assigning values to any of the variables, it is necessary to turn all the variables into apparent variables. Thus if is a matrix, (is a proposition.) We will give the name first-order matrices to such as have only individuals for their arguments, and we will give the name of first-order functions (of any number of variables) to such as either are first-order matrices or are derived from first-order matrices by generalization applied to some (not all) of the arguments to such matrices. First-order propositions will be such as result from applying generalization to all the arguments to a first-order matrix.

As we have already stated, the notation "" is used for any elementary function of one variable. Thus "" represents any value of any elementary function of one variable. It will be seen that "" is a function of two variables, namely and . Since it contains no apparent variable, it is a matrix, but since it contains a variable (namely which is not an individual, it is not a first-order matrix. The same applies to , where a is some definite constant. We can build up a number of new matrices, such as

All these are matrices which involve first-order functions among their arguments. Such matrices we will call second-order matrices. From these matrices, by applying generalization to their arguments, whether to such as are functions or to such (if any) as are individuals, we obtain new functions and propositions. Such functions (together with second-order matrices) will be called second-order functions, and such

propositions will be called second-order propositions. Thus we are led to the following definitions:

A second-order matrix is one which has at least one first-order matrix among its arguments, but has no arguments other than first-order matrices and individuals.

A second-order function is one which either is a second-order matrix or results from one by applying generalization to some (not all) of the arguments to a second-order matrix.

A second-order proposition is one which results from a second-order matrix by applying generalization to all its arguments.

[Pg 171]

In addition to the above illustrations of second-order matrices, we may give the following examples of second-order functions:

(1) Functions in which the argument is , (, , where and are constants, , where is a constant function, and so on.

(2) Functions in which the arguments are and :

where and are constants, and so on.

(3) Functions in which the argument is an individual , , , where is constant, and so on.

(4) Functions in which the arguments are and , , where is constant, (and so on.

Examples of second-order functions might, of course, be multiplied indefinitely, but the above seem sufficient for purposes of illustration.

A second-order matrix of one variable will be called a predicative second-order function of one variable or a predicative function of a first-order matrix. Thus , and are predicative functions of . Similarly a function of several variables of which at least one is a first-order matrix, while the rest are either individuals or first-order matrices, will be called predicative if it is a matrix.

It will be seen, however, that a second-order function may have only individuals for its arguments; instances were given just now under the heading (3). Such functions we shall not call predicative, since predicative functions of individuals have already been defined as being such as are of the first order. Thus the order of a function is not determined by the order of its argument or arguments; indeed, the function may be of any order superior to the order or orders of its arguments.

A variable matrix whose argument is will be denoted by , and generally, a matrix whose arguments are , , ... , , ... (where there is at least one function among the arguments) will be denoted by

Such a matrix is not of the first or second order, since it contains the new variable whose values are second-order matrices. We proceed to construct new matrices as we did with the matrix ; these constitute third-order matrices. These together with the functions derived from them by generalization are called third-order functions,

and the propositions derived from third-order matrices by generalization are called third-order propositions.

[Pg 172]

In this way we can proceed indefinitely to matrices, functions and propositions of higher and higher orders. We introduce the following definition:

A function is said to be predicative when it is a matrix. It will be observed that, in a hierarchy in which all the variables are individuals or matrices, a matrix is the same thing as an elementary function (cf. pp. 132, 133).

"Matrix" or "predicative function" is a primitive idea.

The fact that a function is predicative is indicated, as above, by a note of exclamation after the functional letter.

The variables occurring in the present work, from this point onwards, will all be either individuals or matrices of some order in the above hierarchy. Propositions, which have occurred hitherto as variables, will no longer do so except in a few isolated cases of which no subsequent use is made. In practice, for the reasons explained on p. 169, a function of a matrix may be regarded as capable of any argument which is a function of the same order and takes arguments of the same type.

In practice, we never need to know the absolute types of our variables, but only their relative types. That is to say, if we prove any proposition on the assumption that one of our variables is an individual, and another is a function of order , the proof will still hold if, in place of an individual, we take a function of order , and in place of our function of order we take a function of order , with corresponding changes for any other variables that may be involved. This results from the assumption that our primitive propositions are to apply to variables of any order.

We shall use small Latin letters (other than , , , for variables of the lowest type concerned in any context. For functions, we shall use the letters , , , , , , (except that, at a later stage, will be defined as a constant relation, and will be defined as the order-type of the continuum).

We shall explain later a different hierarchy, that of classes and relations, which is derived from the functional hierarchy explained above, but is more convenient in practice.

When any predicative function, say , occurs as apparent variable, it would be strictly more correct to indicate the fact by placing "(" before what follows, as thus: "(." But for the sake of brevity we write simply "(" instead of "(." Since what follows the in brackets must always contain with arguments supplied, no confusion can result from this practice.

It should be observed that, in virtue of the manner in which our hierarchy of functions was generated, non-predicative functions always result from such as are predicative by means of generalization. Hence it is unnecessary to introduce a special notation for non-predicative functions of a given order[Pg 173] and taking arguments

of a given order. For example, second-order functions of an individual are always derived by generalization from a matrix

where the functions , , , ... are predicative. It is possible, therefore, without loss of generality, to use no apparent variables except such as are predicative.

We require, however, a means of symbolising a function whose order is not assigned. We shall use "" or "" or etc. to express a function (or) whose order, relatively to its argument, is not given. Such a function cannot be made into an apparent variable, unless we suppose its order previously fixed. As the only purpose of the notation is to avoid the necessity of fixing the order, such a function will not be used as an apparent variable; the only functions which will be so used will be predicative functions, because, as we have just seen, this restriction involves no loss of generality.

We have now to state and explain the axiom of reducibility.

It is important to observe that, since there are various types of propositions and functions, and since generalization can only be applied within some one type (or, by means of systematic ambiguity, within some well-defined and completed set of types), all phrases referring to "all propositions" or "all functions," or to "some (undetermined) proposition" or "some (undetermined) function," are prima facie meaningless, though in certain cases they are capable of an unobjectionable interpretation. Contradictions arise from the use of such phrases in cases where no innocent meaning can be found.

If mathematics is to be possible, it is absolutely necessary (as explained in the Introduction, Chapter II) that we should have some method of making statements which will usually be equivalent to what we have in mind when we (inaccurately) speak of "all properties of ." (A "property of " may be defined as a propositional function satisfied by .) Hence we must find, if possible, some method of reducing the order of a propositional function without affecting the truth or falsehood of its values. This seems to be what common-sense effects by the admission of classes. Given any propositional function of whatever order, this is assumed to be equivalent, for all values of , to a statement of the form " belongs to the class ." Now assuming that there is such an entity as the class , this statement is of the first order, since it involves no allusion to a variable function. Indeed its only practical advantage over the original statement is that it is of the first order. There is no advantage in assuming that there really are such things as classes, and the contradiction about the classes which are not members of themselves shows that, if there are classes, they must be something radically different from individuals. It would seem that the sole purpose which classes serve, and one main reason which makes them linguistically convenient, is[Pg 174] that they provide a method of reducing the order of a propositional function. We shall, therefore, not assume anything of what may seem to be involved in the common-sense admission of classes, except this, that every propositional function is equivalent, for all its values, to some predicative function of the same argument or arguments.

This assumption with regard to functions is to be made whatever may be the type of their arguments. Let be a function, of any order, of an argument , which may itself be either an individual or a function of any order. If is a matrix, we write the function in the form ; in such a case we call a predicative function. Thus a predicative function of an individual is a first-order function; and for higher types of arguments, predicative functions take the place that first-order functions take in respect of individuals. We assume, then, that every function of one variable is equivalent, for all its values, to some predicative function of the same argument. This assumption seems to be the essence of the usual assumption of classes; at any rate, it retains as much of classes as we have any use for, and little enough to avoid the contradictions which a less grudging admission of classes is apt to entail. We will call this assumption the axiom of classes, or the axiom of reducibility.

We shall assume similarly that every function of two variables is equivalent, for all its values, to a predicative function of those variables, i.e. to a matrix. This assumption is what seems to be meant by saying that any statement about two variables defines a relation between them. We will call this assumption the axiom of relations or (like the previous axiom) the axiom of reducibility.

In dealing with relations between more than two terms, similar assumptions would be needed for three, four, ... variables. But these assumptions are not indispensable for our purpose, and are therefore not made in this work.

Stated in symbols, the two forms of the axiom of reducibility are as follows:

*12·1.

*12·11.

We call two functions , formally equivalent when , and similarly we call and formally equivalent when

Thus the above axioms state that any function of one or two variables is formally equivalent to some predicative function of one or two variables, as the case may be.

[Pg 175]

Of the above two axioms, the first is chiefly needed in the theory of classes (*20), and the second in the theory of relations (*21). But the first is also essential to the theory of identity, if identity is to be defined (as we have done, in *13·01); its use in the theory of identity is embodied in the proof of *13·101, below.

We may sum up what has been said in the present number as follows:

(1) A function of the first order is one which involves no variables except individuals, whether as apparent variables or as arguments.

(2) A function of the (th order is one which has at least one argument or apparent variable of order , and contains no argument or apparent variable which is not either an individual or a first-order function or a second-order function or ... or a function of order .

(3) A predicative function is one which contains no apparent variables, i.e. is a matrix. It is possible, without loss of generality, to use no variables except matrices and individuals, so long as variable propositions are not required.

(4) Any function of one argument or of two is formally equivalent to a predicative function of the same argument or arguments.

FOOTNOTES:

[52] Cf. Introduction, Chapter II.

[53] Cf. pp. 95, 96.

[Pg 176]

*13. IDENTITY.

Summary of *13.

The propositional function " is identical with " will be written "." We shall find that this use of the sign of equality covers all the common uses of equality that occur in mathematics. The definition is as follows:

*13·01.

This definition states that and are to be called identical when every predicative function satisfied by is also satisfied by . We cannot state that every function satisfied by is to be satisfied by , because satisfies functions of various orders, and these cannot all be covered by one apparent variable. But in virtue of the axiom of reducibility it follows that, if and satisfies , where is any function, predicative or non-predicative, then also satisfies (cf. *13·101, below). Hence in effect the definition is as powerful as it would be if it could be extended to cover all functions of .

Note that the second sign of equality in the above definition is combined with "," and thus is not really the same symbol as the sign of equality which is defined. Thus the definition is not circular, although at first sight it appears so.

The propositions of the present number are constantly referred to. Most of them are self-evident, and the proofs offer no difficulty. The most important of the propositions of this number are the following:

*13·101.

I.e. if and are identical, any property of is a property of .

*13·12.

This includes *13·101 together with the fact that if and are identical any property of is a property of .

*13·15·16·17, which state that identity is reflexive, symmetrical and transitive.

*13·191.

[Pg 177]

I.e. to state that everything that is identical with has a certain property is equivalent to stating that has that property.

*13·195.

I.e. to state that something identical with has a certain property is equivalent to saying that has that property.

*13·22.

This is the analogue of *13·195 for two variables.

*13·01.

The following definitions embody abbreviations which are often convenient.

*13·02.

*13·03.

*13·1.

*13·101.

Dem.

In virtue of this proposition, if , satisfies any function, whether predicative or non-predicative, which is satisfied by . It will be observed that the proof uses the axiom of reducibility (*12·1). But for this axiom, two terms and might agree in respect of all predicative functions, but not in respect of all non-predicative functions. We should thus be led to identities of different degrees, according to the degree of the functions in respect of which and agreed. Strict identity would, in this case, have to be taken as a primitive idea, and *13·101 would have to be a primitive proposition, as would also *13·15·16·17.

*13·11.

Dem.

[Pg 178]

*13·12.

Dem.

*13·13.

*13·14.

*13·15.

*13·16.

*13·17.

Dem.

In the above use of *10·3, , , are regarded as three different functions of , and replaces the of *10·3.

The above three propositions show that identity is reflexive (*13·15), symmetrical (*13·16), and transitive (*13·17). These are the three marks of relations having the formal properties which we associate commonly with the sign of equality.

*13·171.

*13·172.

*13·18.

*13·181.

*13·182.

*13·183.

Dem.

*13·19.

*13·191.

Dem.

This proposition is constantly used in subsequent proofs.

[Pg 179]

*13·192.

Dem.

This proposition is useful in the theory of descriptions (*14).

*13·193.

Dem.

This proposition is very often used.

*13·194.

This proposition is used in *37·65 and *101·14.

*13·195.

Dem.

The use of this proposition in subsequent proofs is very frequent.

*13·196.

*13·21.

Dem.

This proposition is the analogue, for two variables, of *13·191.

[Pg 180]

*13·22.

Dem.

This proposition is the analogue, for two variables, of *13·195. It is frequently used, especially in the theory of couples (*54, *55, *56).

The following proposition is useful in the theory of types. Its purpose is to show that, if is any argument for which "" is significant, i.e. for which we have , then "" is significant when, and only when, is either identical with a or not identical with . It follows (as will be proved in *20·81) that, if "" and "" are both significant, the class of values of for which "" is significant is the same as the class of those for which "" is significant, i.e. two types which have a common member are identical.

In the following proof, the chief point to observe is the use of *10·221. There are two variables, and , to be identified. In the first use, we depend upon the fact that and both occur in both (4) and (5): the occurrence of in both justifies the identification of the two 's, and when these have been identified, the occurrence of in both justifies the identification of the two 's. (Unless the 's had been already identified, this would not be legitimate, because "" is typically ambiguous if neither nor is of given type.) The second use of *10·221 is justified by the fact that both and occur in both (2) and (6).

*13·3.

Dem.

[Pg 181]

*14. DESCRIPTIONS.

Summary of *14.

A description is a phrase of the form "the term which etc.," or, more explicitly, "the term which satisfies ," where , is some function satisfied by one and only one argument. For reasons explained in the Introduction (Chapter III), we do not define "the which satisfies ," but we define any proposition in which this phrase occurs. Thus when we say: "The term which satisfies satisfies ," we shall mean: "There is a term such that is true when, and only when, is , and is true." That is, writing "" for "the term which satisfies ," is to mean

This, however, is not yet quite adequate as a definition, for when (occurs in a proposition which is part of a larger proposition, there is doubt whether the smaller or the larger proposition is to be taken as the "." Take, for example, . This may be either

If "(" is false, the first of these must be true, while the second must be false. Thus it is very necessary to distinguish them.

The proposition which is to be treated as the "" will be called the scope of (. Thus in the first of the above two propositions, the scope of (is , while in the second it is . In order to avoid ambiguities as to scope, we shall indicate the scope by writing "" at the beginning of the scope, followed by enough dots to extend to the end of the scope. Thus of the above two propositions the first is

while the second is

Thus we arrive at the following definition:

*14·01.

It will be found in practice that the scope usually required is the smallest proposition enclosed in dots or brackets in which "(" occurs. Hence[Pg 182] when this scope is to be given to (, we shall usually omit explicit mention of the scope. Thus e.g. we shall have

Of these the first necessarily implies (, while the second does not. We put

*14·02.

This defines: "The satisfying exists," which holds when, and only when, is satisfied by one value of and by no other value.

When two or more descriptions occur in the same proposition, there is need of avoiding ambiguity as to which has the larger scope. For this purpose, we put

*14·03.

It will be shown *14·113 that the truth-value of a proposition containing two descriptions is unaffected by the question which has the larger scope. Hence we shall in general adopt the convention that the description occurring first typographically is to have the larger scope, unless the contrary is expressly indicated. Thus e.g.

will mean

i.e.

By this convention we are able almost always to avoid explicit indication of the order of elimination of two or more descriptions. If, however, we require a larger scope for the later description, we put

*14·04

Whenever we have behaves, formally, like an ordinary argument to any function in which it may occur. This fact is embodied in the following proposition:

*14·18

That is to say, when (exists, it has any property which belongs to everything. This does not hold when (does not exist; for example, the present King of France does not have the property of being either bald or not bald.

If (has any property whatever, it must exist. This fact is stated in the proposition:

*14·21

This proposition is obvious, since "E" is, by the definitions, part[Pg 183] of "." When, in ordinary language or in philosophy, something is said to "exist," it is always something described, i.e. it is not something immediately presented, like a taste or a patch of colour, but something like "matter" or "mind" or "Homer" (meaning "the author of the Homeric poems"), which is known by description as "the so-and-so," and

is thus of the form (. Thus in all such cases, the existence of the (grammatical) subject (can be analytically inferred from any true proposition having this grammatical subject. It would seem that the word "existence" cannot be significantly applied to subjects immediately given; i.e. not only does our definition give no meaning to "," but there is no reason, in philosophy, to suppose that a meaning of existence could be found which would be applicable to immediately given subjects.

Besides the above, the following are among the more useful propositions of the present number.

*14·202.

From the first equivalence in the above, it follows that

*14·204.

I.e. (exists when there is something which (is.

We have

*14·205.

I.e. (has the property when there is something which is (and which has the property .

We have to prove that such symbols as "(" obey the same rules with regard to identity as symbols which directly represent objects. To this, however, there is one partial exception, for instead of having

we only have

*14·28.

I.e. "(" only satisfies the reflexive property of identity if (exists.

The symmetrical property of identity holds for such symbols as (, without the need of assuming existence, i.e. we have

*14·13.

*14·131.

Similarly the transitive property of identity holds without the need of assuming existence. This is proved in *14·14·142·144.

[Pg 184]

*14·01.

*14·02.

*14·03.

*14·04.

*14·1.

In virtue of our conventions as to the scope intended when no scope is explicitly indicated, the above proposition is the same as the following:

*14·101.

*14·11.

*14·111.

Dem.

*14·112.

In the above proposition, we assume the convention explained on p. 182, after the statement of *14·03.

*14·113.

This proposition shows that when two descriptions occur in the same proposition, the truth-value of the proposition is unaffected by the question which has the larger scope.

*14·12.

Dem.

[Pg 185]

*14·121.

Dem.

*14·122.

Dem.

The two following propositions (*14·123·124) are placed here because of the analogy with *14·122, but they are not used until we come to the theory of couples (*55 and *56).

*14·123.

Dem.

[Pg 186]

*14·124.

Dem.

*14·13.

Dem.

This proposition is not an immediate consequence of *13·16, because "" is not a value of the function "." Similar remarks apply to the following propositions.

*14·131.

[Pg 187]

Dem.

In the above proposition, in accordance with our convention, the descriptive expression (is eliminated before (, because it occurs first in "("; but in "(," (is to be first eliminated. The order of elimination makes no difference to the truth-value, as was proved in *14·113.

The above proposition may also be proved as follows:

*14·14.

*14·142.

Dem.

*14·144.

Dem.

*14·145.

[Pg 188]

Dem.

*14·15.

Dem.

*14·16.

Dem.

*14·17.

Dem.

It should be observed that we do not have

for, if is always false, and therefore

holds for all values of . But we do have

[Pg 189]

*14·171.

Dem.

*14·18.

Dem.

The above proposition shows that, provided (exists, it has (speaking formally) all the logical properties of symbols which directly represent objects. Hence when (exists, the fact that it is an incomplete symbol becomes irrelevant to the truth-values of logical propositions in which it occurs.

*14·2.

Dem.

*14·201.

Dem.

*14·202.

Dem.

[The second half is proved in the same way as the first half.]

*14·203.

[Pg 190]

Dem.

*14·204.

Dem.

*14·205.

*14·21.

Dem.

This proposition shows that if any true statement can be made about (, then (must exist. Its use throughout the remainder of the work will be very frequent.

When (does not exist, there are still true propositions in which "(" occurs, but it has, in such propositions, a secondary occurrence, in the sense explained in Chapter III of the Introduction, i.e. the asserted proposition concerned is not of the form , but of the form , in other words, the proposition which is the scope of (is only part of the whole asserted proposition.

*14·22.

Dem.

[Pg 191]

As an instance of the above proposition, we may take the following: "The proposition 'the author of Waverley existed' is equivalent to 'the man who wrote Waverley wrote Waverley.'" Thus such a proposition as "the man who wrote Waverley wrote Waverley" does not embody a logically necessary truth, since it would be false if Waverley had not been written, or had been written by two men in collaboration. For example, "the man who squared the circle squared the circle" is a false proposition.

*14·23.

Dem.

Note that in the second line of the above proof *10·5, not only *3·26, is required. For the scope of the descriptive symbol (is the whole product , so that, applying *14·1, the proposition on the right in the first line becomes

which, by *10·5 and *3·26, implies

*14·24.

Dem.

This proposition should be compared with *14·241, where, in virtue of the smaller scope of (, we get an implication instead of an equivalence.

*14·241.

Dem.

[Pg 192]

*14·242.

*14·25.

Dem.

*14·26.

Dem.

*14·27.

Dem.

*14·271.

Dem.

[Pg 193]

*14·272.

Dem.

The above two propositions show that and are "extensional" properties of , i.e. their truth-value is unchanged by the substitution, for , of any formally equivalent function .

*14·28.

Dem.

This proposition states that (is identical with itself whenever it exists, but not otherwise. Thus for example the proposition "the present King of France is the present King of France" is false.

The purpose of the following propositions is to show that, when , the scope of (does not matter to the truth-value of any proposition in which (occurs. This proposition cannot be proved generally, but it can be proved in each particular case. The following propositions show the method, which proceeds always by means of *14·242, *10·23 and *14·11. The proposition can be proved generally when (occurs in the form , and occurs in what we may call a "truth-function," i.e. a function whose truth or falsehood depends only upon the truth or falsehood of its argument or arguments. This covers all the cases with which we are ever concerned. That is to say, if occurs in any of the ways which can be generated by the processes of *1—*11,

then, provided , the truth-value of is the same as that of

[Pg 194]This is proved in the following proposition. In this proposition, however, the use of propositions as apparent variables involves an apparatus not required elsewhere, and we have therefore not used this proposition in subsequent proofs.

*14·3.

Dem.

The following propositions are immediate applications of the above. They are, however, independently proved, because *14·3 introduces propositions (, namely) as apparent variables, which we have not done elsewhere, and cannot do legitimately without the explicit introduction of the hierarchy of propositions with a reducibility-axiom such as *12·1.

*14·31.

Dem.

The following propositions are proved in precisely the same way as *14·31; hence we shall merely give references to the propositions used in the proofs.

*14·32.

The equivalence asserted here fails when . Thus, for example, let be " is King of France." Then (= the King of France. Let be " is bald." Then the[Pg 195] King of France exists and is not bald; but it is false that the King of France exists and is bald. Of these the first is false, the second true. Either might be meant by "the King of France is not bald," which is ambiguous; but it would be more natural to take the first (false) interpretation as the meaning of the words. If the King of France existed, the two would be equivalent; thus as applied to the King of England, both are true or both false.

*14·33.

*14·331.

*14·332.

*14·34.

This proposition does not require the hypothesis .

Dem.

Propositions of the above type might be continued indefinitely, but as they are proved on a uniform plan, it is unnecessary to go beyond the fundamental cases of , , and

It should be observed that the proposition in which (has the larger scope always implies the corresponding one in which it has the smaller scope, but the converse implication only holds if either (a) we have or (b) the proposition in which (has the smaller scope implies . The second case occurs in *14·34, and is the reason why we get an equivalence without the hypothesis . The proposition in which (has the larger

scope always implies , in virtue of *14·21.

[Pg 196]

SECTION C.

CLASSES AND RELATIONS.

*20. GENERAL THEORY OF CLASSES.

Summary of *20.

The following theory of classes, although it provides a notation to represent them, avoids the assumption that there are such things as classes. This it does by merely defining propositions in whose expression the symbols representing classes occur, just as, in *14, we defined propositions containing descriptions.

The characteristics of a class are that it consists of all the terms satisfying some propositional function, so that every propositional function determines a class, and two functions which are formally equivalent (i.e. such that whenever either is true, the other is true also) determine the same class, while conversely two functions which determine the same class are formally equivalent. When two functions are formally equivalent, we shall say that they have the same extension. The incomplete symbols which take the place of classes serve the purpose of technically providing something identical in the case of two functions having the same extension; without something to represent classes, we cannot, for example, count the combinations that can be formed out of a given set of objects.

Propositions in which a function occurs may depend, for their truth-value, upon the particular function , or they may depend only upon the extension of . In the former case, we will call the proposition concerned an intensional function of ; in the latter case, an extensional function of . Thus, for example, (or (is an extensional function of , because, if is formally equivalent to , i.e. if , we have (and (. But on the other hand "I believe (" is an intensional function, because, even if , it by no means follows that I believe (provided I believe (. The mark of an extensional function of a function is [Pg 197] (We write "" when we wish to speak of the function itself as opposed to its argument.) The functions of functions with which mathematics is specially concerned are all extensional.

When a function of is extensional, it may be regarded as being about the class determined by , since its truth-value remains unchanged so long as the class is unchanged. Hence we require, for the theory of classes, a method of obtaining an extensional function from any given function of a function. This is effected by the following definition:

*20·01.

Here is in reality a function of , which is defined whenever is significant for predicative functions . But it is convenient to regard as though it had an argument), which we will call "the class determined by the function ." It will be proved shortly that is always an extensional function of , and that, applying the definition of identity

(*13·01) to the fictitious objects) and), we have

This last is the distinguishing characteristic of classes, and justifies us in treating) as the class determined by .

With regard to the scope of), and to the order of elimination of two such expressions, we shall adopt the same conventions as were explained in *14 for (. The condition corresponding to

which is always satisfied because of *12·1.

Following Peano, we shall use the notation

to express " is a member of the class determined by ." We therefore introduce the following definition:

*20·02.

In this form, the definition is never used; it is introduced for the sake of the proposition

which results from *20·02 and *20·01, and leads to

by the help of *12·1.

We shall use small Greek letters (other than , , , , , ,) to represent classes, i.e. to stand for symbols of the form or . When a small Greek letter occurs as apparent variable, it is to be understood to stand for a[Pg 198] symbol of the form , where is properly the apparent variable concerned. The use of single letters in place of such symbols as or is practically almost indispensable, since otherwise the notation rapidly becomes intolerably cumbrous. Thus "" will mean " is a member of the class ," and may be used wherever no special defining function of the class is in question.

The following definition defines what is meant by a class.

*20·03.

Note that the expression "" has no meaning in isolation: we have merely defined (in *20·01) certain uses of such expressions. What the above definition decides is that the symbol "Cls" may replace the symbol "," wherever the latter occurs, and that the meaning of the combination of symbols concerned is to be unchanged thereby. Thus "Cls," also, has no meaning in isolation, but merely in certain uses.

The above definition, like many future definitions, is ambiguous as to type. The Latin letter , according to our conventions, is to represent the lowest type concerned; thus is of the type next above this. It is convenient to speak of a class as being of the same type as its defining function; thus is of the type next above that of , and "Cls" is of the type next above that of . Thus the type of "Cls" is fixed relatively to the lowest type concerned; but if, in two different contexts, different types are the lowest concerned, the meaning of "Cls" will be different in these two contexts. The meaning of "Cls" only becomes definite when the lowest type concerned is specified.

Equality between classes is defined by applying *13·01, symbolically unchanged, to their defining functions, and then using *20·01.

The propositions of the present number may be divided into three sets. First, we have those that deal with the fundamental properties of classes; these end with *20·43. Then we have a set of propositions dealing with both classes and descriptions; these extend from *20·5 to *20·59 (with the exception of *20·53·54). Lastly, we have a set of propositions designed to prove that classes of classes have all the same formal properties as classes of individuals.

In the first set, the principal propositions are the following.

*20·15.

I.e. two classes are identical when, and only when, their defining functions are formally equivalent. This is the principal property of classes.

*20·31.

[Pg 199]

I.e. two classes are identical when, and only when, they have the same members.

*20·43.

This is the same proposition as *20·31, merely employing Greek letters in place of and .

*20·18.

I.e. if two classes are identical, any property of either belongs also to the other. This is the analogue of *13·12.

*20·2·21·22, which prove that identity between classes is reflexive, symmetrical and transitive.

*20·3.

I.e. a term belongs to a class when, and only when, it satisfies the defining function of the class.

In the second set of propositions (*20·3—·59), we show that, under suitable circumstances, expressions such as (may be substituted for in *20·3 and various other propositions of the first set, and we prove a few properties of such expressions as "(," i.e. "the class which satisfies the function ." Here it is to be remembered that "" stands for "," and that "" therefore stands for "." This is, in reality, a function of , namely the extensional function associated with by means of *20·01. Thus an expression containing a variable class is always an abbreviation for an expression containing a variable function.

In the third set of propositions, we prove that variable classes satisfy all the primitive propositions assumed for variable individuals or functions, whence it follows, by merely repeating the proofs of the first set of propositions (*20·1—·43), that classes of classes have all the formal properties of classes of individuals or functions. We shall never have occasion explicitly to consider classes of functions, but classes of classes will occur constantly—for example, every cardinal number will be defined as a class of classes. Classes of relations, which will also frequently occur,

will be considered in *21.

*20·01.

*20·02.

*20·03.

The three following definitions serve merely for purposes of abbreviation.

*20·04.

*20·05.

*20·06.

[Pg 200]

The following definitions merely extend to symbols representing classes the definitions which have already been given for other symbols, with the smallest possible modifications.

*20·07. (

*20·071. (

*20·072.

*20·08.

*20·081.

The propositions which follow give the most general properties of classes.

*20·1.

*20·11.

Dem.

This proves that every proposition about a class expresses an extensional property of the determining function of the class, and therefore does not depend for its truth or falsehood upon the particular function selected for determining the class, but only upon the extension of the determining function.

*20·111.

Dem.

*20·112.

Dem.

Thus the axiom of reducibility still holds for classes as arguments.

[Pg 201]

*20·12.

*20·13.

The meaning of "" is obtained by a double application of *20·01 to *13·01, remembering the convention that is to have a larger scope than because it occurs first.

Dem.

*20·14.

Dem.

This proposition is the converse of *20·13.

*20·15.

This proposition states that two functions determine the same class when, and only when, they are formally equivalent, i.e. are satisfied by the same set of values. This is the essential property of classes, and gives the justification of the definition *20·01.

*20·151.

Dem.

In virtue of this proposition, all classes can be obtained from predicative functions. This fact is especially important when classes are used as apparent variables. For in that case, according to the definitions *20·07·071, the apparent variable really involved is a predicative function. In virtue of *20·151, this places no limitation upon the classes concerned, except the limitation which inevitably results from the nature of their membership.

[Pg 202]

A class, therefore, unlike a function, has its order completely determined by the order of its possible members, i.e. of the arguments which render its defining function significant.

*20·16.

*20·17.

*20·18.

*20·19.

Dem.

*20·191.

*20·2.

Dem.

*20·21.

*20·22.

[Pg 203]

The above propositions are not immediate consequences of *13·15·16·17, for a reason analogous to that explained in the note to *14·13, namely because is not a value of , and therefore in particular "" is not a value of "."

*20·23.

*20·24.

*20·25.

Dem.

*20·3.

Dem.

This proposition shows that is a member of the class determined by when, and only when, satisfies .

*20·31.

*20·32.

*20·33.

Dem.

Here is written in place of some expression of the form . The use of the single Greek letter is more convenient whenever the determining function is irrelevant.

*20·34.

Dem.

[Pg 204]

The above proposition and *20·25 illustrate the use of Greek letters as apparent variables.

*20·35.

*20·4.

*20·41.

*20·42.

A Greek letter, such as , is merely an abbreviation for an expression of the form , thus this proposition is *20·32 repeated.

Dem.

*20·43.

The following propositions deal with cases in which both classes and descriptions occur. In such cases, we shall, in the absence of any indication to the contrary, adopt the convention that the descriptions are to have a larger scope than the classes, in applying the definitions *14·01 and *20·01.

*20·5.

Dem.

*20·51.

Dem.

*20·52.

Dem.

*20·53.

This is the analogue of *13·191.

[Pg 205]

Dem.

*20·54.

This proposition is the analogue of *13·195.

Dem.

*20·55.

Dem.

*20·56.

*20·57.

Dem.

*20·58.

Dem.

[Pg 206]

*20·59.

Dem.

In the following propositions, we shall prove that classes have all the formal properties of individuals, and have the same relations to classes of classes as individuals have to classes of individuals. It is only necessary to prove the analogues of our primitive propositions, and of our definitions in cases where their analogues are not themselves definitions. We shall take the propositions *10·1·11·12·121·122, rather than those of *9, and we shall prove the analogue of *10·01. As was pointed out in *10, we shall thus have proved everything upon which subsequent proofs depend. The analogues of *20·01·02 and of *14·01 remain definitions, but those of *10·01 and *13·01 become propositions to be proved. *9·131 must be extended by the definition: Two classes are "of the same type" when they have predicative defining functions of the same type. In addition to these, we have to prove the analogues of *10·1·11·12·121·122, *11·07 and *12·1·11. When these have been proved, the

analogues of other propositions follow by merely repeating previous proofs. These analogues will, therefore, be quoted by the numbers of the original propositions whose analogues they are.

*20·6.

Dem.

This is the analogue of *10·01.

*20·61.

Dem.

This is the analogue of *10·1.

In practice we also need

This is *20·17.

We need further

This is *20·41.

[Pg 207]

*20·62. When is true, whatever possible argument of the form may be, then (is true.

This is the analogue of *10·11.

Dem.

when is true, whatever possible argument may be, then (is true, i.e. (by *20·07), (is true.

*20·63.

This is the analogue of *10·12.

Dem.

*20·631. If "" is significant, then if is of the same type as , "" is significant, and vice versa.

This is the analogue of *10·121.

Dem.

By *20·151, is of the form , and therefore, by *20·01, is a function of . Similarly is of the form , and is a function of . Hence by applying *10·121 to and the result follows.

*20·632. If, for some , there is a proposition , then there is a function , and vice versa.

Dem.

By the definition in *20·01, is a function of . Hence the proposition follows from *10·122.

*20·633. "Whatever possible class may be, is true whatever possible class may be" implies the corresponding statement with and interchanged except in "." (The corresponding exception is to be understood in *11·07.)

This is the analogue of *11·07, and follows at once from *11·07 because is a function of the defining functions of and .

*20·64.

Dem.

[Pg 208]

Observe that "" is merely an abbreviation for any symbol of the form . This is why nothing further is required in the above proof.

The above proposition is the analogue of *10·14. Like that proposition, it requires, for the significance of the conclusion, that and should be functions which take arguments of the same type. This is not required for the significance of the hypothesis. Hence, though the above proposition is true whenever it is significant, it is not true whenever its hypothesis is significant.

*20·7.

This is the analogue of *12·1.

*20·701.

[The proof proceeds as in *20·112, using *12·11 instead of *12·1.]

*20·702.

[Proof as in *20·701.]

*20·703.

Dem.

*20·701·702·703 give the analogues, for classes, of *12·11.

*20·71.

This is the analogue of *13·01.

[Pg 209]

This completes the proof that all propositions hitherto given apply to classes as well as to individuals. Precisely similar reasoning extends this result to classes of classes, classes of classes of classes, etc.

From the above propositions it appears that, although expressions such as have no meaning in isolation, yet those of their formal properties with which we have been hitherto concerned are the same as the corresponding properties of symbols which have a meaning in isolation. Hence nothing in the apparatus hitherto introduced requires us to determine whether a given symbol stands for a class or not, unless the symbol occurs in a way in which only a class can occur significantly. This is an important result, which enables us to give much greater generality to our

propositions than would otherwise be possible.

The two following propositions (*20·8·81) are consequences of *13·3. The "type" of any object will be defined in *63 as the class of terms either identical with or not identical with . We may define the "type of the arguments to " as the class of arguments for which "" is significant, i.e. the class . Then the first of the following propositions shows that if "" is significant, the type of the arguments to is the type of ; the second proposition shows that, if "" and "" are both significant, the type of the arguments to is the same as the type of the arguments to , because each is the type of . *20·8 will be used in *63·11, which is a fundamental proposition in the theory of relative types.

*20·8.

Dem.

*20·81.

Dem.

In the third line of the above proof, the use of *10·121 depends upon the fact that the "" in both (1) and (2) must be such as to render the hypothesis significant, i.e. such as to render [Pg 210] significant. Hence the "" in (1) and the "a" in (2) must be of the same type, by *10·121, and hence by *10·13 we can assert the product of (1) and (2), identifying the two "'s."

Since a type is the range of significance of a function, if is a function which is always true, must be a type. For if a function is always true, the arguments for which it is true are the same as the arguments for which it is significant; hence is the range of significance of , if (holds. Thus any class is a type if (. It follows that, whatever function may be, is a type; and in particular, is a type. Since is a member of this class, this class is the type to which a belongs. In virtue of *20·8, if is significant, the type to which a belongs is the class of arguments for which is significant, i.e. . And if there is any argument a for which and are both significant, then and have the same range of significance, in virtue of *20·81.

[Pg 211]

*21. GENERAL THEORY OF RELATIONS.

Summary of *21.

The definitions and propositions of this number are exactly analogous to those of *20, from which they differ by being concerned with functions of two variables instead of one. A relation, as we shall use the word, will be understood in extension: it may be regarded as the class of couples (for which some given function is true. Its relation to the function is just like that of the class to its determining function. We put

*21·01.

Here "" has no meaning in isolation, but only in certain of its uses. In *21·01 the alphabetical order of and corresponds to the typographical order of and in , so that

This is important in relation to the substitution-convention below.

It will be shown that

i.e. that two relations, as above defined, are identical when, and only when, they are satisfied by the same pairs of arguments.

For substitution in and , we adopt the convention that when a function (as opposed to its values) is represented in a form involving and , or any other two letters of the alphabet, the value of this function for the arguments and is to be found by substituting for and for , while the value for the arguments and is to be found by substituting for and for . That is, the argument mentioned first is to be substituted for the letter which comes first in the alphabet, and the argument mentioned second for the later letter; thus the mode of substitution depends upon the alphabetical order of the letters which have circumflexes and the typographical order of the other letters.

[Pg 212]

The above convention as to order is presupposed in the following definition, where is the first argument mentioned and the second:

*21·02.

Hence, following the convention,

This definition is not used as it stands, but is introduced for the sake of

which results from *21·01·02. We shall use capital Latin letters to represent variable expressions of the form , just as we used Greek letters for variable expressions of the form . If a capital Latin letter, say , is used as an apparent variable, it is supposed that the which occurs in the form "(" or "(" is to be replaced by "(" or "(," while the which occurs later is to be replaced by "." In fact we put

The use of single letters for such expressions as is a practically indispensable convenience.

The following is the definition of the class of relations:

*21·03.

Similar remarks apply to it as to the definition of "Cls" (*20·03).

In virtue of the definitions *21·01·02 and the convention as to capital Latin letters, the notation "" will mean " has the relation to ." This notation is practically convenient, and will, after the preliminaries, wholly replace the cumbrous notation .

The proofs of the propositions of this number are usually omitted, since they are exactly analogous to those of *20, merely substituting *12·11 for *12·1, and propositions in *11 for propositions in *10.

The propositions of this number, like those of *20, fall into three sections. Those of the second section are seldom referred to. Those of the third section, extending to relations the formal properties hitherto assumed or proved for individuals and functions, are not explicitly referred to in the sequel, but are constantly relevant, namely whenever a proposition which has been assumed or proved for individuals

and functions is applied to relations. The principal propositions of the first section are the following.

*21·15.

I.e. two relations are identical when, and only when, their defining functions are formally equivalent.

*21·31.

[Pg 213]

I.e. two relations are identical when, and only when, they hold between the same pairs of terms. The same fact is expressed by the following proposition:

*21·43.

*21·2·21·22 show that identity of relations is reflexive, symmetrical and transitive.

*21·3.

I.e. two terms have a given relation when, and only when, they satisfy its defining function.

*21·151.

I.e. every relation can be defined by a predicative function. Hence when, using *21·07 or *21·071, we have a relation as apparent variable, and are therefore confined to predicative defining functions, there is no loss of generality.

*21·01.

On the convention as to order in *21·01·02, cf. p. 211, and thus relate , to , so that

*21·02.

*21·03.

The following definitions merely extend to relations, with as little modification as possible, the definitions already given for other symbols.

*21·07. (

*21·071. (

*21·072.

*21·08.

*21·081.

The convention as to typographic and alphabetic order is here retained.

*21·082.

*21·083.

*21·1.

*21·11.

This proposition proves that every proposition about a relation expresses an extensional property of the determining function.

*21·111.

[Pg 214]

*21·112.

It is *12·1, not *12·11, which is required in this proposition, because we are concerned with a function (of one variable, namely , although that one variable is itself a function of two variables.

*21·12.

This is the first use of the primitive proposition *12·11, except in *20·701·702·703.

*21·13.

*21·14.

*21·15.

This proposition states that two double functions determine the same relation when, and only when, they are formally equivalent, i.e. are satisfied by the same pairs of arguments. This is a fundamental property of relations as defined above (*21·01).

*21·151.

*21·16.

*21·17.

*21·18.

*21·19.

*21·191.

*21·2.

*21·21.

*21·22.

*21·23.

*21·24.

*21·3.

[Pg 215]

This shows that has to y the relation determined by when, and only when, and satisfy .

Note that the primitive proposition *12·11 is again required here.

*21·31.

*21·32.

*21·33.

Here is written for some expression of the form . The use of a single capital letter for a relation is convenient whenever the determining function is irrelevant.

*21·4.

*21·41.

*21·42.

*21·43.

*20·5·51·52 have no analogues in the theory of relations.

*21·53.

*21·54.

*21·55.

*21·56.

*21·57.

*21·58.

The following propositions are the analogues of *20·6 ff., and have a similar purpose.

*21·6.

*21·61.

*21·62. When is true, whatever possible argument of the form may be, (is true. [Proof as in *20·62]

*21·63.

*21·631. If "" is significant, then if is of the same type as , "" is significant, and vice versa.

[Proof as in *20·631]

[Pg 216]

*21·632. If, for some , there is a proposition , then there is a function , and vice versa.

[Proof as in *20·632]

*21·633. "Whatever possible relation may be, is true whatever possible relation may be" implies "whatever possible relation may be, is true whatever possible relation may be."

[Proof as in *20·633]

*21·64.

*21·7.

*21·701.

*21·702.

*21·703.

*21·704.

*21·705.

*21·71.

From the above propositions it appears that relations, like classes, have all the formal properties which they would have if they were symbols having a meaning in isolation. Hence unless a symbol occurs in a way in which only a relation can occur significantly, we do not need to decide whether it stands for a relation or not. This result, like the corresponding result for classes mentioned at the end of *20, is important as giving greater generality to our propositions than they would otherwise possess. The results obtained in *20 and *21 for classes and relations whose members or terms are neither classes nor relations can be extended, by mere repetition of the proofs, to classes of classes, classes of relations, relations of classes, relations of relations, and so on.

[Pg 217]

*22. CALCULUS OF CLASSES.

Summary of *22.

In this number we reach what was historically the starting-point of symbolic logic. The Greek letters used (except , , ,) are always to stand for expressions of the form , or, where the Greek letters are not apparent variables, . The small Latin letters may either be such as have a meaning in isolation, or may represent classes or relations; this is possible in virtue of the notes at the ends of *20 and *21. We put:

*22·01.

This defines "the class is contained in the class ," or "all 's are 's."

*22·02.

This defines the logical product or common part of two classes and .

*22·03.

This defines the logical sum of two classes; it is the class consisting of all the members of one together with all the members of the other.

*22·04.

This defines the negation of a class. It is read "not-." It does not contain every object concerning which "" is not true, but only those objects concerning which "" is false; i.e. it excludes those objects for which "" is meaningless. Thus it consists of all objects, of the type next below , which are not members of ; but it does not contain objects of any other type but this.

*22·05.

This definition gives an abbreviation which is often convenient.

The postulates required for the algebra of logic have been enumerated by Huntington[54]. In our notation, they are as follows.

[Pg 218]

We assume a class , with two rules of combination, namely and ; and we then require the following ten postulates:

I a. is in the class whenever and are in the class.

I b. is in the class whenever and are in the class.

II a. There is an element such that for every element .

II b. There is an element such that for every element .

III a. whenever , , and are in the class.

III b. whenever , , and are in the class.

IV a. whenever , , , , , , , and are in the class.

IV b. whenever , , , , , , , and (are in the class.

V. If the elements and in postulates II a and II b exist and are unique, then for every element there is an element such that and .

VI. There are at least two elements, and , in the class, such that .

The form of the above postulates is such that they are mutually independent, i.e. any nine of them are satisfied by interpretations of the symbols which do not satisfy the remaining one.

For our purposes, "" must be replaced by "." and will be the null-class and the universal class, which are defined in *24. Then the above ten postulates are proved below, as follows:

I a. in *22·37, namely ""

I b. in *22·36, namely ""

II a. in *24·24, namely ""

II b. in *24·26, namely ""

III a. in *22·57, namely ""

III b. in *22·51, namely ""

IV a. in *22·69, namely ""

IV b. in *22·68, namely ""

V. in *24·21·22, namely "" and ""

VI. in *24·1, namely ""

Hence, assuming Huntington's analysis of the postulates for the formal algebra of logic, the propositions proved in what follows suffice to establish that this algebra holds for classes. The corresponding propositions of *23 and *25 prove that it holds for relations, substituting , , , , for , , , , , .

[Pg 219]

The principal propositions of the present number are the following:

(1) Those embodying the formal rules:

*22·51.

*22·57.

These embody the commutative law.

*22·52.

*22·7.

These embody the associative law.

*22·5.

*22·56.

These embody the law of tautology.

*22·68.

*22·69.

These embody the distributive law. It will be seen that the second results from the first by everywhere interchanging the signs of addition and multiplication.

*22·8.

This is the principle of double negation.

*22·81.

This is the principle of transposition.

(2) Other useful propositions:

*22·44.

*22·441.

These embody the two forms of the syllogism in Barbara.

*22·62.

*22·621.

These two propositions enable us to transform any inclusion (into an equation.

*22·91.

I.e. " or " is identical with " or the part of which is excluded from ."

*22·01.

*22·02.

*22·03.

*22·04.

[Pg 220]

*22·05.

*22·1.

*22·2.

*22·3.

*22·31.

*22·32.

*22·33.

*22·34.

*22·35.

*22·351.

Dem.

This proposition is used in proving that the null-class is not identical with the class containing everything (*24·1), which is used to show that at least two classes exist. Our axioms do not suffice to prove that more than one individual exists, but they prove the existence of at least two classes and at least two relations.

*22·36.

*22·37.

*22·38.

*22·39.

Dem.

*22·391.

*22·392.

*22·4.

Dem.

*22·41.

[Pg 221]

*22·42.

*22·43.

*22·44.

This is one form of the syllogism in Barbara. Another form is the following:

*22·441.

*22·45.

Dem.

*22·46.

*22·47.

*22·48.

*22·481.

Dem.

*22·49.

*22·5.

Dem.

The above is the law of tautology for the logical multiplication of classes.

*22·51.

*22·52.

Thus logical multiplication of classes obeys the commutative and associative laws. References to *22·33·34·35 and to *20·43 will in future often be omitted.

*22·53.

This definition serves merely for the avoidance of brackets.

*22·54.

*22·55.

*22·551.

*22·56.

[Pg 222]

The above is the law of tautology for the logical addition of classes.

*22·57.

*22·58.

*22·59.

Dem.

The analogue of *4·78, i.e.

is false. We have only

A similar remark applies to the analogue of *4·79. Cf. *22·64·65.

*22·6.

Dem.

*22·61.

*22·62.

Dem.

*22·621.

The proof proceeds as in *22·62. The proposition *22·621 is one of the most useful propositions in the present number.

*22·63.

The process of obtaining *22·63 from *4·44 is of the same kind as the process employed in the proofs that have been written out in this number.[Pg 223] Hence only *4·44 is referred to. We shall similarly restrict references for later propositions in this number. The process is always roughly as follows: , , are replaced by , , ; then *10·11 is applied, and such further propositions of *10 as may be required, together with *22·33·34·35.

*22·631.

*22·632.

*22·633.

*22·64.

Dem.

The converse of this proposition does not hold, because the converse of *10·41 does not hold.

*22·65.

Here again the converse is untrue.

*22·66.

*22·68.

Dem.

*22·69.

The above propositions *22·68·69 are the two forms of the distributive law. Note that either results from the other by interchanging the signs of addition and multiplication.

*22·7.

*22·71.

*22·72.

*22·73.

*22·74.

[Pg 224]

Dem.

*22·8.

*22·81.

*22·811.

*22·82.

*22·83.

*22·831.

*22·84.

*22·85.

*22·86.

*22·87.

*22·84·85·86·87 are De Morgan's formulae.

*22·88.

This is a form of the law of excluded middle.

*22·89.

This is a form of the law of contradiction.

*22·9.

*22·91.

Dem.

*22·92.

*22·93.

Dem.

[Pg 225]

*22·94.

Dem.

This proposition is used in connection with mathematical induction, in *90·102, which is required for the proof of *90·132, which is one of the fundamental propositions in the theory of mathematical induction.

*22·95.

Dem.

FOOTNOTES:

[54] Trans. Amer. Math. Soc. Vol. 5, July 1904, p. 292.

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*23. CALCULUS OF RELATIONS.

Summary of *23.

The definitions and propositions of this number are to be exact analogues of those of *22. Properties of relations which have no analogues for classes will not be dealt with till Section D. Proofs will be omitted in the present number, as they are precisely analogous to those of analogous propositions in *22. In this number, as always in future, capital Latin letters stand for expressions of the form , or, where they are not being used as apparent variables, for . The principal propositions of this number are the analogues of those of *22.

*23·01.

*23·02.

*23·03.

*23·04.

*23·05.

Similar remarks apply to these definitions as to those of *22.

*23·1.

*23·2.

*23·3.

*23·31.

*23·32.

*23·33.

*23·34.

*23·35.

*23·351.

*23·36.

*23·37.

*23·38.

[Pg 227]

*23·39.

*23·391.

*23·392.

*23·4.

*23·41.

*23·42.

*23·43.

*23·44.

*23·441.

*23·45.

*23·46.

*23·47.

*23·48.

*23·481.

*23·49.

*23·5.

*23·51.

*23·52.

*23·53.

*23·54.

*23·55.

*23·551.

*23·56.

*23·57.

*23·58.

*23·59.

*23·6.

*23·61.

*23·62.

*23·621.

*23·63.

*23·631.

*23·632.

*23·633.

[Pg 228]

*23·64.

*23·65.

*23·66.

*23·68.

*23·69.

*23·7.

*23·71.

*23·72.

*23·73.

*23·74.

*23·8.

*23·81.

*23·811.

*23·82.

*23·83.

*23·831.

*23·84.

*23·85.

*23·86.

*23·87.

*23·88.

*23·89.

*23·9.

*23·91.

*23·92.

*23·93.

*23·94.

*23·95.

[Pg 229]

*24. THE UNIVERSAL CLASS, THE NULL-CLASS, AND THE EXISTENCE OF CLASSES.

Summary of *24.

The universal class, denoted by , is the class of all objects of the type which, in the given context, is being denoted by small Latin letters, i.e. of the lowest type concerned. Thus , like "," is ambiguous as to type. Its definition is as follows:

*24·01.

Any other property possessed by everything would do as well as "," but this is the only such property which we have hitherto studied.

The null-class, denoted by , is the class which has no members. Like , it is ambiguous as to type. We use the same symbol, , for null-classes of various types; but these null-classes differ. The type of is determined by that of the terms concerning which "" is false: whatever may be, "" will not represent a true proposition, but unless is of the appropriate type, "" will be meaningless, not false. Thus is of the type next above that of an concerning which "" is significant and false. The definition of is

*24·02.

When a class is not null, so that it has one or more members, it is said to exist. (This sense of "existence" must not be confused with that defined in *14·02.) We write "" for " exists." The definition is

*24·03.

In the present number, we shall deal first with the properties of and , then with those of existence. In comparing the algebra of symbolic logic with ordinary algebra, takes the place of , while combines the properties of and of .

Among the more important properties of and which are proved in this number are the following:

*24·1.

I.e. "nothing is not everything." This is useful as giving us the existence of at least two classes. If the monistic philosophers were right in maintaining that only one individual exists, there would be only two classes,[Pg 230] and , being (in that case) the class whose only member is the one individual. Our primitive propositions do not require the existence of more than one individual.

*24·102·103 show that any function which is always true determines the universal class, and any function which is always false determines the null-class.

*24·21·22 give forms of the laws of contradiction and excluded middle, namely "nothing is both and not-" (and "everything is either or not-" (.

*24·23·24·26·27 give the properties of and with respect to addition and multiplication, namely: multiplication by and addition of make no change in a class (*24·26·24); addition of gives , and multiplication by gives (*24·27·23). It will be observed that the properties of and result from each other by interchanging addition

and multiplication.

*24·3.

I.e. " is contained in " is equivalent to "nothing is but not ."

*24·311.

I.e. "no is a " is equivalent to "nothing is both and ."

*24·411.

*24·43.

As a rule, propositions concerning are much less used than the correlative propositions concerning .

The properties of the existence of classes result from those of , owing to the fact that is the contradictory of , as is proved in *24·54. Thus we have, in virtue of *24·3,

*24·55.

I.e. "not all 's are 's" is equivalent to "there are 's which are not 's." This is the familiar proposition of formal logic, that the contradictory of the universal affirmative is the particular negative.

We have

*24·56.

*24·561.

I.e. if a sum exists, then one of the summands exists, and vice versa; and if a product exists, both the factors exist (but not vice versa).

The proofs of propositions in the present number offer no difficulty.

[Pg 231]

*24·01.

*24·02.

*24·03.

*24·1.

*24·101.

*24·102.

Dem.

Thus any function which is always true determines the universal class, and vice versa.

*24·103.

Dem.

*24·104.

Dem.

*24·105.

Dem.

*24·11.

Dem.

*24·12.

Dem.

[Pg 232]

*24·13.

Dem.

*24·14.

Dem.

*24·141.

Dem.

*24·15.

Dem.

*24·17.

*24·21.

*24·22.

*24·23.

*24·24.

The above two propositions (*24·23·24) exhibit the algebraical analogy of to zero.

*24·26.

This exhibits the analogy of to 1.

*24·27.

This exhibits the analogy of to .

*24·3.

Dem.

The above proposition is very frequently used.

[Pg 233]

*24·31.

Dem.

This proposition is the correlative of *24·3, but, unlike that proposition, it is not useful in the sequel. Every proposition concerning has a correlative concerning , but we shall often not give these correlatives, since they are seldom required for subsequent proofs.

*24·311.

Dem.

*24·312.

Dem.

*24·313.

*24·32.

Dem.

*24·33.

Dem.

*24·34.

*24·35.

*24·36.

[Pg 234]

*24·37.

Dem.

*24·38.

Dem.

*24·39.

*24·4.

Dem.

*24·401.

Dem.

*24·402.

Dem.

[Pg 235]

*24·41.)

Dem.

*24·411.)

Dem.

*24·412.

Dem.

This proposition is used in *234·181, in the theory of continuous functions.

*24·42.

Dem.

*24·43.

Dem.

*24·431.)

This and the following proposition are lemmas for *24·44.

Dem.

[Pg 236]

*24·432.

Dem.

*24·44.

*24·45.

Dem.

*24·46.

Dem.

The following propositions, down to *24·495 inclusive, are lemmas inserted for use in much later propositions, most of them being only used a few times.

*24·47.

Dem.

*24·48.

[Pg 237]

Dem.

The above proposition, besides being used in the next two, is used in the theory of couples (*54·6), in the theory of greater and less (*117·632), and in the chapter on the ordering of classes by the principle of first differences (*170·68).

*24·481.

Dem.

The above proposition is used in the theory of selections (*83·74), in the theory of greater and less (*117·582), and in the theory of transfinite induction (*257).

*24·482.

[Pg 238]

The above proposition is used in the theory of convergence (*232·34).

*24·49.

Dem.

*24·491.

Dem.

The above proposition is used in the theory of selections (*83·63·65) and in the theory of segments of a series (*211·84).

*24·492.

Dem.

The above proposition is used fairly frequently, especially in the theory of series. It is first used in *93·273, in the theory of "generations."

*24·493.

Dem.

*24·494.

[Pg 239]

Dem.

This proposition is used in the theory of selections (*83·63 and *88·45).

*24·495.

Dem.

The above proposition is used in the theory of minimum points (*205·83·832·84).

In the remainder of this number we shall be concerned with the existence of classes. Many of the properties of the existence of classes follow from the fact that to say a class exists is equivalent to saying that the class is not equal to the null-class. This is proved in *24·54.

*24·5.

*24*51.

Dem.

*24·52.

This proposition states that the class of all objects of the type in question is not null, but has at least one member. The assumption that there is something, which is equivalent to this proposition, is implicit in the proposition *10·1, that what is true always is true in any instance. This would not hold if there were no instances of anything; hence it implies the existence of something. It will be observed that the above proposition (*24·52) depends on *24·1, which depends on *23·351, which

depends on *10·251, which depends on *10·24, which depends on *10·1 or on *9·1. The assumption that there is something is involved in the use of the real variable, which would otherwise be meaningless. This is made explicit in *9·1, and in the proof of *9·2, which is the same proposition as *10·1.

*24·53.

*24·54.

[Pg 240]

*24·55.

*24·56.

*24·561.

*24·57.

Dem.

*24·571.

Dem.

*24·58.

*24·6.

Dem.

*24·61.

*24·62.

*24·63.

In this proposition, the conditions of significance require that should be a class of classes. The condition "" is one required as hypothesis in many propositions. In virtue of the above proposition, this hypothesis may be replaced by "."

Dem.

This proposition is frequently used in later parts of the work. We often have to deal with classes of existent classes, and the most convenient form in which to state that all the members of a class of classes exist is "."

[Pg 241]

*25. THE UNIVERSAL RELATION, THE NULL RELATION, AND THE EXISTENCE OF RELATIONS.

Summary of *25.

This number contains the analogues, for relations, of the definitions and propositions of *24. Proofs will not be given, as they proceed precisely as in *24.

The universal relation, denoted by , is the relation which holds between any two terms whatever of the appropriate types, whatever these may be in the given context.

The null relation, , is the relation which does not hold between any pair of terms whatever, its type being fixed by the types of the terms concerning which the denial that it holds is significant. A relation is said to exist when there is at least one pair of terms between which it holds; " exists" is written "."

The propositions of this number are much less often referred to than those of *24, but for the sake of uniformity we have given the analogues of all propositions in *24, with the same numeration (except for the integral part).

All the remarks made in *24 apply, mutatis mutandis, in the present number.

*25·01.

*25·02.

*25·03.

*25·1.

*25·101.

*25·102.

*25·103.

*25·104.

*25·105.

*25·11.

*25·12.

[Pg 242]

*25·13.

*25·14.

*25·141.

*25·15.

*25·17.

*25·21.

*25·22.

*25·23.

*25·24.

*25·26.

*25·27.

*25·3.

*25·31.

*25·311.

*25·312.

*25·313.

*25·32.

*25·33.

*25·34.

*25·35.

*25·36.

*25·37.

*25·38.

*25·39.

*25·4.

*25·401.

*25·402.

*25·41.

*25·411.

*25·412.

*25·42.

*25·43.

*25·431.

*25·432.

*25·44.

[Pg 243]

*25·45.

*25·46.

*25·47.

*25·48.

*25·481.

*25·482.

*25·49.

*25·491.

*25·492.

*25·493.

*25·494.

*25·495.

*25·5.

*25·51.

*25·52.

*25·53.

*25·54.

*25·55.

*25·56.

*25·561.

*25·57.

*25·571.

*25·58.

*25·6.

*25·61.

*25·62.

*25·63.

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SECTION D.

LOGIC OF RELATIONS.

In the present section we shall be concerned with such of the general properties of relations as have no analogues in the theory of classes. The notations introduced in this section will be used constantly throughout the rest of the work, and the ideas expressed in the definitions will be found to be of fundamental importance.

[Pg 245]

*30. DESCRIPTIVE FUNCTIONS.

Summary of *30.

The functions hitherto considered, with the exception of a few particular functions such as , have been propositional, i.e. have had propositions for their values. But the ordinary functions of mathematics, such as , , , are not propositional. Functions of this kind always mean "the term having such and such a relation to ." For this reason they may be called descriptive functions, because they describe a certain term by means

of its relation to their argument. Thus "" describes the number ; yet propositions in which occurs are not the same as they would be if were substituted for . This appears e.g. from the proposition "," which conveys valuable information, whereas "" is trivial. Descriptive functions, like descriptions in general, have no meaning by themselves, but only as constituents of propositions[55].

The general definition of a descriptive function is:

*30·01.

That is, "" is to mean "the term which has the relation to ." If there are several terms or none having the relation to , all propositions about , i.e. all propositions of the form "," will be false. The apostrophe in "" may be read "of." Thus if is the relation of father to son, "" means "the father of ." If is the relation of son to father, "" means "the son of "; in this case, all propositions of the form "" will be false unless has one son and no more.

All the functions that occur in ordinary mathematics are instances of the above definition; all are obtained in the above manner from some relation. Thus in our notation "" takes the place of what would commonly be "," this latter notation being reserved for propositional functions. We should write "" in place of "," using "" to express the relation of to when .

A definition such as , where the meaning given to the term defined is a description, must be understood to mean that the term defined (in this case) and the description assigned as its meaning (in[Pg 246] this case (are to be interchangeable in use: the definition is, in a sense, more purely symbolic than other definitions, since the description assigned as the meaning has itself no meaning except in use. It would perhaps be more formally correct to write

But even this definition would not be quite complete, because it omits mention of the scope of the two descriptions and . Thus the complete form would be

But it is unnecessary to adopt this form of definition, provided it is understood that the definition *30·01 means that "" may be written for "(" everywhere, i.e. in indications of scope as well as elsewhere. The use of the definition occurs always in accordance with the proposition:

which is *30·1, below.

It is to be observed that *30·01 does not necessarily involve

For this, by the definition, is equivalent to

which, by *14·28, only holds when , i.e. when there is one term, and no more, which has the relation to .

All the conventions as to scope explained in *14 are to be transferred to , i.e., in the absence of any contrary indication, the scope of is to be the smallest proposition, enclosed in dots or other brackets, in which the in question occurs.

We put

*30·02.

This definition serves merely for the avoidance of brackets. It is to be interpreted as meaning

In future, we shall often define a new expression as having a descriptive phrase for its meaning; in such a case, the definition is always to be interpreted as above. That is, any proposition in which the new expression occurs is to be the proposition which is obtained by substituting the old expression for the new one wherever the latter occurs.

[Pg 247]

, in the above, is to be interpreted by first treating as if it were not a descriptive symbol, and applying *30·01 and *14·01 or *14·02 to , and by then applying *30·01 and *14·01 or *14·02 to .

The majority of the propositions of the present number are immediate consequences of the corresponding propositions in *14. Thus *14·31—·34 and *14·113 lead immediately to *30·12—·16, which show that, either always or when exists, the "scope" of or of and makes no difference to the truth-values of such propositions as we are concerned with. We have

*30·18.

so that what holds of everything holds of , provided exists. This results immediately from *14·18, and shows that, provided exists, the fact that "" is an incomplete symbol does not prevent its being substituted as a value of whenever we have (, or an assertion of the propositional function .

One of the most used propositions of this number is:

*30·3.

which results immediately from *14·202. The following analogous proposition results from the above by means of *14·122:

*30·31.

I.e. "" involves, in addition to "," the statement that whatever has the relation to is identical with .

A proposition constantly referred to is:

*30·37.

In the hypothesis, might be replaced by , but one or other of them is essential. For, by *14·21, "" implies and (these are equivalent when), and therefore cannot be true when and do not exist.

The use of *30·37 is chiefly in cases where or or both are replaced by descriptive functions. Suppose, for example, that is replaced by . By *30·18, we may substitute for if exists. By *14·21, both sides of the implication in *30·37 will become false if does not exist, and therefore the implication will still hold. Hence whether exists or not, we may substitute it for and obtain

In like manner, if we replace by , we obtain

A very important proposition is:

*30·4.

This proposition states that, provided exists, to say that is the term which has the relation to is equivalent to saying that a has the relation to . Thus for example " is the occupier of the house " is equivalent to " occupies the house ," " is the writer of Waverley" is equivalent to[Pg 248] " wrote Waverley," " is the father of " is equivalent to " begot ." But we cannot argue from "John Smith inhabits London" to "John Smith is the inhabitant of London."

We shall introduce in this and subsequent sections many constant relations for which is always true. When is such that is always true, we have, in virtue of *30·4,

for every possible value of . The following proposition is useful in cases where both and are such that and always exist:

*30·41.

Thus if we know that and are always identical, we know not only that and are identical, but also that (and therefore) always exists.

*30·01.

*30·02.

In interpreting , is to be treated as an ordinary symbol until has been eliminated by *30·01 and *14·01 or *14·02, and then the above definitions are to be applied to .

*30·1.

*30·11.

The following propositions are immediate applications of *14·31 ff., made in accordance with *30·1.

*30·12.

*30·13.

*30·14.

*30·141.

*30·142.

*30·15.

The following two propositions are immediate consequences of *14·113·112.

*30·16.

*30·17.

*30·18.

[Pg 249]

*30·19.

*30·2.

In proving *30·2, we have to use the definition *30·01, not *30·1, because is not of the form . This appears if we attempt to apply the definition *14·01 to , which leads to an expression containing the meaningless constituent . But by the definition *30·01, every typographical occurrence of the symbol "" means what results when this symbol is replaced by "(" hence "" means "."

*30·21.

*30·22.

Note that we do not necessarily have

which is only true when .

*30·3.

*30·31.

*30·32.

*30·33.

*30·34.

*30·341.

Dem.

*30·35.

*30·36.

*30·37.

Dem.

This proposition is very frequently used.

*30·4.

This is a very important proposition, of which the use is constant.

[Pg 250]

*30·41.

Dem.

*30·42.

The hypothesis (is fulfilled by a number of important special relations, of which examples will occur in the subsequent numbers of the present section.

*30·5.

Dem.

*30·501.

On the meaning of "," see note to the definition *30·02.

Dem.

*30·51.

*30·52.

FOOTNOTES:

[55] Cf. *14, above.

[Pg 251]

*31. CONVERSES OF RELATIONS.

Summary of *31.

If is a relation, the relation which has to when is called the converse of . Thus greater is the converse of less, before of after, husband of wife. The converse of identity is identity, and the converse of diversity is diversity. The converse of is written (read "-converse"). When , is called a symmetrical relation, otherwise it is called not-symmetrical. When is incompatible with , is called asymmetrical. Thus "cousin" is symmetrical, "brother" is not-symmetrical (because when is the brother of , may be either the brother or the sister of , and "husband" is asymmetrical.

The relation of to is called "." It will be shown that every relation has one, and only one, converse; hence, applying the notation of *30, that one is . Thus . We have thus two notations for the converse of ; the second is more convenient for the converse of a relation not denoted by a single letter.

The more important propositions of the present number are the following:

*31·13.

I.e. any relation has a converse. Hence the relation "" verifies the hypothesis , i.e. we have .

*31·32.

I.e. two relations are identical when, and only when, their converses are identical.

*31·33.

I.e. any relation is the converse of its converse.

[Pg 252]

Very many of the subsequent uses of the notion of the converse of a relation require only the propositions which embody the definitions of and , namely

*31·11.

and

*31·131.

*31·01.

*31·02.

*31·1.

*31·101.

Dem.

*31·11.

*31·111.

*31·12.

Dem.

*31·13.

*31·131.

*31·132.

*31·14.

Dem.

*31·15.

[Pg 253]

*31·16.

Dem.

*31·17.

*31·18.

*31·21.

Dem.

*31·22.

*31·23.

Dem.

*31·24.

*31·32.

Dem.

*31·33.

Dem.

[Pg 254]

*31·34.

Dem.

*31·4.

*31·41.

*31·5.

*31·51.

Dem.

*31·52.

[Pg 255]

*32. REFERENTS AND RELATA OF A GIVEN TERM WITH RESPECT TO A GIVEN RELATION.

Summary of *32.

Given any relation , the class of terms which have the relation to a given term are called the referents of , and the class of terms to which a given term has the relation are called the relata of . We shall denote by the relation of the class of referents of to , and by the relation of the class of relata of to . It is convenient also to have a notation for the relations of and to . We shall denote the relation of to by "," where "" stands for "sagitta." Similarly we shall denote by "" the relation of to , to suggest an arrow running from right to left instead of from left to right. and are chiefly useful for the sake of the descriptive functions to which they give rise; thus and . Thus e.g. if is the relation of parent to son, the parents of , the sons of . If is the relation of less to greater among numbers of any kind, numbers less than , and numbers greater than . When exists, is the class whose only member is . But when there are many terms having the relation to , , which is the class of those terms, supplies a notation which cannot be supplied by . And similarly if there are many terms to which has the relation , supplies the notation for these terms. Thus for example let be the relation "," i.e. the relation which has to when . Then "" represents all values of such that , i.e. all values of or . Unlike the usual symbol, it is not ambiguous, since instead of representing some one of these values, it represents the class of them.

The definitions of , , , are as follows:

*32·01.

*32·02.

[Pg 256]

*32·03.

*32·04.

In virtue of the above definitions, we shall have , . This gives an alternative notation which is convenient in dealing with a relation not represented by a single letter.

It should be observed that if is a homogeneous relation (i.e. one in which referents and relata are of the same type), then and are not homogeneous, but relate a class to objects of the type of its members.

In virtue of the definitions of and , we shall have

*32·13.

*32·131.

Thus by *14·21, we always have and . Thus whatever relation may be, we have and . We do not in general have or (. Thus taking to be the relation of parent and child, and . Thus , i.e. when is childless, and , i.e. , when is Adam or Eve. The two sorts of existence, and , can both be significantly predicated of , because "" is a descriptive function whose value is a class; and the same applies to . It will be seen that (by *14·21) , but the converse implication does not hold in general.

We have

*32·16.

Also by *32·18·181,

Thus by the use of or , every statement of the form "" can be reduced to a statement asserting membership of a class. Since, however, the class in question is given by a descriptive function, and descriptive functions are defined by means of relations, we do not thus obtain a method of reducing the theory of relations to the theory of classes.

*32·01.

*32·02.

[Pg 257]

*32·03.

*32·04.

*32·1.

*32·101.

*32·11.

*32·111.

*32·12.

*32·121.

"" must not be confounded with "." The former means that there is such a class as , which, as we have just seen, is always true; the latter means that is not null, which is only true if is a term to which some other term has the relation . Note that, by *14·21, both and imply . The contradictory of is not , but . This last would not imply , but for the fact that is always true.

*32·13.

*32·131.

*32·132.

*32·133.

The use of *20·57 will in general be tacit. It happens constantly that we have propositions such as *32·13, in which a descriptive expression is shown to be identical with a class. In such cases, whenever the properties of the class are asserted of the descriptive expression, *20·57 is relevant.

*32·14.

Dem.

[Pg 258]

*32·15.

*32·16.

*32·18.

*32·181.

*32·182.

The transformation from "" to "" is one commonly effected in language. E.g. suppose "" is " loves ," then "" is " is a lover of ."

*32·19.

Dem.

*32·2.

*32·201.

*32·21.

*32·211.

*32·22.

*32·221.

*32·23.

*32·231.

*32·24.

Dem.

[Pg 259]

*32·241.

*32·25.

*32·251.

*32·3.

Notice that we do not have

Dem.

*32·31.

*32·32.

*32·33.

*32·34.

*32·35.

The proofs of the above propositions are similar to that of *32·3.

*32·4.

*32·41.

Dem.

*32·42.

[Pg 260]

*33. DOMAINS, CONVERSE DOMAINS, AND FIELDS OF RELATIONS.

Summary of *33.

If is any relation, the domain of , which we denote by , is the class of terms which have the relation to something or other; the converse domain, , is the class of terms to which something or other has the relation ; and the field, , is the sum of the domain and the converse domain. (Note that the field is only significant when is a homogeneous relation.)

The above notations , , are derivative from the notations , , for the relations, to a relation, of its domain, converse domain, and field respectively. We are to have

hence we define , , as follows:

*33·01.

*33·02.

*33·03.

The letter is chosen as the initial of the word "campus." We require one other definition, namely of the relation of to when is a member of the field of . This relation, which we will call , is defined as follows:

*33·04.

We shall find that . will be the relation of a relation to its domain, will be the class of relations having for their domain. Similar remarks apply to and . The field of a

relation is specially important in connection with series.

The propositions of this number are constantly used throughout the remainder of the work. The ideas of the domain, converse domain, and field are very general, and have somewhat different uses for relations of different[Pg 261] kinds. Consider first the sort of relation that gives rise to a descriptive function . For this we require that should exist whenever there is anything having the relation to , i.e. that there should never be more than one term having the relation to a given term . In this case, the values of for which exists will constitute the "converse domain" of , i.e. , and the values which assumes for various values of will constitute the "domain" of , i.e. . Thus the converse domain is the class of possible arguments for the descriptive function , and the domain is the class of all values of the function. Thus, for example, if is the relation of the square of an integer to , then =the square of , provided is an integer. In this case, is the class of integers, and is the class of perfect squares. Or again, suppose is the relation of wife to husband; then =the wife of , =married men, =married women. In such cases, the field usually has little importance; and if the values of the function are not of the same type as its arguments, i.e. if the relation is not homogeneous, the field is meaningless. Thus, for example, if is a homogeneous relation, and are not homogeneous, and therefore "" and "" are meaningless.

Let us next suppose that is the sort of relation that generates a series, say the relation of less to greater among integers. Then = all integers that are less than some other integer = all integers, = all integers that are greater than some other integer = all integers except 0. In this case, = all integers that are either greater or less than some other integer = all integers. Generally, if generates a series, = all members of the series except the last (if any), = all members of the series except the first (if any), and = all members of the series. In this case, "" expresses the fact that is a member of the series. Thus when generates a series, becomes important, and the relation is likely to be useful.

We shall have occasion to deal with many relations having some of the properties of series, and with many propositions which, though only important in connection with serial relations, hold much more generally. In such cases, the field of a relation is likely to be important. Thus in the section on Induction (Part II, Section E), where we are preparing the way for the construction of serial relations by means of a certain kind of non-serial relation, and throughout relation-arithmetic (Part IV), the fields of relations will occur constantly. But in the earlier parts of the work, it is chiefly domains and converse domains that occur.

Among the more important properties of domains, converse domains and fields, which are proved in the present number, are the following.

[Pg 262]

We have always , , (*33·12·121·122). (The last of these, however, is only significant when is homogeneous.)

*33·13.

*33·131.

*33·132.

*33·14.

*33·16.

*33·2·21·22. The converse domain of a relation is the domain of its converse, the domain of a relation is the converse domain of its converse, and the field of a relation is the field of its converse.

*33·24.

*33·4.

with corresponding propositions (*33·41·42) for and .

*33·43.

*33·431.

*33·5.

*33·51.

The proofs of propositions concerning and are usually similar to those for , and are therefore often omitted.

*33·01.

*33·02.

*33·03.

*33·04.

*33·1.

*33·101.

*33·102.

*33·103.

*33·11.

*33·111.

*33·112.

*33·12.

*33·121.

*33·122.

*33·123.

*33·124.

[Pg 263]

*33·125.

*33·13.

*33·131.

*33·132.

*33·14.

Dem.

*33·15.

Dem.

*33.151.

*33·152.

*33·16.

Dem.

*33·161.

*33·17.

*33·18.

Dem.

*33·181.

Dem.

*33·182.

If is the sort of relation which generates a series, so that "" may be read " precedes ," then is the condition that the series may[Pg 264] have no last term, since it states that every term which follows some term precedes some other term, and is therefore not the last of the series.

*33·2.

Dem.

*33·21.

*33·22.

Dem.

*33·24.

Dem.

*33·241.

*33·25.

Dem.

[Pg 265]

*33·251.

*33·252.

*33·26.

Dem.

*33·261.

*33·262.

*33·263.

Dem.

*33·264.

*33·265.

*33·27.

Dem.

*33·271.

*33·272.

*33·28.

Dem.

[Pg 266]

*33·29.

*33·3.

Dem.

*33·31.

The three following propositions are used in the theory of selections (*80, *83 and *85). The second of them is also used in the theory of greater and less (*117) and in the theory of transitive relations (*201).

*33·32.

The converse of this proposition is not true.

Dem.

*33·33.

*33·34.

Dem.

*33·35.

Dem.

*33·351.

*33·352.

Dem.

[Pg 267]

The two following propositions (*33·4·41) are very frequently used.

*33·4.

Dem.

*33·41.

*33·42.

Dem.

*33·43.

Dem.

*33·431.

Dem.

*33·432.

Dem.

*33·44.

Dem.

*33·45.

[Pg 268]

Note that by our conventions as to denoting expressions, the scope of both and in the above is "," and is to have the larger scope.

Dem.

*33·46.

*33·47.

Dem.

*33·48.

*33·5.

Dem.

*33·51.

is useful in ordinal arithmetic, where we are concerned with a series generated by a relation , and "" expresses the fact that is a member of this series. The above two propositions (*33·5·51) will be much used in Part IV, where we deal with the

foundations of ordinal arithmetic, but will not often be referred to elsewhere.

*33·6.

Dem.

*33·61.

*33·62.

[Pg 269]

*34. THE RELATIVE PRODUCT OF TWO RELATIONS.

Summary of *34.

The relative product of two relations and is the relation which holds between and when there is an intermediate term such that has the relation to and has the relation to . Thus e.g. the relative product of brother and father is paternal uncle; the relative product of father and father is paternal grandfather; and so on. The relative product of and is denoted by ""; the definition is:

*34·01.

This definition is only significant when and belong to the same type.

The relative product of and is called the square of ; we put

*34·02.

*34·03.

The most useful propositions in the present number are the following:

*34·2.

I.e. the converse of a relative product is obtained by turning each factor into its converse and reversing the order of the factors.

*34·21.

I.e. the relative product obeys the associative law.

*34·25.

*34·26.

I.e. the relative product obeys the distributive law with respect to the logical addition of relations. (For logical multiplication instead of logical addition, we only get inclusion instead of identity; cf. *34·23·24.)

*34·34.

*34·36.

*34·41.

[Pg 270]

*34·01.

*34·02.

*34·03.

*34·1.

*34·11.

Dem.

*34·12.

*34·2.

Dem.

*34·202.

Dem.

*34·203.

*34·21.

Dem.

*34·22.

This definition serves merely for the avoidance of brackets.

[Pg 271]

*34·23.

Dem.

The converse of the above is not true.

*34·24.

*34·25.

Dem.

*34·26.

The above two forms of the distributive law, and the associative law (*34·21), are the only ones of the usual formal laws that hold for the relative product. The commutative law, in particular, does not hold in general.

*34·27.

Dem.

*34·28.

*34·29.

Dem.

[Pg 272]

In proving the equality of two relations, say and , we usually establish first an asserted proposition of the form

We then proceed by *11·11 (together with *11·3 in the second case) to

whence the result follows by *21·43. We shall in future omit these steps, and write "" after we have established

A similar ellipsis will be made in proving the equality of classes.

*34·3.

Dem.

*34·301.

*34·302.

Dem.

*34·31.

Dem.

*34·32.

[Pg 273]

*34·33.

Dem.

*34·34.

Dem.

*34·35.

Dem.

*34·351.

*34·36.

Dem.

The following proposition is a lemma for *95·31.

*34·361.

Dem.

[Pg 274]

*34·37.

*34·38.

*34·4.

Dem.

*34·41.

Dem.

The above proposition is no longer true if we change the hypothesis into , since (may exist when does not. Suppose, e.g., that is the relation of child to father, and the relation of daughter to father. Then (= the granddaughter of , but = the daughter of the child of . The first exists whenever has only one granddaughter, while the second requires further that should have only one child.

For the same reason we do not have

This will hold if , are one-many relations (cf. *71), but not in general otherwise.

*34·42.

Dem.

[Pg 275]

*34·5.

*34·51.

Dem.

*34·52.

*34·53.

*34·531.

*34·54.

Dem.

*34·55.

*34·56.

*34·6.

Dem.

*34·62.

Dem.

The above proposition is a lemma for *160·51, as is also *34·73, which employs the above proposition.

*34·63.

Dem.

[Pg 276]

*34·7.

Dem.

Thus is always a symmetrical relation, i.e. one which is equal to its converse.

*34·701.

*34·702.

Dem.

*34·703.

*34·73.

Dem.

*34·8.

Dem.

[Pg 277]

The hypothesis of the above proposition is the hypothesis that is symmetrical () and transitive (. These are the formal properties of those relations which can suitably be regarded as expressing equality in some respect.

*34·81.

The following propositions are lemmas for *34·85, which is used in *72·64:

*34·82.

Dem.

*34·83.

Dem.

*34·84.

Dem.

*34·841.

Dem.

*34·85.

[Pg 278]

*35. RELATIONS WITH LIMITED DOMAINS AND CONVERSE DOMAINS.

Summary of *35.

In this section, we have to consider the relation derived from a given relation by limiting either its domain or its converse domain to members of some assigned class. A relation with its domain limited to members of is written ""; with its converse domain limited to members of , it is written ""; with both limitations, it is written "." Thus e.g. "brother" and "sister" express the same relation (that of a common parentage), with the domain limited in the first case to males, in the second to females. "The relation of white employers to coloured employees" is a relation limited

both as to its domain and as to its converse domain. We put

*35·01.

with similar definitions for and .

A particularly important case is the case in which the same limitation is imposed on the domain and on the converse domain, i.e. where we have a relation of the form "." In this case, the limitation to members of may be more briefly stated as being imposed on the field. For this case, it is convenient to adopt "" as an alternative notation. This case will be considered in *36.

It is convenient to consider in the present connection the relation between and which is constituted by x being a member of and being a member of . This relation will be denoted by "." Thus we put

*35·04.

The chief importance of relations with limited fields arises in the theory of series. Given a series generated by a relation , let be a class consisting of part of this series. Then is the field of the relation [Pg 279]or , and it is this relation which is the generating relation of the series of members of in the same order which they have as parts of the original series. Thus parts of a series, considered not merely as classes but as series, are dealt with by means of serial relations with limited fields.

Relations with limited domains are not nearly so much used as relations with limited converse domains. Relations with limited converse domains play a great part in arithmetic, especially in establishing the formal laws. What is wanted in such cases is a one-one relation correlating two classes or two series. That is, we want a relation such that not only does exist whenever , but also exists whenever . The kind of relation which is most frequently found to effect such a correlation is some such relation as or or , or some other constant relation for which we always have , with its converse domain so limited that, subject to the limitation, only one value of gives any given value of . Thus for example let be a class of relations no two of which have the same domain; then will give a one-one correlation of these relations with their domains: if , , we shall have

We shall also have and . Moreover the converse domain of is , and the domain of is the class of domains of members of . Thus gives a one-one correlation of with the domains of members of . It is chiefly in such ways that relations with limited converse domains are useful.

For purposes of reference, a great many propositions are given in the present number, but the propositions that will be used frequently are comparatively few. Among these are the following:

*35·21.

*35·31.

*35·354.

I.e. in a relative product it makes no difference whether we limit the converse domain of the first factor, or the domain of the second.

*35·412.

*35·452.

*35·48.

*35·52.

*35·61.

*35·64.

*35·65.

The hypothesis is fulfilled in the great majority of cases in which we have occasion to use .

*35·66.

*35·7.

[Pg 280]

This proposition is used very frequently, owing to the fact that limitation of the converse domain is chiefly applied to such relations as give rise to descriptive functions (e.g. , ,).

*35·71.

This proposition is useful for a reason similar to that which makes *35·7 useful.

*35·82.

Owing to this proposition, the properties of can be deduced from the already proved properties of , by putting .

The relation "" is what may be called an "analysable" relation, i.e. it holds between and y when and , i.e. when has a property independent of , and has a property independent of .

*35·85.

*35·86.

If either or is null, so is (*35·88).

*35·01.

*35·02.

*35·03.

*35·04.

*35·05.

The last definition serves merely for the avoidance of brackets.

*35·1.

*35·101.

*35·102.

*35·103.

*35·11.

Dem.

*35·12.

Dem.

[Pg 281]

*35·13.

Dem.

*35·14.

*35·15.

Dem.

*35·16.

*35·17.

*35·18.

*35·21.

Dem.

*35·22.

Dem.

*35·23.

*35·24.

*35·25.

[Pg 282]

*35·26.

Dem.

*35·27.

*35·31.

Dem.

*35·32.

*35·33.

*35·34.

*35·35.

Dem.

*35·351.

*35·352.

*35·354.

Dem.

*35·41.

*35·412.

[Pg 283]

*35·413.

*35·42.

*35·421.

*35·422.

*35·43.

Dem.

*35·431.

*35·432.

*35·44.

Dem.

*35·441.

*35·442.

*35·451.

Dem.

*35·452.

*35·453.

*35·454.

*35·46.

Dem.

*35·461.

[Pg 284]

*35·462.

*35·471.

Dem.

*35·472.

*35·473.

*35·474.

*35·48.

Dem.

*35·481.

*35·51.

Dem.

*35·52.

*35·53.

[Pg 285]

*35·61.

Dem.

*35·62.

*35·63.

Dem.

*35·64.

*35·641.

*35·642.

*35·643.

*35·644.

*35·65.

*35·66.

*35·671.

Dem.

*35·672.

*35·68.

Dem.

[Pg 286]

*35·7.

This proposition is very often used in the later parts of the work.

Dem.

*35·71.

Dem.

*35·75.

Dem.

*35·76.

Dem.

[Pg 287]

The rest of this number, down to *35·93 exclusive, is concerned with , except *35·81·812.

*35·81.

*35·812.

*35·82.

Dem.

*35·822.

Dem.

*35·83.

Dem.

*35·831.

Dem.

[Pg 288]

*35·832.

*35·834.

Dem.

*35·84.

*35·85.

Dem.

*35·86.

*35·87.

Dem.

*35·88.

*35·881.

Dem.

*35·882.

[Pg 289]

*35·89.

Dem.

*35·891.

Dem.

*35·892.

*35·895.

*35·9.

Dem.

*35·91.

Dem.

*35·92.

[Pg 290]

*35·93.

Dem.

*35·931.

*35·932.

*35·94.

*35·941.

*35·942.

[Pg 291]

*36. RELATIONS WITH LIMITED FIELDS.

Summary of *36.

In this number we are concerned with the special case in which the same limitation is imposed upon the domain and the converse domain of a relation. In this case, the same result is achieved by imposing the limitation on the field. It is convenient to be able to regard as a descriptive function of or of , which we secure by the notation , whence, as will be explained in *38, and will both mean . If is a serial relation, and , "" will stand for "the terms of arranged in the order determined by ," or, as we may call it briefly, " in the -order." is defined as follows:

*36·01.

We thus have

*36·13.

Most of the propositions concerning demand that should have some at least of the characteristics of a serial relation. Hence the propositions concerning which can be given in the present number are, for the most part, not the most useful propositions concerning . The most useful propositions in the present number are the following:

*36·25.

*36·29.

*36·3.

*36·33.

*36·01.

*36·11.

*36·13.

The following propositions are obtained from those of *35 by means of *36·11, which, as it is used in each case, is not referred to again.

*36·2.

*36·201.

*36·202.

*36·203.

*36·21.

[Pg 292]

*36·22.

Dem.

*36·23.

*36·24.

*36·241.

*36·25.

Dem.

*36·26.

*36·27.

*36·28.

*36·29.

*36·3.

Dem.

*36·31.

*36·32.

*36·33.

*36·34.

*36·35.

*36·4.

Dem.

[Pg 293]

*37. PLURAL DESCRIPTIVE FUNCTIONS.

Summary of *37.

In this number, we introduce what may be regarded as the plural of . "" was defined to mean "the term which has the relation to ." We now introduce the notation "" to mean "the terms which have the relation to members of ." Thus if is the class of great men, and is the relation of wife to husband, will mean "wives of great men." If is the class of fractions of the form for integral values of , and is the relation "less than," will be the class of fractions each of which is less than some member of this class of fractions, i.e. will be the class of proper fractions. Generally, is the class of those referents which have relata that are members of .

We require also a notation for the relation of to . This relation we will call . Thus is the relation which holds between two classes and when consists of all terms which have the relation to some member of .

A specially important case arises when always exists if . In this case, is the class of all terms of the form when . We will denote the hypothesis that always exists if by the notation , meaning "the 's of 's exist."

The definitions are as follows:

*37·01.

*37·02.

*37·03.

This definition serves merely for the avoidance of brackets. Without it, "" would be ambiguous as between (and , which are not equal. In all cases in which a suffix occurs, we shall adopt the same convention, i.e. we shall always put

[Pg 294]

*37·04.

Thus consists of all classes which have the relation to some member of . is only significant when is a class of classes relatively to members of the converse domain of

; in this case, is a class of classes relatively to members of the domain of .

*37·05.

Here the symbol "" must be treated as a whole, i.e. we must not regard it as making an assertion about . If , we must not suppose that we shall be able to put "," which would be nonsense, just as "" is nonsense even when and .

The notation , introduced in the present number, is extremely useful, and embodies a very important idea. Its use is somewhat different according to the kind of relation concerned. Consider first the kind of relation which leads to a descriptive function, say . If is a class of relations, is the class of the domains of these relations. In this case, is a class each of whose members is of the form , where . Again, let us denote by "" the relation of to ; then if we denote by "" the class of cardinal numbers, will denote all numbers that result from multiplying a cardinal number by , i.e. all multiples of . Thus e.g. will be the class of even numbers. If is a correlation between two classes and , i.e. a relation such that, if , exists and is a member of , while conversely, if , exists and is a member of , then , and we may regard as a transformation applied to each member of and giving rise to a member of . It is by means of such transformations that two classes are shown to be similar, i.e. to have the same (cardinal) number of terms.

[Pg 295]

In the case of serial relations, the utility of the notation is somewhat different. Suppose, for example, that is the relation of less to greater among real numbers. Then if is any class of real numbers, will be the segment of real numbers determined by , i.e. the class of real numbers which are less than the limit or maximum of . In any series, if is a class contained in the series and is the generating relation of the series, is the segment determined by . If has either a limit or a maximum, say , will be . But if has neither a limit nor a maximum, will be what we may call an "irrational" segment of the series. We shall see at a later stage that the real numbers may be identified with the segments of the series of rationals, i.e. if is the relation of less to greater among rationals, the real numbers will be all classes such as , for different values of . The real numbers which correspond to rationals will be those resulting from a which has a limit or maximum; the irrationals will be those resulting from a which has no limit or maximum.

The present number may be divided into various sections, as follows: (1) First, we have various elementary properties of the terms defined at the beginning of the number; this section ends with *37·29. (2) We have next a set of propositions dealing with relative products, and with such symbols as , , and so on. The central proposition here is

*37·33.

By the definition, . Thus This connects propositions concerning such symbols as with propositions concerning relative products. This second section consists of the propositions from *37·3 to *37·39. (3) We have next a set of propositions on relations with limited domains and converse domains. The chief of these are

*37·401.

*37·412.

*37·41.

These propositions on relations with limited domains and converse domains, together with certain others naturally connected with them, extend from *37·4 to *37·52. (4) We next have a number of very important propositions on the consequences of the hypothesis , i.e. the hypothesis that, for any argument which is a member of , gives rise to a descriptive function . The chief proposition in this section is

*37·6.

Propositions with the hypothesis are applied to the cases of and , in which the hypothesis is verified. This section extends from *37·6 to *37·791. (5) Finally, we have three propositions on the relative product of with other relations. These propositions are useful in relation-arithmetic (Part IV).

The propositions of the present number which are most used in the sequel, apart from those already mentioned, are the following (omitting such as merely embody definitions):

*37·15.

*37·16.

*37·2.

*37·22.

*37·25.

*37·26.

*37·265.

[Pg 296]

*37·29.

*37·32.

*37·45.

*37·46.

*37·61.

For example, let be the relation of father to son, the class of Etonians, the class of rich men; then "" states "all fathers of Etonians are rich," while "" states "if a boy is an Etonian, his father must be rich." In virtue of the above proposition, these two statements are equivalent.

*37·62.

*37·63.

*37·01.

*37·02.

*37·03.

*37·04.

*37·05.

*37·1.

*37·101.

*37·102.

*37·103.

*37·104.

*37·105.

*37·106.

Dem.

*37·11.

*37·111.

[Pg 297]

*37·12.

*37·13.

Dem.

*37·131.

Dem.

*37·14.

Dem.

*37·15.

Dem.

*37·16.

*37·17.

Dem.

[Pg 298]

*37·171.

Dem.

*37·18.

Dem.

*37·181.

*37·2.

Dem.

The above proposition (*37·2) is one of the forms of asyllogistic inference due to Leibniz's teacher Jungius. The instance given by Jungius is: "Circulus est figura; ergo qui circulum describit, is figuram describit[56]." Here the class of circles is our , the class of figures is our , and the relation of describing is our .

*37·201.

*37·202.

*37·21.

Dem.

*37·211.

*37·212.

*37·22.

[Pg 299]

This proposition is very frequently used. The fact that here we have identity, while in *37·21 we only have inclusion, is due to the fact that *10·42 states an equivalence, while *10·5 only states an implication.

Dem.

*37·221.

*37·222.

*37·23.

*37·231.

The type of "" here is that type whose members are of the same type as . In the proof, use is made of the convention that a Greek letter always stands for an expression of the form .

Dem.

As appears in the above proof, it is necessary, when a proposition containing "" is to be proved, to abandon the notation with Greek letters, and revert to the explicit functional notation.

*37·24.

Dem.

[Pg 300]

*37·25.

Dem.

*37·26.

Dem.

*37·261.

*37·262.

*37·263.

*37·264.

Dem.

*37·265.

Dem.

[Pg 301]

*37·27.

*37·271.

*37·28.

*37·29.

Dem.

*37·3.

Dem.

*37·301.

*37·302.

*37·31.

Dem.

*37·311.

*37·32.

Dem.

[Pg 302]

*37·321.

*37·322.

*37·323.

*37·33.

Dem.

*37·34.

Dem.

*37·341.

*37·35.

Dem.

*37·351.

[Pg 303]

*37·352.

*37·353.

Dem.

*37·354.

*37·355.

*37·36.

*37·37.

*37·371.

This definition serves merely for the avoidance of brackets. Like *37·03, this definition will be extended to all suffixes.

*37·38.

*37·39.

*37·4.

Dem.

*37·401.

*37·402.

Dem.

*37·41.

[Pg 304]

*37·411.

Dem.

*37·412.

Dem.

*37·413.

Dem.

*37·42.

*37·421.

*37·43.

Dem.

*37·431.

*37·44.

*37·441.

*37·45.

*37·451.

*37·46.

*37·461.

*37·462.

*37·47.

Dem.

[Pg 305]

*37·5.

Dem.

*37·501.

Dem.

*37·502.

*37·51.

Dem.

*37·52.

The following propositions, down to *37·7 exclusive, are concerned with the special properties of which result from the hypothesis , defined in *37·05. The hypothesis is important, because it has many consequences and is satisfied in many cases with which we wish to deal.

*37·6.

This proposition is very important, and is used constantly.

Dem.

[Pg 306]

*37·601.

Dem.

*37·61.

Dem.

*37·62.

Dem.

[Pg 307]

The above is the type of inference concerning which Jevons says[57]: "I remember the late Prof. De Morgan remarking that all Aristotle's logic could not prove that 'Because a horse is an animal, the head of a horse is the head of an animal.'" It must be confessed that this was a merit in Aristotle's logic, since the proposed inference is fallacious without the added premiss " the head of the horse in question." E.g. it does not hold for an oyster or a hydra. But with the addition , the above proposition gives an important and common type of asyllogistic inference.

*37·63.

Dem.

This proposition is very frequently used.

*37·64.

Dem.

*37·65.

Dem.

*37·66.

Dem.

[Pg 308]

*37·67.

Dem.

*37·68.

Dem.

*37·69.

Dem.

A specially important case of is or . This case will be further studied later (in *70); for the present, we shall only give a few preliminary propositions about it. It will be observed that the hypothesis or is always verified, in virtue of *32·12·121. Hence the following applications of *37·6 ff.:

*37·7.

*37·701.

[Pg 309]

*37·702.

*37·703.

*37·704.

*37·705.

*37·706.

*37·707.

*37·708.

*37·709.

*37·71.

*37·711.

*37·712.

*37·713.

*37·72.

Dem.

*37·721.

*37·73.

*37·731.

[Pg 310]

Observe that the 's which occur in this proposition will not be all of the same type. E.g. if relates individuals to individuals, the first will be the class of no individuals, while the second and third will be the class of no classes. Thus the ambiguity which attaches to the type of must be differently determined for different occurrences of in this proposition. In general, when this is the case with our ambiguous symbols, we shall adopt a notation which indicates the fact. But when the ambiguous symbol is , it seems hardly worth while.

*37·74.

Dem.

*37·75.

*37·76.

Dem.

*37·761.

*37·77.

*37·771.

*37·772.

*37·773.

*37·78.

*37·781.

*37·79.

*37·791.

*37·8.

Dem.

*37·81.

*37·82.

FOOTNOTES:

[56] We quote from Couturat, La Logique de Leibniz, Chapter III, § 15 (p. 75 n.).

[57] Principles of Science, chap. I. (p. 18 of edition of 1887).

[Pg 311]

*38. RELATIONS AND CLASSES DERIVED FROM A DOUBLE DESCRIPTIVE FUNCTION.

Summary of *38.

A double descriptive function is a non-propositional function of two arguments, such as , , , , , , , . The propositions of the present number apply to all such functions, assuming the notation to be (as in the above instances) a functional sign placed between the two arguments. In order to deal with all analogous cases at once, we shall in this number adopt the notation

where "" stands for any such sign as , , , , , , , , or any functional sign to be hereafter defined and satisfying the condition

The derived relations and classes with which we shall be concerned may be illustrated by taking the case of . The relation of to will be written , and the relation of to will be written . Thus we shall have

The utility of this notation is chiefly due to the possibility of such notations as and . For example, take such a phrase as "the foreign members of English Clubs." Then if we put = foreigners, = English Clubs, we have

Or again, let be a conic, and a pencil of lines; then

In this case, since , we have . But when the function concerned is not commutative, this does not hold. Thus for example we do not have .

The notations of this number will be frequently applied hereafter to . In accordance with what was said above, we write for the relation of to , and for the relation of to .

Hence we have

Hence will be the class of relations obtained by taking members of [Pg 312] and relatively multiplying them by . Thus if were the class of relations first cousin, second cousin, etc., and were the relation of parent to child, would be the class of relations first cousin once removed, second cousin once removed, etc., taken in the sense which goes from the older to the younger generation.

It is often convenient to be able to exhibit and kindred expressions as descriptive functions of the first argument instead of the second. For this purpose we put

with similar notations for other descriptive double functions. We then have, just as in the case of ,

This enables us to form the class . This class is chiefly useful because the members of its members (i.e. , as we shall define it in *40) constitute the class of all products that can be formed of a member of and a member of .

Thus we are led to three general definitions for descriptive double functions, namely (if be any such function)

Since is again a descriptive double function, the first two of the above definitions can be applied to it. The third definition, for typographical reasons, cannot be applied conveniently, though theoretically it is of course applicable. The relations and represent the general idea contained in some of the uses in mathematics of the term 'operation,' e.g. +1 is the operation of adding 1.

The uses of the notations introduced in the present number occur chiefly in arithmetic (Parts III and IV). Few propositions can be given at this stage, since most of the important uses of the notation here introduced depend upon the substitution of some special function for the general function "" here used. In the present number, the propositions given are all immediate consequences of the definitions.

*38·01.

*38·02.

*38·03.

*38·1.

[Pg 313]

*38·101.

*38·11.

*38·12.

*38·13.

*38·131.

*38·2.

*38·21.

*38·22.

*38·23.

*38·24.

Dem.

*38·3.

*38·31.

[Pg 314]

NOTE TO SECTION D.

General Observations on Relations. The notion of "relation" is so general that it is important to realize the different sorts of relations to which the notations defined in the preceding section may be applied. It often happens that a proposition which holds for any relation is only important for relations of certain kinds; hence it is desirable that the reader should have in mind some of the principal kinds of relations. Of the various uses to which different sorts of relations may be put, there are three which are specially important, namely (1) to give rise to descriptive functions, (2) to establish correlations between different classes, (3) to generate series. Let us consider these in succession.

(1) In order that a relation may give rise to a descriptive function, it must be such that the referent is unique when the relatum is given. Thus, for example, the relations , , , , , , , defined above, all give rise to descriptive functions. In general, if gives rise to a descriptive function, there will be a certain class, namely , to which the argument of the function must belong in order that the function may have a value for that argument. For example, taking the sine as an illustration, and writing "" instead of "," must be a number in order that may exist. Then is the relation of to when . If we put numbers between and , both included, will be the relation of to when and . The converse of this relation, which is , will also give rise to a descriptive function; thus that value of which lies between and . This illustrates a case which arises very frequently, namely, that a relation does not, as it stands, give rise to a descriptive function, but does do so when its domain or converse domain is suitably limited. Thus for example the relation "parent" does not give rise to a descriptive function, but does do so when its domain is limited to males or limited to females. The relation "square root," similarly, gives rise to a descriptive function when its domain is limited to positive numbers, or limited to negative numbers. The relation "wife" gives rise to a descriptive function when its converse domain is limited to Christian men, but not when Mohammedans are included. The domain[Pg 315] of a relation which gives rise to a descriptive function without limiting its domain or converse domain consists of all possible values of the function; the converse domain consists of all possible arguments to the function. Again, if gives rise to a descriptive function, will be the class of those arguments for which the value of the function is . Thus consists of all numbers whose sine is , i.e. all values of . Again, will be the sines of the various members of . If is a class of numbers, then, by the notation of *38, will be the doubles of those numbers, the trebles of them, and so on. To take another illustration, let be a

pencil of lines, and let be the intersection of a line with a given transversal. Then will be the intersections of lines belonging to the pencil with the transversal.

(2) Relations which establish a correlation between two classes are really a particular case of relations giving rise to descriptive functions, namely the case in which the converse relation also gives rise to a descriptive function. In this case, the relation is "one-one," i.e. given the referent, the relatum is determinate, and vice versa. A relation which is to be conceived as a correlation will generally be denoted by or . In such cases, we are as a rule less interested in the particular terms and for which , than in classes of such terms. We generally, in such cases, have some class contained in the converse domain of our relation , and we have a class such that . In this case, the relation correlates the members of and the members of . We shall have also , so that, for such a relation, the correlation is reciprocal. Such relations are fundamental in arithmetic, since they are used in defining what is meant by saying that two classes (or series) have the same cardinal (or ordinal) number of terms.

[Pg 316]

(3) Relations which give rise to series will in general be denoted by or , and in propositions whose chief importance lies in their application to series we shall also, as a rule, denote a variable relation by or . When is used, it may be read as "precedes." Then may be read "follows," may be read "predecessors of ," may be read "followers of ." will be all members of the series generated by except the last (if any), will be all members of the series except the first (if any), will be all the members of the series. will consist of all terms preceding some member of . Suppose, for example, that our series is the series of real numbers, and that is the class of members of an ascending series , , , ... , ... Then will be the segment of the real numbers defined by this series, i.e. it will be all the predecessors of the limit of the series. (In the event of the series , , , ... , ... growing without limit, will be the whole series of real numbers.)

It very often happens that a relation has more or less of a serial character, without having all the characteristics necessary for generating series. Take, for example, the relation of son to father. It is obvious that by means of this relation series can be generated which start from any man and end with Adam. But these series are not the field of the relation in question; moreover this relation is not transitive, i.e. a son of a son of is not a son of . If, however, we substitute for "son" the relation "descendant in the direct male line" (which can be defined in terms of "son" by the method explained in *90 and *91), and if we limit the converse domain of this relation to ancestors of in the direct male line, we obtain a new relation which is serial, and has for its field and all his ancestors in the direct male line. Again, one relation may generate a number of series, as for example the relation " is east of ." If and are points on the earth's surface, and in the eastern hemisphere, this relation generates one series for every parallel of latitude. By confining the field of the relation further to one parallel of latitude, we obtain a relation which generates a series. (The reason for confining and to one hemisphere is to insure that the relation shall be transitive, since otherwise we might have east of and east of , but west of .)

A relation may have the characteristics of all the three kinds of relations, provided we include in the third kind all those which lead to series by some such limitations as those just described. For example, the relation , i.e. (in virtue of the notation of *38) the relation of to , where is supposed to be a finite cardinal integer, has the characteristics of all three kinds of relations. In the first place, it leads to the descriptive function (, i.e. . In the second place, it correlates with any class of numbers the class obtained by adding 1 to each member of , i.e. (. This correlation may be used to prove that the number of finite integers is infinite (in one of the two senses of the word "infinite"); for if we take as our class all the natural numbers including 0, the class (consists of all the natural numbers except 0, so that the natural numbers can be correlated with a proper part[58] of themselves. Again the relation may be used, like that of father to son, to generate a series, namely the usual series of the natural numbers in order of magnitude, in which each has to its immediate predecessor the relation . Thus this relation partakes of the characteristics of all three kinds of relations.

FOOTNOTES:

[58] I.e. a part not the whole. On this definition of infinity, see *124.

[Pg 317]

SECTION E.

PRODUCTS AND SUMS OF CLASSES.

Summary of Section E.

In the present section, we make an extension of , , , . Given a class of classes, say , the product of (which is denoted by is the common part of all the members of , i.e. the class consisting of those terms which belong to every member of . The definition is

If has only two members, and say, . If has three members, , , , then ; and so on. But this process can only be continued to a finite number of terms, whereas the definition of does not require that should be finite. This notion is chiefly important in connection with the lower limits of series. For example, let be the class of rational numbers whose square is greater than 2, and let "" mean ", where and are rationals." Then if , will be the class of rationals less than x. Thus will be the class of such classes as , where . Thus the product of , which we call , will be the class of rationals which are less than every member of , i.e. the class of rationals whose squares are less than 2. Each member of is a segment of the series of rationals, and is the lower limit of these segments. It is thus that we prove the existence of lower limits of series of segments.

Similarly the sum of a class of classes is defined as the class consisting of all terms belonging to some member of ; i.e.

i.e. belongs to the sum of if belongs to some . This notion plays the same part for upper limits of series of segments as plays for lower limits. It has, however, many more other uses than , and is altogether a more important conception. Thus in

cardinal arithmetic, if no two members of have any term in common, the arithmetical sum of the numbers of members possessed by the various members of is the number of members possessed by .

[Pg 318]

The product of a class of relations (say) is the relation which holds between and when and have every relation of the class . The definition is

The properties of are analogous to those of , but its uses are fewer.

The sum of a class of relations (say) is the relation which holds between and whenever there is a relation of the class which holds between and . The definition is

This conception, though less important than , is more important than . The summation of series and ordinal numbers depends upon it, though the connection is less immediate than that of the summation of cardinal numbers with .

Instead of defining , , , , it would be formally more correct to define , , and , which are the relations giving rise to the above descriptive functions. Thus we should have

whence we should proceed to

But in cases where the relation, as opposed to the descriptive function, is very seldom required, it is simpler and easier to give the definition of the descriptive function in the first instance. In such cases, the relation is always tacitly assumed to be also defined; i.e. when we give a definition of the form

where is some previously defined relation, we always assume that this definition is to be regarded as derived from

In addition to products and sums, we deal, in the present section, with certain properties of the relations and , the meanings of which result from the notation introduced in *38. Such relations are very useful in arithmetic. The reason for dealing with them in the present section is that a large proportion of the propositions to be proved involve sums of classes of classes or relations.

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*40. PRODUCTS AND SUMS OF CLASSES OF CLASSES.

Summary of *40.

In this number, we introduce the two notations (explained above)

Both these notions will be found increasingly useful as we proceed, but remains more useful than throughout. It is required for the significance of and that should be a class of classes.

In the present number, the most useful propositions are the following:

*40·12.

I.e. the product of is contained in every member of .

*40·13.

I.e. every member of is contained in the sum of .

*40·15.

I.e. is contained in the product of if is contained in every member of , and vice versa.

*40·151.

I.e. the sum of is contained in if every member of is contained in , and vice versa.

*40·2.

I.e. the product of the null-class of classes is the universal class. This may seem paradoxical at first sight, but it is really not so. The fewer members has, the larger, speaking generally, becomes. If has no members, then has no members to which a given term does not belong, and therefore belongs to .

*40·23.

I.e. unless is null, its product is contained in its sum.

*40·38.

This proposition is very often used in arithmetic. What it states is as follows: Given a class of classes , take its sum, , and then consider all the[Pg 320] terms that have the relation to some member of ; this gives the class ; next, take each separate member of , say , and form the class , consisting of all terms having the relation to some member of . The class of all such classes as , for various 's which are members of , is ; the sum of this class, by the above proposition, is the same as .

*40·4.

This proposition requires, for significance, that should always be a class. The proposition states that, if always exists when , then the sum of all classes which have the relation to some member of consists of all members of such classes as , where .

*40·5.

This proposition results from *40·4 by substituting for in that proposition.

*40·51.

In virtue of *40·5, is correlative to . Thus if is a serial relation, consists of terms preceding the whole of , and consists of terms preceding part of . If has a lower limit, it will be the upper limit or maximum of ; if has an upper limit, it will be the upper limit of .

*40·61.

In this proposition the hypothesis is essential, since, if , and .

*40·01.

*40·02.

*40·1.

*40·11.

*40·12.

Dem.

*40·13.

Dem.

[Pg 321]

*40·14.

*40·141.

*40·15.

Dem.

*40·151.

Dem.

This proposition is frequently used.

*40·16.

Dem.

*40·161.

Dem.

*40·17.

Dem.

[Pg 322]

*40·171.

Dem.

*40·18.

Dem.

*40·181.

Dem.

*40·19.

This proposition is the extension of *22·6.

Dem.

[Pg 323]

*40·2.

Dem.

*40·21.

Dem.

In the above proposition, the two 's are of different types, since is of the type next above that of . Thus it would be more correct to write

But in the case of it is not very important to keep the types distinct.

*40·22.

Dem.

In this proposition, the two 's are of the same type.

*40·221.

Dem.

Observe that the hypothesis is essential to this proposition, since when , and . Thus

*40·23.

Dem.

Observe that the hypothesis is essential to this proposition, since when , and . Thus

[Pg 324]

*40·24.

Dem.

The above proposition is used in the proof of *215·25.

*40·25.

Dem.

*40·26.

Dem.

The following proposition is used in the proof of *216·51.

*40·27.

Dem.

The following propositions are only significant when is a relation whose domain consists of classes, for they concern or , and therefore require that should be a class of classes.

*40·3.

*40·31.

*40·32.

Dem.

[Pg 325]

*40·33.

The following propositions no longer require that the domain of should be composed of classes.

*40·35.

Dem.

*40·36.

*40·37.

Dem.

*40·38.

Dem.

This proposition is frequently used in the proofs of arithmetical propositions.

*40·4.

This proposition is only significant when .

Dem.

[Pg 326]

*40·41.

*40·42.

Dem.

This proposition is used in *40·57, where we take , , .

*40·43.

Dem.

*40·44.

Dem.

The following proposition is used in the proof of *84·44.

*40·45.

Dem.

The following proposition is used in the proof of *94·402.

*40·451.

Dem.

[Pg 327]

*40·5.

Dem.

*40·51.

is the class of terms each of which has the relation to every member of , just as is the class of terms each of which has the relation to some member of . In the theory of series, plays an important part, correlative to that played by (which is , by *40·5). If is a class contained in a series whose generating relation is , then will be the predecessors of all members of , while will be the predecessors of some .

*40·52.

*40·53.

*40·54.

*40·55.

From this point onwards to *40·69, the propositions are inserted on account of their use in the theory of series.

*40·56.

In the above proposition, the conditions of significance require that should be a class of relations.

*40·57.

*40·6.

*40·61.

Dem.

[Pg 328]

*40·62.

The two following propositions (*40·63·64) are used in proving *40·65, which is used in *204·63.

*40·63.

Dem.

*40·64.

*40·65.

*40·66.

Dem.

*40·67.

*40·68.

Dem.

This proposition is used in the theory of series (*206·2).

*40·681.

The following proposition is used in *211·56.

[Pg 329]

*40·682.

Dem.

*40·69.

Dem.

The above propositions concerning and of course have analogues for and . But owing to *40·5, these analogues are more simply stated as properties of and . Thus, for example, *37·264 is the analogue of *40·67. The above propositions concerning and will be used in the theory of series, but until we reach that stage they will seldom be referred to.

*40·7.

Dem.

[Pg 330]

This proposition is of considerable importance, since it gives a compact form for the class of all values of the function obtained by taking in the class and in the class. Thus, for example, suppose is the class of numbers which are multiples of 3, and is the class of numbers which are multiples of 5, and represents the arithmetical product of and , then will be the class of products of multiples of 3 and multiples of 5, i.e. the class of multiples of 15. Again suppose and are both classes of relations; then will be all relative products obtained by choosing in the class and in the class .

*40·71.

Dem.

The hypothesis , which appears in *40·8·81, is one which plays an important part at a later stage. In the theory of induction (Part II, Section E) it characterizes a hereditary class, and in the theory of series it characterizes an upper section (when combined with).

*40·8.

Dem.

*40·81.

Dem.

[Pg 331]

*41. THE PRODUCT AND SUM OF A CLASS OF RELATIONS.

Summary of *41.

The propositions to be given in this number, down to *41·3 exclusive, are the analogues of those of *40, excluding those from *40·3 onwards, which have no

analogues. Proofs will not be given, in this number, when they are exactly analogous to those of propositions with the same decimal part in *40. The smaller importance of and , as compared with and , is illustrated by the smaller number of propositions in *41 as compared with *40.

Our definitions are

*41·01.

*41·02.

Of the propositions preceding *41·3, which are analogues of propositions in *40, the only two that are frequently used are

*41·13.

*41·151.

Of the remaining propositions of this number, which have no analogues in *40, the most important are *41·43·44·45, namely

These propositions are constantly required in the theory of selections (Part II, Section D) and in relation-arithmetic. Most of the other propositions of this number are used only once or not at all.

*41·01.

*41.02.

*41·1.

*41·11.

*41·12.

*41·13.

[Pg 332]

*41·14.

*41·141.

*41·15.

*41·151.

*41·16.

*41·161.

*41·17.

*41·171.

*41·18.

*41·181.

*41·19.

*41·2.

*41·21.

*41·22.

*41·221.

*41·23.

*41·24.

*41·25.

*41·26.

*41·27.

*41·3.

Dem.

*41·31.

*41·32.

*41·33.

*41·34.

Dem.

[Pg 333]

*41·341.

*41·342.

Dem.

The following proposition is used in *85·22.

*41·35.

Dem.

*41·351.

*41·4.

Dem.

*41·41.

*41·42.

Dem.

*41·43.

Dem.

[Pg 334]

*41·44.

*41·45.

Dem.

*41·5.

Dem.

*41·51.

Dem.

The above proposition, which is used in *92·31, states that, if and are classes of relations, the relative product of the relational sum of and the relational sum of is the relational sum of all the relative products formed of a member of and a member of .

The following proposition is used in *96·111.

[Pg 335]

*41·52.

Dem.

The following proposition is used in *162·32 and in *166·461.

*41·6.

Dem.

[Pg 336]

*42. MISCELLANEOUS PROPOSITIONS.

Summary of *42.

The present number contains various propositions concerning products and sums of classes. They are concerned chiefly with classes of classes of classes, or with relations of relations of relations. These are required respectively in cardinal and in ordinal arithmetic. Thus *42·1 is used in *112 and *113, which are concerned with cardinal addition and multiplication, while *42·12·2 are used in *160 and *162, which are concerned with ordinal addition. *42·22, though not explicitly referred to, is useful in facilitating the comprehension of propositions on series of series of series, or rather on relations between relations between relations, which are required in connection with the associative law of multiplication in relation-arithmetic.

*42·1.

Here must, for significance, be a class of classes of classes. The proposition states that if we take each member, , of , and form , and then form the sum of all the classes so obtained, the result is the same as if we form the sum of the sum of . This is the associative law for , and is (as will appear later) the source of the associative law of addition in cardinal arithmetic. The way in which this proposition comes to be the associative law for may be seen as follows: Suppose consists of two classes, and ; suppose in turn consists of the two classes and , and of the two classes and . Then .

(This will be proved later.) Thus has two members, one of which is , while the other is . Thus

But has four members, namely , , , . Thus .

Thus our proposition leads to

which is obviously a case of the associative law.

[Pg 337]

Our proposition states the associative law generally, including the case where the number of brackets, or of summands in any bracket, is infinite. The proof is as follows.

Dem.

*42·11.

Dem.

This is the associative law for products. Supposing again, for illustration, that consists of the two classes , , while consists of the two classes , and \beta of the two classes , , then consists of the two classes and , so that , while . Thus our proposition becomes

A descriptive function whose arguments are classes or classes of classes may be said to obey the associative law provided

This equation may be interpreted as follows: Given a class , divide it into any number of subordinate classes, so that no member is left out, though one member may belong to two or more classes. Let the classes into which is divided make up the class , so that is a class of classes, and . Then the above equation asserts that if we first form the of the various sub-classes of , and then the of the resulting class, the result is the same as if we formed the of directly.

In some cases—for example, that of arithmetical addition of cardinals—the above equation holds only when no two members of have a common term, i.e. when the parts into which is divided are mutually exclusive.

[Pg 338]

For a descriptive function whose arguments are relations of relations, we shall find another form for the associative law; this form plays in ordinal arithmetic a part analogous to that played by the above form in cardinal arithmetic.

*42·12.

Dem.

*42·13.

Dem.

*42·2.

This proposition assumes that is a relation between relations. For example, suppose we have a series of series, whose generating relations are ordered by the relation . Then is the class of these generating relations; is the relation "one or other of the generating relations which compose ," and is the class of all the terms occurring in any of the series. is the fields of the various series, and is again all the terms occurring in any of the series. is all the terms belonging to fields of series which are members of , and is all members of fields of members of the field of ; each of these again is all the terms occurring in any of the series. The proof is as follows:

Dem.

[Pg 339]

The following propositions apply to a relation of relations of relations. These propositions are useful for proving associative laws in ordinal arithmetic, since these laws deal with series of series of series, and series of series of series are most simply constituted by supposing the generating relations of the constituent series to be ordered by relations which are themselves ordered by a relation .

*42·21.

Dem.

*42·22.

If , in the above proposition, is a relation which generates a series of series of series, the above gives various forms for the class of ultimate terms of these series. Thus suppose ; then is a relation between generating relations of series. If now , is the generating relation of a series which we may regard as composed of individuals. The class of individuals so obtainable may be expressed in any of the above forms, as well as in others which are not given above.

*42·3.

Dem.

*42·31.

[Pg 340]

*43. THE RELATIONS OF A RELATIVE PRODUCT TO ITS FACTORS.

Summary of *43.

The purpose of the present number is to give certain propositions on the relation which holds between and whenever , or whenever , or whenever , where and are fixed. In virtue of the general definitions of *38, these relations are respectively , , and (. Such relations are of great utility both in cardinal and in ordinal arithmetic; they are also much used in the theory of induction (Part II, Section E). In place of the notation (, which is cumbrous, we adopt the more compact notation . If is a class of relations, will be the class of relations where , will be the class of relations where , and (will be the class of relations where . These classes of relations are often required in subsequent work.

In virtue of our definitions, we have

*43·112.

The propositions most used in the present number (except such as merely embody definitions) are the following:

*43·302.

*43·411.

*43·421.

The remaining propositions are used seldom, but their uses, when they are used, are important.

*43·01.

[Pg 341]

At a later stage (in *150) we shall introduce a simpler notation for the special case of . The following propositions are for the most part immediate consequences of the definitions, and proofs are therefore usually omitted.

*43·1.

*43·101.

*43·102.

*43·11.

*43·111.

*43·112.

*43·12.

*43·121.

*43·122.

*43·2.

Dem.

*43·201.

*43·202.

*43·21.

*43·211.

*43·212.

*43·213.

*43·22.

*43·3.

*43·301.

*43·302.

*43·31.

Dem.

*43·311.

*43·312.

*43·34.

*43·4.

*43·401.

*43·41.

[Pg 342]

*43·411.

*43·42.

Dem.

*43·421.

*43·43.

Dem.

*43·48.

*43·481.

*43·49.

Dem.

*43·491.

*43·5.

*43·51.

Dem.

The above proposition is used in the proof of *74·773.

[Pg 343]

PART II.

PROLEGOMENA TO CARDINAL ARITHMETIC.

[Pg 345]

SUMMARY OF PART II.

THE objects to be studied in this Part are not sharply distinguished from those studied in Part I. The difference is one of degree, the objects in this Part being of somewhat less general importance than those of Part I, and being studied more on account of their bearing on cardinal arithmetic than on their own account. Although cardinal arithmetic is the goal which determines our course in Part II, all the objects studied will be found to be also required in ordinal arithmetic and the theory of series. As this Part advances, the approach to cardinal arithmetic becomes gradually more marked, until at last nothing is lacking except the definition of cardinal numbers, with which Part III opens.

Section A of this Part deals with unit classes and couples. A unit class is the class of terms identical with a given term, i.e. the class whose only member is the given term. (As explained in the Introduction, Chapter III, pp. 80 to 83, the class whose only member is is not identical with .) We define as the class of all unit classes, leaving it to Part III to show that , so defined, is a cardinal number. In like manner, we define a (cardinal or ordinal) couple, and then define as the class of all couples. The propositions on couples will not be much referred to in the remainder of the present Part, since their use belongs chiefly to arithmetic (Parts III and IV). On the other hand, the properties of unit classes are constantly required in Sections C, D, E of this Part.

Section B deals, first, with the class of sub-classes of a given class, i.e. of classes contained in a given class. The sub-classes of a given class are often important in arithmetic. Next we consider the class of sub-relations of a given relation, i.e. relations contained in a given relation. The propositions on this subject are analogous to those on sub-classes, but less important. Next we consider the question of "relative types," i.e. taking any object , and calling its type , we give a notation for expressing in terms of the type of classes of which is a member, or of relations in which may be either referent or relatum, and so on. The notations introduced in this connection are very useful in arithmetic, especially in connection with existence-theorems. But the propositions of Section B are very seldom required in the later sections of the present Part.

[Pg 346]

Section C, which deals with one-many, many-one and one-one relations, is very important, and is constantly relevant in the sequel. A relation is one-many when no term has more than one referent, many-one if no term has more than one relatum, and one-one if it is both one-many and many-one. In this section, we define the notion of similarity, upon which all cardinal arithmetic is based: two classes are said to be similar when there is a one-one relation whose domain is the one and whose converse domain is the other. We prove the elementary properties of similarity, including the Schröder-Bernstein theorem, namely: If is similar to part of , and is similar to part of , then is similar to .

Section D deals with the notion of selections, upon which both cardinal and ordinal multiplication are based. A selection from a set of classes is a class consisting of one member from each class of the set. Thus a selective relation may be defined as one

which, for a given class of classes , makes a member of whenever is a member of . More exactly, a selective relation for a class of classes is one which is one-many, which has for its converse domain, and is such that, if , then . Such a relation may be called an -selector from . More generally, we may define a -selector from as a relation which is one-many, which has for its converse domain, and which is contained in . The theory of selectors is very important in arithmetic. But until we come to cardinal multiplication in Part III, Section B, the propositions of this fourth section will seldom be relevant.

Section E deals with mathematical induction, not in the special form in which it applies to finite integers (this is considered in Part III, Section C), but in a general form in which it applies to all relations. The propositions of this section are of very great importance, primarily in the theory of finite and infinite (Part III, Section C, and Part V, Section E), but also in many other subjects, and especially in the derivation of series from one-many, many-one or one-one relations—for example, in ordering the "rational" points of a projective space by means of successive constructions of harmonic points. The ideas involved in this section are somewhat complicated, and we must refer the reader to the section itself for an account of them.

[Pg 347]

SECTION A.

UNIT CLASSES AND COUPLES.

Summary of Section A.

In this section we begin (*50) by introducing a notation for the relation of identity, as opposed to the function ""; that is, calling the relation of identity I, we put

The purpose of this definition is chiefly convenience of notation. The definition enables us to speak of , , , , , etc., which we could not otherwise do.

At the same time we introduce diversity, which is defined as the negation of identity, and denoted by the letter . The properties of and result immediately from *13, since

We next introduce a very important notation, due to Peano, for the class whose only member is . If we took a strictly and purely extensional view of classes, we should naturally suppose this class to be identical with . But in view of the theory of classes explained in *20, it is plain that can never be identical with a class of which it is a member, even when it is the only member of that class. Peano uses the notation "" for the class whose only member is ; we shall alter this to "," following our general notation for descriptive functions. Thus we are to have

Hence we take as our definition

since this definition gives the desired value of . The properties of are many and important.

It is important to observe that "" means "the only member of ." Thus it exists when, and only when, has one member and no more, in which case is of the form , if x is its only member. Thus "" means the same as "(," and "" means the same as "(." What we call "" is denoted, in Peano's notation, by "."

[Pg 348]

Classes of the form are called unit classes, and the class of all such classes is called 1. This is the cardinal number 1, according to the definition of cardinal numbers which will be given in *100. The properties of 1, so far as they do not depend upon other cardinals, or upon the fact that 1 is a cardinal, will be studied in *52.

After a number (*53) containing various propositions involving 1 or , we pass to the consideration of cardinal couples (*54) and ordinal couples (*55). A cardinal couple is a class , where . The class of such couples is defined as 2, and will be shown at a later stage (*101) to be a cardinal number. An ordinal couple, which, unlike a cardinal couple, involves an order as between its members, is defined as a relation (cf. *35·04), where we may either add or not. The properties of ordinal couples are in part analogous to those of unit classes, in part to those of cardinal couples. In *56, we define the ordinal number 2 (which we denote by , to distinguish it from the cardinal 2) as the class of all ordinal couples , where . It will be shown at a later stage that this is an ordinal number according to our definition of ordinal numbers (*153 and *251).

[Pg 349]

*50. IDENTITY AND DIVERSITY AS RELATIONS.

Summary of *50.

The purpose of the present number is primarily notational. For notational reasons, we must be able to express identity and diversity as relations, and not merely as propositional functions, i.e. we require a notation for and . We therefore put

In spite of the fact that diversity is merely the negation of identity, the kinds of propositions which employ diversity are quite different from the kinds that employ identity. Identity as a relation is required, to begin with, in the theory of unit classes, which is our reason for treating of it at this stage. It is next required, constantly, in the theory of mathematical induction (Part II, Section E). It is required also in showing that cardinal and ordinal similarity are reflexive. These are its principal uses.

Diversity, on the other hand, is required almost exclusively in the theory of series (Part V), and the first number in that theory will be devoted to diversity. Until that stage, diversity will seldom be referred to, with one important exception, namely in proving the associative law of multiplication in relation-arithmetic (*174).

The most important propositions on identity in the present number are the following:

50·16.

50·4.

50·5.

50·51.

50·52.

50·62.

50·63.

[Pg 350]

The most important propositions on diversity in the present number are the following:

*50·23.

*50·24.

*50·43.

*50·45.

*50·47.

It will be observed that all these propositions are concerned with or , both of which are satisfied if is a serial relation. The hypothesis or characterizes an asymmetrical relation, i.e. one which, if it holds between and , cannot hold between and .

*50·01.

*50·02.

Most of the propositions of this number are obvious, and call for no comment.

*50·1.

*50·11

*50·12.

*50·13.

*50·14.

*50·15.

*50·16.

Dem.

*50·17.

Dem.

*50·2.

Dem.

[Pg 351]

*50·21.

Dem.

*50·22.

*50·23.

*50·24.

Dem.

*50·3.

*50·31.

Dem.

*50·32.

*50·33.

Dem.

In the above proposition (*50·33), the hypothesis is equivalent to the hypothesis that more than one object exists of the type in question. This can be proved for all except the lowest type. For the lowest type, we can only prove the existence of at least one object: this is proved in *24·52. For the next type, we can prove the existence of at least two objects, namely and ; these are distinct, by*24·1. For a class of order , we can prove the existence of objects. But for the class of individuals we cannot prove,[Pg 352] from our primitive propositions, that there is more than one object in the universe, and therefore we cannot prove . We might, of course, have included among our primitive propositions the assumption that more than one individual exists, or some assumption from which this would follow, such as

But very few of the propositions which we might wish to prove depend upon this assumption, and we have therefore excluded it. It should be observed that most philosophers, being monists, deny this assumption.

*50·34.

Dem.

*50·35.

*50·4.

Dem.

*50·41.

Dem.

[Pg 353]

*50·42.

Dem.

*50·43.

This proposition is useful in the theory of series. "" is the characteristic of an asymmetrical relation.

*50·44.

Dem.

*50·45.

*50·46.

*50·47.

Dem.

This proposition is used in the theory of series. If is a serial relation, we shall have and .

*50·5.

Dem.

[Pg 354]

*50·51.

*50·52.

Dem.

*50·53.

Dem.

*50·54.

Dem.

*50·55.

Dem.

*50·56.

Dem.

*50·57.

Dem.

[Pg 355]

*50·58.

Dem.

*50·59.

Dem.

*50·6.

Dem.

*50·61.

Dem.

*50·62.

*50·63.

*50·64.

*50·65.

*50·7.

*50·71.

*50·72.

*50·73.

*50·74.

Dem.

*50·75.

*50·76.

Dem.

*50·761.

[Pg 356]

*51. UNIT CLASSES.

Summary of *51.

In this number we introduce a new descriptive function , meaning "the class of terms which are identical with ," which is the same thing as "the class whose only member is ." We are thus to have

But . Hence we secure what we require by the following definition:

*51·01.

As a matter of notation, it might be thought that would do as well as , and that this definition is superfluous. But we need also the converse of this relation, and "" is not a sufficiently convenient symbol.

The propositions of this number are constantly used in what follows. It should be observed that the class whose members are and is , the class whose members are , , is , the class formed by adding to is , and the class formed by taking away from is . (If is not a member of , this is equal to .)

The distinction between and is one of the merits of Peano's symbolic logic, as well as of Frege's. On the basis of our theory of classes, the necessity for the distinction is of course obvious. But apart from this, the following consideration makes the necessity apparent. Let be a class; then the class whose only member is has only one member, namely , while may have many members. Hence the class whose only member is cannot be identical with [59].

The propositions of the present number which are most used are the following:

*51·15.

*51·16.

*51·2.

[Pg 357]

This proposition is useful because it enables us to replace membership of a class () by inclusion in the class ().

*51·211.

*51·221.

*51·222.

*51·23.

*51·4.

I.e. an existent class contained in a unit class must be identical with the unit class. From this proposition it will follow that 0 is the only cardinal which is less than 1.

*51·51.

For classes, has the same uses that (has for functions; "" means "the only member of ." We have

*51·59.

*51·01.

*51·1.

Dem.

*51·11.

*51·12.

*51·13.

*51·131.

*51·14.

*51·141.

*51·15.

*51·16.

*51.161.

*51·17.

Dem.

The above proposition is used in the theory of selections (*83·71).

[Pg 358]

*51·2.

Dem.

The above proposition shows how to replace membership of a class by inclusion in a class; thus for example it gives:

Before Peano and Frege, the relation of membership () was regarded as merely a particular case of the relation of inclusion (). For this reason, the traditional formal logic treated such propositions as "Socrates is a man" as instances of the universal affirmative , "All is ," which is what we express by "." This involved a confusion of fundamentally different kinds of propositions, which greatly hindered the development and usefulness of symbolic logic. But by means of the above proposition (*51·2), we can always obtain a proposition stating an inclusion (namely "") which is equivalent to a given proposition stating membership of a class (namely "").

*51·21.

Dem.

*51·211.

Dem.

*51·22.

Dem.

*51·221.

Dem.

[Pg 359]

*51·222.

*51·23.

Dem.

*51·231.

Dem.

*51·232.

This proposition states that a member of must be either or , and vice versa, i.e. that is the class whose only members are and .

*51·233.

*51·234.

Dem.

*51·235.

Dem.

*51·236.

*51·237.

[Pg 360]

*51·238.

Dem.

*51·239.

Dem.

*51·24.

Dem.

*51·25.

*51·3.

*51·31.

Dem.

*51·34.

*51·35.

*51·36.

*51·36 is frequently used.

*51·37.

[Pg 361]

*51·4.

Dem.

*51·401.

Dem.

This proposition shows that unit classes are the smallest existent classes.

*51·41.

Dem.

The two following propositions are lemmas for *51·43.

*51·42.

Dem.

*51·421.

*51·43.

[Pg 362]

The following propositions are concerned with , i.e. with the relation of the only member of a unit class to that class. If is a unit class, is its only member. (and are equal whenever either exists, and any proposition about the one is equivalent to the same proposition about the other.

*51·51.

Dem.

*51·511.

*51·52.

*51·53.

*51·54.

*51·55.

Dem.

*51·56.

Dem.

*51·57.

Dem.

*51·58.

*51·59.

FOOTNOTES:

[59] This argument is due to Frege. See his article "Kritische Beleuchtung einiger Punkte in E. Schröder's Vorlesungen über die Algebra der Logik," Archiv für Syst. Phil., vol. I. p. 444 (1895).

[Pg 363]

*52. THE CARDINAL NUMBER 1.

Summary of *52.

In this number, we introduce the cardinal number 1, defined as the class of all unit classes. The fact that 1 so defined is a cardinal number is not relevant at present, and cannot of course be proved until "cardinal number" has been defined. For the present, therefore, 1 is to be regarded simply as the class of all unit classes, unit classes being such classes as are of the form for some .

Like and , 1 is ambiguous as to type: it means "all unit classes of the type in question." The symbol "," where is a type, will mean "all unit classes whose sole members belong to the type " (cf. *65). Thus e.g. ")" will mean " is a class consisting of one individual," if "" stands for the class of individuals.

The properties of 1 to be proved in the present number are what we may call logical as opposed to arithmetical properties, i.e. they are not concerned with the arithmetical operations (addition, etc.) which can be performed with 1, but with the relations of 1 to unit classes. The arithmetical properties of 1 will be considered later, in Part III.

The propositions of the present number which are most used are the following:

*52·16.

I.e. is a unit class if, and only if, it is not null, and all its members are identical.

*52·22.

*52·4.

We shall define 0 as . Thus the above proposition states that a class has one member or none when, and only when, all its members are identical.

*52·41.

[Pg 364]

This proposition is obtainable from *52·4 by transposition, i.e. by negating each side of the equivalence.

*52·46

I.e. two unit classes are identical when, and only when, one is contained in the other, and when and only when they have a common part.

*52·01

*52·1

*52·11

*52·12

Dem.

*52·13

Dem.

*52·14

*52·15

*52·16

*52·17

*52·171

*52·172

*52·173

*52·18

Dem.

*52·181

*52·2

Dem.

[Pg 365]

*52·21.

Dem.

*52·22.

*52·23.

Dem.

*52·24.

*52·3.

Dem.

*52·31.

Dem.

*52·4.

Dem.

[Pg 366]

This proposition is frequently useful. We shall define the number 0 as ; thus the above proposition states that a class has one member or none when, and only when, all its members are identical. It will be seen that does not imply , and therefore allows the possibility of having no members.

*52·41.

Dem.

*52·42.

Dem.

*52·43.

*52·44.

Dem.

*52·45.

Dem.

52·46.

Dem.

[Pg 367]

*52·6.

Dem.

*52·601.

Dem.

*52·602.

*52·61.

*52·62.

Dem.

*52·63.

*52·64.

Dem.

*52·7.

Dem.

[Pg 368]

*53. MISCELLANEOUS PROPOSITIONS INVOLVING UNIT CLASSES.

Summary of *53.

The propositions to be given in this number are mostly such as would have come more naturally at an earlier stage, but could not be given sooner because they involved unit classes. It is to be observed that is the class consisting of the members and , while is the relation which holds only between and . If and are classes, is a class of classes, its members being and . If and are relations, is a relation of relations; and so on.

The present number begins by connecting products and sums , , , , in cases where the members of or are specified, with the products or sums , , , . We have

*53·01.

*53·1.

*53·14.

with similar propositions for , and .

We have next a set of propositions on sums and products of classes of unit classes. The most important of these is

*53·22.

We have next a proposition showing that the sum of is null when, and only when, is either null or has the null-class for its only member, i.e.

*53·24.

(Here we write "," to show that the "" in question is of the next type above that of the other two 's.)

We have next various propositions on the relations of and and in various cases, first for a general relation , and then for the particular relation defined in *40. Three of these propositions are very frequently used, namely:

*53·3.

*53·301.

[Pg 369]

*53·31.

The remaining propositions of this number are of less importance, and are seldom referred to.

*53·01.

Dem.

*53·02.

Dem.

*53·03.

*53·04.

*53·1.

Dem.

This proposition can be extended to , etc. It shows the connection (for finite classes of classes) between the product and the product of the members

*53·11.

Dem.

Similar remarks apply to this proposition as to *53·1.

*53·12.

This proposition shows the connection between the product for a class consisting of two relations and , and the product . The proposition can be extended to the product of any given finite class of relations.

*53·13.

Similar remarks apply to this proposition as to *53·12.

*53·14.

Dem.

[Pg 370]

*53·15.

*53·16.

*53·17.

The above proposition and the next are both used in connection with mathematical induction (*91·55 and *97·46 respectively).

*53·18.

Dem.

*53·181.

*53·2.

This proposition requires, for significance, that should be a class of classes. It is used in *88·47, in the number on the existence of selections and the multiplication axiom.

Dem.

*53·21.

This proposition requires, for significance, that should be a class of relations.

*53·22.

Dem.

*53·221.

Dem.

[Pg 371]

*53·222.

Dem.

*53·23.

Dem.

*53·231.

Dem.

*53·24.

Dem.

In the enunciation and the last line of the proof of the above proposition, we write "" rather than "," because this must be of the type next above that of the in "."

The following proposition is used in the theory of selections (*83·731).

*53·25.

Dem.

[Pg 372]

*53·3.

Dem.

The above proposition is very frequently used.

*53·301.

Dem.

*53·302.

The above proposition is used in the cardinal theory of exponentiation (*116·71).

*53·31.

The above proposition is one of which the subsequent use is frequent.

Dem.

*53·32.

Dem.

*53·33.

*53·34.

*53·35.

Dem.

[Pg 373]

The above proposition may also be proved as follows:

*53·4.

Dem.

*53·5.

Dem.

In the above proof, as usually where "Cls" or other type-symbols occur, it is necessary to abandon the notation by Greek letters and revert to the explicit notation.

*53·51.

*53·52.

Dem.

*53·53.

The following propositions are inserted because of their connection with the definition of in *70. and (are both important classes.

*53·6.

Dem.

*53·601.

Dem.

*53·602.

*53·603.

*53·604.

*53·61.

Dem.

*53·611.

*53·612.

*53·613.

*53·614.

Dem.

*53·615.

[Pg 375]

The two following propositions are used in *70·12.

*53·62.

Dem.

*53·621.

*53·63.

*53·631.

*53·64.

*53·641.

[Pg 376]

*54. CARDINAL COUPLES.

Summary of *54.

Couples are of two kinds, namely (1) , in which there is no order as between and , and (2) , in which there is an order. We may distinguish these two kinds of couples as cardinal and ordinal respectively, since (as will be shown hereafter) the class of all couples of the form (where) is the cardinal number 2, while the class of all couples of the form (where) is the ordinal number 2, to which, for the sake of distinction, we assign the symbol "," where the suffix "" stands for "relational," because the ordinal 2 is a class of relations. In the present and the following numbers, we shall define 2 and as the classes of cardinal and ordinal couples respectively, leaving it to a later stage to show that 2 and , so defined, are respectively a cardinal and an ordinal number. An ordinal couple will also be called an ordered couple or a couple with sense. Thus a couple with sense is a couple of which one comes first and the other second.

We introduce here the cardinal number 0, defined as . That 0 so defined is a cardinal number, will be proved at a later stage; for the present, we postpone the proof that 0 so defined has the arithmetical properties of zero.

Cardinal couples are much less important, even in cardinal arithmetic, than ordinal couples, which will be considered in the two following numbers (*55 and *56). It is necessary, however, to prove some of the properties of cardinal couples, and this will be done in the present number. Some properties of cardinal couples which have been already proved are here repeated for convenience of reference. The definitions of 0 and 2 are:

*54·01.

*54·02.

Most of the propositions of the present number, except those that merely embody the definitions (*54·1·101·102) are used very seldom. The following are among the most important.

*54·26.

*54·3.

[Pg 377]

*54·4.

*54·53.

*54·56.

*54·01.

*54·02.

*54·1.

*54·101.

*54·102.

The two following propositions have already occurred in *51, but are here repeated, because they belong to the subject of the present number.

*54·21.

*54·22.

*54·25.

Dem.

*54·26.

Dem.

*54·27.

*54·271.

Dem.

[Pg 378]

*54·3.

Dem.

*54·4.

Dem.

This proposition shows that a class contained in a couple is either the null-class or a unit class or the couple itself, whence it will follow that 0 and 1 are the only numbers which are less than 2.

*54.41.

Dem.

*54·411.

[Pg 379]

*54·42.

Dem.

*54·43.

Dem.

From this proposition it will follow, when arithmetical addition has been defined, that 1 + 1 = 2.

*54·44.

Dem.

*54·441.

Dem.

[Pg 380]

This proposition is used in *163·42, in the theory of relations of mutually exclusive relations.

*54·442.

*54·443.

*54·45.

*54·451.

*54·452.

*54·46.

*54·5.

Dem.

*54·51.

Dem.

*54·52.

*54·53.

Dem.

[Pg 381]

*54·531.

Dem.

*54·54.

Dem.

In the above proposition, "" secures that has not more than two members, while "(" secures that has not fewer than two members.

*54·55.

Dem.

[Pg 382]

*54·56.

Dem.

In virtue of this proposition, a class which is neither null nor a unit class nor a couple contains at least three distinct members. Hence it will follow that any cardinal number other than 0 or 1 or 2 is equal to or greater than 3. The above proposition is used in *104·43, which is an existence-theorem of considerable importance in cardinal arithmetic.

*54·6.

Dem.

The above proposition is useful in dealing with sets of couples formed of one member of a class and one member of a class , where and have no members in common. It is used in the theory of cardinal multiplication (*113·148).

[Pg 383]

*55. ORDINAL COUPLES.

Summary of *55.

Ordinal couples, which are now to be considered, are much more important, even in cardinal arithmetic, than cardinal couples. Their properties are in part analogous to those of cardinal couples, but in part also to those of unit classes; for they are the smallest existent relations, just as unit classes are the smallest existent classes. The properties which are analogous to those of unit classes do not demand that the two terms of the couple should be distinct, i.e. they hold for as well as for (where); on the other hand, the properties which are analogous to those of cardinal couples do in general demand that the two terms of the ordinal couple should be distinct.

The notation is cumbrous, and does not readily enable us to exhibit the couple as a descriptive function of for the argument , or vice versa. We therefore introduce a new symbol, "," for the couple. In a couple , we shall call the referent of the couple, and the relatum. In virtue of the definitions in *38, this gives rise to two relations, and ; hence we obtain the notations , , , and so on, which will be much used in the sequel. It should be observed that means the class of ordinal couples in which is referent and a member of is relatum, while or denotes the class of couples having as relatum and a member of as referent; denotes all such classes of couples as , where y is any member of ; and in virtue of *40·7, denotes all ordinal couples of which the referent is a member of , while the relatum is a member of . This is a very important class, which will be used to define the product of two cardinal numbers; for it is evident that the number of members of is the product of the number of members of and the number of members of .

[Pg 384]

The first few propositions of the present number are immediate consequences of the definition of and the notations introduced in *38. We then proceed to various elementary properties of the relation , of which the most used are the following:

*55·13.

*55·15.

*55·16.

*55·202.

This proposition should be contrasted with *54·22, as giving one reason why ordinal couples are more useful in arithmetic than cardinal couples. In virtue of the

above proposition, when two ordinal couples are identical, their referents are identical, and their relata are identical.

We proceed next to various properties of the relations and . These relations play a great part in arithmetic. It will be observed that if two terms have the relation , the referent is a couple whose relatum is the relatum in the relation , i.e. when we have , we have (cf. *55·122). Similar remarks apply to the relation . The class , consisting of all couples whose referent is a member of , while the relatum is , is important. We have

*55·232.

This proposition is frequently useful.

We proceed next (*55·3—·51) to give various properties of which are analogous to the properties of unit classes. Among the more important of these properties are the following:

*55·3.

This is the analogue of *51·31.

*55·34.

This is the analogue of *51·4.

*55·5.

This is the analogue of *54·4.

We then proceed to such properties of ordinal couples as are not analogous to those of unit classes. For connecting the cardinal number 2 with the ordinal number , we have the proposition

*55·54.

[Pg 385]

This proposition shows that the only asymmetrical relations which have a given cardinal couple for their field are the two corresponding ordinal couples and . We have next a set of propositions on the relative products of couples and other relations, i.e. on , , and . These propositions are very useful in arithmetic. The chief of them is

*55·61.

Finally we have four propositions which belong, by their subject, to *43, but could not be given there, because the proofs make use of ordinal couples.

*55·01.

*55·02.

This definition serves merely for the avoidance of brackets.

*55·1.

*55·11.

*55·12.

*55·121.

*55·122.

*55·123.

*55·13.

Dem.

*55·132.

*55·134.

*55·14.

*55·15.

*55·16.

Dem.

The above proposition is important, and will be frequently used.

[Pg 386]

*55·161.

Dem.

*55·17.

*55·2.

Dem.

*55·201.

*55·202.

Dem.

The above proposition is important.

*55·21.

*55·22.

*55·221.

[Pg 387]

*55·222.

Dem.

*55·223.

*55·224.

Dem.

*55·23.

*55·231.

*55·232.

Dem.

*55·233.

The above two propositions are frequently useful in arithmetic.

*55·24.

Dem.

*55·241.

[Pg 388]

*55·25.

Dem.

*55·251.

This proposition is used in the theory of cardinal multiplication (*113·142).

*55·26.

*55·261.

*55·262.

*55·27.

*55·28.

*55·281.

*55·282.

*55·283.

*55·29.

*55·291.

*55·292.

The following propositions, down to *55·51 inclusive, give properties of ordinal couples which are analogous to the properties of unit classes.

*55·3.

The first half of this proposition is the analogue of *51·2; like that proposition, it gives a means of reducing propositions to the form of inclusions. For the second half, compare *51·31.

*55·31.

This proposition is the analogue of *51·23.

Dem.

[Pg 389]

*55·32.

Dem.

*55·33.

*55·34.

Dem.

*55·341.

Dem.

*55·35.

Dem.

*55·36.

Dem.

*55·37.

Dem.

The following proposition is the analogue of *51·232.

*55·4.

[Pg 390]

*55·41.

Dem.

The above proposition is the analogue of *51·234. The following proposition (*55·42) is the analogue of *51·235.

*55·42.

Dem.

*55·43.

This proposition is the analogue of *51·41.

Dem.

*55·431.

[Pg 391]

Dem.

*55·44.

Dem.

The above proposition is the analogue of *51·43.

*55·5.

Dem.

The above proposition is the analogue of *54·4.

[Pg 392]

*55·51.

Dem.

In the remainder of the present number, we are concerned with properties of ordinal couples which have no analogues for unit classes.

*55·52.

*55·521.

*55·53.

Dem.

*55·54.

[Pg 393]

Dem.

*55·57.

*55·571.

*55·572.

*55·573.

*55·58.

*55·581.

*55·582.

*55·583.

The above propositions are frequently useful in arithmetic. Their use arises as follows. Let , , , be classes of which is correlated with by the relation , and with by the relation . Then if , the couple consisting of the correlate of and the correlate of is (, i.e., by the above, , i.e. (. Thus the relation correlates the couples, in and , composed of the correlates of terms in and . The most useful form, in practice, of *55·583, is that given below in *55·61.

*55·6.

*55·61.

*55·62.

Dem.

[Pg 394]

*55·621.

The four following propositions belong to *43, but are inserted here because the proof uses *55·13.

*55·63.

Dem.

*55·631.

*55·632.

Dem.

*55·64.

[Pg 395]

*56. THE ORDINAL NUMBER .

Summary of *56.

In this number, we have to consider the class of those relations which are each constituted by a single couple. In case the two members of this couple are not identical, the class of such relations is (as will be shown later) the ordinal number 2, which, to distinguish it from the cardinal number 2, we denote by "." (Here the suffix is intended to suggest "relational.") The class of all relations consisting of a single couple, without the restriction that the two members of the couple are to be distinct, will be denoted by "." This is not an ordinal number. It will be observed that there is no ordinal number 1, because ordinal numbers apply to series, and series must have more than one member if they have any members. This will appear more fully when we come to deal with series.

The properties of are largely analogous to those of 1, while the properties of are more analogous to those of 2.

Most of the propositions of the present number are seldom referred to in the sequel, but such references as occur are important. The most useful propositions in the present number are the following.

*56·111.

*56·112.

*56·113.

Observe that "" means "relations whose fields have two terms."

*56·13.

*56·37.

I.e. is the class of asymmetrical relations whose fields have two terms.

*56·381.

*56·39.

I.e. the relations which are couples whose referent and relatum are identical are the relations whose fields consist of a single term.

[Pg 396]

*56·01.

*56·02.

*56·03.

*56·1.

*56·101.

Dem.

*56·102.

Dem.

*56·103.

Dem.

*56·104.

*56·11.

*56·111.

Dem.

[Pg 397]

*56·112.

Dem.

*56·113.

Dem.

*56·114.

*56·12.

Dem.

*56·121.

*56·122.

*56·13.

Dem.

might be defined as the ordinal number 1, since it is what we shall call a relation number (cf. *153). But we wish our ordinal numbers to be[Pg 398] classes of serial relations, and such relations have the property of being contained in diversity. Hence

if we were to define as the ordinal number 1, we should introduce a tiresome exception, from which trivial complications would be introduced into ordinal arithmetic. We have, therefore, not adopted this course.

*56·14.

Dem.

*56·141.

*56·15.

Dem.

*56·151.

*56·16.

Dem.

[Pg 399]

*56·17.

Dem.

*56·18.

Dem.

*56·19.

Dem.

*56·191.

*56·2.

*56·21.

*56·22.

*56·24.

*56·25.

*56·26.

This proposition is the analogue of *52·4.

[Pg 400]

Dem.

*56·261.

Dem.

*56·262.

Dem.

*56·27.

Dem.

*56·28.

Dem.

*56·281.

Dem.

[Pg 401]

*56·29.

Dem.

*56·3.

Dem.

The steps from (2) to the conclusion are analogous to those from (2) of *56·29 to the conclusion of *56·29. Analogous steps in succeeding proofs will be merely indicated as above.

*56·31.

*56·32.

Dem.

*56·33.

Dem.

[Pg 402]

*56·34.

Dem.

*56·35.

Dem.

*56·36.

Dem.

The following proposition, in addition to being used in *56·38, is used in the elementary theory of series (*204·463).

*56·37.

*56·38.

Dem.

[Pg 403]

This proposition is important as establishing the connection between the cardinal and ordinal 2. It shows that the ordinal 2 consists of those asymmetrical relations whose fields have (cardinal) 2 terms. It is used in the theory of well-ordered series (*250·44).

The following proposition, in addition to being used in *56·39, is used in relation-arithmetic (*165·38) and in the theory of series (*205·4).

*56·381.

Dem.

*56·39.

Dem.

This proposition establishes the connection between and 1, showing that is the class of those relations whose fields consist of a single term. It is used in the discussion of and and as relation-numbers (*153·301).

*56·4.

Dem.

This proposition is the analogue of *53·23. It is used in the number on exponentiation in relation-arithmetic (*176·19).

[Pg 404]

SECTION B.

SUB-CLASSES, SUB-RELATIONS, AND RELATIVE TYPES.

Summary of Section B.

In this section, we consider first the classes contained in a given class and the relations contained in a given relation. If is any class, the classes contained in are the members of ; these are also called the sub-classes of , or (sometimes) the "parts" of . In this last usage, they are called "proper parts" when they are not coextensive with , this phrase being formed on the analogy of "proper fractions." The sub-classes of are all the classes that can be formed from members of ; they are the same thing as the "combinations" of members of taken any number at a time. If is the number of members of , is the number of sub-classes of , whether be finite or infinite. The number of sub-classes of is always greater than the number of members of . On account of these and other propositions, the class of sub-classes of a given class is an important function of the class. If the class is , we denote the class of its sub-classes by "." This is a descriptive function, derived from the relation "," defined as follows:

The sub-relations of a given relation are all the relations contained in the given relation, i.e. all relations which imply the given relation for all possible arguments. That is, if is the given relation, is a sub-relation of if . Thus denoting the class of sub-relations of by "," we are to have

hence we take as the definition of "" the following:

Sub-relations have properties analogous to those of sub-classes, but they are of somewhat less importance. It should, however, be observed that when one series is contained in another, i.e. is obtained by selecting some of the terms of the other series without changing their order, then the generating relation of the one series is a sub-relation of the generating relation of the other series. (It is not the case that a sub-relation of the[Pg 405] generating relation of a series must generate a contained series, for its field may fall apart into detached portions, or otherwise fail of being serial.)

We shall also consider in this section (*62) the relation of membership of a class, i.e. the relation which has to when . This relation bears the same relation to "" as "" bears to "." Strictly speaking, we ought to introduce a new notation for it, putting (say)

But as , unlike "," is a letter, and capable of being conveniently used alone, it seems more desirable, from the point of view of avoiding unnecessary duplication of symbols, to put

Strictly speaking, this definition is faulty, since it gives two different meanings to "." But practically this does not matter, since the above definition gives

where the first has the meaning just defined, while the second has the old meaning. Thus all that is really required of the above definition, namely to give a meaning to formulae in which occurs without referent or relatum, is effected without the danger of any confusion that could lead to errors.

The chief importance of as a relation arises from the fact that relations contained in play a very important part in arithmetic. Take, for example, the problem of selecting one term out of each member of a class of classes: in this case we require a selecting relation which is such that whenever , is a member of , i.e. such that . (This condition is only part of the definition of a selecting relation; the complete definition is given in *80.)

Three numbers in this section (*63, *64, *65) are devoted to the discussion of relative types. Given a variable , we often want to define the relative types of other variables, or of ambiguous symbols, occurring in the same context; that is, we wish to express the types of these other symbols in terms of that of . We use "" for the type of , "" for the type in which is contained. Then , , and . Also we introduce a notation (*65) for giving typical definiteness, relatively to , to typically ambiguous symbols. This notation is very useful in cardinal and ordinal arithmetic, since numbers are typically ambiguous, and the failure to take account of this fact has led to the contradictions concerning the greatest cardinal and the greatest ordinal.

[Pg 406]

*60. THE SUB-CLASSES OF A GIVEN CLASS.

Summary of *60.

Our definitions in this number are as follows:

*60·01.

This defines the relation to a class of the class of all its sub-classes.

*60·02.

This defines the relation to a class of the class of all its existent sub-classes, i.e. of all its sub-classes except . This is often required, as, for example, in the statement of Zermelo's axiom: "Given any class , there is a relation such that, if is any existent sub-class of , is a member of ," i.e.

This axiom, or its equivalent the multiplicative axiom, plays (as will appear hereafter) an important part as the hypothesis to many propositions in cardinal arithmetic.

*60·03.

A is a class whose members are classes.

*60·04.

A is a class whose members are classes whose members are classes, i.e. a is a class of classes of classes.

Apart from propositions which merely embody the definitions, the most useful propositions in this number are the following:

*60·3.

*60·32.

*60·34.

*60·362.

I.e. and are the only sub-classes of a unit class .

*60·5.

*60·57.

*60·6.

[Pg 407]

The propositions of this number are chiefly useful in cardinal and ordinal arithmetic, but uses also occur in the theory of series; hardly any uses occur before cardinal arithmetic.

*60·01.

*60·02.

*60·03.

*60·04.

*60·1.

*60·11.

*60·12.

*60·13.

*60·14.

*60·15.

*60·2.

*60·21.

*60·22.

*60·23.

*60·24.

*60·3.

*60·31.

*60·32.

Dem.

*60·321.

Dem.

[Pg 408]

*60·33.

We write "" on the right, to indicate that the concerned is of higher type than the on the left.

Dem.

*60·34.

*60·35.

*60·36.

*60·361.

*60·362.

*60·37.

Dem.

*60·371.

Dem.

*60·38.

Dem.

[Pg 409]

*60·39.

*60·391.

This proposition is used in the theory of the continuity of functions (*234·202).

*60·4.

*60·41.

The following proposition is used in the theory of well-ordered series (*250·14).

*60·42.

*60·43.

*60·44.

The following proposition is required in the theory of "first differences" (*170·65).

*60·45.

Dem.

*60·5.

Dem.

[Pg 410]

*60·501.

Dem.

The above proposition is used in the theory of cardinal multiplication (*115·17).

*60·51.

The following proposition is used in the cardinal theory of finite and infinite (*124·541).

*60·52.

*60·53.

Dem.

*60·54.

*60·55.

Dem.

*60·56.

The following proposition is used frequently.

*60·57.

Dem.

*60·6.

[Pg 411]

The following proposition is used in connection with cardinal multiplication and with greater and less (*115·17 and *117·66).

*60·61.

*60·62.

*60·7.

Dem.

*60·71.

*60·72.

[Pg 412]

*61. THE SUB-RELATIONS OF A GIVEN RELATION.

Summary of *61.

The propositions of this number (except that *61·371·372·373 imperfectly correspond to *60·371) are the analogues of those with the same decimal part in *60. Proofs are omitted, as they are exactly analogous to those in *60. There are very few subsequent references to the propositions of this number.

*61·01.

*61·02.

*61·03.

*61·04.

*61·1.

*61·11.

*61·12.

*61·13.

*61·14.

*61·15.

*61·2.

*61·21.

*61·22.

*61·23.

*61·24.

*61·3.

*61·31.

*61·32.

*61·321.

*61·33.

[Pg 413]

*61·34.

*61·35.

*61·36.

*61·361.

*61·362.

*61·37.

*61·371.

*61·372.

*61·373.

*61·38.

*61·39.

*61·391.

*61·4.

*61·41.

*61·42.

*61·43.

*61·44.

*61·5.

*61·501.

*61·51.

*61·52.

*61·53.

*61·54.

*61·55.

*61·56.

*61·6.

The analogue of *60·61 is not given, because we have no suitable notation for expressing it.

*61·62.

*61·7.

[Pg 414]

*62. THE RELATION OF MEMBERSHIP OF A CLASS.

Summary of *62.

When "" was defined, in *20, it was defined as a propositional function; and this mode of definition was necessary, because we had to treat of this function before treating of relations. But for many purposes it is desirable to regard as a relation, so that "" becomes an instance of the notation "." This requires, strictly speaking, a change in the meaning of "," but it is a change which does not falsify any of the previous propositions in which "" occurs; for if we call the new meaning "," i.e. if we put

Hence it is unnecessary in practice to have a new notation for the new meaning, and we put simply

This definition, though strictly incorrect, is recommended by its convenience, and by the fact that it cannot lead to any harmful confusions. The new meaning of may be taken as replacing the old throughout the remainder of this work.

The uses of the propositions of the present number occur almost exclusively in the theory of selections from a class of classes (*83, *84, *85 and *88). Such selections are effected by means of selective relations, part of whose definition is that they are contained in . Hence the uses of the present number. If is the class of classes from which a selection is to be made, a selective relation will in fact be contained in ; hence the properties of become important. Some of these properties are given in *62·4 ff.

The most important propositions of the present number are the following:

*62.2.

*62.231.

*62.26.

*62.3.

*62.42.

[Pg 415]

*62·43.

*62·55.

*62·01.

*62·1.

In the above proposition, the first has the newly-defined meaning, while the second has the old meaning. In virtue of the above proposition, the new meaning may be substituted for the old in all propositions hitherto proved concerning , and may take the place of the old meaning in all that follows.

*62·2.

Dem.

*62·21.

Thus consists of the classes of which is a member.

*62·22.

Dem.

*62·23.

Dem.

*62·231.

*62·24.

Dem.

*62·25.

Dem.

[Pg 416]

*62·26.

Dem.

*62·3.

Dem.

*62·31.

Note that, since is not a homogeneous relation, i.e. not one in which referent and relatum belong to the same type, is strictly meaningless. For if we have , the two 's have different meanings, and do not therefore properly give . But it is convenient to allow , on the understanding that the ambiguity of is to be differently determined for the two factors in the product , namely the second must make both referent and relatum belong to the next type above that to which they respectively belong for the first .

Dem.

*62·32.

*62·33.

Dem.

The use of *20·41 in the above proof depends upon the fact that is merely an abbreviation for an expression of the form .

*62·34.

Dem.

[Pg 417]

*62·4.

The relation is very important in cardinal arithmetic, in connection with the problem of selection from the members of , i.e. of extracting one term out of each of the members of . A relation which is to effect this selection must be contained in .

*62·41.

Dem.

*62·42.

Dem.

*62·43.

Dem.

*62·44.

Dem.

*62·45.

Dem.

[Pg 418]

This proposition is useful in the theory of selections. It is used in the proof of *83·27, and thence of *83·28.

*62·5.

Dem.

*62·51.

Dem.

*62·52.

Dem.

*62·53.

*62·54.

*62·55.

Dem.

*62·56.

Dem.

*62·57.

Dem.

[Pg 419]

*63. RELATIVE TYPES OF CLASSES.

Summary of *63.

The notations introduced in this and the two following numbers serve to express the type of one variable in terms of the type of another. They are very useful in arithmetic, where it is necessary to take account of types in order to avoid contradictions. The two chief notations are "," for the type in which is contained, and "," for the type of which is a member. We put

*63·02.

This defines "the type of members of ," or "the type which is of the same type as ." The characteristic of a type is that if is a type, we have

and conversely, if , then is a type. For in that case, "" is true whenever it is significant, i.e. whenever belongs to the type which is the range of significance of in "." Consequently is this range of significance, i.e. is a type.

Since we have (, it follows that is a type. It is not "the type of ," but "the type of the members of ." (In case is null, "the type of the members of " may be interpreted as meaning "the type to which belongs when '' is significant.") "The type of ," i.e. the type of which is a member, is defined as follows:

*63·01.

By what was said above, "" is the type of the members of , i.e. the type of . By combining the definitions of and , we obtain

Thus

In short, consists of everything either identical or not identical with , that is, every for which there is such a proposition, whether true or false, as "." We put "" here instead of "," because need not be a class, and is in fact subject to no limitation whatever, whereas "" is not significant unless is a class, and therefore we write "" rather [Pg 420]than "."

We put also

*63·011.

This definition serves merely to bring notationally into line with and the types defined below.

In virtue of *20·8, we have

i.e. if "" is significant, then the range of significance of the function is the type of . It follows that two ranges of significance which overlap are identical, and two

different ranges of significance have no member in common.

It will be seen that is always of the next type above that of , and (if is a class of classes) is of the next type below that of . We put

*63·03.

so that is the type next below that in which is contained. Thus if is a class of classes of individuals, is the class of individuals. We put also

*63·04.

*63·041.

*63·05.

*63·051.

Thus given any two objects which are members of any one of the following: the type of , the type of the classes to which belongs, the type of the classes to which these classes belong, and so on, we can express the type of either of our two objects by means of its relation to the other object.

The propositions of this and the two following numbers will hardly ever be used until we come to cardinal arithmetic. They are used constantly in the first section on cardinal arithmetic, and they are constantly relevant in the first section on relation-arithmetic. Moreover they are usually required for cardinal and ordinal existence-theorems.

Among the most useful propositions of the present number are the following:

*63·103.

*63·105.

*63·11.

I.e. if either is or is not a member of , then the type of is the type which contains . This proposition uses *20·8.

*63·13.

I.e. if there is any function satisfied by both and , then is of the type of . It is necessary to the use of this proposition that, if is a typically[Pg 421] ambiguous function, it should receive the same typical determination for and for . For example, we have always and ; but we must not regard these as values of one function , because such a function is typically ambiguous. On the other hand, and are values of one function , because here the presence of a renders the function typically determinate.

*63·15.

*63·19.

*63·16.

This proposition, which depends upon *63·11, and thence upon *20·8 and *13·3, and thence upon *9·14·15, is vital to the whole theory of types.

*63·32.

*63·371.

*63·383.

We shall have generally , where we may count suffixes as negative indices, so that or according as or is the greater.

*63·5.

This proposition is used constantly.

*63·51.

*63·52.

*63·53.

The above four propositions, together with four similar ones (*63·54·55·56·57), give transformations which enable us to express any relation of type, as between class and members or members of members or etc., that is likely to occur in practice.

*63·64.

This proposition is often used in the first section on cardinal arithmetic.

*63·66.

*63·01.

*63·011.

*63·02.

*63·03.

*63·04.

*63·041.

[Pg 422]

*63·05.

*63·051.

*63·1.

*63·101.

*63·102.

*63·103.

*63·104.

*63·105.

*63·106.

*63·107.

Dem.

*63·108.

*63·109.

*63·11.

Dem.

*63·12.

Dem.

*63·13.

*63·14.

*63·15.

*63·151.

[Pg 423]

*63·152.

*63·16.

Dem.

*63·17.

*63·18.

*63·181.

Dem.

*63·182.

*63·19.

Dem.

[Pg 424]

*63·191.

*63·2.

Dem.

*63·21.

Dem.

*63·22.

Dem.

*63·23.

Propositions of the same kind as the above can obviously be extended to , etc.

*63·3.

Dem.

*63·31.

Dem.

Note that the use of *10·221 in the above proof depends upon the fact that occurs both in (2) and in (3), so that these are both of the form .

[Pg 425]

*63·32.

*63·321.

Dem.

*63·33.

*63·34.

Dem.

*63·35.

*63·36.

*63·361.

*63·37.

*63·371.

Dem.

*63·38.

Dem.

*63·381.

Dem.

[Pg 426]

*63·382.

*63·383.

Dem.

*63·384.

*63·39.

*63·391.

Dem.

*63·392.

Dem.

*63·4.

Dem.

*63·41.

Dem.

*63·42.

*63·43.

*63·44.

It is obvious that the analogues of the above propositions will hold for and , and , etc. We shall not prove these analogues, but if occasion arises we shall assume them, referring to the corresponding propositions for and .

*63·5.

Dem.

[Pg 427]

*63·51.

Dem.

*63·52.

Dem.

*63·53.

Dem.

*63·54.

Dem.

*63·55.

[Pg 428]

*63·56.

Dem.

*63·57.

*63·61.

*63·62.

Dem.

*63·621.

*63·63.

Dem.

*63·64.

Dem.

*63·65.

*63·66.

*63·661.

*63·67.

*63·68.

[Pg 429]

*64. RELATIVE TYPES OF RELATIONS.

Summary of *64.

In the present number, we introduce notations defining the type of a relation relatively to the types of its domain and converse domain, when these types are given relatively to some fixed class . If is any relation, it is of the same type as . If and are both of the same type as , is of the same type as , which is of the same type as . The type of we call , and the type of we call , and the type of we call , and the type of we call , and the type of we call . We thus have a means of expressing the type of any relation in terms of the type of , provided the types of the domain and converse domain of are given relatively to .

The most useful propositions of the present number are the following:

*64·16.

*64·201.

*64·231.

Here "" will only be significant if and are homogeneous relations, which is not required by the rest of the proposition. When and are homogeneous relations we have

*64·24.

This proposition is useful in connecting ordinal and cardinal existence-theorems.

*64·312.

*64·5.

This proposition is frequently used. It states that the class of relations whose referents are of the type of members of while its relata are of the type of members of (i.e. the class of all relations contained in) is the type of and is also the type of .

*64·55.

[Pg 430]

*64·57.

The propositions of the present number are mostly obvious, though formal proofs are sometimes not very easily found. The use of the propositions of this number occurs chiefly in the first section on relation-arithmetic and in the proofs of existence-theorems in ordinal arithmetic and the theory of ratio.

*64.01.

*64.011.

*64.012.

*64.013.

*64.014.

*64.02.

*64.021.

*64.022.

*64.03.

*64.031.

*64.04.

*64.041.

*64.1.

Dem.

*64.11.

*64.12.)

Dem.

[Pg 431]

*64·13.

*64·14.

*64·15.

*64·16.)

Dem.

By putting (where and are some index and suffix which have been defined) for and for , the above propositions give results applicable to any of the types defined at the beginning of this number, because of .

*64·2.

*64·201.

Dem.

*64·21.)

Dem.

*64·22.

*64·23.

Dem.

*64·231.

Dem.

[Pg 432]

*64·24.

This proposition is only significant when and are homogeneous relations.

Dem.

*64·3.

Dem.

*64·31.

*64·311.

*64·312.

*64·313.

*64·32.

Dem.

Similarly the other equivalences are proved.

*64·33.

Dem.

Similarly the other equivalences are proved.

*64·34.

[Pg 433]

*64·5.

*64·51.

*64·52.

*64·53.

Dem.

This proposition is used in connection with cardinal addition (*110·18).

*64·54.

*64·55.

Dem.

*64·56.

Dem.

*64·57.

*64·6.

Dem.

*64·61.

Dem.

*64·62.

Dem.

*64·63.

Dem.

[Pg 434]

*65. ON THE TYPICAL DEFINITION OF AMBIGUOUS SYMBOLS.

Summary of *65.

In this number we are concerned with definitions and propositions in which an ambiguous symbol is determined as belonging to some assigned type. If "" is an ambiguous symbol representing a class (such as or for example), "" is to denote what becomes when its members are determined as belonging to the type of , while "" denotes what becomes when its members are determined as belonging to the type of . Thus e.g. "" will be everything of the same type as , i.e. ; will be . Similarly if "" stands for a relation of ambiguous type, such as or , will denote what becomes when its domain is confined within the type of ; will denote what becomes when its domain and converse domain are confined respectively within the types of and ; will have the domain and converse domain confined respectively to the types of and ; with analogous meanings for and . Throughout this number, and do not stand for proper variables, but for typically ambiguous symbols.

The notations of the present number are used in the elementary parts of the theory of cardinals and ordinals, i.e. in Part III, Section A, and in Part IV, Section A. The only proposition, however, which is much used, is

*65·13.

Here is supposed to be a typically ambiguous symbol. The first equivalence, "," merely embodies the definition of (*65·01). It is the second equivalence that is

important. Let us, for the sake of illustration, put 1 in place of . Then we are to have

(Since 1 is a class of classes, we shall have to suppose that is a class.) Consider . If , . But we have . Hence , whence . Also if , of course . Thus . The converse implication follows from *22·621. The reason for the proposition is that a symbol such as "1," if it occurs in such a proposition as , must, for significance, be determined as meaning that 1 which is of the same type as , i.e. the class of all[Pg 435] unit classes which are of the same type as members of . And similarly, when we put , that does not mean that is the class of all unit classes, but only that it is the class of all unit classes of the appropriate type, which, if , will be . The proposition "" is true whenever it is significant, but is typically definite when is given, whereas 1 is typically ambiguous. The use of the above proposition lies in its enabling us to substitute typically definite symbols for such as are typically ambiguous.

Another useful proposition is

*65·2.

Here is supposed to be a typically ambiguous symbol; the proposition states that if is typically defined as going from objects of type to objects of type , then must go from objects of type to objects of type . This proposition is only used twice (*102·3 and *154·2), but both uses are of great importance, the one in cardinal and the other in ordinal arithmetic.

The only other proposition of this number which is subsequently used is

*65·3.

This proposition is used in *102·84.

*65·01.

*65·02.

*65·03.

*65·04.

*65·1.

*65·11.

*65·12.

*65·13.

Dem.

*65·14.

*65·15.

[Pg 436]

*65·16.

*65·2.

Dem.

*65·21.

Dem.

*65·22.

This and the following three propositions are proved as *65·21 is proved.

*65·23.

*65·24.

*65·25.

*65·3.

Dem.

[Pg 437]

SECTION C.

ONE-MANY, MANY-ONE, AND ONE-ONE RELATIONS.

Summary of Section C.

In the present section we have to consider three very important classes of relations, of which the use in arithmetic is constant. A one-many relation is a relation such that, if is any member of , there is one, and only one, term which has the relation to , i.e. . Thus the relation of father to son is one-many, because every son has one father and no more. The relation of husband to wife is one-many except in countries which practise polyandry. (It is one-many in monogamous as well as in polygamous countries, because, according to the definition, nothing is fixed as to the number of relata for a given referent, and there may be only one relatum for each given referent without the relation ceasing to be one-many according to the definition.) The relation in algebra of to is one-many, but that of to is not, because there are two different values of that give the same value of .

When a relation is one-many, exists whenever , and vice versa; i.e. we have

Thus relations which give descriptive functions that are existent whenever their arguments belong to the converse domains of the relations in question are one-many relations. Hence , , , , , , , , , , , , , , , , , are all of them one-many relations.

When is a one-many relation, is a one-valued function; conversely, every one-valued function is derivable from a one-many relation. A many-valued function of is a member of , where is not a unit class, and any one of its members is regarded as a value of the function for the argument y; but a one-valued function of is the single term which is obtained when is one-many. Thus for example the sine would, in our notation, appear as a relation, i.e. we should put [Pg 438] so that "" has the usual meaning of . Then instead of , we should have , which would be the class of values of ; and instead of "," which is a misleading notation because and do not imply , we should

have . Similar remarks would apply to any of the other functions that occur in analysis.

A relation is called many-one when, if is any member of , there is one, and only one, term to which has the relation , i.e. . Thus many-one relations are the converses of one-many relations. When a relation is many-one, exists whenever .

A relation is called one-one when it is both one-many and many-one, or, what comes to the same, when both it and its converse are one-many. Of the one-many relations above enumerated, , , , , , , , are one-one.

Two classes , are said to be similar when there is a one-one relation such that , i.e. when their terms can be connected one to one, so that no term of either is omitted or repeated. We write "" for " is similar to ." When two classes are similar, the cardinal numbers of their terms are the same; it is this fact chiefly that makes one-one relations of fundamental importance in cardinal arithmetic.

According to the above, a relation is one-many when

Similarly a relation is many-one when

and a relation is one-one when both conditions are fulfilled. The classes , , which appear here, are often important; some of their properties have already been given in *37·77·771·772·773 and in *53·61 to *53·641.

It is convenient to regard one-many, many-one and one-one relations as particular cases of relations which, for some given and , have

Hence, without a new definition, "" becomes the class of one-one relations; also, as will be shown, "" becomes the class of one-many relations, and "" becomes the class of many-one relations. Although it is chiefly these three special values of that are important, we shall begin by a general study of classes of relations of the form .

[Pg 439]

*70. RELATIONS WHOSE CLASSES OF REFERENTS AND OF RELATA BELONG TO GIVEN CLASSES.

Summary of *70.

If and are two given classes of classes, a relation is said to belong to the class if whenever , and whenever . If only one of these conditions is to be imposed, this result is secured by replacing the class involved in the other condition by "," since "" always holds, and so does "," and therefore neither imposes any limitation on . In the most important cases, and are either both cardinal numbers, or one is a cardinal number while the other is .

In virtue of *37·702·703, the conditions above mentioned as imposed upon by membership of are equivalent to

This form is used in the definition (*70·01).

The propositions of the present number are hardly ever used except in *71, where and are both replaced by or . The most useful propositions are

*70·1.

(This merely embodies the definition.)

*70·13.

*70·22.

*70·4.

*70·41.

*70·42.

*70·54.

with similar propositions for and .

*70·62.

with a similar proposition for .

[Pg 440]

*70·01.

*70·1.

*70·11.

*70·12.

*70·13.

Dem.

*70·14.

*70·15.

*70·16.

*70·17.

Dem.

*70·171.

*70·18.

[Pg 441]

*70·2.

Dem.

*70·21.

Dem.

*70·22.

Dem.

[Pg 442]

*70·3.

Dem.

*70·31.

Dem.

*70·32.

Dem.

*70·4.

Dem.

*70·41.

*70·42.

*70·43.

*70·431.

*70·44.

*70·441.

*70·45.

[Pg 443]

*70·451.

*70·46.

*70·461.

*70·47.

*70·471.

*70·48.

*70·481.

*70·5.

*70·51.

Dem.

*70·52.

*70·53.

Dem.

[Pg 444]

*70·54.

Dem.

*70·55.

*70·56.

*70·57.

Dem.

*70·6.

Dem.

[Pg 445]

*70·61.

*70·62.

Dem.

*70·63.

[Pg 446]

*71. ONE-MANY, MANY-ONE, AND ONE-ONE RELATIONS.

Summary of *71.

In this number we shall be concerned with the more elementary properties of one-many, many-one, and one-one relations. These properties are very numerous and very important. The properties of many-one relations (i.e. of relations belonging to the class) result from those of one-many relations by means of *70·5, whence it follows that many-one relations are the converses of one-many relations. It is thus only necessary to interchange and , and , and in order to obtain a property of a many-one relation from a property of a one-many relation. Or we may repeat the various steps of any proof, making the above interchanges at every step, and the analogous proposition will result. For this reason, in what follows, we shall omit all proofs of properties of many-one relations, confining ourselves to proving the analogous properties of one-many relations.

In virtue of *70·42, one-one relations (i.e. relations belonging to the class) are the relations which are both one-many and many-one; hence their properties result from combining the properties of one-many and many-one relations. We shall omit the proofs when they consist merely in such combinations.

A one-many relation gives rise to a descriptive function which is existent whenever its argument belongs to the converse domain of the relation. That is, if , we have whenever . Conversely, if a descriptive function exists for the argument , then is one-many so far as that argument is concerned, i.e. . Thus we find

The descriptive function derived from a one-many relation has thus a definite value whenever , and not otherwise. Thus the class of arguments for which such a function exists is the converse domain of the relation which gives rise to the function, i.e.

and the converse implication also holds.

It often happens that a relation which is not in general one-many becomes so when its domain, converse domain, or field is subjected to some limitation. For example, let be the relation of parent to child, the class of males, and[Pg 447] the class of females. Then is not one-many, but and are one-many, and in fact (= the father of , (= the mother of . We shall often have occasion to deal with relations obtained by limitations imposed on or ; thus belongs to the class , and has for its domain. The class may be so constituted that only one relation fulfils this condition; in that case, . Since , we find . This type of condition, or or , is one which frequently occurs in subsequent work. Another condition which often occurs is . When this condition is realized, a term which belongs to the field of one relation of the class does not belong to the field of any other relation of this class, i.e. the fields of relations of this class are mutually exclusive.

For purposes of realizing imaginatively the properties of one-many relations, it is often convenient to picture their structure as in the accompanying figure.

Here , , , ... form the domain of , and all the points in the oval marked are such that has the relation to each of them, with similar conditions for and . What characterizes as a is the absence of overlapping in the ovals. For if and had a point in common, this would be a relatum both to and , and both and would be referents to it; whereas in a , no term has more than one referent.

[Pg 448]

The above figure illustrates a very important property of one-many relations, namely

In the above figure, is the relation of identity confined to , , ,.... If were not a , we could sometimes go from to some term of by the relation , and thence back to by the relation . But when , must bring us back to the point from which we started.

When , each of the ovals , , , ... in the above figure shrinks to a single point, so that . Thus when is given as a , it will be a if . This proposition is constantly used, and so is the consequence that is a if . (These propositions are *71·54·55 below.)

The hypothesis is equivalent to the hypothesis

(cf. *71·17, below), and the hypothesis is equivalent to

These are for many purposes the most convenient hypotheses to use.

The most useful propositions in the present number are the following. (We omit here propositions concerning or which are mere analogues of propositions concerning .)

*71·16.

This gives the connection of one-many relations with descriptive functions. We have also

*71·163.

For many of the constant relations defined from time to time, such as or , the following proposition is useful:

*71·166.

*71·17.

This might have been taken as the definition of one-many relations, if we had not wished to derive them from the more general notion of . In proving that a relation is one-many, *71·17 is more often employed than any other proposition.

*71·22.

*71·25.

*71·36.

*71·381.

[Pg 449]

(This proposition is more useful than the corresponding property of)

*71·55.

This proposition is constantly used. For example, putting for , it gives

Most of the relations used to establish correlations in arithmetic are obtained from a one-many relation, such as , by imposing some limitation on the converse domain which makes the relation one-one.

*71·571.

Here "" is , which has already played a large part as a hypothesis, e.g. in *37·6 ff.

*71·7.

Thus for example we shall have .

*71·01.

*71·02.

*71·03.

*71·04.

*71·1.

*71·101.

*71·102.

*71·103.

*71·11.

*71·111.

*71·112.

*71·12.

*71·121.

*71·122.

*71·13.

*71·131.

[Pg 450]

*71·132.

*71·14.

*71·141.

*71·142.

*71·15.

*71·151.

*71·152.

*71·16.

Dem.

This proposition is very important; it exhibits the connection of descriptive functions with one-many relations.

*71·161.

*71·162.

*71·163.

Dem.

*71·164.

*71·165.

*71·166.

Dem.

*71·167.

[Pg 451]

*71·168.

*71·17.

This proposition is constantly used in the sequel.

Dem.

*71·171.

*71·172.

*71·18.

Dem.

*71·181.

*71·182.

*71·19.

Dem.

*71·191.

*71·192.

[Pg 452]

*71·2.

*71·21.

Dem.

*71·211.

*71·212.

*71·22.

Dem.

*71·221.

*71·222.

*71·223.

*71·224.

*71·225.

*71·23.

*71·231.

*71·232.

*71·233.

Dem.

[Pg 453]

*71·234.

*71·235.

*71·24.

*71·241.

*71·242.

*71·243.

*71·244.

Dem.

*71·245.

*71·25.

Dem.

*71·251.

*71·252.

*71·25 may also be deduced from *70·6, as follows:

[Pg 454]

Alternative Dem. of *71·25.

Similarly *71·251 may be deduced from *70·61.

*71·26.

*71·261.

*71·27.

*71·271.

*71.28.

*71·281.

*71·29.

*71·31.

*71·311.

*71·312.

*71·32.

*71·321.

*71·33.

Dem.

[Pg 455]

*71·331.

*71·332.

*71·333.

*71·34.

*71·341.

*71·35.

Dem.

*71·351.

*71·352.

*71·36.

Dem.

*71·361.

*71·362.

*71·37.

Dem.

*71·371.

[Pg 456]

*71·38.

Dem.

*71·381.

*71·4.

*71·401.

*71·41.

*71·411.

*71·42.

*71·421.

*71·43.

*71·431.

*71·44.

*71·441.

*71·45.

Dem.

*71·451.

[Pg 457]

*71·46.

Dem.

*71·461.

*71·47.

Dem.

*71·471.

*71·48.

Dem.

*71·481.

The following proposition is used in the theory of derivatives of a series (*216·411).

*71·49.

Dem.

*71·491.

This proposition is used in the theory of derivatives of a series (*216·4) and in the theory of ordinal numbers (*251·11).

*71·5.

Dem.

[Pg 458]

*71·501.

*71·51.

Dem.

*71·511.

*71·52.

Dem.

*71·521.

*71·53.

Dem.

*71·531.

*71·532.

*71·54.

This proposition and the next (*71·55) are very often used.

Dem.

[Pg 459]

*71·55.

Dem.

*71·56.

Dem.

*71·561.

*71·57.

Dem.

*71·571.

Dem.

*71·572.

[Pg 460]

*71·58.

Dem.

*71·59.

Dem.

The following proposition is used in the theory of selections (*80·91).

*71·6.

Dem.

*71·61.

[Pg 461]

Dem.

*71·611.

*71·612.

*71·613.

*71·613 is used in the theory of series (*206·6), and in the theory of "similarity of position" (*272·131).

*71·7.

Dem.

*71·701.

[Pg 462]

*72. MISCELLANEOUS PROPOSITIONS CONCERNING ONE-MANY, MANY-ONE, AND ONE-ONE RELATIONS.

Summary of *72.

In this number we shall prove various propositions involving , , or , but not embodying fundamental properties of these classes of relations.

The present number begins with various propositions (*72·1-·191) showing that various special relations are one-many or one-one. The most useful of these are

*72·182.

*72·184.

We have next a set of propositions concerning when and are one-many, or when is one-one, and kindred matters. The most useful of these is

*72·241.

We have next a set of propositions (*72·3-·341) concerning products and sums of classes of relations; of these the one most used is

*72·32.

which is an extension of *71·24.

We have next a set of propositions (*72·4—·481) giving various relations of and when , or of and when . The more useful propositions of this set are those that have the hypothesis ; these are occasionally useful in arithmetic. We have

*72·401.

*72·411.

[Pg 463]

For example, the relation of son to father is many-one. Let = Cabinet Ministers, = fools; then assuming , it will follow that the sons of Cabinet Ministers and the sons of (male) fools have no common member. If we make the relation of son to parent (which is not many-one), it no longer follows that the sons of Cabinet Ministers and the sons of fools have no common member.

We have

*72·451.

The effect of this proposition is that if and are both contained in , and , then (using .

We next have a set of propositions concerned with the relations of and (, or, what comes to the same thing, with the circumstances under which and under which . We have

*72·502.

Thus for example the fathers of the children of wise fathers are the class of wise fathers; but the fathers of the children of wise parents are not all wise, and the parents of the children of wise parents are not all wise—the first because "" fails, the second because "" fails.

We have also

*72·52.

We have next a set of propositions (*72·59—·66) in which the relative product occurs if , or if . The most useful propositions in this set are

*72·591.

*72·601.

*72·66.

This is the "principle of abstraction." It shows that every relation which has the formal properties of equality, i.e. which is transitive and symmetrical, is equal to the relative product of a many-one relation into its converse; i.e. whenever the relation holds between and , there is a term such that , where is a many-one relation; and *72·64 shows that this term may be taken to be , which is equal to . This principle embodies a great part of the reasons for our definitions of the various kinds of numbers; in seeking these definitions, we always have, to begin with, some transitive symmetrical relation which we regard as sameness of number; thus by *72·64, the desired properties of the numbers of the kind in question are secured by taking the number of an object to be the class of objects to which the said object has the transitive symmetrical relation in question. It is in this way that we are led to define cardinal numbers as classes of classes, and ordinal numbers as classes of relations.

[Pg 464]

The remaining propositions of this number are of less importance, with the exception of

*72·92.

This proposition shows that every relation contained in a one-many relation is obtainable by a limitation of the converse domain. Thus e.g. every relation contained in that of father to son can be specified by specifying the class of sons who are to be its converse domain; for then all the fathers of these sons must be included to provide referents. But if we take the relation of parent and child, which is not one-many or many-one, a contained relation is not determinate even when both its domain and its converse domain are given; for the relation may relate some of the children in any one family to the father and some to the mother, and so long as all the children and both parents are each related to some one by the relation, the domain and converse domain remain unchanged by permutations within the family.

*72·1.

Dem.

*72·11.

Dem.

*72·12.

*72·121.

Dem.

*72·13.

*72·131.

*72·132.

*72·14.

This proposition applies to a great many of the relations we have to deal with, for example , , , , , , , etc.

*72·15.

[Pg 465]

In *72·16 below, has the meaning defined in *40·01, and does not represent a variable proposition. Similarly s in *72·161 has the meaning defined in *40·02.

*72·16.

Dem.

*72·161.

*72·162.

*72·163.

*72·17.

Dem.

*72·18.

*72·181.

*72·182.

Dem.

*72·184.

*72·185.

*72·19.

*72·191.

*72·192.

*72·193.

[Pg 466]

*72·2.

Dem.

*72·201.

*72·202.

*72·21.

Dem.

*72·211.

*72·22.

Dem.

*72·221.

*72·23.

Dem.

*72·24.

Dem.

*72·241.

[Pg 467]

*72·242.

Dem.

*72·243.

Dem.

The above proposition is used in *272·4·41, which are used in the theory of "rational series," i.e. series ordinally similar to the series of rationals.

*72·25.

Dem.

The propositions and , which have been previously proved, are particular cases of the above; the former is a particular case because .

*72·26.

[Pg 468]

In this proposition, the conditions of significance require that the domain of should consist of classes. This proposition is used in *72·27.

Dem.

*72·27.

*72·27 is used in *74·63·631 and again in *163·15.

*72·3.

Dem.

*72·301.

*72·302.

*72·303.

*72·31.

Dem.

*72·311.

*72·312.

*72*32.

Dem.

[Pg 469]

*72·321.

*72·322.

*72·323.

Dem.

*72·34.

Dem.

*72·341.

This proposition should be compared with *40·37 and *40·38.

*72·4.

Dem.

When is not a , we only have in general (cf. *37·21)

*72·401.

*72·41.

*72·411.

[Pg 470]

*72·42.

*72·421.

*72·43.

Dem.

*72·431.

*72·44.

*72·441.

*72·441 is used in the theory of cardinal exponentiation (*116·659).

*72·45.

Dem.

*72·451.

*72·46.

*72·461.

*72·47.

Dem.

*72·471.

[Pg 471]

*72·48.

Dem.

*72·481.

*72·49.

Dem.

*72·491.

*72·492.

*72·5.

Dem.

*72·501.

*72·502.

*72·503.

*72·504.

Note that means , not (. *72·504 is used in the theory of segments of a series (*211·64).

*72·51.

*72·511.

*72·512.

Dem.

*72·513.

*72·52.

[Pg 472]

*72·53.

Dem.

*72·54.

Dem.

*72·541.

*72·55.

Dem.

*72·551.

*72·57.

Dem.

[Pg 473]

*72·59.

Dem.

*72·591.

*72·6.

*72·601.

*72·61.

*72·611.

The following propositions lead up to the "principle of abstraction" (*72·66), which, though not explicitly referred to in the sequel, has a certain intrinsic interest, and generalizes a type of reasoning frequently employed by us.

*72·62.

Dem.

*72·621.

Dem.

*72·622.

*72·63.

Dem.

[Pg 474]

*72·64.

Dem.

*72·65.

*72·66.

*72·7.

Dem.

*72·71.

*72·72.

*72·8.

The above proposition is used in *72·62.

*72·81.

*72·9.

Dem.

[Pg 475]

*72·91.

Dem.

*72·911.

*72·92.

Dem.

*72·921.

*72·93.

Dem.

*72·931.

*72·94.

Dem.

[Pg 476]

*73. SIMILARITY OF CLASSES.

Summary of *73.

Two classes and are said to be similar when there is a one-one relation whose domain is and whose converse domain is . We express " is similar to " by the notation "." When two classes are similar, they have the same cardinal number of terms: it is this fact which gives importance to the relation of similarity.

We have

The relation of similarity is that of the domain of a to the converse domain, i.e. it is the relative product of and , or, what comes to the same thing, it is the relative product of and .

Most of the properties of similarity result immediately from those of one-one relations and offer no difficulty of any kind.

When there are relations which correlate 's with 's so as to make similar to , we denote the class of such relations by "." Thus we have

When, as in this case, we have a descriptive double function closely connected with a relation, we shall make it a practice to distinguish the descriptive double function by a bar.

It is to be observed that "," like and and and , is ambiguous as to type, and only acquires a definite meaning when the types of its domain and converse domain are specified. The domain and the converse domain may or may not be of the same type, i.e. "" may or may not be a homogeneous relation. This enables us to speak of two classes of different types as having the same number of terms. We shall return to this point in connection with cardinal numbers (cf. especially *102—*106).

The propositions of the present number are important, and are very frequently referred to throughout cardinal arithmetic. In order to prove that two classes and have the same cardinal number of terms, it is[Pg 477] generally necessary, in the fundamental arithmetical propositions with which we are concerned, actually to construct a relation such that . Such relation will be called a correlator of and . It will usually be obtained by taking some relation for which we have (, and limiting the converse domain to , so that is the required correlator. Very frequently we shall have , not , but will be such that .

Among the more important propositions of the present number are the following:

*73·142.

I.e. is a correlator of and if (1) is one-one, (2) is contained in the converse domain of , (3) is the class of those terms which have the relation to members of .

*73·2.

This results immediately from the definition.

*73·22.

*73·3.

*73·31.

*73·32.

The above three propositions show that similarity is reflexive, symmetrical and transitive.

*73·36.

*73·41.

Thus every class is similar to a class of higher type, and consisting wholly of unit classes.

*73·45.

Thus 1 is the class of all classes similar to any unit class.

*73·48.

Thus 0 is the class of all classes similar to the null-class.

*73·611.

This proposition is very often useful. For arithmetical purposes, we often wish to obtain mutually exclusive classes. Now whether or not and be mutually exclusive, and are mutually exclusive provided . Thus by means of the above proposition we can always construct mutually exclusive classes each similar to a given class, i.e. each having some assigned number of members.

*73·71.

This proposition is fundamental in the theory of addition.

[Pg 478]

*73·88.

I.e. "if is similar to a part of , and is similar to a part of , then is similar to ." This is the Schröder-Bernstein theorem. The proof given below is due to Zermelo.

*73·01.

*73·02.

*73·03.

*73·04.

*73·1.

*73·11.

Dem.

*73·12.

*73·13.

[Pg 479]

Dem.

*73·131.

*73·14.

Dem.

The use of this proposition in proving similarity is very frequent.

*73·141.

*73·142.

Dem.

*73·15.

Dem.

[Pg 480]

*73·2.

Dem.

The following propositions, down to *73·241, are deduced from preceding propositions of this number just as "" was deduced in *73·2 from *73·1. The proofs are therefore merely indicated by references to the previous propositions of this number which are used.

*73·21.

*73·22.

*73·23.

*73·231.

*73·24.

*73·241.

*73·25.

Dem.

This proposition will be convenient in such cases as the following: Let be a class of relations whose domains are mutually exclusive, i.e. such that no two members of have domains which have a member in common, and suppose we wish to prove that the class of these domains is similar to .[Pg 481] The class of domains is , and we have (. Hence we have only to prove (putting in place of the of *73·25)

which, in the case supposed, is proved immediately.

*73·26.

Dem.

*73·27.

*73·28.

Dem.

*73·3.

Dem.

This is the reflexive property of similarity. The conditions of significance require that should be a class of some type, but impose no restriction as to the type of class.

*73·301.

Dem.

*73·31.

This proposition shows that similarity is a symmetrical relation.

*73·311.

Dem.

[Pg 482]

*73·32.

This proposition shows that similarity is a transitive relation. Thus we have now proved that similarity is reflexive, symmetrical, and transitive.

*73·33.

*73·34.

Dem.

*73·35.

Dem.

*73·36.

Dem.

*73·37.

Dem.

*73·4.

[Pg 483]

*73·41.

This proposition is useful, because it gives a class (similar to but of higher type. Thus if is a cardinal number, and it is known that in a certain type there are classes having terms, it follows that there will be classes having terms in the next higher type, and therefore in the next type above that, and so on. No corresponding means exist for lowering the type.

*73·42.

Dem.

This proposition gives a means of lowering the type without altering the cardinal number, provided our class is composed wholly of unit classes; for is of the type next below the type of . But when is not composed wholly of unit classes, this construction fails.

*73·43.

*73·44.

Dem.

*73·45.

Dem.

*73·46.

[Pg 484]

*73·47.

Dem.

*73·48.

The following proposition is used in the theory of double similarity (*111·111).

*73·5.

Dem.

*73·501.

Dem.

[Pg 485]

*73·51.

Dem.

*73·511.

*73·52.

Dem.

*73·521.

*73·53.

*73·531.

*73·61.

*73·611.

*73·62.

*73·621.

*73·63.

Dem.

[Pg 486]

The above proposition is used once in connection with cardinal addition (*112·231), and once in connection with cardinal multiplication (*114·561).

The following proposition (*73·69) is a lemma for *73·7.

*73·69.

Dem.

*73·7.

*73·701.

Dem.

*73·71.

*73·72.

Dem.

The following propositions give the proof of the Schröder-Bernstein theorem, namely: If one class is similar to part of another, and the other is similar to part of the one, then the two classes are similar. The proof here given is due to Zermelo[60]. An explanation of the following proof is given in connection with another proof in the summary of *94.

*73·8.

Dem.

*73·801.

Here "Hp*73·8" means "the hypothesis of *73·8."

Dem.

*73·802.

Dem.

*73·81.

Dem.

*73·811.

Dem.

[Pg 488]

*73·812.

Dem.

*73·82.

Dem.

*73·821.

Dem.

*73·83.

Dem.

*73·84.

Dem.

*73·841.

[Pg 489]

Dem.

*73·85.

*73·86.

Dem.

*73·87.

Dem.

*73·88.

Dem.

This is the Schröder-Bernstein theorem.

FOOTNOTES:

[60] Math. Annalen, vol. LXV. Heft 2, February 1908.

[Pg 490]

*74. ON ONE-MANY AND MANY-ONE RELATIONS WITH LIMITED FIELDS.

Summary of *74.

The purpose of the present number is to collect together various propositions in which we have such hypotheses as

or in which such hypotheses are shown to be deducible from others. Hypotheses of this kind occur very frequently, and it is important to be able to deal with them easily. For the sake of completeness, we shall here repeat propositions previously proved on this subject.

The propositions of this number are mostly of the nature of lemmas, to be used in the theory of selections (Part II, Section D), and in cardinal and ordinal arithmetic. The most useful of them are *74·772·773·774·775. These propositions are concerned with circumstances under which or , with or without some limitation of the converse domain, is a one-one relation. The reason they are important is that the correlators by means of which many of the fundamental theorems of cardinal and ordinal arithmetic are proved are such relations as (with the converse domain limited) for suitable values of and . The above-mentioned propositions are as follows:

*74·772.

The hypothesis of this proposition will be verified if we put, for example, . Thus . This proposition is used in *116·531, which is used in proving one of the formal laws of exponentiation, namely .

*74·773.

This proposition is used in connection with both cardinal and ordinal multiplication and exponentiation. If and correlate with and with , then if we take for the class of all ordinal couples that can be formed of an and a , (will be the class of all couples[Pg 491] that can be formed of a and a . Thus in virtue of the above proposition, if is similar to and is similar to , the class of ordinal couples formed of a and a is similar to the class of ordinal couples formed of an and a . This result is useful because we define the product of the number of members of and the number of members of as the number of ordinal couples formed of an and a .

*74·774.

This proposition is useful when, for example, is .

*74·775.

This is a particular case of *74·773, and has similar uses.

*74·1.

Dem.

*74·11.

*74·12.

*74·13.

*74·131.

*74·14.

*74·141.

*74.15.

*74·151.

*74·16.

*74·161.

*74·17.

*74·171.

*74·2.

Dem.

[Pg 492]

*74·201.

*74·21.

*74·211.

*74·22.

*74·221.

*74·23.

*74·231.

*74·24.

*74·25.

*74·251.

*74·26.

Dem.

*74·27.

Dem.

[Pg 493]

*74·271.

*74·3.

Dem.

*74·301.

*74·31.

Dem.

*74·311.

*74·32.

Dem.

*74·4.

Dem.

[Pg 494]

*74·41.

Dem.

*74·42.

*74·43.

*74·44.

*74·5.

Dem.

*74·51.

Dem.

*74·511.

*74·52.

Dem.

[Pg 495]

*74·521.

*74·53.

Dem.

*74·531.

*74·6.

Dem.

*74·61.

Dem.

*74·62.

Dem.

[Pg 496]

*74·63.

*74·631.

*74·632.

*74·7.

Dem.

*74·701.

*74·71.

*74·711.

*74·72.

Dem.

*74·721.

*74·73.

*74·731.

*74·74.

*74·741.

*74·75.

Dem.

*74·751.

[Pg 497]

*74·76.

*74·761.

*74·77.

Dem.

*74·771.

*74·772 and its immediate successors are of very great use in cardinal and ordinal arithmetic.

*74·772.

*74·773.

Dem.

*74·774.

Dem.

[Pg 498]

*74·775.

*74·8.

Dem.

*74·801.

*74·81.

Dem.

*74·811.

*74·82.

Dem.

*74·821.

[Pg 499]

*74·822.

*74·823.

*74·83.

*74·831.

*74·832.

*74·833.

*74·84.

Dem.

*74·841.

*74·842.

*74·843.

[Pg 500]

SECTION D.

SELECTIONS.

Summary of Section D.

The subject to be considered in this section is important chiefly in connection with multiplication, both cardinal and ordinal. In order to get a definition of multiplication which is not confined to the case where the number of factors is finite, we have to seek a construction by which, from a given class of classes, say, we construct another class which, when is finite, has that number of terms which, in the usual elementary sense, is the product of the numbers of terms in the various classes which are members of , and which, whether is finite or not, obeys as many as possible of the formal laws of multiplication. The usual elementary sense of multiplication is derived from addition; that is to say, is to be the number of terms in , where is a class of mutually exclusive classes each having members, or vice versa. This sense can be extended to any finite number of factors, but not to an infinite number of factors; hence for a number of factors which may be infinite we require a different definition, and this is derived from the theory of selections.

Selections are of two kinds, selections from classes of classes, and selections from relations. The latter is the more general notion, from which the former is derived. But as the former is an easier notion, we will begin by explaining selections from classes of classes.

Given a class of classes , a class is called a selected class of when is formed by choosing one term out of each member of . For example, if consists of two members, and , and if and , then is a selected class of . If every constituency elects a local man, Parliament is a selected class of the constituencies. If is a class of mutually exclusive classes, i.e. a class no two of whose members have any member in common, then a selected class consists of only one term from each member of ; i.e. is a selected class if

But if is not a class of mutually exclusive classes, this does not hold necessarily; for a term which is a member of both and (where) may be chosen as the representative of , while some other term may be[Pg 501] chosen as the representative of , so that two members of may belong to the selected class. Again, if is a class of mutually exclusive classes, the relation of the representative to its class must be one-one, because, since no term belongs to two classes which are members of , no term can be the representative of two classes. But when is not a class of mutually exclusive classes, a term which belongs to two classes and may be chosen as the representative

of both. Thus the relation of the representative to its class may be only one-many, not one-one.

The relation of the representative to its class may be called a selective relation. A selective relation of is one which selects, from every class which is a member of , a certain member as the representative of ; that is, we have, if is the selective relation,

This condition is equivalent to

If is a selective relation, is a selected class; and if is a selected class, there is a selective relation such that . Thus the study of selections from classes of classes is wholly contained in the study of selective relations.

The class of selective relations from a class is called . Thus

Then is the class of selected classes.

It will be seen that, if , may be any member of , and we get a different for each different member of . Thus if we keep the representatives of all the other members of unchanged, the number of selective relations to be obtained by varying the representative of is the number of members of . Hence the number of selective relations altogether may be fitly defined as the product of the numbers of terms possessed by the various members of . In case is finite, this agrees with the usual definition of multiplication; and whether is finite or infinite, the product so defined obeys all the formal laws of multiplication.

To illustrate the notion of selective relations, let us take a very simple case, the case where consists of two classes and , each of which has two members. Let and be the members of , and the members of . We assume , , . Then the selective relations of are the following: [Pg 502] Thus they are four in number, i.e. the number of members of is the product of the number of members of and the number of members of . A similar process would show that our definition of the product agrees with the usual definition in any case in which all the numbers concerned are finite.

Selections from relations are an obvious generalization of selections from classes of classes. We had above

We put, generally,

which we derive from the definition

This is the fundamental definition in the subject of selections. We have, in virtue of this definition,

When , we may call the class of selections from . Thus generally, is the class of selections from provided ; and if this condition is not fulfilled, . We may call the class the class of "-selections from ." The class of "-selections from " will be what we previously called the class of "selective relations of ." It will be observed that we have

Thus if is a class of mutually exclusive classes, selects one from each of these classes, and is therefore a selective class of ; hence in this case

In Cardinal Arithmetic, is the important notion, and the more general notion is seldom required. In Ordinal Arithmetic, is the important notion. It will be seen that

Thus is only significant when is a class of relations; in this case we have

[Pg 503]Thus chooses a representative member of the field of every member of . The most important case is when is of the form , where is a serial relation whose field consists of serial relations. Then becomes the field of a relation which may be defined as the ordinal product of the relations composing ; in this way we get an infinite ordinal product analogous to the infinite cardinal product. This will be explained at a later stage (*172).

Although it is chiefly and that will be required in the sequel, we shall treat generally, because this introduces little extra complication, and most of the theorems which hold for or have exact analogues for .

, as above defined, is the class of one-many relations contained in and having for their converse domain. We know of no proof that there always are such relations when . In fact, the proposition

is equivalent to the "multiplicative axiom," i.e. to the axiom that, given any class of mutually exclusive classes, none of which is null, there is at least one class formed of one member from each of these classes. (This equivalence is proved in *88·36, below.) It is also equivalent to Zermelo's axiom[61], which is

hence also it is equivalent to the proposition that every class can be well-ordered. In the absence of evidence as to the truth or falsehood of these various propositions, we shall not assume their truth, but shall explicitly introduce them as hypotheses wherever they are relevant.

In the present section, we shall begin (*80) by considering such properties of as do not depend upon any hypothesis as to . We shall then (*81) proceed to consider such further properties of as result from the hypothesis . This hypothesis is important, because it is verified in many of the applications we wish to make, and because it leads to important properties of which are not true in general when is not subject to any hypothesis. These special properties are mostly due to the fact that when is a many-one relation, consists of one-one relations (not merely of one-many relations, as it does in the general case). This is proved in *81·1. We then (*82) proceed to consider the case of relative products, i.e. (. It will appear that, with a suitable hypothesis, (and . In the following number (*83) we apply the results of *80 to the particular case where is replaced by , which is the important case for cardinal arithmetic. In *84 we apply the propositions of *81 to the case where is replaced by , and where, therefore, we have the hypothesis . This hypothesis is equivalent to the hypothesis that no two members of have any members in common, i.e. that [Pg 504] When fulfils this hypothesis, it is a class of mutually exclusive classes. For classes of mutually exclusive classes we adopt the notation "." It is shown in *84·14 that a is one for which we have . When is a , is a one-one relation, and . Also in this case consists of all classes formed of one member from each member of , i.e. all classes such that

In *85, we prove various important propositions, of which the chief is a form of the associative law[62], namely

Finally, in *88, we consider the question of the existence of selections. This cannot in general be proved when is an infinite class. The assumption that is never null unless one member of is null is equivalent to various other assumptions, for example to the assumption that every class can be well-ordered. One of these equivalent assumptions is called the "multiplicative axiom." This axiom is equivalent to the assumption that an arithmetical product cannot be zero unless one of its factors is zero, and is regarded by some mathematicians as a self-evident truth. This can be proved when the number of factors is finite, i.e. when is a finite class, but not when the number of factors is infinite. We have not assumed its truth in the general case where it cannot be proved, but have included it in the hypotheses of all propositions which depend upon it.

FOOTNOTES:

[61] See his "Beweis, dass jede Menge wohlgeordnet werden kann," Math. Annalen, Vol. LIX. pp. 514-516.

[62] Cf. notes to *42·1·11.

[Pg 505]

*80. ELEMENTARY PROPERTIES OF SELECTIONS.

Summary of *80.

In this number, we shall give such properties of as follow most directly from the definition, without any restrictive hypothesis as to .

If , selects one member of , whenever , as the selected referent of . For, since , we have ; and since , we have , i.e. . Calling the selected referent of , it is evident that we may replace by any other member of , and still have a member of . (This is proved in *80·4.) Thus if has any members at all, we can get as many members as there are members of by merely altering the selected referent of , leaving the other selected referents unchanged.

In the present section, we first prove various simple properties of . Most of these are almost immediate consequences of

*80·14.

The most useful of them are

*80·2.

*80·291.

*80·3.

*80·33.

We then have various propositions (*80·4—·46) concerned with when . Of these the most important are the following:

*80·41.

I.e. given a selective relation , the selected referent of (where) may be replaced by any other term having the relation to , and we shall still have a selective relation.

*80·45.

We then have a set of propositions (*80·5—·54) connecting (with and . These are chiefly useful as leading to the next set

[Pg 506]

(*80·6—·69), connecting with and . The most useful of these are the following:

*80·6.

*80·65.

*80·66.

We have next a set of propositions (*80·7—·78) dealing with the relations of and when (e.g.) and . These propositions are seldom used, but they would be useful in considering division.

We next have a set of propositions (*80·8—·84) dealing with the relations of and . The most useful are

*80·81.

*80·82.

Finally, we have four propositions (*80·9—·93) on and one on . The most useful of these is

*80·9.

*80·01.

*80·1.

*80·11.

*80·12.

*80·13.

*80·14.

*80·15.

*80·16.

Dem.

*80·17.

Dem.

[Pg 507]

This proposition is used in the theory of ordinal multiplication (*172·162).

*80·2.

Dem.

*80·21.

*80·22.

Dem.

*80·23.

Dem.

*80·24.

*80·25.

*80·26.

Dem.

Note that is a unit class, not the null-class. It is owing to this fact (as will appear later) that, if is any cardinal, . See the note to *83·15.

[Pg 508]

*80·27.

Dem.

*80·28.

Dem.

*80·29.

Dem.

*80·291.

Dem.

*80·3.

Dem.

*80·31.

Dem.

*80·32.

Dem.

[Pg 509]

*80·33.

Dem.

*80·34.

Dem.

*80·35.

*80·36.

Dem.

This proposition is used in dealing with greater and less among cardinals (*117·68).

*80·4.

This proposition is important. It shows that, if and is the selected referent of (i.e. is , then may be replaced by any other member of without our ceasing to have a member of .

Dem.

[Pg 510]

*80·41.

Dem.

*80·42.

Dem.

*80·43.

Dem.

*80·44.

Dem.

*80·45.

Dem.

[Pg 511]

*80·46.

*80·5.

Dem.

*80·51.

Dem.

*80·511.

Dem.

*80·52.

Dem.

[Pg 512]

*80·53.

Dem.

*80·54.

Dem.

*80·6.

Dem.

*80·61.

Dem.

[Pg 513]

*80·62.

*80·621.

Dem.

*80·63.

*80·64.

Dem.

*80·65.

*80·651.

Dem.

*80·66.

Dem.

*80·661.

Dem.

[Pg 514]

*80·67.

Dem.

*80·68.

Dem.

*80·69.

Dem.

*80·7.

Dem.

*80·71.

Dem.

[Pg 515]

*80·72.

*80·73.

Dem.

*80·731.

Dem.

*80·732.

Dem.

*80·74.

Dem.

*80·75.

*80·76.

Dem.

[Pg 516]

*80·761.

Dem.

*80·77.

Dem.

*80·771.

Dem.

*80·78.

Dem.

*80·8.

Dem.

*80·81.

Dem.

*80·82.

Dem.

[Pg 517]

The following proposition is used in *80·84 and in the theory of double similarity (*111·3).

*80·83.

Dem.

*80·84.

Dem.

The three following propositions are useful both in cardinal and in ordinal multiplication (*113 and *172).

*80·9.

Dem.

*80·91.

Dem.

[Pg 518]

80·9·91 can be extended, by precisely similar proofs, to any finite number of variables , , ... They will, on occasion, be assumed for three or four variables, without fresh proofs.

*80·92.

Dem.

*80·93.

*80·94.

From this proposition, together with *80·26 (which gives , we shall obtain an inductive proof that exists whenever is a finite class contained in (cf. *120·611).

[Pg 519]

*81. SELECTIONS FROM MANY-ONE RELATIONS.

Summary of *81.

When is a many-one relation, has many important properties which do not hold in the general case. In the first place, consists wholly of one-one relations. In the second place, if , takes one term and no more out of each member of . Again, if , is determinate when is given; i.e. . It follows that is similar to ; hence the number of members of is the number of ways of choosing one member out of each class belonging to . It should be remembered that when is many-one, is a class of mutually exclusive classes, i.e. no two different members of have any common member. This follows immediately from *71·181.

As explained in the introduction to this section, the propositions of this number are chiefly useful on account of their application to the case of . This application is made in *84. The most important propositions in this number are:

*81·1.

*81·14.

This proposition, by exhibiting as a function of , leads immediately to

*81·21.

This is the principal proposition of this number. The following also is important:

*81·22.

*81·1.

Dem.

[Pg 520]

*81·11.

Dem.

*81·12.

Dem.

*81·13.

Dem.

*81·14.

This proposition, by exhibiting as a function of , shows that a member of is determinate when its domain is given, provided .

*81·15.

Dem.

[Pg 521]

*81·2.

Dem.

*81·21.

This proposition is very important. The class , when , is formed, as we shall prove later, by making every possible selection of one term out of each member of , each such selection giving us one member of . The fact that, with the above hypothesis, the class of classes has the same number of terms as (which results from the above proposition), is of great utility in the theory of cardinal multiplication and exponentiation.

*81·211.

Dem.

*81·212.

[Pg 522]

Dem.

*81·22.

*81·221.

Dem.

*81·23.

Dem.

*81·24.

Dem.

[Pg 523]

*81·25.

Dem.

*81·26.

Dem.

*81·3.

Dem.

*81·31.

Dem.

[Pg 524]

*82. SELECTIONS FROM RELATIVE PRODUCTS.

Summary of *82.

The propositions contained in this number are not much used except in connection with the associative law for cardinal multiplication, but they have a certain intrinsic interest. We prove in this number that, with a suitable hypothesis, (results from by multiplying each member by , i.e.

*82·272.

Also under a suitable hypothesis the domains of (are the domains of , i.e.

*82·32.

In the applications of propositions of the present number in *85, and are replaced by and . By *62·26, ; thus we obtain relations between and .

*82·2.

Dem.

[Pg 525]

*82·21.

Dem.

*82·22.

Dem.

*82·221.

Dem.

*82·23.

Dem.

[Pg 526]

*82·231.

Dem.

*82·24.

Dem.

*82·241.

Dem.

*82·25.

*82·251.

[Pg 527]

*82·26.

*82·261.

*82·27.

*82·271.

*82·272.

Dem.

*82·28.

*82·29.

*82·291.

*82·3.

Dem.

*82·31.

Dem.

[Pg 528]

*82·32.

Dem.

*82·33.

Dem.

The following propositions (*82·4·41·411·42) are lemmas for *83·43, which is used in the proof of *114·5, in the theory of cardinal multiplication.

*82·4.

Dem.

*82·41.

Dem.

[Pg 529]

*82·411.

*82·42.

*82·43.

Dem.

*82·45.

Dem.

*82·5.

*82·51.

*82·52.

Dem.

[Pg 530]

*82·53.

Dem.

[Pg 531]

*83. SELECTIONS FROM CLASSES OF CLASSES.

Summary of *83.

In this number, the general propositions which have been proved for are to be applied to the important special case where is . In this case, we have selections from classes of classes: if , picks out a representative from each class which is a member of ; i.e. we have

The propositions of this number result from those of previous numbers either immediately, by the substitution of for , or by the use of propositions of *62, notably (*62·2), and (*62·3).

The propositions of the present number follow, in the main, the same course as those of *80, with substituted for (except that the special forms of propositions before *80·2 are not given). We have first a set of propositions resulting immediately from early propositions of *80. Of these the most used are:

*83·11.

This leads to the proposition that an arithmetical product is null if one of its factors is null. (We cannot prove the converse universally without assuming the multiplicative axiom.)

*83·15.

Thus is a unit class. This is the source of the proposition , where is a cardinal (cf. note to *83·15).

*83·2.

Here is the "representative" of .

*83·21.

We have next a set of propositions (*83·4—·44) on selections from unit classes and classes of unit classes. We have

[Pg 532]

*83·41.

This leads to the proposition that a product of one factor is equal to that factor.

*83·43.

This leads to

*83·44.

whence it follows that a product of factors, each of which is one, is one. This holds even if the number of factors is infinite or zero.

We have next a set of propositions (*83·5—*·58) on changing the representative of a class, and on selections from a class of classes some of which are unit classes. These propositions are seldom referred to in the sequel.

We have next (*83·6—·74) a set of propositions on the domains of selections, i.e. on the class . We have

*83·66.

(The hypothesis here cannot be dispensed with unless we assume the multiplicative axiom.)

*83·7.

*83·71.

We have next two propositions (*83·8·81) on the types of and . The type of is the same as that of (*83·81).

The last set of propositions in this number (*83·9—·904) deals with the existence of selections. We have

*83·9.

*83·901.

*83·904.

From these propositions we shall deduce by mathematical induction that whenever is a finite class, exists unless (cf. *120·62). Thus a product consisting of a finite number of factors (which may themselves be either finite or infinite) can only vanish if one of the factors vanishes.

*83·1.

Dem.

*83·11.

[Pg 533]

*83·12.

*83·13.

*83·14.

*83·15.

In virtue of this proposition, the product of 0 cardinal numbers is 1—a proposition of which a particular case, namely , is familiar. This arithmetical proposition results from the above as follows. We shall define the product of the numbers of members of as the number of members of . Thus when , the number of members of is a product of 0 factors. Now by the above proposition, has one member, namely . Hence a product of 0 factors is 1.

*83·16.

*83·2.

*83·21.

*83·22.

*83·23.

*83·24.

*83·25.

*83·26.

*83·27.

*83·271.

*83·28.

*83·29.

*83·3.

*83·31.

*83·4.

*83·41.

[Pg 534]

This proposition shows that a cardinal product of one factor is equal to that one factor. For the number of members of is the product of the numbers of members of members of , i.e. it is a product whose only factor is the number of members of . By the above proposition, this product is equal to the number of members of .

*83·42.

Dem.

This proposition shows that a cardinal product whose factors are all 1 is 1. For is a class whose members are all unit classes, and thus the number of members of is the product of a number of 1's; and by the above proposition, is a unit class, its sole member being . This result is rendered more explicit by *83·43·44.

*83·43.

Dem.

*83·44.

*83·5.

Dem.

It follows from this proposition that if is a class of classes for which there are selections, and if one member (not null) be added to , there are still selections from the resulting class of classes.

*83·51.

[Pg 535]

*83·52.

*83·54.

Dem.

*83·55.

Dem.

*83·56.

Dem.

The following proposition is used in the theory of cardinal multiplication (*114·41).

*83·57.

Dem.

[Pg 536]

*83·58.

Dem.

This proposition shows that in a product any number of factors each equal to 1 may be omitted without altering the value of the product.

The following propositions, down to *83·74, are concerned with the domains of selective relations, i.e. with the selected classes.

*83·6.

Dem.

*83·61.

Dem.

*83·62.

*83·63.

Dem.

[Pg 537]

*83·64.

Observe that the hypothesis required here is , not as in *83·63.

Dem.

The following proposition is used in connection with cardinal multiplication (*115·14).

*83·641.

Dem.

[Pg 538]

*83·65.

Dem.

*83·66.

Dem.

*83·7.

*83·71.

Dem.

*83·72.

Dem.

*83·73·731 are lemmas for *83·74.

*83·73.

Dem.

[Pg 539]

*83·731.

Dem.

*83·74.

Dem.

*83·8.

Dem.

*83·81.

Dem.

*83·9.

*83·901.

*83·902.

*83·903.

*83·904.

*83·9·904 leads to an inductive proof (to be given later) of whenever is a finite class of classes none of which is .

[Pg 540]

*84. CLASSES OF MUTUALLY EXCLUSIVE CLASSES.

Summary of *84.

A class of mutually exclusive classes is one such that, if and are two different members of , and have no common members; i.e. it is a class composed of non-overlapping classes. Classes of mutually exclusive classes have many important properties. They are important in cardinal arithmetic, among other reasons, because if is a class of mutually exclusive classes, the cardinal number of is the sum of the cardinal numbers of the members of . Also if is a class of mutually exclusive classes, the number of selected classes of (i.e.) is the same as the number of selective relations (i.e.)

" is a class of mutually exclusive classes" is written "."

An important case is when no member of is null; in this case we write

For a which is contained in a class of classes , we write

on the analogy of the notation .

The definitions are as follows:

*84·01.

*84·02.

*84·03.

The propositions of this number begin (*84·1—·14) with various equivalent forms for the definitions. Of these the most useful are:

*84·11.

*84·13.

*84·14.

The last of these is specially important, because it renders the propositions of *81 applicable to when [Pg 541].

We have next (*84·2—·28) a set of propositions dealing with various special cases, such as and 1. The most useful of these are

*84·23.

*84·241.

*84·25.

We next have a set of propositions (*84·3—·37) which are immediate consequences of propositions in *81, by means of *84·14. The most useful of these is

*84·3.

We next have a set of propositions (*84·4—·43) dealing with the domains of selections from a . These are for the most part still immediate consequences of propositions in *81, in virtue of *84·14. The most useful are

*84·41.

*84·412.

*84·43.

This proposition applies to such cases as the relations of rows and columns. Imagine any set of terms arranged in rows and columns so as to form a rectangle. Then each column is a selection from the rows, and each row is a selection from the columns. This is a particular case of the above proposition.

We next have a set of propositions on , , and (*84·5—·55). The most important of these are

*84·51.

*84·53.

Finally we have a set of propositions (*84·59—·62) showing circumstances under which is a . The only one of these which is used subsequently is

*84·62.

*84·01.

*84·02.

*84·03.

*84·1.

[Pg 542]

*84·11.

*84·12.

*84·121.

*84·13.

Dem.

*84·131.

*84·132.

*84·133.

*84·134.

Dem.

*84·135.

Dem.

[Pg 543]

*84·14.

Dem.

This proposition is important, since it enables us to apply the propositions of *81 to when .

*84·2.

Dem.

*84·21.

Note. is the class of all unit classes whose members are classes; this results from *65·02. Thus "" is equivalent to " consists of one class."

Dem.

*84·22.

Dem.

*84·23.

*84·24.

Dem.

*84·241.

Dem.

[Pg 544]

*84·242.

*84·25.

Dem.

*84·26.

Dem.

*84·28.

Dem.

The following propositions are concerned with selections from a . In virtue of *84·14, the propositions of *81 which have the hypothesis become applicable when is and is a . Thus has many important properties when is a which it does not have in the general case.

*84·3.

*84·31.

*84·32.

*84·33.

*84·34.

*84·341.

[Pg 545]

*84·342.

*84·35.

Dem.

*84·37.

*84·4.

*84·41.

This is an important proposition, since it shows that, when is a , the number of classes that can be selected from is the product of the numbers of the various classes that are members of .

*84·411.

*84·412.

This proposition gives what might be taken as the definition of the class of selected classes, namely

We might, starting with this as our definition, deal with the class of selected classes without first considering selective relations. The disadvantages of this method would be, first, that it requires that should be a if it is to give the results desired in arithmetic; secondly, that it is much more cumbrous technically than the method which proceeds by selective relations; thirdly, that it does not enable us to deal with selection from a class of classes as a particular case of selection from a relation (namely from , and therefore does not yield theorems of such generality as those obtained by the method adopted above.

*84·42.

*84·421.

[Pg 546]

*84·422.

*84·43.

Dem.

*84·5.

Dem.

It might be supposed that the converse of the above would also hold. But this is not the case; for although

secures that and cannot overlap when they are unequal, yet we may have without having , so that if , we shall have , whence, if , it follows that is not a even if .

*84·51.

Dem.

[Pg 547]

*84·52.

Dem.

*84·521.

Dem.

The above proposition is a lemma for *84·522, which is used in an important proposition on relations of mutually exclusive relations (*163·17).

*84·522.

Dem.

*84·53.

Dem.

[Pg 548]

*84·54.

*84·55.

*84·59.

Dem.

*84·6.

*84·61.

*84·62.

[Pg 549]

*85. MISCELLANEOUS PROPOSITIONS.

Summary of *85.

In this number certain important propositions are proved, and the other propositions of this number are mainly lemmas. The most important propositions are the following:

*85·1 and *85·14, which show that if is a , then the domains of are the same as the domains of , and is similar to , thus reducing the problem of selections from many-one relations to that of selections from classes of classes.

*85·27 and *85·43, which show that if , consists of the relational sums of the domains of and is similar to ; i.e. the class of -selections from is similar to the class obtained as follows: take the members of one by one, and form the -selections of each; we thus obtain a class of classes, each class being of the form , where ; we then make a selection from this class of classes; this selection is a member of ; the number of such selections is the same as the number of .

*85·28 and *85·44, which are special cases of *85·27 and *85·43, but more useful than these. *85·44 is the source of the associative law in cardinal multiplication; it states that, if is a , has the same number of members as . (On associative laws in general, see the notes to *42·1·11.) That is to say, if we form the class of selective relations (for every which is a member of , and then form the class of selective relations for , we get the same number of terms as if we proceeded to form the class of selective relations for . The way in which this proposition yields the associative law of multiplication may be explained as follows. We shall define the product of the numbers of members of as the number of . Thus e.g. if the numbers of the members of are , the number of is . Suppose the other members of are and , and that and again have three members each. Then the number of is the product of the numbers of , , , i.e. it is the product of , and [Pg 550].

But the numbers of the members of are

Thus the number of is

Hence *85·44 enables us to conclude that

which is a case of the associative law. In fact *85·44 gives us this law in its general form, when the number of brackets, and of factors in each bracket, may be infinite or finite indifferently.

Another important pair of propositions is *85·53·54. These enable us to reduce the problem of selections for any relation to the problem of selections from a class of classes. The method is as follows: Given any term , form the class of ordered couples of which is relatum while the referent is a term having the relation to . Call this class of couples . Form this class for every which is a member of ; we thus obtain a class of classes, namely . Then the number of selections from this class of classes is the same as the number of .

We have one other important pair of propositions in this number, namely *85·61·63. These show that what is called "Zermelo's axiom" is equivalent to what is called the "multiplicative axiom." Zermelo's axiom[63] is to the effect that if is any class, is never null, i.e. . The "multiplicative axiom" is to the effect that if , there is at least one class formed by taking one representative from each member of , which is equivalent to

In *85·63, these two axioms are shown to be equivalent. From Zermelo's theorem[64] it follows that both are equivalent to the assumption that every class can be well-ordered. This will be proved later (*258).

The above-mentioned propositions, stated symbolically, are as follows:

*85·1.

*85·14.

*85·27.

*85·28.

[Pg 551]

*85·43.

*85·44.

The following propositions depend upon the definition

*85·5.

I.e. is the class of all couples whose relatum is y while the referent has the relation to . We then have

*85·53.

giving a construction for by means of , and

*85·54.

which reduces the question of the existence of -selections to that of the existence of -selections.

*85·61.

This proposition gives a construction for any -selection in terms of an -selection from a , and reduces the question of the existence of the former to that of the existence of the latter. A particularly important case is when . This is considered in

*85·63.

*85·1.

Dem.

*85·11.

Dem.

*85·111.

*85·112.

*85·12.

Dem.

[Pg 552]

This proposition is used in connection with ordinal multiplication (*173·14).

*85·13.

Dem.

In the above proposition, the hypothesis required as to by *82·231 is only ; but since , .

The above proposition is used in connection with "families" (*97·31).

*85·14.

Dem.

*85·21·22 are lemmas for *85·24, which, with *85·26, is required for *85·27.

*85·21.

*85·22.

Here can also be written . The brackets are omitted because no other meaning is possible.

Dem.

[Pg 553]

*85·24.

Dem.

The following propositions are lemmas for *85·26.

*85·241.

Dem.

*85·243.

Dem.

*85·244.

Dem.

*85·245.

Dem.

*85·25.

[Pg 554]

*85·26.

Dem.

*85·27.

*85·28.

The following proposition is a lemma for *85·31.

*85·3.

The conditions of significance here and in *85·31·32·33·34 require .

Dem.

The following propositions, down to *85·42 inclusive, deal with circumstances under which we can infer from . *85·32·33·34 are not subsequently used; the remainder are used in proving *85·43.

*85·31.

Dem.

[Pg 555]

*85·32.

Dem.

*85·33.

The proof proceeds exactly as in *85·32.

*85·34.

The following propositions, *85·4·41·42, are lemmas for *85·43·44, which latter are of fundamental importance, since they are the source of the associative law in cardinal arithmetic.

*85·4.

*85·41.

Dem.

*85·42.

Dem.

[Pg 556]

*85·43.

Dem.

*85·44.

The following proposition is used in connection with cardinal multiplication (*114·301).

*85·45.

Dem.

The purpose of the following propositions, down to *85·55, is to show how to get from a class of classes a class of selections having the same number of terms as . For this purpose we introduce a new notation, representing a rather important analysis of the couples contained in a given relation. A couple is contained in a relation when ; thus if, keeping fixed, we form the class of couples , all these couples are contained in . We put

*85·5.

[Pg 557]

Then . Also is the class of all couples contained in , and . We shall now prove that , so that every member of can be derived from a member of , and the problem of the existence of is reduced to that of the existence of selections from a class of mutually exclusive existent classes.

*85·51.

*85·52.

*85·53.

Dem.

*85·54.

Dem.

The following proposition is frequently useful.

*85·55.

Dem.

*85·56.

*85·6.

Dem.

The following proposition is frequently employed.

[Pg 558]

*85·601.

Dem.

*85·61.

*85·62.

*85·63.

Dem.

Note. (is "Zermelo's axiom." The above proposition shows that this is true if [Pg 559] which again is true if

in virtue of *84·412. The last of these is the "multiplicative axiom," which is thus shown to imply "Zermelo's axiom."

The following propositions lead up to *85·72, which is used in the theory of double similarity (*111·3).

*85·7.

Dem.

*85·701.

*85·702.

*85·71.

This proposition asserts that if we can select one sub-class out of each member of (where is a class of classes), then selections from the sub-classes so obtained are selections from .

*85·72.

Dem.

[Pg 560]

The following proposition is a lemma employed in the theory of double similarity (*111·313).

*85·81.

Dem.

FOOTNOTES:

[63] See Math. Annalen, Vol. LIX.

[64] loc. cit.

[Pg 561]

*88. CONDITIONS FOR THE EXISTENCE OF SELECTIONS.

Summary of *88.

The existence of selections cannot, so far as is known at present, be proved in general. That is, we cannot prove any of the following:

These various propositions can be shown to be all equivalent inter se; and in virtue of Zermelo's theorem (cf. *258), they are equivalent to the proposition "every class can be well-ordered." In the present number we have to prove the above equivalences, as well as certain propositions giving the existence of selections in various particular cases.

The most apparently obvious of the above propositions is the last, namely: "If is a class of mutually exclusive classes, no one of which is null, there is at least one class which takes one and only one member from each member of ." This we shall define as the "multiplicative axiom."

We will call a multipliable relation (denoted by "Rel Mult") if exists, or, what is equivalent, if

Thus we put

We will call a multipliable class of classes if exists, i.e. we put

The multiplicative axiom will be denoted by "." Thus we put

In the present number, we shall first give various equivalent forms of the assumption that is a multipliable relation (*88·1—·15); we shall then do the same for multipliable classes of classes (*88·2—·26); next we shall give various equivalent forms of the multiplicative axiom (*88·3—·39).[Pg 562] (Some important equivalent forms cannot be given at this stage, as they depend upon definitions not yet given, such as the definitions of cardinal multiplication and of well-ordered series. Cf. *114·26 and *258·37.) Finally we shall give propositions showing that various special classes of classes are multipliable. Most of these propositions will not be used in the sequel, but they illustrate the nature of the difficulties involved in proving that a class of classes is multipliable, and some of them show that mere size does not prevent a class from being multipliable. For example, *88·48 shows that, given any class of classes , if each member is replaced by , the result is a multipliable class of classes; but the only effect of this change is to increase the number of members of each member of our class of classes by one.

The chief propositions in this number which are afterwards referred to are the following:

*88·22.

*88·32.

*88·33.

*88·361.

*88·37.

The above is usually the most convenient form of the multiplicative axiom.

*88·372.

This proposition is used in *114, to prove that the multiplicative axiom is equivalent to the proposition that a cardinal product vanishes when, and only when, one of its factors vanishes.

*88·01.

*88·02.

*88·03.

*88·1.

*88·11.

Dem.

[Pg 563]

*88·12

Dem.

*88·13

*88·14

Dem.

*88·15.

*88·2.

*88·21.

*88·22.

Dem.

*88·23

*88·24

Dem.

*88·25

Dem.

[Pg 564]

*88·26.

Dem.

*88·3.

*88·31.

Dem.

*88·32.

*88·33.

Note that (is Zermelo's axiom.

Dem.

[Pg 565]

*88·34.

Dem.

*88·35.

Dem.

*88·36.

*88·361.

*88·37.

Dem.

[Pg 566]

*88·371.

*88·372.

This proposition shows that the multiplicative axiom is equivalent to the assumption that a cardinal product is zero when, and only when, one of its factors is zero.

*88·373.

Dem.

*88·38.

*88·39.

Dem.

The following propositions are concerned with certain cases in which a construction exists by which the existence of selections can be proved.

*88·4.

Dem.

[Pg 567]

*88·41.

*88·411.

Dem.

*88·42.

In virtue of this proposition, as will be proved later, every finite class of existent classes is a . For we have ; and, by the above, a remains a when one existent class is added as an additional member; hence the result follows by induction.

*88·43.

Dem.

*88·431.

*88·44.

*88·441.

*88·45.

Dem.

[Pg 568]

*88·46.

Dem.

*88·47.

Dem.

*88·48.

The proof proceeds as in *88·46.

*88·5.

*88·51.

*88·52.

*88·53.

[Pg 569]

SECTION E.

INDUCTIVE RELATIONS.

Summary of Section E.

The subjects to be treated in this section are certain general ideas of which a particular instance is afforded by mathematical induction. Mathematical induction is, in fact, the application to the number-series of a conception which is applicable to all

relations, and is often very important. The conception in question is that which we shall call the ancestral relation with respect to a given relation. If is the given relation, we denote the corresponding ancestral relation by ""; the name is chosen because, if is the relation of parent and child, will be the relation of ancestor and descendant—where, for convenience of language, we include among his own ancestors if is a parent or a child of anything.

It would commonly be said that has to the relation of ancestor to descendant if there are a certain number of intermediate people , , , ... such that in the series , , , , ... each term has to the next the relation of parent and child. But this is not an adequate definition, because the dots in

represent an unanalysed idea. We may then try to amend this definition by saying that there is a finite class of intermediate terms such that one member () of is a child of , one () is a parent of , every member of except is a child of one (and only one) member of , and every member of except is a parent of one (and only one) member of . This definition is open to several objections. In the first place, it is very complicated; in the second place, there will, in regard to a general relation, be difficulty in securing the uniqueness of the member of which is to be a parent (or a child) of a given member of ; in the third place (and this is the really fatal objection) the proposed definition states that is to be a finite class, and we shall find that finitude, in the relevant sense, is only defined by means of the very conception of the ancestral relation which we are here engaged in defining. In fact, if denotes the relation of to , where is a cardinal number, then a finite cardinal (in the sense we require) is [Pg 570]one to which 0 has the relation , i.e. one of which 0 is an ancestor with respect to the relation

Hence we must not use the notion of finitude in defining the ancestral relation. In fact, the ancestral relation is defined as follows.

Let us call a hereditary class with respect to if , i.e. if successors of 's (with respect to) are 's. Thus, for example, if is the class of persons named Smith, is hereditary with respect to the relation of father to son. If is the Peerage, is hereditary with respect to the relation of father to surviving eldest son. If is numbers greater than 100, is hereditary with respect to the relation of to ; and so on. If now is an ancestor of , and is a hereditary class to which belongs, then also belongs to this class. Conversely, if belongs to every hereditary class to which belongs, then (in the sense in which a is one of his own ancestors if is anybody's parent or child) must be an ancestor of . For to have for one's ancestor is a hereditary property which belongs to , and therefore, by hypothesis, to . Hence is an ancestor of when, and only when, belongs to the field of the relation in question and belongs to every hereditary class to which belongs. This property may be used to define the ancestral relation; i.e. since we have

we put

We then have

Here may be called "the descendants of ." It is the class of terms of which is an ancestor.

To make plain the relation of the above to mathematical induction, put 0 for , and for . Then, since 1=0+1, we have . Again

Thus we find

Thus if is a descendant of 0, belongs to every class to which 0 belongs and to which belongs whenever belongs. Hence mathematical induction, starting from 0, will prove properties of . In elementary mathematics it is customary to speak as if this held of all integers, i.e. as if (as above defined) included all integers; but in fact only finite integers (in[Pg 571] one of the two senses which the word finite may have) belong to the class , and they belong to it by definition, being defined as the class

i.e. as in the above sense. To infinite numbers, inductive proofs of this kind starting from 0 cannot be applied.

The study of will occupy *90. The relation holds between and if or or or etc. The study of this "etc." occupies *91, "on the powers of a relation." We may, for many technical purposes, regard as the 0th power of ; the other powers are , , etc. If is a power of , so is . Now is , according to the definition in *38. Thus if we have

must be a power of , because the class of powers of is a value of which satisfies the hypothesis

Conversely, if is a power of , then is reached by repetitions of the process of turning into , starting this process with . Hence if is a power of , we shall have

Consequently, if we denote the class of powers of by , we have

We might use this as the definition of ; but we can get a somewhat simpler form. For the above is shown, without much difficulty, to be equivalent to

that is, belongs to the ancestry of with respect to , in other words, is reached from by proceeding along the series

which is the same as the series

The relation is important on its own account. We put

and then we put

We often want to include among the powers of ; the class consisting of together with we call . The definition is [Pg 572] whence we easily prove

The relation of being related by some power of (other than) is a very important one. We denote it by , and put

Thus when , we have one of , , , etc. It is easy to prove that

In a series in which every term (except the first, if there is a first) has an immediate predecessor, and every term (except the last, if there is a last) has an immediate successor, if is the relation of a term to its immediate successor, is the relation of any earlier term to any later one.

The next number (*92) concerns itself with some special properties of the powers of one-many, many-one and one-one relations.

The next number (*93) analyses the field of a relation into successive generations; e.g. if the relation is that of parent and child, the first generation will consist of Adam and Eve, the second of their children, the third of their grandchildren, and so on, taking always the longest route from Adam and Eve when there have been intermarriages between generations. That is, taking any relation , the first generation is , the second is , the third is , and so on. Generally, if is a power of (including), the corresponding generation is

In order to express this more conveniently, we introduce a new symbol , which is required also on other grounds, especially in series. "" may be read "minimum with respect to ." We regard "" as " precedes "; then in a class , the "minima of " will be those members of which belong to and are not preceded by any other members of , i.e. . We put therefore

Hence we have

i.e. consists of those members of which are not preceded by any other members of . (If has a single first term, this term is .) Thus we have, when is a power of , [Pg 573] Thus , where is any power of (including) is the generation of corresponding to ; thus the whole class of generations is . Hence we put

where "" stands for "generation."

The notation "" will not be much used until we come to series, but then it will be constantly used. At present, we shall only give such properties of as are necessary for our immediate purposes, but in Part V (on series) we shall devote a number (*205) to its properties.

In this number we also introduce the notation "" for "." "" may be read " begins ." If there is a single beginning of , this is ; otherwise the class of beginnings is , which =. Thus if is the relation of father and son, = Adam; if is the relation of parent and child, = Adam and Eve. will be the end of , if there is one; generally, will be the class of ends, i.e. . The first generation of is . If , any generation of is , where is the corresponding power of .

The field of a relation consists, in general, not only of the generations of , but also of another part, the part in which, however far we go backwards, we never reach a beginning. This part is . The two parts and are mutually exclusive, and together exhaust .

The two next numbers, *94 and *95, are hardly ever relevant in subsequent propositions, and may therefore be omitted by any reader who is not interested in their subject-matter. *94 deals with powers of relative products. It is only used in the following number (*95), on "equi-factor relations." The matter to be dealt with in this number may be explained as follows. In dealing with correlations and similar topics, we often wish to consider the series of relations

Now we have not yet at our command a definition of , where is any finite number; thus we cannot define a general term of this series as . We need therefore a different method of definition. We have

and so on. Thus if is any power of (, a general term of our series is . For convenience of notation, we put [Pg 574] Then our series consists of (. The sum of all relations of this class is considered in this number.

The principal propositions proved in *94 and *95 are two which have the same hypothesis as the Schröder-Bernstein theorem, namely

These two propositions state that, with the above hypothesis,

The two combined reconstitute the Schröder-Bernstein theorem, since

Thus they present, so to speak, an itemized account of the equality proved by the Schröder-Bernstein theorem.

*96, on the posterity of a term, is concerned with the properties of , chiefly when . In this case, in general, consists of two parts, first an open series and then a cyclic series. Either of these may vanish, or may reduce to a single term. If we call the two parts and , the whole of precedes the whole of , and , . Thus if either or vanishes, . If vanishes, the series never returns into itself, that is, . If exists, there is a definite power of , say , such that . If and both exist, there is one term, namely the successor of the last term of , which has just two immediate predecessors, one in and one in ; every other term of has only one immediate predecessor in . Thus is shaped like a , with at the tip of the tail.

[Pg 575]

*97 deals with the analysis of the field of a relation into families. Taking any member of , the family of with respect to is , which we write . Thus the class of families is . Those families which contain a member of are . If we regard as arranged in a rectangle, in which the generations are the successive rows, then

will be the columns. Thus the relation of to may be regarded as a generalized form of the relation of rows and columns. Under a suitable hypothesis, each row is a selection from the columns, and each column a selection from the rows. This is expressed in the following proposition:

whence we derive existence-theorems for selections in the cases concerned.

The importance of the ideas dealt with in the present section is very great. These ideas dominate the treatment of finite and infinite, the theory of progressions and , and the transition from series generated by one-one or many-one relations of consecutive terms to series generated by transitive relations of before and after. Wherever, in short, mathematical induction is used the ideas treated in this section are required. The portions of our subsequent work in which this section is most referred to are the two sections on finite and infinite cardinals and ordinals (Part III, Section C and Part V, Section E). In the general theory of cardinals, i.e. in Part III, Sections A and B, before the distinction of finite and infinite has been introduced, the present section will be seldom if ever referred to[65].

FOOTNOTES:

[65] The present section is based on the work of Frege, who first defined the ancestral relation. See his Begriffsschrift (Halle, 1879), Part III., pp. 55-87. Cf. also his Grundgesetze der Arithmetik, Vol. I. (Jena, 1893), §§ 45, 46 (pp. 59, 60). In this work the ancestral relation is used to prove the properties of finite cardinals and .

[Pg 576]

*90. ON THE ANCESTRAL RELATION.

Summary of *90.

If is any relation, "" is to mean " is an ancestor of with respect to ," where a term counts as its own ancestor provided it belongs to the field of . The definition of is as follows:

90·01.

That is, is to hold when belongs to the field of , and belongs to every hereditary class to which belongs; a hereditary class being a class such that , i.e. such that all successors of 's are 's.

*90·02.

This definition serves merely to decide the ambiguity between and , either of which might be meant of . It will be shown, however, that the two are equal (*90·132).

The most important propositions of this number are the following:

90·112.

I.e. if and if is a hereditary property belonging to , then it belongs to .

*90·12.

I.e. is reflexive throughout the field of , but not elsewhere.

*90·14

*90·15.

*90·151.

*90·16.

*90·163.

I.e. is a hereditary class.

[Pg 577]

*90·17.

*90·21.

*90·22.

I.e. the classes that are hereditary with respect to are the same as those that are hereditary with respect to .

*90·31.

*90·32.

*90·33.

*90·4.

*90·01.

*90·02.

*90·1.

*90·101.

Dem.

*90·102 is a lemma for *90·11.

*90·102.

Dem.

*90·11.

*90·111.

*90·112.

Dem.

[Pg 578]

*90·12.

Dem.

*90·13.

Dem.

The following proposition is a lemma for *90·132.

*90·131.

Dem.

*90·132.

Dem.

[Pg 579]

In accordance with our general convention as regards suffixes, and with the definition *90·02, means , not (.

*90·14.

Dem.

*90·141.

*90·15.

Dem.

Note that may be conveniently regarded as the 0th power of . By *50·64·65, when multiplied by it gives ; also it is contained in , , etc. I has properties, as regards relational multiplication, analogous to those of 1 in ordinary multiplication; thus to regard as the 0th power of is analogous to regarding 1 as the 0th power of , where is a number.

*90·151.

Dem.

*90·16.

Dem.

[Pg 580]

*90·161

Dem.

*90·162

*90·163

This proposition is important, since it proves that is a hereditary class.

*90·164

This proposition shows that is a hereditary class.

*90·17

Note that means (.

Dem.

*90·171

*90·172

Dem.

*90·18

Dem.

[Pg 581]

*90·21.

Dem.

*90·22.

Dem.

*90·23.

*90·23 is useful in the theory of sections of a series (*211). A section of the series generated by is defined as a class satisfying

*90·24.

Dem.

This proposition shows that if is a hereditary class which contains , then contains all the descendants of 's.

*90·25.

Dem.

[Pg 582]

*90·26.

Dem.

*90·27.

Dem.

*90·31.

Dem.

In the last line of the above proof, the process is as follows. Writing for , (2) becomes , while (3) becomes . Hence, by (2) and (3),

Hence, by *90·112, , which is the proposition to be proved.

*90·311.

Dem.

[Pg 583]

*90·32.

Dem.

*90·33.

Dem.

*90·331.

*90·34.

*90·341.

*90·35.

Dem.

[Pg 584]

*90·351.

Dem.

*90·36.

*90·4.

Dem.

*90·41.

Dem.

*90·42.

Dem.

[Pg 585]

*91. ON POWERS OF A RELATION.

Summary of *91.

In the present number, we consider the class of relations

Each of these has to its predecessor the relation ; we have

Thus every term of the series has to the relation (; hence the powers of may be defined as those relations which have to the relation (. The series of powers starting with instead of with is similarly composed of those relations which have to the relation . (This class consists of the previous class together with .) To say that the relation holds between and turns out to be equivalent to saying that one of the relations

holds between and ; and to say that the relation holds between and turns out to be equivalent to saying that one of the relations

holds between and . Thus we might have begun by defining powers of , and proceeded to define as their sum.

For notational convenience we put

Then the definition of powers of excluding is

and the definition of powers of including is

(Here the letters "id" are added to suggest that identity is to be added to .)

We put also

[Pg 586]

Many of the propositions in this number are very often used. Among the more important propositions are the following:

*91·17.

*91·171.

*91·373.

These are formulae of induction. The first two state that if the property is hereditary with respect to , then if belongs to it belongs to any member of , while if belongs to it belongs to any member of . The third gives a form of induction which is sometimes more powerful than the second. It states that if is hereditary provided its argument is a power of , and if , then every power of satisfies , and vice versa.

*91·23.

*91·24.

These two propositions are very useful as giving relations of and .

*91·27.

*91·271.

We do not have in general . If is the sort of relation which generates a series (i.e. is either itself serial, or such that is serial), the above would characterize a series without a first or last term. To illustrate the matter, consider a series of four terms, , , , , and let be the relation of immediately preceding in this series. Thus holds between and , and , and . Then holds between and , and ; thus , which belongs to , does not belong to . holds only between and ; thus neither nor belongs to . All powers of beyond the third are null. On the other hand, if we take a cyclic relation, such as that of left-hand neighbour at a dinner-table, we shall always have , whatever power of may be.

*91·282.

This proposition shows that is a hereditary class with respect to .

*91·34.

This proposition states that the relative product is commutative when each factor is or a power of .

We come next to propositions concerning . We have

*91·502.

[Pg 587]

*91·504.

*91·511.

*91·52.

*91·54.

*91·52·54 are fundamental in the theory of inductive relations.

*91·542.

This proposition is particularly useful when (as often happens) we have . In that case, it gives .

*91·55.

*91·56.

Thus is always transitive, which is one of the three characteristics of serial relations (cf. *204). We shall find that is often serial when is not so.

*91·574.

*91·602.

*91·01.

*91·02.

*91·03.

*91·04.

*91·05.

The first two of the above definitions are introduced merely for notational convenience. The other three represent ideas of great importance. The last is especially useful when a series is given as the field of a one-one relation between consecutive terms—as, e.g., when the series of natural numbers is given as the field of the relation of to . Then is the relation of any earlier term to any later term—e.g., in the above case of the natural numbers, the relation of a less integer to a greater.

*91·1.

Dem.

[Pg 588]

*91·11.

*91·12.

*91·13.

*91·14.

*91·15.

*91·16.

*91·17.

*91·171.

These propositions are of great importance, because they enable us to prove that a property belongs to every power of if it belongs to (or) and also belongs to whenever it belongs to .

*91·2.

Dem.

*91·201.

*91·204.

Dem.

*91·205.

*91·21.

Dem.

[Pg 589]

*91·211

*91·212

Dem.

*91·213

*91·22

*91·221

*91·23

Dem.

*91·231

*91·24

Dem.

*91·241

Dem.

The last line of the above proof is obtained as follows: writing for (1) becomes

But by *91·11, writing for the of *91·11, and for the , we have

Hence, by (1) and (2), , i.e.

which is the proposition to be proved.

[Pg 590]

*91·242.

Dem.

*91·25.

Dem.

*91·251.

*91·26.

*91·261.

*91·262.

*91·263.

*91·264.

*91·27.

Dem.

[Pg 591]

*91·271.

Dem.

*91·28.

*91·281.

*91·282.

*91·283.

The following propositions show that the relative product of two powers of is commutative, i.e. (cf. *91·34)

We also have (cf. *91·341)

It is these propositions (as will appear in the sequel) which are the source of the commutative law for the addition of finite ordinals. Ordinals in general are not commutative, just as relative products in general are not commutative; but owing to the fact that relative products whose factors are powers of a given relation are commutative, finite ordinals are commutative.

*91·3.

Dem.

*91·301.

*91·302.

Dem.

[Pg 592]

*91·303.

*91·304.

*91·31.

*91·33.

Dem.

*91·331.

Dem.

*91·34.

Dem.

This is the commutative law for the relative product of two powers of .

*91·341.

Dem.

[Pg 593]

*91·342.

Dem.

*91·343.

*91·35.

*91·351.

*91·352.

*91·36.

*91·37.

Dem.

*91·371.

*91·372.

*91·373.

*91·41.

*91·411.

*91·42.

*91·421.

*91·43.

Dem.

[Pg 594]

*91·431.

*91·44.

Dem.

*91·45.

Dem.

*91·46.

The remainder of this number is concerned with and its relations to .

*91·502.

*91·503.

*91·504.

Dem.

The following propositions are concerned mainly with the relations of and . These relations are embodied in the propositions

[Pg 595]

*91·51.

Dem.

*91·511.

*91·512.

Dem.

*91·513.

Dem.

*91·514.

Dem.

[Pg 596]

*91·52.

*91·521.

Dem.

*91·522.

*91·53.

Dem.

*91·54.

*91·541.

*91·542.

*91·543.

Dem.

*91·544.

*91·545.

[Pg 597]

*91·546.

*91·55.

Dem.

*91·56.

Dem.

*91·561.

*91·562.

*91·57.

*91·571.

*91·572.

*91·573.

*91·574.

Dem.

*91·575.

Dem.

*91·58.

*91·581.

*91·59.

Dem.

[Pg 598]

*91·6.

Dem.

*91·601.

Dem.

*91·602.

Dem.

*91·603.

Dem.

*91·62.

[Pg 599]

This formula should be compared with *90·11, in which an analogous formula is given for . It will be observed that here we do not require to add , for if , the above formula leads to , i.e. to . Hence , i.e. . It will be observed that holds whenever belongs to every hereditary class which contains the immediate successors of , whereas holds whenever belongs to every hereditary class to which itself belongs.

*91·7.

*91·71.

Dem.

*91·711.

Dem.

The above proposition is used in the theory of minimum points in a series (*205·68).

*91·72.

Dem.

*91·721.

*91·73.

Dem.

*91·731.

By means of *91·73 or *91·731, the powers of can often be arranged in a series, the rule of arrangement being that comes earlier than if[Pg 600] , and later in the converse case. But we shall only get an open series from this arrangement if ; otherwise the powers from a certain point onwards form a cyclic series.

*91·732.

Dem.

*91·74.

*91·75.

Dem.

[Pg 601]

*92. POWERS OF ONE-MANY AND MANY-ONE RELATIONS.

Summary of *92.

If , it follows that, starting from a given term , there is only one series of terms such that

Thus for example the relation of son to father is a ; and starting from a given man, the series of ancestors in the direct male line (which is the above series , , ,...) is unique and determinate. A result of this property of many-one relations is that if, starting from a term , we go backwards a certain number of steps to a term , and then forward a greater number of steps to a term , we must pass through in going from to ; while if the number of steps from to is less than that from to , must lie on the road from to . These facts are expressed by the proposition:

In the present number, we have to establish various propositions of this kind.

We prove in this number various propositions which are used in the discussion of "families" in *96 and *97, and some which are used in the theory of finite and infinite. But on the whole the propositions of this number are not much used. The most important of them are the following:

*92·11.

with a similar proposition (*92·111) for .

*92·132.

with a similar proposition (*92·133) for .

*92·14.

On this proposition, compare the remarks on *91·271 in the introduction to *91. If is a serial relation, is the condition that the series may have no last term.

*92·31.

*92·311.

[Pg 602]

*92·1.

Dem.

*92·101.

*92·102.

*92·11.

Dem.

*92·111.

*92·112.

*92·113.

*92·12.

*92·121.

*92·13.

Dem.

*92·131.

In this number, when proofs have been given for , we shall omit the proofs of corresponding propositions for , as these are always exactly analogous to the proofs for .

*92·132.

*92·133.

[Pg 603]

*92·14.

Dem.

*92·141.

*92·142.

Dem.

*92·143.

*92·144.

Dem.

*92·145.

*92·146.

Dem.

*92·147.

*92·15.

[Pg 604]

*92·151.

*92·152.

*92·153.

*92·16.

Dem.

*92·161.

*92·17.

Dem.

*92·171.

*92·18.

Dem.

*92·181.

[Pg 605]

*92·19.

Dem.

*92·191.

*92·3.

Dem.

*92·301.

*92·31.

Dem.

*92·311.

[Pg 606]

*92·312.

*92·32.

Dem.

*92·33.

Dem.

*92·34.

Dem.

[Pg 607]

*93. INDUCTIVE ANALYSIS OF THE FIELD OF A RELATION.

Summary of *93.

For this number, we introduce three new notations, of which the first two will be used constantly, especially in the theory of series, while the third will be seldom used except in the present section. The two which are constantly used are

i.e. is a member of and of , and no member of precedes in .

The letter may be regarded as standing for "begins." Thus if we take any member of , and proceed backwards and forwards as far as possible by -steps, we obtain a series which may be called the "family" of : this series, if it has a first term, has one which is a member of ; thus the members of are the beginners of families. For example, if is the relation of a peer to his heir, "" will mean " is a peer who is not the heir of a peer"; thus is the first of his family. If is the relation of parent and child, "" will be satisfied only by Adam and Eve; and so for other relations.

The definition of is

Hence . If is the generating relation of a series which has a first term, that first term is ; if there is a last term it is .

If is any class, we may call a term a minimum of with respect to if it is a member of and of , but does not follow any member of , i.e. is not a member of . We denote this relation of to by ""; thus we have

and the definition of is

[Pg 608]

We shall also, when convenient, write "" in place of "."

We have .

If is serial, reduces to a single term if it is not null; thus if a class has a first term, this term is . We also put

and then , if it exists, is the last term of in the -series. Thus if is the class of peers, and is the relation of father to son, consists of those peers who are the first of their line, while consists of those peers who are the last of their line. If is a class of numbers, and is the relation of less to greater, is the smallest member of (if it exists), and is the largest (if it exists).

and "" and "" will be used constantly in connection with series, where the two latter will be considered in detail, but the present number is more specially concerned with a less general idea, namely that of generations. Take, e.g., the relation of parent and child; let us call it . Then the first generation consists of those who are parents but not children, i.e. ; the second consists of those who are children but not grandchildren, i.e. , i.e. , i.e. ; the third consists of those who are grandchildren but not great-grandchildren, i.e. , i.e. , i.e. ; and so on. Also we have

hence the generations of are . Thus we put

where "" stands for "generation."

When is a one-many relation, such as that of father and son, every generation is of the form , where is a power of (including). When is not a one-many relation, this is not in general the case.

The generations of do not in general exhaust the field of . For will only belong to a generation of if can be reached by successive -steps starting from a member of . If some of the families constituting the field of have no beginning, the members of these families will not belong to any generation of . Such terms together constitute the class

which is the same class.

[Pg 609]

Thus the field of may be divided into two mutually exclusive portions, and .

The present number begins with some elementary properties of and and . We then (*93·2—·275) consider such properties of generations as do not demand any hypothesis as to . We prove

*93·25.

*93·261.

and we prove (*93·274·275) that and are mutually exclusive, and together constitute . We then proceed to a set of propositions (*93·3—·41) demanding that P should be one-many or many-one or one-one. We prove

*93·32.

*93·36.

*93·381.

and various other properties of and when .

The propositions of this number are used throughout the rest of this section; they are also used in the cardinal theory of finite and infinite. The early propositions, down to *93·12 inclusive, are also used in the theory of series.

*93·01.

*93·02.

*93·021.

*93·03.

*93·1.

*93·101.

*93·102.

*93·103.

Dem.

[Pg 610]

*93·104.

Dem.

*93·11.

*93·111.

*93·112.

Dem.

*93·113.

*93·114.

*93·115.

*93·116.

*93·117.

*93·118.

*92·12.

*93·13.

*93·131.

Dem.

[Pg 611]

*93·132.

Dem.

*92·2.

*93·21.

*93·22.

*93·221.

*93·23.

Dem.

*93·231.

Dem.

*93·24.

Dem.

[Pg 612]

*93·25.

Dem.

*93·26.

Dem.

*93·261.

Dem.

*93·27.

[Pg 613]

Dem.

*93·271.

Dem.

*93·272.

Dem.

*93·273.

*93·274.

*93·275.

*93·3.

Dem.

*93·31.

Dem.

[Pg 614]

*93·32.

*93·33.

*93·34.

*93·35.

Dem.

*93·36.

Dem.

*93·37.

*93·38.

Dem.

[Pg 615]

*93·381.

*93·382.

*93·4.

Dem.

*93·41.

*93·412.

Dem.

*93·42.

Dem.

*93·431.

Dem.

[Pg 616]

The following propositions, not being needed in subsequent propositions, are here inserted without proof, merely for the sake of their intrinsic interest.

*93·5.

*93·51.

*93·52.

*93·53.

*93·54.

*93·55.

*93·56.

[Pg 617]

*94. ON POWERS OF RELATIVE PRODUCTS.

Summary of *94.

In this number we shall be chiefly concerned with propositions connecting powers of with powers of . If is a power of , will be a power of . If is a power of , it is a product of the form

If we transfer the initial to the end, we get a power of . Thus there is a power of , say , such that

If , we find

by rearranging and observing that . Thus

Expressions of the form are constantly needed. They will be specially dealt with in *150, and will occur constantly in the sequel.

The above connections of and are embodied in the following propositions:

*94·14.

*94·21.

*94·31.

From *94·4 to *94·54, the propositions are all concerned with and . We prove

*94·5.

*94·51.

[Pg 618]

Finally we prove (*94·53·54) that if either is one-one and , or is one-one and , then is similar to .

The only proposition of this number which is ever subsequently referred to is the last, *94·64, which, owing to the fact that the Schröder-Bernstein theorem has been already proved (*73·88), is only used in *95·23. But *95·23 itself is never referred to again. The reader may therefore omit the reading of the propositions of this number (as also of *95) without detriment to the understanding of what follows; he should, however, read the summaries.

The chief importance of the propositions in the present number is when and fulfil the hypothesis of the Schröder-Bernstein theorem, i.e.

In this case, gives what we may call a "reflexion" of into part of itself; this part may be again reflected by into a part of itself, and so on. The terms in which are eliminated sooner or later by this process of reflexion constitute , since any one reflexion eliminates terms which constitute one generation of . The terms not

eliminated by any number of reflexions constitute . These two sets of terms together constitute , i.e. . In this number and *95 we shall prove that, with the Schröder-Bernstein hypothesis,

These two propositions together yield a proof of the Schröder-Bernstein theorem, in virtue of *93·274·275. This proof is essentially the same as Bernstein‘s published originally by Borel[66].

The nature of the two proofs of the Schröder-Bernstein theorem, namely Zermelo's (that given in *73) and Bernstein's (that to be given in this number and *95) will be best apprehended by means of figures.

In Zermelo's proof, we first prove that if is one-one, and is a class contained in and containing , then is similar both to and to . In the figure, the points of the outer rectangle form , those of the[Pg 619] inner rectangle form , and those of the outer oval form . Thus the shaded portion of the figure is . We now define a class of classes by the following characteristics: is a member of if (1) is contained in , (2) contains the whole of the shaded area, (3) , i.e. if is a member of , so is any term to which has the relation . Our proposition is obtained by considering , i.e. the area common to all the members of . We prove (*73·81) that and (*73·811) that does not contain any of the shaded area. In the figure, is the smaller oval. We then prove (*73·83) that consists entirely of the shaded portion and the smaller oval. Hence (the larger oval) consists of two mutually exclusive parts, namely and , the latter being that part of the inner rectangle which lies outside the inner oval. Assuming now that is one-one, is similar to ; hence, adding , it follows that is similar to , and therefore to .

In order to obtain hence the Schröder-Bernstein theorem, it is only necessary to replace by and by , and to assume further that is a one-one whose domain contains . Then , and we obtain (*73·87) , and therefore , which was to be proved.

In Bernstein's proof, we have the two relations and from the beginning. In the left-hand part of the figure, the outer rectangle is , which = , the oval is , and the second rectangle is . Thus the points of the outer but not the second rectangle form the first generation of . Within we can form a third rectangle, which will be , i.e. . The points belonging to the second rectangle but not to the third form the second generation of . We can proceed in this way to continually smaller rectangles. The points which sooner or later are left outside some rectangle form ; those which are common to all the rectangles form . A similar analysis,[Pg 620] exhibited in the right-hand part of the figure, may be applied to , which is thus divided into and . We prove in this number (*94·53) that, with a hypothesis which is part of the hypothesis of the Schröder-Bernstein theorem, ; in the next number (*95·71) we prove that with the hypothesis of the Schröder-Bernstein theorem, . Hence by addition, .

*94·12.

Dem.

*94·13.

*94·14.

Dem.

*94·2.

Dem.

[Pg 621]

*94·201.

Dem.

*94·21.

Dem.

*94·22.

Dem.

[Pg 622]

*94·3.

Dem.

*94·31.

The following series of propositions lead up to the proof that when , or , we have

*94·4.

Dem.

[Pg 623]

*94·401.

Dem.

*94·402.

Dem.

*94·41.

Dem.

*94·42.

Dem.

*94·43.

*94·441.

*94·442.

*94·5.

Dem.

[Pg 624]

*94·51.

*94·52.

*94·53.

Dem.

*94·54.

*94·6.

Dem.

*94·61.

Dem.

*94·62.

Dem.

[Pg 625]

*94·63.

Dem.

*94·64.

Dem.

FOOTNOTES:

[66] Leçons sur la théorie des fonctions (Paris, 1898), Note I (pp. 102-7).

[Pg 626]

*95. ON THE EQUI-FACTOR RELATION.

Summary of *95.

The purpose of this number may be explained as follows. Consider the series of relations

it is required to find a means of defining this series without the use of numbers. If we used numbers, and had the definition given later (*301) of where is any finite integer, the general term of the series would be . But we have not yet defined numbers, and we therefore desire some means, not involving numbers, of expressing what is intended when we say that, in a given term of the series, the same power of and of is to be involved. This we do as follows. Using the definition of in *43, we have

Thus the general term of our series is got by taking any power of (, and forming . The whole of the terms of the series are therefore constituted by the terms which have to the relation (; i.e. they are . For convenience of notation we put[67]

Thus the class of relations we wish to consider is (.

To illustrate the nature of (, suppose is the relation "first cousin," while is the relation of child to parent and is the relation of parent to child. Then is the relation "second cousin," is the relation "third cousin," and so on. Thus (is the class of all relations of cousinship which do not involve a difference of generation; and "" will mean " is a cousin of in the same generation."

Most of the propositions in this number are inserted because they are required in the proof of *95·52, which states that, under suitable circumstances, . This proposition itself is proved mainly because it is required in the proof of *95·63, which states that, if , are one-one's each of which has its converse domain contained in its domain, and if the[Pg 627] first generation of is similar to the first generation of , then the sum of the generations of is similar to the sum of the generations of . This leads immediately to a proposition (*95·71) which is half of the Schröder-Bernstein theorem (the other half being *94·53 or *94·54), namely: "If and are one-one's each of which has its converse domain contained in the domain of the other, then the sum of the generations of is similar to the sum of the generations of ."

*95·01. (

*95·1.

Dem.

*95·11.

Dem.

*95·12.

Dem.

*95·13.

*95·131.

Dem.

[Pg 628]

*95·132.

*95·14.

Dem.

The use of *95·11 in the last line of the above proof proceeds by substituting for .

*95·21.

Dem.

*95·211.

Dem.

*95·212.

*95·22.

[Pg 629]

*95·221.

Dem.

*95·222.

*95·23.

Dem.

*95·24.

*95·3.

Dem.

*95·301.

*95·302.

Dem.

*95·303.

*95·304.

*95·305.

[Pg 630]

*95·31.

Dem.

*95·32.

*95·33.

Dem.

*95·34.

*95·35.

*95·351.

Dem.

*95·352.

*95·36.

Dem.

[Pg 631]

*95·361.

*95·37.

*95·38.

Dem.

*95·381.

*95·382.

Dem.

*95·383.

*95·4.

Dem.

[Pg 632]

*95·41.

*95·411.

*95·42.

Dem.

*95·43.

Dem.

*95·431.

Dem.

*95·44.

[Pg 633]

Dem.

*95·45.

Dem.

*95·46.

Dem.

*95·47.

Dem.

[Pg 634]

*95·471.

*95·51.

Dem.

*95·511.

*95·52.

Dem.

*95·6.

[Pg 635]

Dem.

*95·601.

*95·61.

*95·62.

*95·63.

Dem.

*95·64.

*95·65.

The following example may illustrate the scope of *95·65. Let , be the generating relations of two well-ordered series, neither of which has a last term. Put . Then is the relation of immediately preceding in the -series, and is the relation of immediately preceding in the -series. We shall have [Pg 636] Also, except in certain exceptional cases, , are the first derivatives of the two series (including the first terms of the two series).

states that, starting from any term of the series and going backwards, a finite number of steps will bring us to a member of the first derivative, which is true. Hence, by *95·65, neglecting certain exceptional cases, we arrive at the result that if the first derivatives of two well-ordered series have the same cardinal number of terms, then the series themselves have the same cardinal number of terms. This proposition can of course be proved otherwise; the above is merely mentioned as an illustration of the results of *95·65.

*95·7.

Dem.

*95·71.

Dem.

This proposition and *94·53 or *94·54 together reconstitute the Schröder-Bernstein theorem (*73·88). For, in virtue of *93·274·275 and *73·71, they together give

and with this hypothesis

FOOTNOTES:

[67] This notation is used in the present number only. In *257, we shall introduce a different and wholly unconnected meaning for (. A temporary definition is indicated by the letters "" followed by a reference in square brackets to the number or numbers in which the definition is used.

[Pg 637]

*96. ON THE POSTERITY OF A TERM.

Summary of *96.

By the "posterity" of a term with respect to a relation we mean the class . In the present number, we shall be chiefly concerned with the relation (, i.e. the relation confined to the posterity of . We shall also be concerned with (and (, which, as is proved in *96·13, are respectively

The most interesting case is when . In this case, is in general shaped like a , with at the tip of the tail; that is, may be divided into two parts, the first an open series, the second a closed series. If is the junction of the two, we shall have

We have also, when ,

It thus appears that is divided into two parts, the first consisting of those terms for which , the second of those for which . The first wholly precedes the second; the first exists if , the second if . Every term in has one and only one immediate predecessor, except the term (if it exists) at the junction of the tail and circle of the ; this term has just two immediate predecessors, one in the tail and one in the circle. But if either the tail or the circle is null, then every term in has only one immediate predecessor, and therefore [Pg 638]

(these definitions being only to apply within *96). Then is the open part of the series , and is the circular part. The open part wholly precedes the circular part, provided ; i.e.

If and both exist, has a last term, say . The successor of this term, , is the only term in which has two immediate predecessors in , namely and .

The most important applications of the propositions of the present number are in the theory of finite and infinite, both cardinal and ordinal. When is many-one, then if exists, or, more generally, if has a last term, is a finite class, i.e. what we shall call a "" (cf. *120). That is, we have

If exists, but has no last term, is a progression (cf. *122) when its terms are arranged in the order generated by . That is, giving to and the meanings given by Cantor (cf. *123 and *263), and using "" for the class of one-one relations which generate progressions, we have

Another very important proposition in the proof of which the present number is useful is *121·47, which proves that if is either one-many or many-one, and and are any two terms whatever, then (which we call the "interval" from to) is always a finite class. The proof that progressions are well-ordered series depends upon the propositions of this number, since it uses *122·23, which depends upon *96·52.

The present number begins with a series of propositions (ending with *96·16) on and , both in general and when . We then proceed to a few propositions (*96·2-·25) on (when ; with the exception of *96·24, these propositions are all used in the cardinal theory of finite and infinite. They are, however, less important than the subsequent

propositions, which are concerned with [Pg 639] when .

If is a many-one relation, and x is a member of , the relation in general arranges (i.e. the posterity of) in a figure such as is here given. The relation holds between each dot and the next, starting from , and travelling round the circle in the sense indicated by the arrow. The dots from to constitute , and the dots in the circle constitute . is the last term of , i.e. ; is , and is , or, what comes to the same thing, . is the only term which has more than one immediate predecessor in ; always exists if neither nor is null, and conversely, if exists, neither nor is null. The proof of these propositions is long; the following are useful stages in the proof.

If , the whole posterity of is itself (*96·33); if and , and constitute the whole posterity of (*96·331), and so on. The successors of members of belong to (*96·341), and the predecessors of members of , if they belong to , belong to (*96·351). (It should be observed that, since is only assumed to be many-one, not one-one, every member of may have any number of predecessors which do not belong to . We have a series of propositions, beginning with *96·4, which deal with the hypothesis . We prove (*96·42) that if and , then , i.e. belongs to . We prove (*96·431) that wholly precedes ; that (and (are both one-one (*96·45), so that if , one of and must belong to and the other to (*96·441). Hence it follows (*96·453) that if either (in which case or ((in which case , then (is a one-one relation. (This proposition is used twice in the cardinal theory of finite and infinite, namely in *121·43 and *122·17.) Hence we arrive at the proposition (*96·47) that if two different members and of both immediately precede a term , then one of and (say) is the last term of , is its immediate successor and is the immediate predecessor of in , i.e. we have

Thus , , are unique if they exist. We prove next (*96·475) that , , exist when, and only when, neither nor is null.

It follows from the above propositions that if is one-one, either or must be null (*96·491), i.e. the posterity of a term is either an open series or a cycle, and cannot have the -shape.

[Pg 640]

*96·01.

*96·02.

*96·1.

*96·101.

*96·102.

*96·103.

Dem.

*96·104.

Dem.

*96·11.

Dem.

*96·111.

Dem.

[Pg 641]

*96·112.

Dem.

*96·121.

*96·122.

*96·13.

*96·131.

*96·14.

*96·141.

Dem.

*96·142.

*96·143.

Dem.

*96·144.

Dem.

[Pg 642]

*96·15.

Dem.

*96·151.

Dem.

*96·152.

*96·153.

*96·154.

*96·155.

Dem.

*96·156.

Dem.

*96·157.

*96·158.

Dem.

[Pg 643]

*96·159.

*96·16.

Dem.

*96·2.

Dem.

*96·21.

Dem.

*96·22.

Dem.

*96·23.

Dem.

[Pg 644]

*96·24.

Dem.

*96·25.

Dem.

The following propositions lead up to *96·32, i.e.

which is a proposition used in the following number (*97).

*96·3·301·302·303 are also frequently used elsewhere.

*96·3.

*96·301.

*96·302.

*96·303.

*95·31.

*96·311.

*96·32.

Dem.

[Pg 645]

*96·33.

Dem.

*96·331.

Dem.

This process of proof can obviously be extended to any finite cycle of terms.

*96·34.

Dem.

*96·341.

Dem.

[Pg 646]

*96·342.

*96·35.

*96·351.

Dem.

*96·352.

The following propositions are lemmas for *96·45·47.

*96·4.

Dem.

*96·401.

Dem.

*96·402.

Dem.

*96·403.

Dem.

[Pg 647]

*96·41.

Dem.

*96·42.

Dem.

*96·421.

Dem.

*96·431.

Dem.

*96·432.

Dem.

[Pg 648]

*96·44.

*96·441.

Dem.

*96·442.

The following proposition (*96·45) is important.

*96·45.

*96·451.

*96·452.

Dem.

*96·453.

Dem.

*96·46.

[Pg 649]

Dem.

*96·461.

Dem.

*96·462.

Dem.

[Pg 650]

The above proposition, since it exhibits , , as functions of and , shows that there is at most one in having more than one immediate predecessor, and that this one has exactly one immediate predecessor in and one in . (These results require *96·441, in addition to *96·462.) Thus we arrive at the following proposition:

*96·47.

We still have to prove

or, what comes to the same thing because of *96·441,

This is effected in the following propositions.

*96·472.

Dem.

*96·473.

*96·474.

Dem.

[Pg 651]

*96·475.

This proposition and *96·45·47 embody the main results of this number.

*96·48.

Dem.

In the above proposition we write "" rather than "," because the latter implies the existence of .

*96·49.

Dem.

*96·491.

Dem.

[Pg 652]

*96·492.

Dem.

The above proposition is used in *122·52.

The following propositions, not being needed in the sequel, are merely stated:

*96·5.

Dem.

*96·501.

*96·502.

Dem.

*96·51.

[Pg 653]

Dem.

*96·52.

Dem.

The proposition is used in *122·23.

[Pg 654]

*97. ANALYSIS OF THE FIELD OF A RELATION INTO FAMILIES.

Summary of *97.

In this number, we consider not only the posterity of a term, but the ancestry and posterity together, i.e. . We put

Thus the whole family of a term, i.e. its ancestry and posterity together, is . The most important case here is when ; in this case families are mutually exclusive, i.e. we have

In case and belongs to a family which has a beginning, i.e. in case , the whole family of consists of the posterity of the beginning, i.e. we have

whence

*97·21.

When , the relation of to may be pictured as the relation of rows to columns. E.g. let the field of consist of the dots in the accompanying rectangle, and let each dot have the relation to the dot below it. Then the top row is , the second [Pg 655]row is , the third is , and so on; thus the rows are the generations of . Again, if is any dot in the top row, the column beginning with is , and if is any member of this column, the column is . Thus the columns are the families of . It will be seen that in the case represented by the above figure, every family consists of a selection from the generations, and every generation consists of a selection from the families, i.e.

The circumstances under which this occurs will be considered in the present number (*97·3—·47). The results are summed up in *97·47.

The remaining propositions (*97·5—·58) are concerned with circular families of one-one relations. If , is a circular family if . In that case, we have ; moreover there is a definite power of , say , such that every member of the family of has the relation to itself (*97·54). (The same will hold, of course, of all powers of .) The families of a are all either circular or open, i.e. we have (*97·55) either , or . The -shaped families considered in *96 are not possible for a , since in such families the term at the junction of the tail and the circle has two predecessors. The family of any member of must be open (*97·57). The family of a member of need not be closed, but cannot have a beginning; if open, it forms a series of type or , according as it has or has not an end[68]. Finite open families are contained in ; families of type are contained in ; those of type , in ; those of type and circular families are contained in . Those of type are distinguished from circular families by the fact that in the former we do not have , while in the latter we do have this.

In addition to the propositions already mentioned, the most useful propositions of the present number are the following:

*97·13.

*97·17.

*97·5.

*97·501.

[Pg 656]

*97·01.

Observe that "" means that is to be included if it is a member of , but not otherwise; for if , and otherwise .

*97·1.

Dem.

*97·101.

Dem.

*97·11.

Dem.

*97·111.

\text{D}em.

*97·12.

Dem.

[Pg 657]

*97·13.

Note. is to mean (, not (. The latter is unmeaning, since is never a homogeneous relation, and therefore its square and higher powers are unmeaning.

Dem.

*97·14.

*97·15.

Dem.

*97·16.

Dem.

*97·17.

Dem.

*97·18.

[Pg 658]

Dem.

*97·2.

Dem.

*97·21.

Dem.

*97·22.

*97·23.

[Pg 659]

Dem.

*97·231.

Dem.

*97·24.

Dem.

*97·241.

Dem.

*97·242.

[Pg 660]

The remaining propositions of this number (except *97·5 ff.) are concerned with proving that, under certain hypotheses,

These propositions have the merit of proving the existence of selections in the cases to which they apply.

*97·3.

Dem.

*97·301.

Dem.

*97·31.

Dem.

[Pg 661]

*97·32.

*97·33.

Dem.

*97·34.

Dem.

*97·341.

*97·35.

[Pg 662]

Dem.

*97·36.

*97·37.

Dem.

*97·38.

Dem.

*97·4.

Dem.

*97·401.

Dem.

*97·402.

Dem.

[Pg 663]

*97·403.

Dem.

*97·41.

*97·42.

Dem.

*97·43.

Dem.

[Pg 664]

*97·44.

Dem.

*97·45.

Dem.

*97·46.

Dem.

*97·47.

[Pg 665]

Dem.

*97·5.

Dem.

*97·501.

*97·51.

*97·52.

[Pg 666]

*97·53.

*97·54.

*97·55.

Dem.

*97·56.

*97·57.

*97·58.

Dem.

It follows from this proposition that every family is either wholly contained in the generations of or wholly contained in , which may be called the residue of the field of .

FOOTNOTES:

[68] Here the type "" is the type of converses of relations of type , i.e. the type of the negative integers in order of magnitude, ending with , being the type of the positive integers in order of magnitude, and therefore being the type of negative and positive integers in order of magnitude.

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TRANSCRIBER’S NOTES

All items in the Errata, from all three volumes, have been added and corrected accordingly.

The author's notation as ‘*2·37·38’ is an abbreviation for *2·37 and *2·38. For this reason, these numbers were not considered in the cross-references.

The lemma *84.44 cited on page 326 was not described by the authors in the corresponding section.